

Triangle Centers

Inspired by Physics

A survey, computational atlas, and reproducibility companion

Research reference manuscript

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Abstract. Physical laws and measurement protocols select natural points of a triangle through equilibria, maxima, probability means and singular limits. We organize the investigated quantum, conformal, electrostatic, magnetic, thermal, fluid, mechanical, reaction and optical constructions, including parameter families and exact classical reductions. Accessible derivations are followed by numerical methods, individual contour atlases, ETC location comparisons, four shape fingerprints and vertex-motion attractivity evidence. We distinguish analytic certificates, validated finite computations, refined numerical observations and unresolved selections. A companion archive supplies source, datasets, figures, checksums, certificate verification and demonstrations. Some constructions lack convenient closed forms; no blanket transcendence or ETC nonmembership assertion is made.

Prepared from the project's supplied draft and computational studies.
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Reader's guide

Begin with the physical catalogue in Chapter 3 for familiar examples. Chapters 4–9 develop the investigated families and contain individual result sheets. A result sheet states its selector, algorithm, location certificate and attractivity status, then gives its own field atlas and ETC-comparison fingerprints. The final chapters collect cross-family conclusions, remaining constructions and the reproduction guide.

The 66 numerical entries are parameter instances, not 66 claimed distinct new centers. The original 58 comprise six quantum selectors, fourteen electromagnetic/heat entries, thirty-four mechanical/reaction/optical variants, and four supplementary illumination entries. The September 30 extension adds M2–M7 and L3–L4, and updates the electrostatic and magnetic attractivity evidence. S2 is an explicit alias of E2; E4, S3 and S4 reduce to classical centers under the adopted limiting convention. Additional families and conditional proposals are catalogued separately. The count is an organizational aid, not a mathematical invariant.

The manuscript is self-contained for definitions, derivations and interpretation; large coefficient arrays, individual sample records and executable verifiers reside in the companion bundle. The supplied third-party ETC database and source PDFs are identified but are not redistributed. No submission to arXiv or other service has been made.

Chapter 1

From physical experiments to triangle centers

A triangle is a remarkably effective laboratory. It is simple enough that its shape is described by two numbers, but sufficiently asymmetric that the balance point of one physical process need not agree with that of another. A wire frame balances at one point; the filled lamina balances at another. A freely diffusing particle can be asked where it survives longest, where its mean lifetime is largest, or where its three possible exit sides are equally likely. These questions refer to the same domain and produce different centers.

This survey assembles the physical definitions and the computational investigations carried out for this project. It is a reference manuscript with a reproducibility archive, rather than a claim that every proposed construction is already a theorem. The original reference draft [2] is the starting inventory. Subsequent calculations correct several definitions and numerical conclusions. In particular, an interior Fresnel zero is impossible in the stated model; individual excited quantum states require a degeneracy convention; a thin conductor requires its ambient dimension and thickness convention; and a mechanically stable equilibrium can fail vertex-motion attractivity.

1.1 What is required of a center?

Write $T = \text{conv}\{V_1, V_2, V_3\}$, with noncollinear vertices. Set $a_i = |V_{i+1} - V_{i+2}|$, with cyclic indices, and

$$P = a + b + c, \quad A_T = \text{area}(T), \quad L^2 = (a^2 + b^2 + c^2)/3, \quad D = \text{diam}(T).$$

The length L fixes dimensionless physical parameters; D normalizes errors. They must not be interchanged silently.

A shape-defined point $C(T)$ must commute with translations, rotations, reflections and positive scalings, and must not depend on how the vertices are numbered. An externally fixed magnetic direction, a material anisotropy axis, a dimensional time or an arbitrary choice among two equal maxima can violate these requirements. Such additional information defines an augmented experiment, not automatically an intrinsic triangle center.

The normalized barycentric coordinates w_i of a point satisfy

$$C = \sum_{i=1}^3 w_i V_i, \quad \sum_i w_i = 1.$$

Each w_i is an affine function of the observation point. For an interior point, $w_i > 0$. A geometric center therefore has implicit barycentric functions even when no convenient formula in a, b, c is

available. Lack of a known algebraic formula is not a proof of transcendence, and is not by itself an exclusion rule for the ETC [1]. We use “implicit” or “PDE-defined” when that is what has actually been established. The magnetic center, for example, is implicitly algebraic despite its inconvenient explicit representation.

The catalogue distinguishes four selectors. A *mode* maximizes a field. A *moment* is the barycenter of a positive measure. An *equilibrium point* solves force or potential equations. A *limit center* is obtained from a specified limiting family. The uniqueness of a field does not prove the uniqueness of its maximum. An eigenspace may be unique while an eigenvector inside it is not. These distinctions are part of each definition, rather than technical afterthoughts.

1.2 Evidence vocabulary

Exact identity or analytic proposition. An equation follows from the stated model, with a derivation here or a cited theorem whose assumptions apply.

Validated computation. An interval enclosure, together with a mathematical existence/uniqueness argument, certifies a finite assertion. The M1 magnetic and E1/E2 conformal counterexamples have this status.

Refined numerical evidence. A calculation survives the stated mesh, quadrature or precision changes. This is not an enclosure of the exact continuum solution.

Screen. A finite set of shapes and perturbations has been tested. Passing a screen never establishes a global sign condition.

Conditional construction. A physical model or selection rule still needs a uniqueness, branch or constitutive convention. It is retained in the survey without being promoted to an established center.

Digits in machine-readable files record the computation, not the number of correct continuum digits. The individual contour atlases use a moderate common discretization for clarity; rankings use the archived refined study data. The two purposes are deliberately identified in their captions.

1.3 Positive moments: the basic location certificate

Proposition 1.1. *If μ is a positive probability measure supported on T , its barycenter belongs to T . If its support is not contained in any one supporting side, the barycenter is strictly interior.*

Proof. Let $\ell_i(p)$ be the affine barycentric coordinate opposite side i . Then $\ell_i \geq 0$ on T and

$$\ell_i \left(\int_T p \, d\mu \right) = \int_T \ell_i(p) \, d\mu(p).$$

The right side is nonnegative, and is positive unless the measure is supported on the zero set of ℓ_i . This proves every claimed inequality without approximating the integral. $\square$

This applies to charge centroids, quantum expectations, positive concentration means, optical weighted means and the compliance-density mean. It does not prove that all these centers are close to one another, nor does it give a quantitative distance from the boundary. Later chapters give stronger bounds when available.

Chapter 2

Fingerprints, similarity and attractivity

2.1 Four complementary shape tests

The two original families are

$$T_{\text{iso}}(h) = \{(-1, 0), (1, 0), (0, h)\}, \quad T_{\text{right}}(h) = \{(0, 1), (0, 0), (h, 0)\}, \quad h > 0.$$

For the first we plot the axial coordinate $y(h)$ and its derivative. For the second we plot both coordinates and both derivatives. The horizontal orientation of the right family's varying leg is a convention, not a different experiment. Figures in this article use d/dh ; a derivative with respect to $\log h$ equals $h d/dh$.

Two further tests are essential:

$$T_{\text{thin}}(x, \epsilon) = \{(-1, 0), (1, 0), (x, \epsilon)\}, \quad 0 \leq x \leq 1,$$

and $T_{\text{eq}}(x, y) = \{(-1, 0), (1, 0), (x, \sqrt{3} + y)\}$. In the thin family, record C_x and C_y/ϵ , not only the collapsing Cartesian point. Take $\epsilon \rightarrow 0$ with fixed $x < 1$ before discussing the separate endpoint $x = 1$. Joint limits can give different answers.

For a smooth center, reflection and threefold symmetry force the equilateral expansion

$$C_x = \alpha x + \beta xy + O(\rho^3), \tag{2.1}$$

$$C_y = 1/\sqrt{3} + \gamma y + \delta x^2 + \eta y^2 + O(\rho^3), \quad \alpha + \gamma = 2/3, \tag{2.2}$$

where $\rho^2 = x^2 + y^2$. The trace relation can also be obtained by adding the three vertex Jacobians: translation covariance gives their sum as the identity, and symmetry makes their traces equal. The displayed polynomial form assumes sufficient differentiability. It is invalid for energy-ordered quantum branches at a degeneracy.

2.2 Uniform sampling of triangle shape

Angles satisfy $A + B + C = \pi$. The angle simplex is an equilateral shape diagram when distances to its three sides are proportional to A, B, C . Its ordered fundamental region $A \leq B \leq C$ has vertices

$$(0, 0, \pi), \quad (0, \pi/2, \pi/2), \quad (\pi/3, \pi/3, \pi/3).$$

Divide this triangle into 100 equal-area cells and use their centroids. This gives uniform area sampling in the requested angle geometry. It is not uniform sampling in side lengths or in the

Cartesian apex coordinates. Forty additional random shapes, sampled independently in this same region, are held out from fitting.

Given the angles, put $c = 1$, $a = \sin A / \sin C$, $b = \sin B / \sin C$ and

$$V_1 = (0, 0), \quad V_2 = (1, 0), \quad V_3 = ((b^2 + 1 - a^2)/2, \sqrt{b^2 - ((b^2 + 1 - a^2)/2)^2}).$$

The samples avoid exactly degenerate triangles. Boundary strata and nearly repeated eigenvalues are examined separately. Sample coordinates, seeds, convergence flags and full outputs accompany the article.

2.3 What a nearest ETC neighbor means

For a candidate X_n , the principal location score is

$$E_{\text{loc}}(n) = \left(\frac{1}{N} \sum_{k=1}^N \frac{|C(T_k) - X_n(T_k)|^2}{D(T_k)^2} \right)^{1/2}.$$

Also retain the maximum sampled error, independent holdout error, family derivative discrepancies, thin-family errors and vertex-response discrepancies. The supplied snapshot has 72,807 entries; 70,853 evaluate finitely on the complete 100-shape grid in the restricted importer. Exclusions include unsupported symbols as well as singular or nonreal values. No statement about the unevaluated entries or a later live ETC catalogue follows from this scan.

The tables identify *location-shortlist neighbors*. Mechanical and electromagnetic shortlists additionally store derivative and response-margin comparisons. These are useful diagnostics, but a scalar minimum response loses eigendirection information. A final full-fingerprint search should compare the full symmetric response matrices, with uncertainty-aware weights. We do not label the location winner the best attractivity match. In particular a nonattractive target may be best approximated by another center with a similar failure, rather than by a globally attractive center.

A finite fingerprint cannot prove identity. For example, the symmetric dimensionless function

$$d(a, b, c) = \frac{\prod_{\text{cyc}}(a^2 - b^2)^2 \prod_{\text{cyc}}(a^2 + b^2 - c^2)^2}{(a + b + c)^{24}}$$

vanishes identically on the isosceles and right families. The center $C + \eta d(G - I)$ has exactly the same curves and tangential derivatives as C there, despite usually differing on scalene triangles. High powers of d also conceal any prescribed finite jet. This is why scalene validation and analytic reasoning are indispensable.

2.4 Vertex-motion attractivity

The definition used throughout follows the project's *Attractive Centers* papers [3]. Move only V_i by h . The finite condition is

$$h \cdot [C(V_1, \dots, V_i + h, \dots) - C(T)] \geq 0.$$

For a differentiable center, write $J_i = D_{V_i}C$. The infinitesimal condition is

$$S_i = (J_i + J_i^T)/2 \succeq 0, \quad \lambda_{\min}(S_i) \geq 0.$$

Indeed $h^T J_i h = h^T S_i h$. If this holds throughout a convex admissible region for the moving vertex, integration along a straight segment proves the finite condition within that region. A negative

eigenvalue gives a local direction of failure; a sufficiently small resolved finite displacement should then be checked directly.

At the equilateral triangle, the apex Jacobian is $\text{diag}(\alpha, 2/3 - \alpha)$. Hence $0 \leq \alpha \leq 2/3$ is necessary and sufficient there, but says nothing about all shapes. The magnetic center passes this local test and fails globally.

There are two different kinds of stability. Physical stability concerns variation of a state or observation point inside a *fixed* domain. Attractivity concerns variation of the *domain's vertices*. Differentiating a stationarity equation $\nabla_p E(p, T) = 0$ gives

$$D_{V_i} C = -[D_p^2 E]^{-1} D_{V_i} \nabla_p E.$$

A positive observation Hessian does not constrain the sign of the mixed derivative on the right. No physical energy-minimum argument alone establishes vertex attractivity.

2.5 Numerical algorithms and their limitations

Second-order Dirichlet problems use conforming quadratic finite elements, or a Bernstein polynomial Ritz space multiplied by the boundary bubble $b = \ell_1 \ell_2 \ell_3$. Clamped fourth-order problems use b^2 times Bernstein polynomials, so both the displacement and normal derivative vanish on every side. Mass and stiffness matrices are assembled by Gaussian quadrature after an affine map from a reference triangle. Orthogonalizing the basis improves conditioning; it does not remove approximation error on slender triangles.

For an eigenproblem, solve $Ku = \lambda Mu$ and integrate first moments using coordinate-weighted mass matrices. For a source problem, solve $Ku = f$. A nonlinear problem minimizes the discretized energy, then checks the residual of the original PDE. Extrema are refined inside elements or in the polynomial field rather than selected from nodes alone. Multiple starting points detect some competing extrema but do not certify global uniqueness.

Charge models use panels and analytically integrated singular self interactions. Optical models use smooth finite-domain quadrature, with increased order for oscillatory phases. Finite differences of a numerically located maximum amplify position errors; derivative signs require finer checks than pictures do. The archive retains failed target flags instead of suppressing difficult shapes.

2.6 Symbolic approximations with controlled interpretation

A useful approximation should respect permutation symmetry and scaling before coefficients are fitted. Put $\ell_i = a_i/P$, $u_i = \ell_i - 1/3$, $v_i = \ell_i^2 - \frac{1}{3} \sum_j \ell_j^2$, and $z = 12\sqrt{3}A_T/P^2$. One family used for the mechanical and optical means is

$$\hat{w}_i = \frac{b_i \exp\{\sum_{k=0}^2 q^{k+r}(A_k u_i + B_k v_i)\}}{\sum_j b_j \exp\{\sum_{k=0}^2 q^{k+r}(A_k u_j + B_k v_j)\}}, \quad \hat{C} = \sum_i \hat{w}_i V_i.$$

Here q is a declared power of z , $b_i > 0$ is a geometrically chosen baseline, and r controls the endpoint behavior. Every finite-coefficient surrogate is strictly interior. This is a certificate for the surrogate's location, not a bound on its error or a proof of its attractivity. The quantum fits use a different additive, branch-aware ansatz and do not inherit this positivity certificate. Coefficients, model conventions and validation errors are supplied in the archive.

Chapter 3

Physical realizations of classical ETC centers

This chapter separates direct homogeneous experiments, limiting experiments, and deliberately calibrated constructions. The last category is useful for teaching but cannot support a claim that a classical center arose naturally from an independent physical law. Replacing a prescribed barycentric formula by three masses of those same weights is a realization, not an explanation of that formula.

3.1 The centroid $X(2)$: one point, many experiments

Let $G = (V_1 + V_2 + V_3)/3$. The following scenarios have different observables or physical mechanisms. Several share the same uniform-measure calculation, and are identified as such rather than counted as independent mathematical theorems.

1. **Balance of a uniform lamina.** A constant areal density gives $\int_T p dA/A_T = G$. An affine change of coordinates reduces the integral to the reference triangle, whose three barycentric means are $1/3$.
2. **Three equal vertex masses.** Static gravitational torque vanishes about their arithmetic mean G . This is a discrete measure, unlike the lamina.
3. **Three equal-mass straight rods along the sides.** Each rod's mass is the same, although their linear densities differ. Their midpoints average to G . Uniform linear density on all three rods instead gives $X(10)$.
4. **Three equal zero-rest-length Hookean springs.** A junction at p has energy $k \sum_i |p - V_i|^2/2$. Its force balance is $3kp = k \sum_i V_i$, hence $p = G$. A spring with nonzero natural length is a different model.
5. **Distributed quadratic attraction.** A junction coupled uniformly to the lamina has energy $\int_T |p - q|^2 dq/2$. Differentiation gives $A_T(p - G)$; the unique equilibrium is G .
6. **Uniform pressure on a horizontal triangular hatch.** The resultant normal force acts through G , since the first moment of a constant load is the area first moment. A tilted submerged plate has varying pressure and generally a different center of pressure.
7. **Buoyancy of a straight triangular prism.** A homogeneous, fully submerged right prism has displaced-volume centroid above G . Its projection is G ; a partially submerged or tilted prism requires a different calculation.

8. **Uniform axial DC conduction.** In a homogeneous straight prism with insulating side-walls and equipotential end faces, the axial electric field and current density are uniform. Their flux centroid in the triangular section is G . Contact and skin effects are excluded.
9. **A reflecting ideal gas at equilibrium.** With no external potential, its equilibrium spatial marginal is uniform in the container. The mean position is G , while the instantaneous spread is not zero.
10. **Reflecting Brownian motion.** Neumann diffusion conserves mass and has constant equilibrium density on a connected triangle. Its stationary expectation is G .
11. **Specular billiard equilibrium.** The invariant interior position-direction measure has uniform spatial marginal. Its spatial mean is G . Individual time averages require the ergodic qualifications in Chapter 4.
12. **Linear imaging of a uniform luminous aperture.** Convolution with a normalized point-spread function of zero first moment preserves the image centroid. Thus any symmetric, integrable blur with a finite first moment yields G . Nonlinear detection leads to the optical family in Chapter 9.
13. **Far-field aperture phase slope.** If $\hat{\chi}_T(k) = \int_T e^{-ik \cdot q} dq$, then $\nabla_k \log \hat{\chi}_T(0) = -iG$. The phase derivative identifies an aperture point, unlike the central Fraunhofer bright direction itself.
14. **Leading far-field current-loop translation.** A planar uniform loop is equivalent, in its solid-angle potential, to its spanning area. The first area moment in the far-distance expansion is at G ; this is an asymptotic multipole convention, not the near-field minimum M1.
15. **High-temperature quantum equilibrium.** The heat-kernel-diagonal spatial mean tends to G as inverse temperature tends to zero. Complete eigenspaces must be included.
16. **High-energy spectral averaging.** Averaging all quantum expectations below an energy cutoff tends to G by the local Weyl law; this is not an assertion about every individual high state.
17. **Strong bulk absorption with uniform production.** The concentration is nearly constant away from a boundary layer when $\mu \rightarrow \infty$. Its normalized mean tends to G ; see the absorption family.
18. **Strong logistic growth with a lethal boundary.** The positive population approaches its uniform carrying capacity in the interior as the dimensionless growth parameter grows. Its mean tends to G .
19. **A uniformly thickened plate in four dimensions.** The leading divergent electrostatic energy penalizes $\int_T \rho^2$ and selects uniform projected charge. Under the stated uniform-thickness convention the limiting center is G ; the geometric limiting argument is discussed with E4.
20. **Large-height solid-angle illumination.** Expanding $h(|p - q|^2 + h^2)^{-3/2}$ at large h leaves a position-dependent term proportional to $-\int_T |p - q|^2 dq$. Its limiting maximizer is G .
21. **Weak darkness preference.** When $\beta \rightarrow 0$, the darkness weights $e^{-\beta J}$ become uniform. The optical moment tends to G .
22. **Very narrow or very broad Gaussian nonlinear fluorescence.** Both endpoint limits of the quadratic-detection family are G , although intermediate widths need not give G .

Each identity involving a uniform measure follows from the same affine moment integral. The physical distinctions matter because nonuniform forcing, different boundary conditions or different detection laws break the equivalence. For $X(2)$, $D_{V_i}G = I_2/3$ proves global vertex attractivity directly.

3.2 The incenter $X(1)$: clearance, arrival and survival

Let $I = (aV_1 + bV_2 + cV_3)/P$, with inradius $r = 2A_T/P$. A disk contained in the triangle has radius at most the smallest distance from its center to a side. Since $\sum_i a_i d_i(p) = 2A_T$, that smallest distance is at most r , with equality precisely when all three distances equal r . Thus the largest inscribed disk has center I .

This gives several clean experiments. An expanding circular balloon constrained to remain circular first reaches maximal size at I . A wave propagating inward from all sides at equal speed reaches its last point at I : its arrival time is $d(p, \partial T)/v$. A uniform inward erosion of the triangle collapses at I . The short-time maximizer of Dirichlet heat content tends to I , because escape probability is exponentially sensitive to the nearest boundary distance. The limiting infinite-exponent torsion profile is the distance function, whose unique triangular maximum is I . These last two statements concern limits; they do not identify a finite-time heat maximum or a finite- p membrane maximum with I .

The incenter's global attractivity is part of the classical attractive-segment result in the project's Paper IV [3]. The simple fact that a largest disk is stable under an optimization algorithm is not an alternative proof.

3.3 The Spieker center $X(10)$: perimeter rather than area

For uniform linear density on the three sides,

$$S = \frac{a(V_2 + V_3) + b(V_3 + V_1) + c(V_1 + V_2)}{2P} = \frac{(P-a)V_1 + (P-b)V_2 + (P-c)V_3}{2P} = \frac{3G - I}{2}.$$

This is $X(10)$. A balanced wire frame, a uniform perimeter heat source's first moment, and an incoherent uniformly luminous wire's image centroid all select this point. The last two require linear observation and uniform emission per unit length. A reflecting diffusion constrained to the closed wire also has uniform arclength equilibrium when its diffusion coefficient and vertex transmission law are uniform.

Uniform-radius thin conducting wires in ambient dimensions three and four have the same limiting projected charge center; finite-radius corrections need not agree. A regularized magnetic-field area moment with a uniform near-edge cutoff has leading perimeter weights and the same limit, whereas the unregularized divergent integral is not a center. The Spieker point is also the incenter of the medial triangle, giving a largest-disk demonstration after physically constructing that smaller triangular chamber. The project's Paper IV establishes attractivity of $X(10)$ for triangles.

3.4 The symmedian point $X(6)$: three normal spring forces

Place a frictionless guide on each side line and connect a movable junction to its orthogonal projection by an identical zero-rest-length spring. Let $d_i(p)$ be signed distance to line i . The energy is

$$E(p) = \frac{k}{2} \sum_i d_i(p)^2.$$

For interior distances the affine constraint is $\sum_i a_i d_i = 2A_T$. Minimizing a sum of squares subject to this constraint gives $d_i = 2A_T a_i / \sum_j a_j^2$. These positive distances correspond to an interior point, and $w_i = a_i d_i / (2A_T) = a_i^2 / \sum_j a_j^2$. Thus the unique equilibrium is the symmedian point $X(6)$. The Hessian is the sum of three rank-one normal projections and is positive definite. This is a direct mechanical derivation of a nontrivial classical center. Its vertex attractivity remains a separate property and is not certified here.

3.5 The circumcenter $X(3)$, orthocenter $X(4)$, and nine-point center $X(5)$

Three simultaneous identical point emitters have equal direct travel time to the circumcenter, since it is the unique point equidistant from the three vertices. A rigid circular track passing through three pins has its rotation center there. These equal-range constraints determine $X(3)$ even when it lies outside the triangle.

The center of the smallest disk containing the three vertices agrees with $X(3)$ only for acute triangles (and the usual right-triangle boundary case). For an obtuse triangle it is the midpoint of the longest side. A minimax illumination or range experiment must retain this piecewise distinction.

There is an explicitly constructed spring experiment for $X(4)$. First construct each altitude line. Attach normal zero-rest-length springs to those three lines. Their quadratic energy is the sum of squared distances to the altitudes, which vanishes only at their common intersection, the orthocenter. This is physically realizable but uses preconstructed geometric rails; it is less intrinsic than using the original sides for $X(6)$. Similarly, three synchronized emitters at the side midpoints have equal travel times at the circumcenter of the medial triangle, the nine-point center $X(5)$. These are labelled *derived-geometry realizations*, not unexplained new equilibrium laws.

3.6 The first Fermat center $X(13)$: tension, networks and transport

For a junction connected to three vertex pulleys with equal hanging loads, the potential is proportional to $E(p) = \sum_i |p - V_i|$. At an interior equilibrium the three unit tension vectors sum to zero. Their pairwise angles are therefore 120° . If every triangle angle is less than 120° , this selects the first Fermat point $X(13)$. The same junction minimizes the total length of a three-terminal Steiner network and the mean linear travel cost to equally weighted vertex destinations.

If a vertex angle is at least 120° , the convex geometric-median problem is minimized at that vertex. Its global physical selector is consequently not the unqualified analytic $X(13)$ construction over every triangle. A soap-film triple junction has the same local force law for equal interfacial tensions, but boundary contact angles and the possibility of other film topologies must be controlled. These qualifications are essential in an experimental demonstration.

3.7 Contact-derived and calibrated realizations

The incircle and nine-point circle are tangent at the Feuerbach point $X(11)$, away from cases where the usual construction loses uniqueness. Their contact point can be observed by fitting the two geometric circles, but it is a derived circle-contact realization, not a center selected by an unconstrained homogeneous sheet.

More generally, positive barycentric weights w_i can be realized by vertex masses w_i or spring

constants w_i , since $\sum_i w_i |p - V_i|^2$ is minimized at their weighted mean. Examples include the Gergonne weights $1/(s - a)$ for X(7) and Nagel weights $s - a$ for X(8). Tangent lengths provide a way to calibrate those masses, but the weighting is part of the apparatus. Signed barycentrics cannot be implemented by positive masses or passive positive-stiffness springs in this way.

3.8 Scope of the first-100 catalogue

The direct and limiting constructions above concentrate on X(1), X(2), X(3), X(6), X(10) and X(13); the derived or calibrated examples add X(4), X(5), X(7), X(8) and X(11). This is not a claim that the other first-100 entries have no physical interpretation. An explicit inventory in the bundle records all 100 identifiers, with “no independent natural realization established in this survey” for the others. It prevents an unexamined entry from being mistaken for a negative theorem. The catalogue is broad in mechanisms, but does not manufacture a natural explanation for every number by restating its barycentric formula as prescribed masses.

For demonstrations, the most informative first comparison is a plate, a uniform wire frame, and the three side-normal springs cut to the same scalene outline. Their balance/equilibrium points are respectively X(2), X(10), and X(6). Add an inscribed expanding disk for X(1). All four are easy to measure and need no numerical PDE solution.

Chapter 4

Quantum confinement, heat spectra and billiards

4.1 From the infinite well to six spatial selectors

A particle confined to a triangular infinite well cannot have a nonzero wavefunction on the wall. The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\Delta\psi = E\psi, \quad \psi|_{\partial T} = 0.$$

Dividing by $\hbar^2/(2m)$ leaves the geometric Dirichlet eigenproblem

$$A\psi_j = \lambda_j\psi_j, \quad A = -\Delta_D, \quad \int_T |\psi_j|^2 = 1, \quad \lambda_0 \leq \lambda_1 \leq \dots$$

The physical constants rescale the energies, but not the normalized spatial densities. The probability of finding the particle in a small area is $|\psi_j|^2 dA$. Its expected location is therefore

$$Q_j = \int_T p |\psi_j(p)|^2 dA, \quad j = 0, 1, 2, 3, 4.$$

The most probable ground-state location is instead $Q_{0m} = \arg \max \psi_0$, equivalently the maximum of ψ_0^2 . An expectation averages all the density; a maximum reads only its highest peak. They need not coincide in a scalene triangle.

The ground state is positive and simple. For a simple excited eigenvalue, changing the sign or complex phase of the eigenfunction leaves Q_j unchanged. For a repeated eigenvalue, different linear combinations in its eigenspace may have different densities and expectations. At the equilateral triangle, levels 1 and 2 form a doublet; level 4 belongs to a doublet with level 5. The individual energy-ordered expectations have direction-dependent limiting branches near this shape. Arbitrarily assigning G at the equilateral point does not repair continuity. The plots leave undefined values missing rather than drawing an invented eigenbasis-independent center.

The canonical replacement for a repeated eigenspace E is its projector density

$$\rho_E(p) = \frac{1}{\dim E} \sum_{j \in E} |\psi_j(p)|^2, \quad Q_E = \int_T p \rho_E(p) dA,$$

using any L^2 -orthonormal basis. A unitary change of basis leaves the sum unchanged. These projector means are natural additional centers; they do not retroactively make the requested individual Q1, Q2 and Q4 globally unambiguous.

4.2 How the computation works

Approximate a wavefunction by $\sum_k c_k \phi_k$ in a quadratic finite-element space with zero boundary trace. Testing the differential equation against each ϕ_l and integrating by parts yields

$$K_{lk} = \int_T \nabla \phi_l \cdot \nabla \phi_k, \quad M_{lk} = \int_T \phi_l \phi_k, \quad Kc = \lambda Mc.$$

For the x expectation use M_x with entries $\int_T x \phi_l \phi_k$, and compute $c^\top M_x c / (c^\top M c)$. The y component is analogous. This integrates the density rather than approximating the mean by a sum of nodal positions. To locate Q0m, test the stationary points of the quadratic polynomial on every element, together with element edges and vertices. Refine the mesh, monitor energy gaps, and repeat. The archive contains both mesh changes and gap flags.

4.3 Thin and almost-equilateral behavior

In a thin triangle the transverse ground-mode energy is proportional to the inverse square of the local width. It is smallest below the apex, where the strip is widest. Thus the first fixed number of states localize near x , rather than becoming modes of a uniform one-dimensional box on $[-1, 1]$. Balancing the longitudinal kinetic term with the linear change of transverse energy gives an $\epsilon^{2/3}$ longitudinal scale and an Airy-type effective problem. This supplies a useful noninteger-power fingerprint. It also explains why a generic polynomial fit can fail near degeneration.

Near the equilateral triangle, simple-state perturbations can be computed by differentiating the stiffness form and solving on the orthogonal complement of the eigenstate. In a multiple eigenspace, first diagonalize the perturbation matrix restricted to that eigenspace. Its eigenvectors depend on the direction of the shape perturbation; their moments explain the observed branch dependence. This is a spectral issue, not numerical noise. The independent second-eigenvalue simplicity literature [8] is relevant to gap monitoring, but is not used to suppress a known equilateral multiplicity.

4.4 A quantitative location bound

All quantum expectations are strictly interior by Proposition 1.1. There is a stronger bound. Let $d_i(p)$ be distance to a supporting side, and let r be the inradius. The half-plane Hardy inequality, applied to the zero extension of ψ_j , gives

$$\int_T \frac{|\psi_j|^2}{d_i^2} \leq 4 \int_T |\nabla \psi_j|^2 = 4\lambda_j.$$

Since $t \mapsto t^{-2}$ is convex, Jensen's inequality gives

$$d_i(Q_j) = \int_T d_i |\psi_j|^2 \geq \frac{1}{2\sqrt{\lambda_j}}.$$

The incircle is a domain contained in T , so the min-max principle yields $\lambda_j(T) \leq \zeta_j^2/r^2$, where ζ_j is the j th unit-disk Dirichlet wave number, counted with multiplicity. Consequently

$$\boxed{d_i(Q_j) \geq r/(2\zeta_j), \quad i = 1, 2, 3.}$$

For levels 0–4 these wave numbers are approximately 2.40483, 3.83171, 3.83171, 5.13562, 5.13562. This is an analytic inner-parallel-triangle certificate; the numerical slacks in the archive are checks, not its proof. It concerns expectations, not Q0m.

4.5 Thermal quantum means and spectral averaging

The canonical quantum thermal density is the diagonal heat kernel normalized by its trace:

$$\rho_\tau(p) = \frac{\sum_j e^{-\tau L^2 \lambda_j} |\psi_j(p)|^2}{\sum_j e^{-\tau L^2 \lambda_j}}, \quad Q_{\text{th}}(\tau) = \int_T p \rho_\tau(p) dA.$$

This is a moment, whereas the later heat center is the maximum of heat evolved from constant initial data. Complete eigenspaces make Q_{th} basis independent. As $\tau \rightarrow \infty$ it tends to Q_0 ; as $\tau \downarrow 0$ the local Weyl law gives G [11]. The corresponding energy-cutoff statement is

$$\frac{1}{N(\Lambda)} \sum_{\lambda_j \leq \Lambda} Q_j \rightarrow G.$$

For rational polygons, spatial quantum ergodicity gives convergence to G along a density-one subsequence [10]. Neither statement proves convergence of every individual Q_j in every triangle. This survey records the thermal and projector constructions as well-defined extensions; their dedicated all-shape ETC and attractivity scans have not been performed.

4.6 Classical billiards: the center and its fluctuations

For unit-speed specular motion $X_z(s)$ launched from boundary state z , define

$$C_z(t) = t^{-1} \int_0^t X_z(s) ds, \quad D_z(t) = |C_z(t) - G|,$$

and report the mean distance μ_D , its standard deviation σ_D , and $R = (\mathbb{E}D_z^2)^{1/2}$. They satisfy $R^2 = \sigma_D^2 + \mu_D^2$. A small SD alone does not imply approach to the centroid: all distances could be nearly the same positive number.

Each straight flight from p with velocity v contributes $p\Delta t + v\Delta t^2/2$ to the time integral. This permits exact integration between numerical collision events, including partial flights at observation times. Averaging collision locations without time weights is a different measure. The experiment uses 2,048 independent uniform-arclength launches and uniform inward angles per triangle, plus a cosine-flux comparison ensemble. Diameters and speeds are one; the final time is 10,000.

For rational triangles, almost every direction is uniquely ergodic on its directional surface [9]. Projecting to the triangle gives $C_z(t) \rightarrow G$ for almost every continuously sampled nonsingular launch. Because $D_z(t) \leq D$, dominated convergence proves $R, \mu_D, \sigma_D \rightarrow 0$. Exceptional periodic paths remain: the recorded orthic orbit in a generic acute triangle has a time mean about $0.05505D$ away from G . No corresponding all-irrational-triangles theorem is asserted.

At $t = 10,000$, the recorded SDs are 0.000443 (equilateral), 0.000255 (right isosceles), 0.010637 (rational scalene 36° – 60° – 84°), and 0.001013 (generic scalene). The slow rational-scalene tail shows why a universal $t^{-1/2}$ rate should not be inferred. Under a stationary law,

$$R(t)^2 = \frac{2}{t^2} \int_0^t (t-s) \mathbb{E}[(X(0) - G) \cdot (X(s) - G)] ds.$$

A square-root law needs appropriate integrability of this correlation, which is not automatic for polygonal billiards. The bundle includes trajectories, uncertainty summaries, independent unfolding checks, and an interactive demonstration.

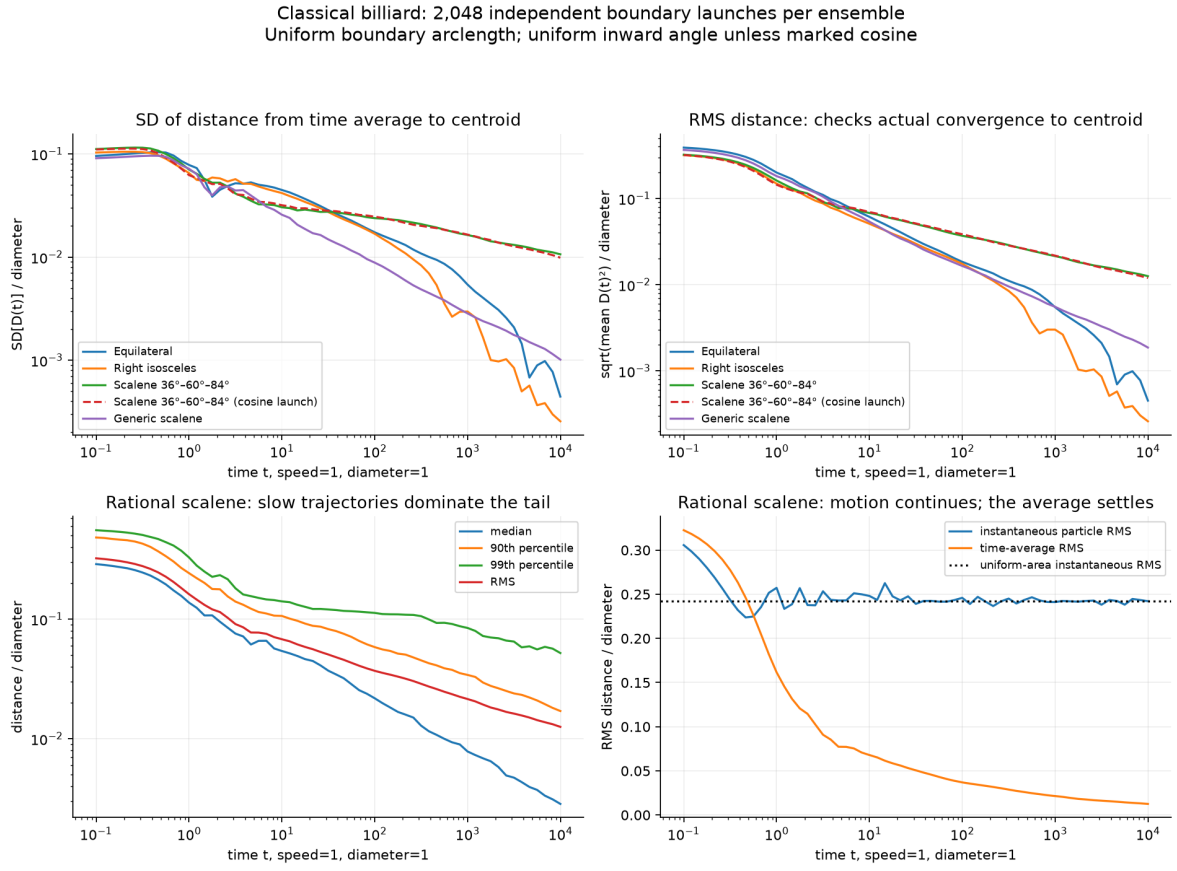


Figure 4.1: Archived billiard experiment: means and fluctuations of time-averaged positions. Finite-time decay is not an asserted universal rate.

4.7 Q0m: Quantum ground-state maximum

Physical scenario and selector. Confine a quantum particle to an infinite triangular well. Select the most probable location in the positive ground state.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_{0m} = \arg \max \psi_0$$

Derivation and algorithm. Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Search stationary points within quadratic elements for the maximum, then refine its location. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive ground state vanishes on the boundary, so its maximum is interior.

Attractivity. A candidate negative finite response appears only after substantial refinement; its small magnitude and maximum-location sensitivity leave the rigorous conclusion open.

Distinguishing behavior. At equilateral the center is unambiguous. The thin correction has the ground-mode Airy scale.

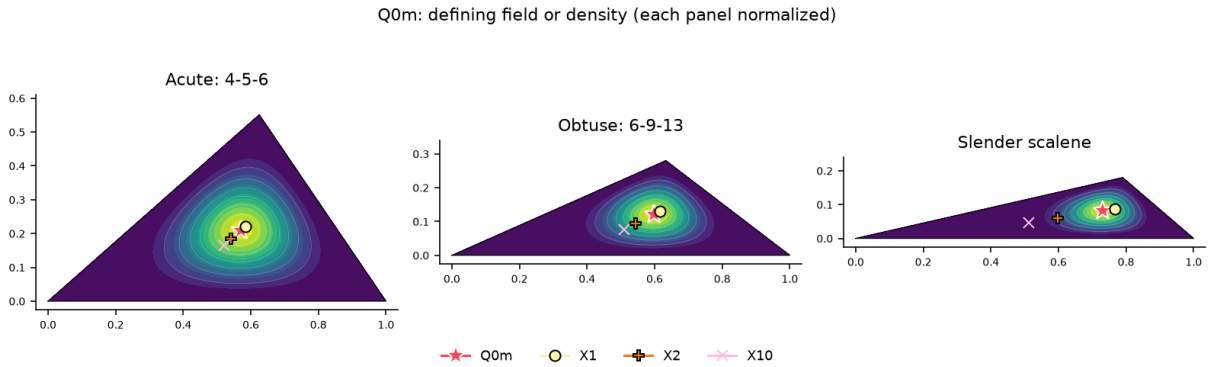
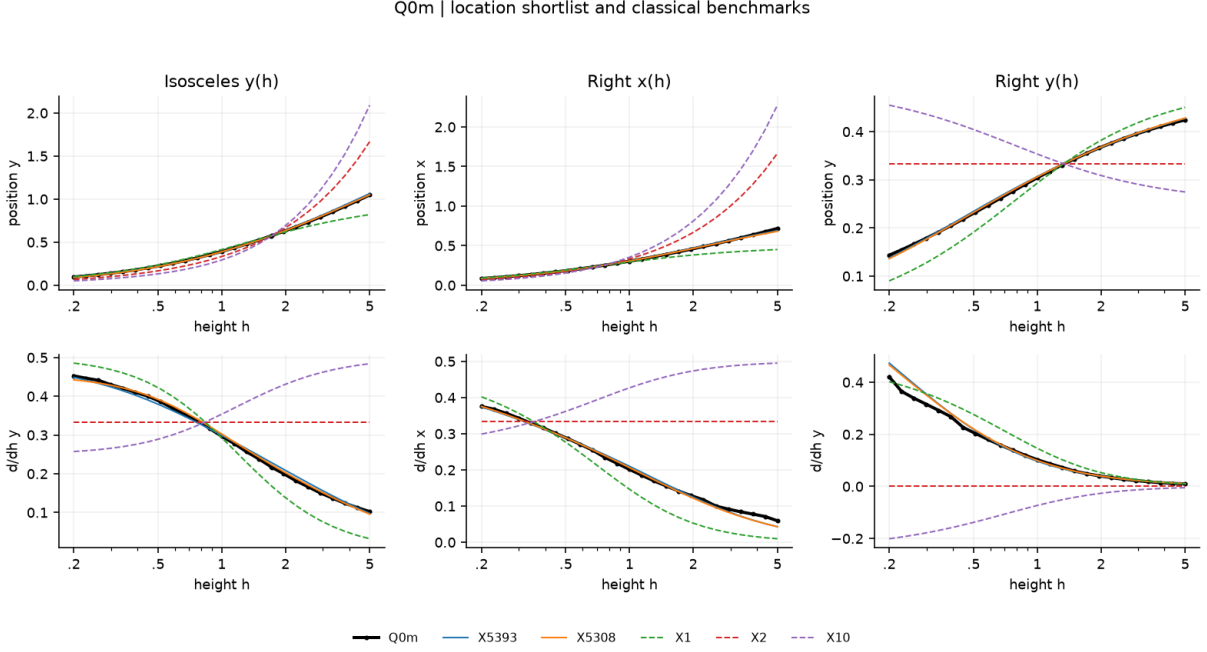


Figure 4.2: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(5393)	0.699%	1.734%	0.690%	1.518%
X(5308)	0.721%	2.459%	0.614%	1.385%
X(29678)	0.722%	2.017%	0.613%	1.259%

Resolution record. The joint six-center quantum targets were met on 99/100 training and 39/40 holdout shapes. Individual mesh changes and spectral gaps remain in the data; a repeated state is not resolved by merely refining the mesh.

Reproducibility. The bundle directory `output/quantum/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Q0m_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

4.8 Q0: Quantum ground-state expectation

Physical scenario and selector. Confine a quantum particle to an infinite triangular well. Prepare energy level 0 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_0 = \int_T p|\psi_0(p)|^2 dp$$

Derivation and algorithm. Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Distinguishing behavior. At equilateral the center is unambiguous. The thin correction has the ground-mode Airy scale.

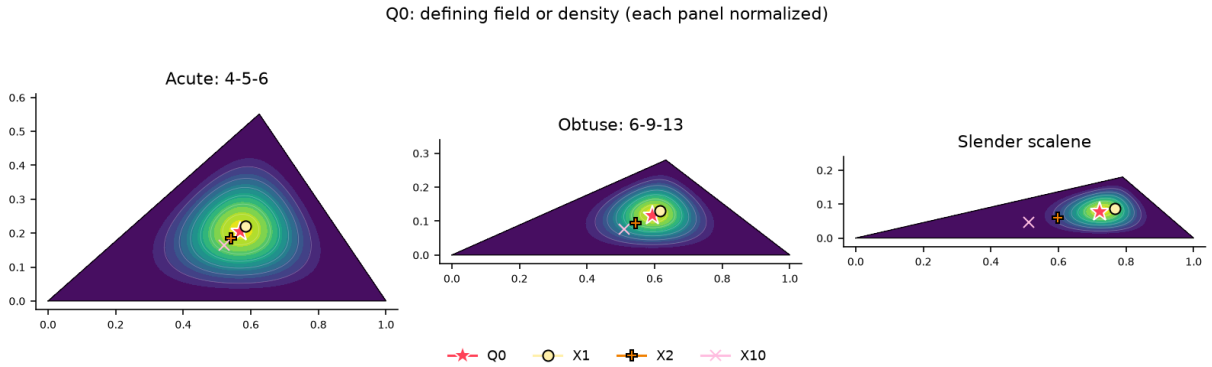
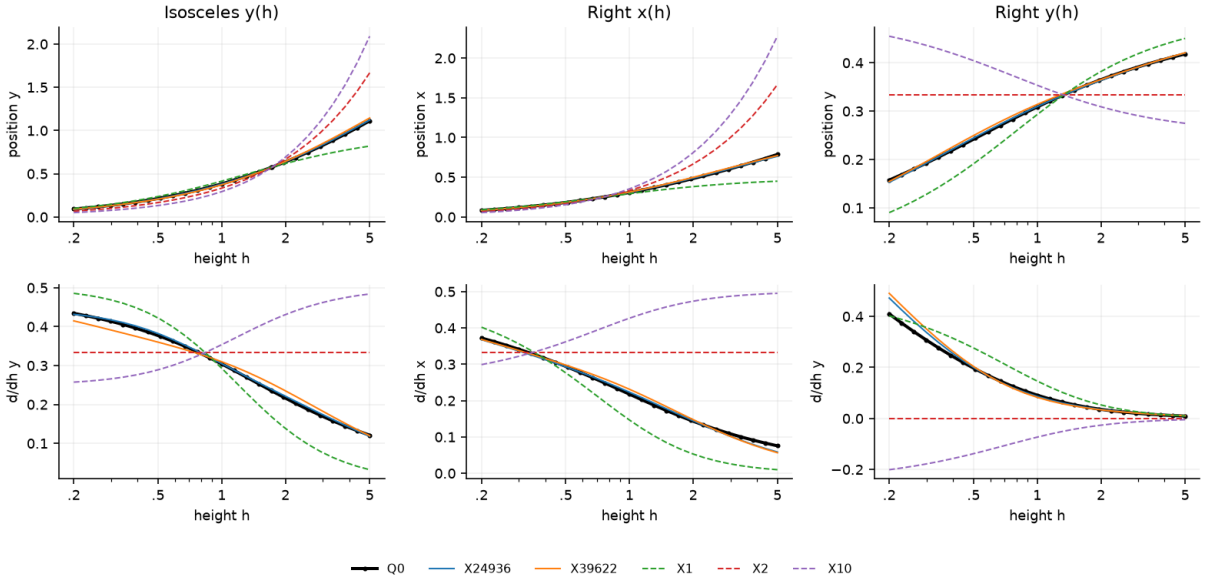


Figure 4.3: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Q0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(24936)	0.771%	2.986%	0.626%	1.472%
X(39622)	0.777%	1.858%	0.756%	1.638%
X(27253)	0.785%	2.140%	0.705%	1.740%

Resolution record. The joint six-center quantum targets were met on 99/100 training and 39/40 holdout shapes. Individual mesh changes and spectral gaps remain in the data; a repeated state is not resolved by merely refining the mesh.

Reproducibility. The bundle directory `output/quantum/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Q0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

4.9 Q1: Quantum first excited-state expectation

Physical scenario and selector. Confine a quantum particle to an infinite triangular well. Prepare energy level 1 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_1 = \int_T p|\psi_1(p)|^2 dp$$

Derivation and algorithm. Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Distinguishing behavior. At equilateral this state belongs to a doublet. Individual-state means are branch dependent; use a full eigenspace projector for a canonical alternative. Missing curve values are intentional.

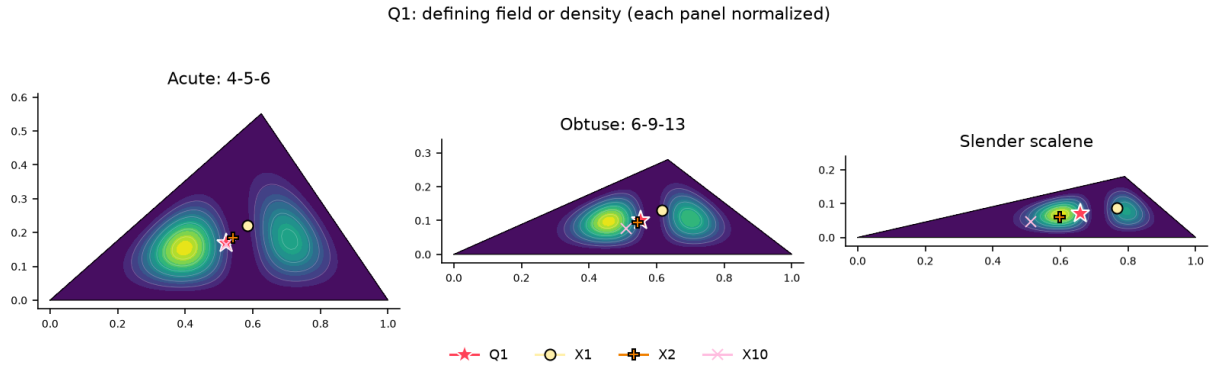
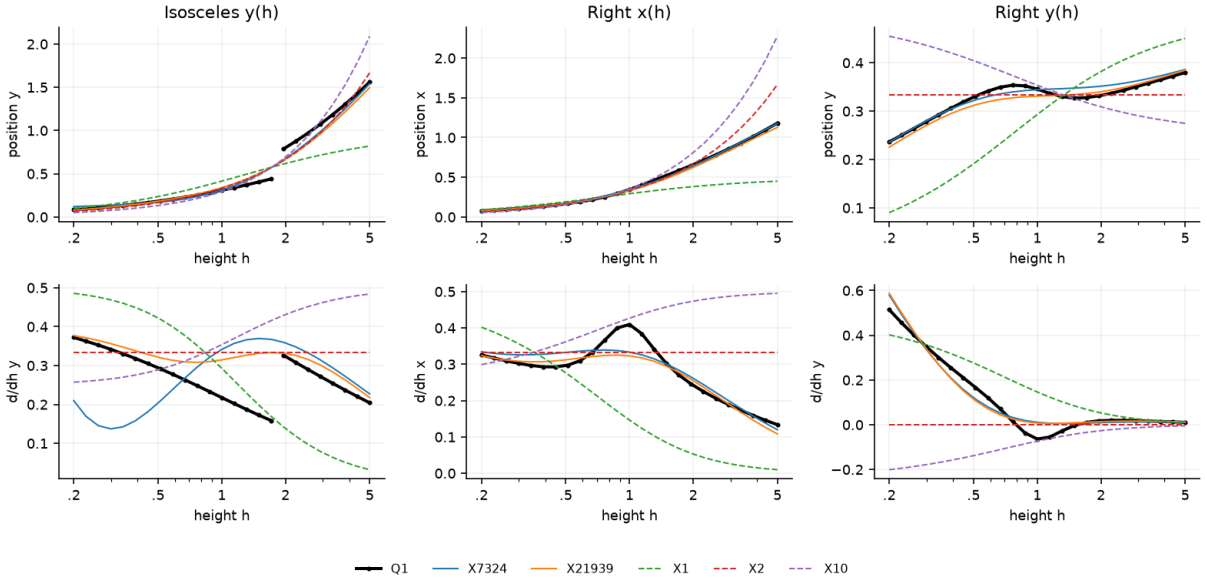


Figure 4.4: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Q1 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(7324)	1.701%	5.175%	1.421%	2.690%
X(21939)	2.015%	5.469%	1.604%	3.456%
X(25061)	2.083%	5.874%	1.712%	3.682%

Resolution record. The joint six-center quantum targets were met on 99/100 training and 39/40 holdout shapes. Individual mesh changes and spectral gaps remain in the data; a repeated state is not resolved by merely refining the mesh.

Reproducibility. The bundle directory `output/quantum/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Q1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

4.10 Q2: Quantum second excited-state expectation

Physical scenario and selector. Confine a quantum particle to an infinite triangular well. Prepare energy level 2 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_2 = \int_T p|\psi_2(p)|^2 dp$$

Derivation and algorithm. Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Distinguishing behavior. At equilateral this state belongs to a doublet. Individual-state means are branch dependent; use a full eigenspace projector for a canonical alternative. Missing curve values are intentional.

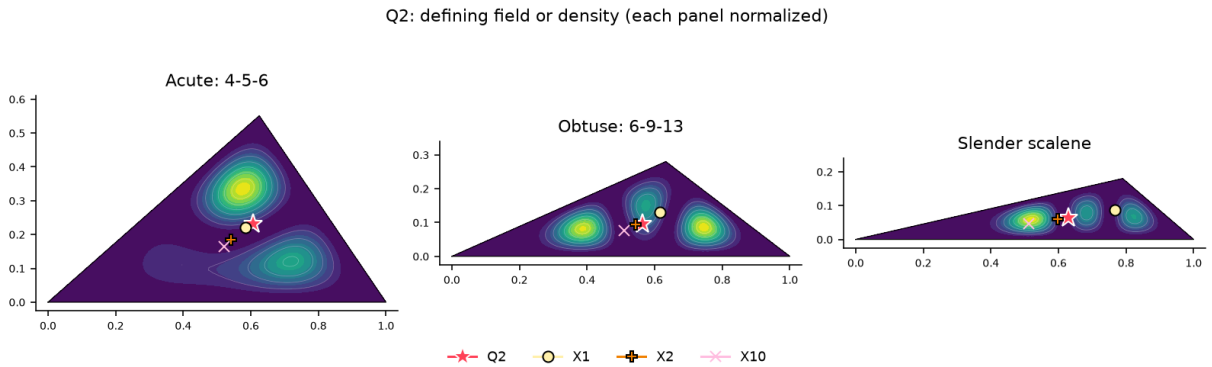
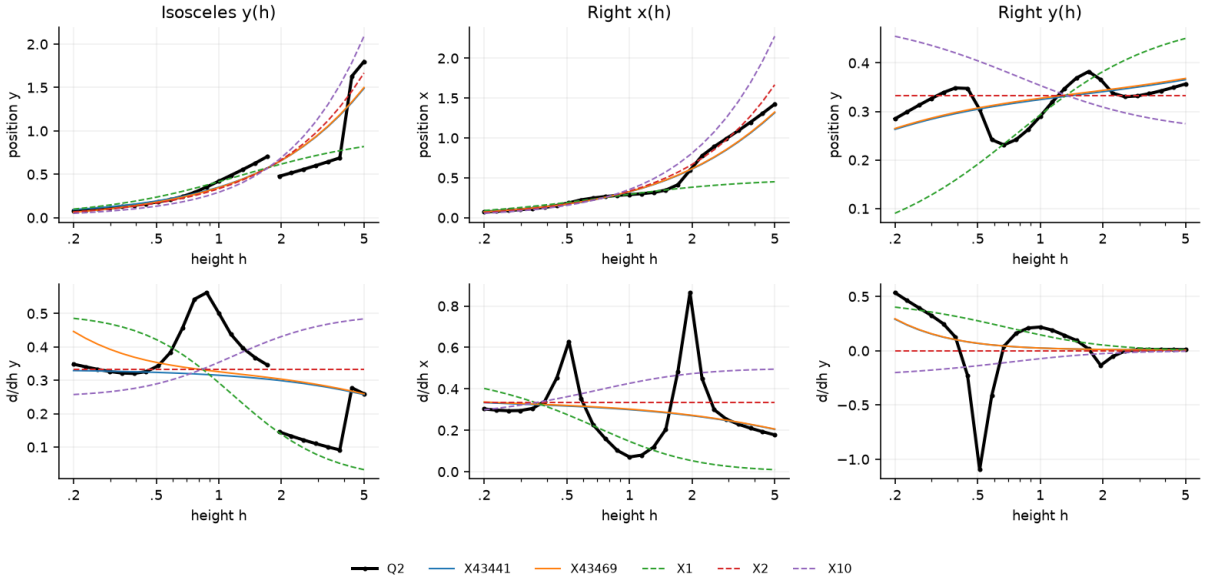


Figure 4.5: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Q2 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(43441)	3.908%	11.415%	3.281%	10.824%
X(43469)	3.933%	11.592%	3.298%	10.993%
X(72043)	4.138%	10.501%	3.634%	9.964%

Resolution record. The joint six-center quantum targets were met on 99/100 training and 39/40 holdout shapes. Individual mesh changes and spectral gaps remain in the data; a repeated state is not resolved by merely refining the mesh.

Reproducibility. The bundle directory `output/quantum/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Q2_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

4.11 Q3: Quantum third excited-state expectation

Physical scenario and selector. Confine a quantum particle to an infinite triangular well. Prepare energy level 3 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_3 = \int_T p|\psi_3(p)|^2 dp$$

Derivation and algorithm. Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Distinguishing behavior. Q3 is simple at equilateral and has a numerically negative horizontal response there (about -0.7206).

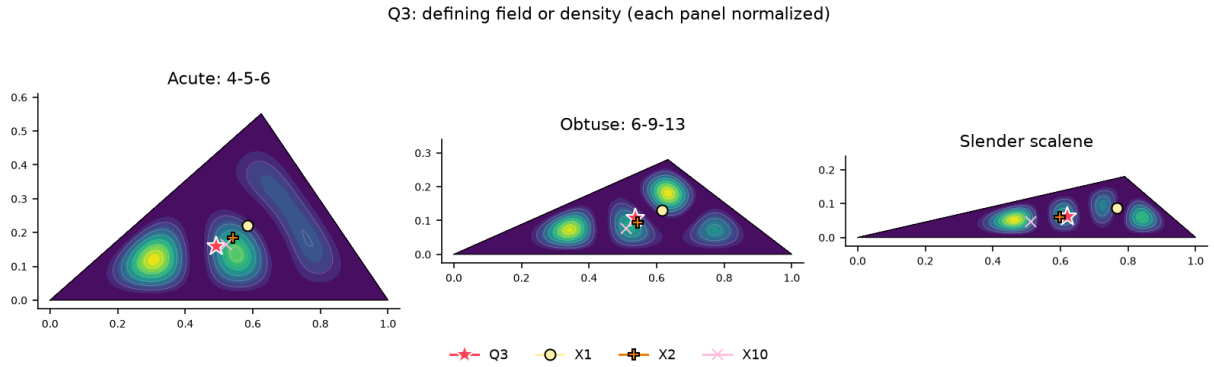
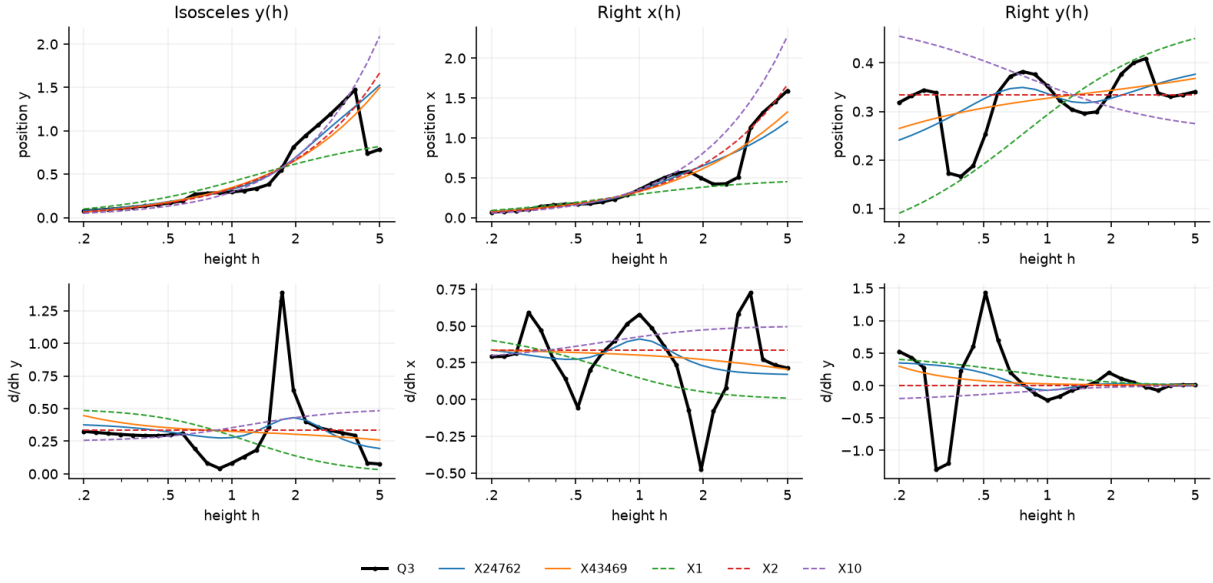


Figure 4.6: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Q3 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(24762)	4.909%	14.051%	5.539%	13.434%
X(43469)	4.933%	14.232%	4.702%	13.401%
X(4179)	4.936%	15.182%	4.843%	14.366%

Resolution record. The joint six-center quantum targets were met on 99/100 training and 39/40 holdout shapes. Individual mesh changes and spectral gaps remain in the data; a repeated state is not resolved by merely refining the mesh.

Reproducibility. The bundle directory `output/quantum/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Q3_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

4.12 Q4: Quantum fourth excited-state expectation

Physical scenario and selector. Confine a quantum particle to an infinite triangular well. Prepare energy level 4 and average repeated position measurements. The selected point is a probability mean, not necessarily a density peak.

$$-\Delta\psi_j = \lambda_j\psi_j, \quad \psi_j|_{\partial T} = 0, \quad Q_4 = \int_T p|\psi_4(p)|^2 dp$$

Derivation and algorithm. Use quadratic FEM eigenpairs and coordinate-weighted mass matrices. Normalize in the mass matrix, integrate the first moment, and retain the adjacent spectral gaps. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Every nonzero state mean is strictly interior; $d_i(Q_j) \geq r/(2\zeta_j)$ gives the stronger inner-parallel-triangle bound.

Attractivity. Refined finite-displacement computations have stable negative signs. These are numerical counterexamples, not interval-certified continuum statements.

Distinguishing behavior. At equilateral this state belongs to a doublet. Individual-state means are branch dependent; use a full eigenspace projector for a canonical alternative. Missing curve values are intentional.

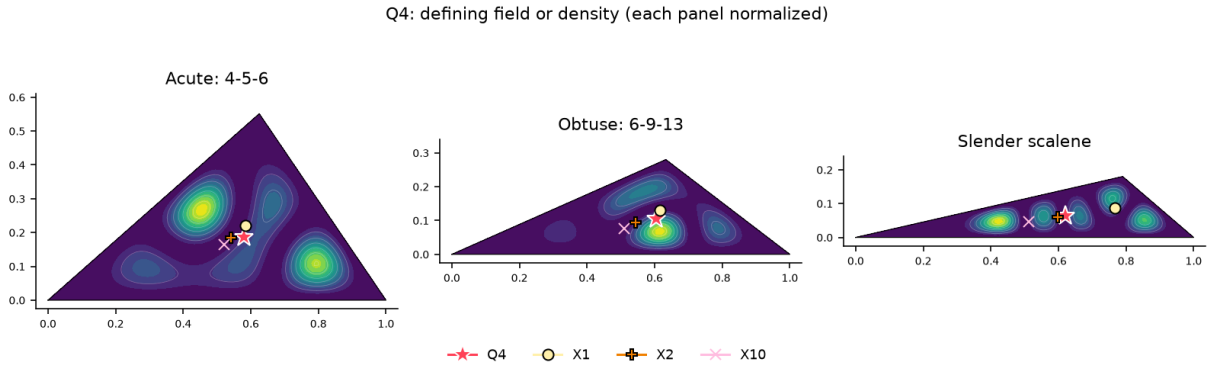
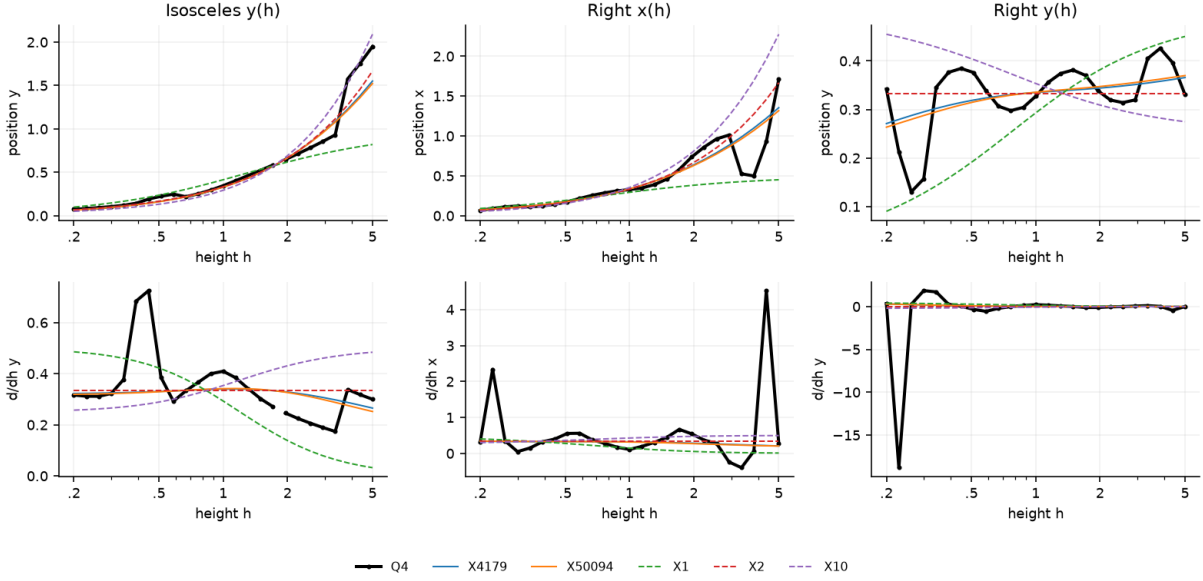


Figure 4.7: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Q4 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(4179)	4.688%	15.967%	4.855%	15.228%
X(50094)	4.688%	15.184%	5.391%	14.451%
X(31323)	4.699%	14.845%	5.259%	14.123%

Resolution record. The joint six-center quantum targets were met on 99/100 training and 39/40 holdout shapes. Individual mesh changes and spectral gaps remain in the data; a repeated state is not resolved by merely refining the mesh.

Reproducibility. The bundle directory `output/quantum/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Q4_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

Chapter 5

Conformal diffusion, electrostatics and magnetism

5.1 One diffusion process, three different questions

A Brownian particle started at p eventually reaches the boundary. Its exit side, its exit time and its finite-time survival probability are different random variables. The equal-exit center E0 equalizes the probabilities of the three sides. The torsion center F1 maximizes mean lifetime. The heat center H maximizes survival at a specified time. These are three clean experimental definitions, with no need to invent a barycentric formula first.

For E0, let h_i be harmonic in T , equal to one on side i and zero on the other two sides. The corner values do not affect interior harmonic measure. Then $h_i(p)$ is the exit probability and $\sum_i h_i = 1$. The desired point satisfies $h_1 = h_2 = h_3 = 1/3$. A Brownian demonstration colors paths by exit side and adjusts the starting point until the proportions balance; Monte Carlo error should be shown as confidence intervals.

5.2 E0 and E1 through a conformal map

Set $\alpha = A/\pi$, $\beta = B/\pi$, $\gamma = C/\pi$. A Schwarz–Christoffel map from the upper half-plane, with prevertices $0, 1, \infty$, is

$$F(z) = V_1 + (V_2 - V_1) \frac{B_z(\alpha, \beta)}{B(\alpha, \beta)}.$$

The powers in $F'(z)$ change boundary direction by the triangle's turning angles. Integrating them gives the incomplete beta function; normalization makes $F(0) = V_1$ and $F(1) = V_2$. Harmonic measure is conformally invariant, so equal angles subtended by the real intervals give

$$z_{E0} = \frac{1}{2} + i \frac{\sqrt{3}}{2}.$$

This supplies an explicit special-function evaluation of E0 [4]. It is not necessary to approximate random paths to compute the center accurately.

For E1, the conformal radius at $F(z)$ is $r_c(F(z)) = 2 \operatorname{Im}(z) |F'(z)|$. It measures the local scale of the disk mapped conformally to the triangle with its origin sent to that point. Maximizing its logarithm gives

$$z_{E1} = \frac{\alpha}{\alpha + \beta} + i \frac{\sqrt{\alpha\beta/\gamma}}{\alpha + \beta}.$$

The least-capacity or inner conformal center is $F(z_{E1})$ [5]. This is a self-interaction/conformal-radius extremum, not an equal-exit condition. Numerically evaluate the incomplete beta function

in the upper half-plane with consistent branches; use high precision and atan2 -based angles near degeneration.

There is a useful E0 location certificate. Brownian position is a bounded stopped martingale, so $p = \mathbb{E}[X_{\text{exit}}]$. On side i , its opposite barycentric coordinate vanishes; on the other sides it lies between zero and one. Equal exit probabilities therefore imply $0 < w_i(E0) < 2/3$. E0 lies in the open hexagon obtained by cutting off the three barycentric corners $w_i \geq 2/3$.

For fixed $x < 1$, the incomplete-beta expansion on the thin family gives

$$\begin{aligned} E0_x &= x + O(\epsilon^2), & E0_y &= 2\epsilon/3 + O(\epsilon^2), \\ E1_x &= x - \epsilon \operatorname{atanh}(x)/\pi + O(\epsilon^2), & E1_y &= \epsilon/2 + O(\epsilon^2). \end{aligned}$$

At $x = 1$ this expansion is not uniform: the E0 vertical ratio instead tends to $1/2$, while the E1 horizontal correction divided by ϵ is unbounded. These endpoint differences are stronger discriminants than two nearly overlapping finite-shape curves.

5.3 What does a charge centroid measure?

Fix total charge one. The conductor chooses the probability measure minimizing its electrostatic self energy. The potential is constant on the conductor, but the charge density is generally nonuniform. A contour plot of that constant surface potential would hide the relevant information. Our electrostatic atlas therefore shows charge density on the plate or along the boundary, and marks its first moment.

For planar electrostatics, use the logarithmic energy

$$\mathcal{E}_0(\mu) = \iint -\log |p - q| d\mu(p) d\mu(q).$$

The equilibrium measure lives on the outer boundary. Its first moment is E2, also the constant term of the normalized exterior conformal map. Filling the triangular boundary with its interior does not change this external equilibrium measure: S2 equals E2 exactly.

For a plate embedded in three-dimensional space, the Newtonian kernel is $|p - q|^{-1}$. Its equilibrium charge is a measure on the two-dimensional plate, with edge enhancement. E3 is its first moment. It differs from the potential maximum of a uniformly charged plate and from the planar logarithmic boundary measure. Positivity gives strict interiority of both E2 and E3 by Proposition 1.1.

5.4 Charge algorithms and singular self panels

Partition each edge into cosine-graded line segments for E2. Integrate the logarithmic kernel exactly over a segment, impose equal potential at panel midpoints, and normalize the panel charges to sum to one. For E3, subdivide the plate into small triangular panels with constant density and impose equal potential at their centroids.

The self interaction must not be replaced by an arbitrary missing constant. In a panel plane,

$$\operatorname{div}_q \frac{q - p}{|q - p|} = \frac{1}{|q - p|}.$$

The divergence theorem converts the panel integral into edge integrals of $1/|q - p|$, multiplied by signed distances to the oriented edges. These have logarithmic antiderivatives and remain finite for a self panel. Solving the resulting dense linear system gives panel masses and a center. Positivity, potential residuals and refinement changes are retained. A tiny linear-system residual is not a bound on the continuum panel error.

5.5 Why E4, S3 and S4 need thickness limits

In ambient dimension $N \geq 3$ the Newtonian kernel is $|p - q|^{-(N-2)}$. A two-dimensional zero-thickness set is critical at $N = 4$; a one-dimensional wire is already critical at $N = 3$. Their exactly zero-thickness finite-energy equilibrium measures do not exist in the naive formulation.

The convention here is a spatially uniform small thickness/radius, followed by its limit to zero. The leading local energies penalize respectively

$$\log(1/\epsilon) \int_T \rho^2 dA, \quad \log(1/\epsilon) \int_{\partial T} \rho^2 ds, \quad \epsilon^{-1} \int_{\partial T} \rho^2 ds.$$

At total mass one, a spatially constant coefficient makes uniform density the leading minimizer. This yields E4= G and S3=S4=X(10) in the adopted uniform-thickness limit. The thin-wire analysis is consistent with [6]; a theorem obtained by varying a Riesz exponent [7] must not be substituted for a theorem about arbitrary geometric thickenings. The archive's softened-kernel experiment illustrates the mechanism, but is not a finite-radius conductor boundary solver. These limiting arguments have no certified finite-thickness remainder here.

As the triangular shape itself collapses, E2 tends to the mean of the segment's arcsine equilibrium measure, namely zero on $[-1, 1]$. E3 has uniform-line leading asymptotics in this critical dimension drop and the same horizontal limit, approached slowly. E4 has horizontal limit $x/3$; S3/S4 have limit zero. Thus the dimension of the field and the order of limits matter even when two endpoint means coincide.

5.6 M1: the least magnetic field inside a current loop

Let a uniform steady current traverse the triangular wire. In its plane, the field is normal to the plane and has one sign in the interior. Suppressing the common positive factor $\mu_0 I/(4\pi)$, an edge of length l , longitudinal projection s and inward distance d contributes

$$B_e(p) = \frac{1}{d} \left(\frac{s}{\sqrt{s^2 + d^2}} + \frac{l-s}{\sqrt{(l-s)^2 + d^2}} \right).$$

Add the three contributions and minimize. The overall current changes magnitude but not the minimizing point. The observation field diverges at the wire, so the minimum is interior.

A useful exact representation is

$$B(p) = \int_{\mathbb{R}^2 \setminus T} |q - p|^{-3} dq = \int_{S^1} \frac{d\theta}{R_p(\theta)}.$$

If $d_i(p)$ is the distance to side i and n_i its outward unit normal, then

$$R_p(u)^{-1} = \max_i \frac{(n_i \cdot u)_+}{d_i(p)}.$$

Each term is convex in p . In any nonzero direction of motion, one active side contributes a strictly positive second variation on an open set of ray directions. Integration makes the Hessian positive definite. Thus M1 is unique and analytic on nondegenerate triangles. Its finite-edge formulas involve only radicals, so stationary equations can be encoded by polynomial equations with positive auxiliary distances: this center is implicitly algebraic.

An exterior half-plane gives $B(p) \geq 2/d_i(p)$. For any interior trial point p_0 ,

$$\boxed{d_i(M1) \geq 2/B(p_0) \quad (i = 1, 2, 3).}$$

For the 6–9–13 triangle, the retained interval calculation proves $B(7.76, 1.57) < 2.857881120466727$, hence all three distances exceed 0.6998. At equilateral, the exact apex Jacobian is $\text{diag}(4/9, 2/9)$.

Nevertheless M1 is globally nonattractive. The bundle encloses two stationary roots by strict Krawczyk inclusion using outward-rounded intervals, proves their uniqueness by the convexity argument above, and bounds the vertex-motion dot product strictly below zero. The approximate dot product is -7.4180×10^{-8} . This is a validated finite counterexample, not a sign inferred from a plotted derivative.

5.7 E0: Equal-exit conformal center

Physical scenario and selector. Release Brownian particles from a common point. Choose that point so each side has first-exit probability one third.

$$E0 = F\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right), \quad h_1 = h_2 = h_3 = 1/3$$

Derivation and algorithm. Compute triangle angles and the normalized incomplete-beta map in high precision. The contour plots show squared harmonic-measure mismatch. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. An exact certificate is $0 < w_i(E0) < 2/3$ for all three normalized barycentric coordinates.

Attractivity. No violation was found in the 100-shape vertex-response screen. Global attractivity remains open here.

Distinguishing behavior. Thin limit: $C_x \rightarrow x$, $C_y/\epsilon \rightarrow 2/3$ for fixed $x < 1$; the endpoint $x = 1$ instead has vertical limit $1/2$.

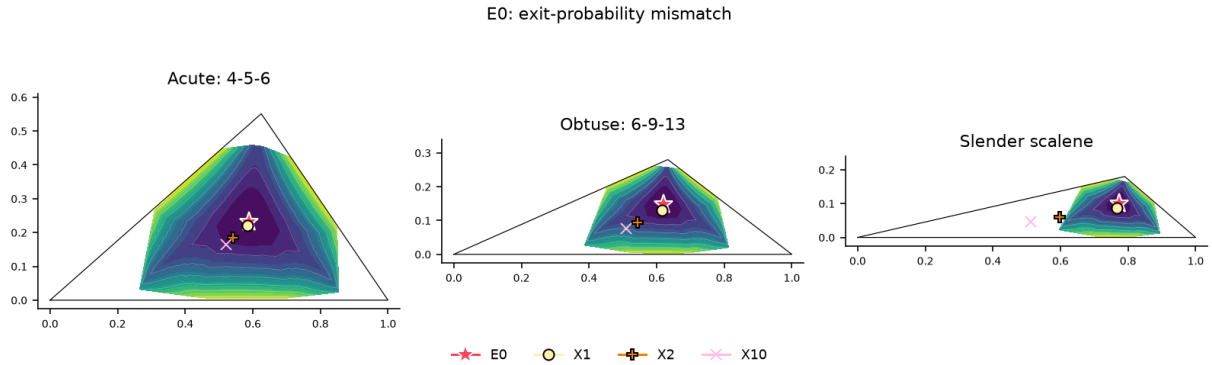
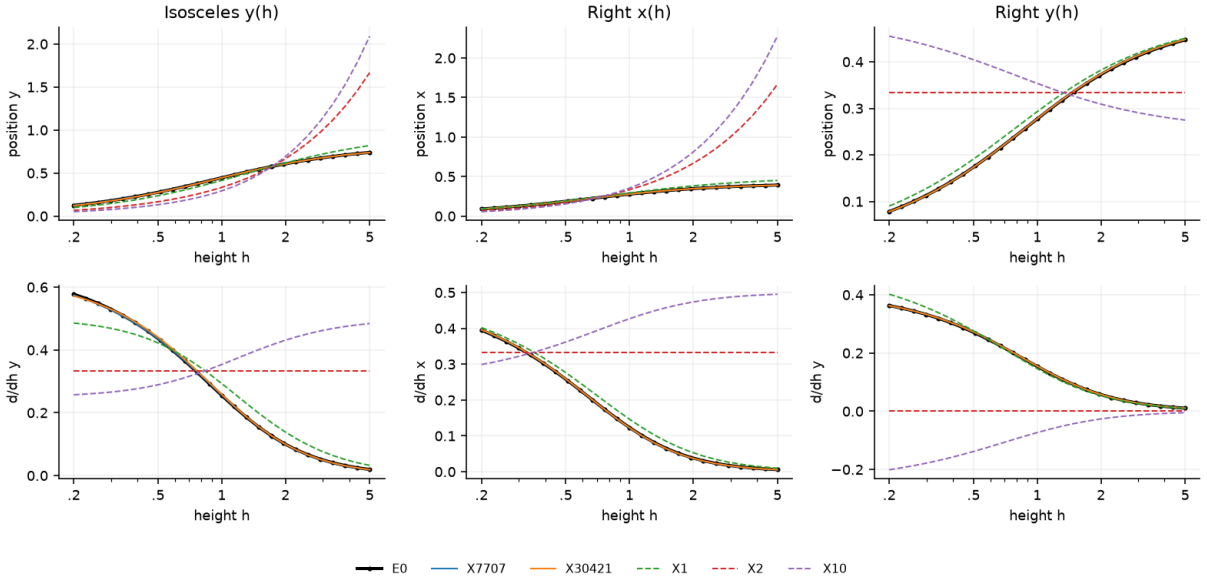


Figure 5.1: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures. White areas are outside the sampled conformal parameter patch, not zero-field regions.

Fingerprints and comparison evidence

E0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(7707)	0.042%	0.060%	0.042%	0.059%
X(30421)	0.085%	0.181%	0.081%	0.173%
X(483)	0.090%	0.141%	0.090%	0.136%

For the location winner, the stored family-derivative RMS discrepancy is 0.00081549 and response-margin RMS discrepancy is 0.0013551. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `E0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

5.8 E1: Inner conformal / least-capacity center

Physical scenario and selector. Choose the point of greatest conformal radius, the local scale of a conformal disk centered at that point.

$$E1 = F(z_*), \quad z_* = \frac{\alpha + i\sqrt{\alpha\beta/\gamma}}{\alpha + \beta}$$

Derivation and algorithm. Evaluate the explicit upper-half-plane maximizer and the incomplete-beta map. Plot conformal radius, not exit-probability mismatch. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

Attractivity. Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.02086913).

Distinguishing behavior. Thin correction: $C_x = x - \epsilon \operatorname{atanh}(x)/\pi + O(\epsilon^2)$ and $C_y/\epsilon \rightarrow 1/2$, with $x < 1$ fixed.

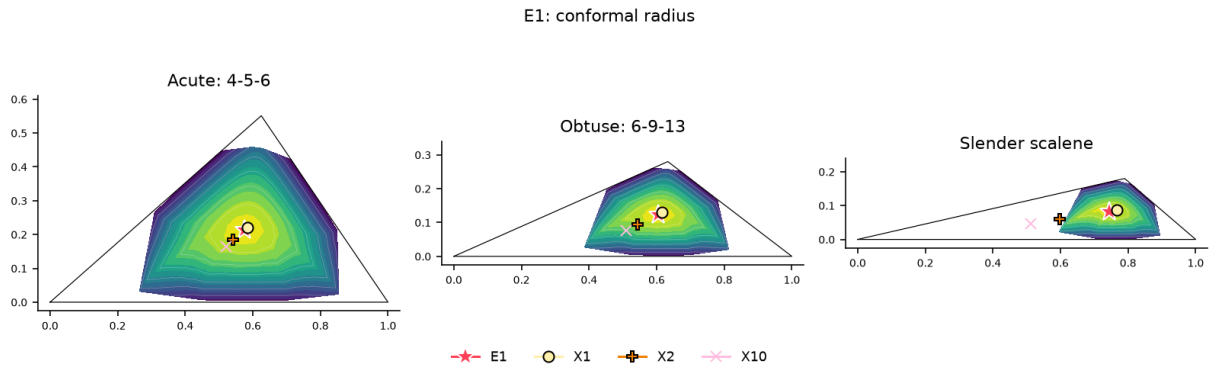
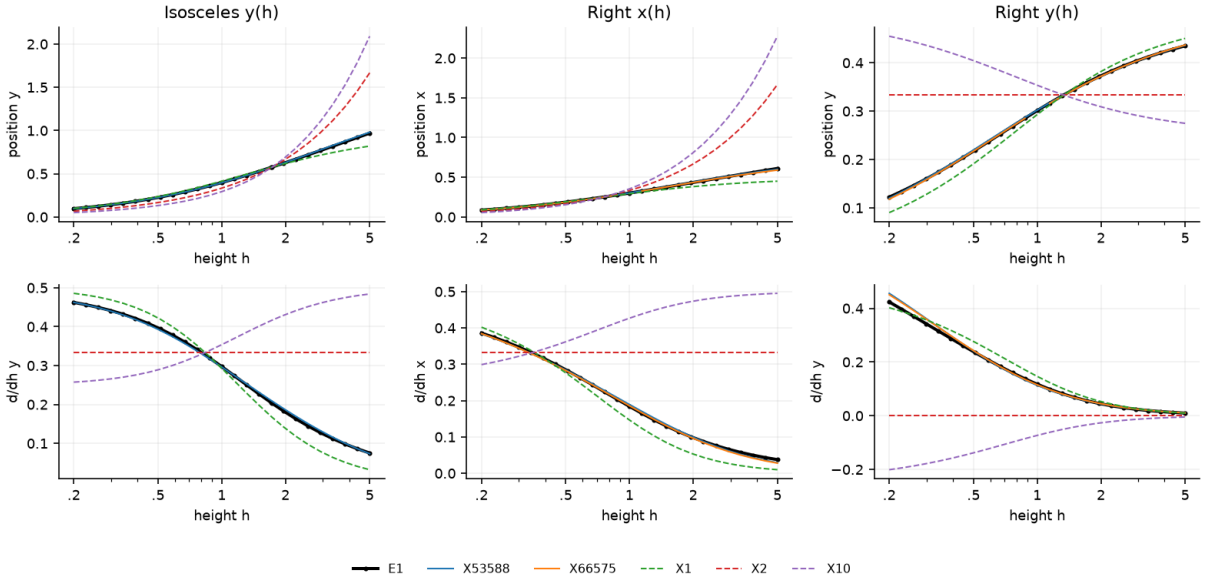


Figure 5.2: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures. White areas are outside the sampled conformal parameter patch, not zero-field regions.

Fingerprints and comparison evidence

E1 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(53588)	0.254%	0.623%	0.246%	0.531%
X(66575)	0.293%	0.730%	0.297%	0.671%
X(66560)	0.404%	0.682%	0.400%	0.583%

For the location winner, the stored family-derivative RMS discrepancy is 0.0079384 and response-margin RMS discrepancy is 0.0088346. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `E1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

5.9 E2: Outer conformal / planar charge centroid

Physical scenario and selector. Let planar logarithmic charge equilibrate on the triangular conductor and measure its charge centroid.

$$\mu_* = \arg \min_{\mu} \iint -\log |p - q| d\mu(p) d\mu(q), \quad E2 = \int p d\mu_*$$

Derivation and algorithm. Use cosine-graded boundary panels with exact logarithmic segment integrals, constant-potential collocation and unit-charge normalization. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

Attractivity. Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.04665096), independently evaluated with the exterior Schwarz-Christoffel map.

Distinguishing behavior. S2 equals E2 exactly. On a collapsing segment the equilibrium density is arcsine and its horizontal mean is zero. The atlas shows boundary charge, since the surface potential is constant.

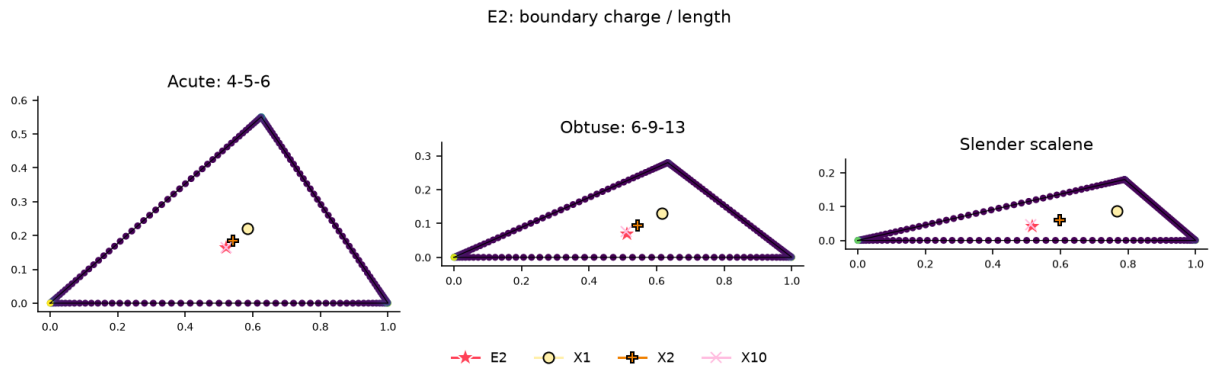
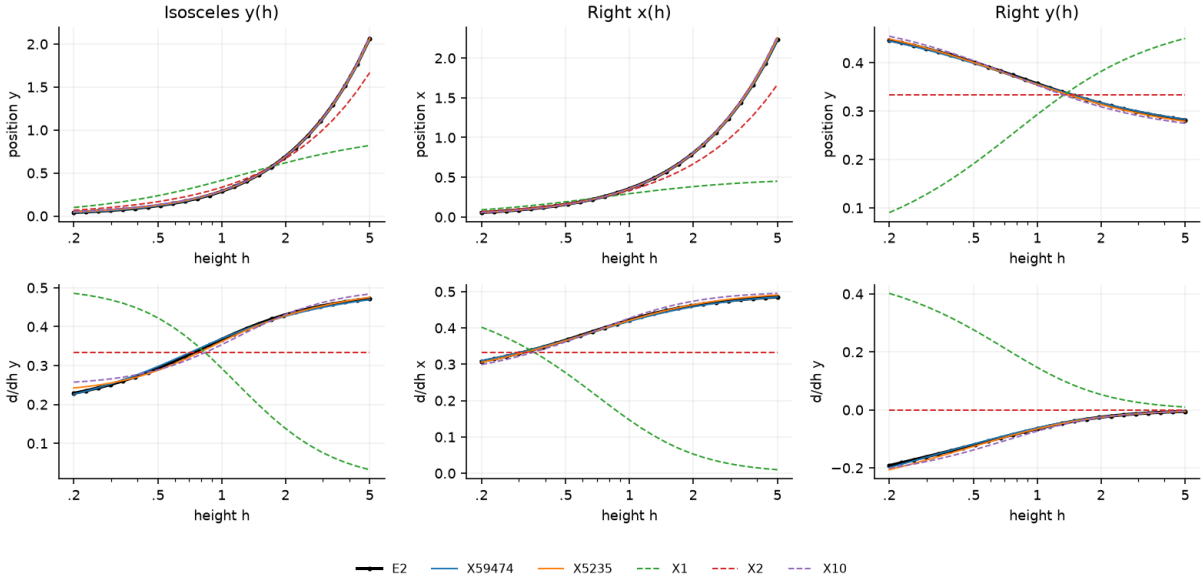


Figure 5.3: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures. A boundary density replaces an interior contour plot for this boundary-supported measure.

Fingerprints and comparison evidence

E2 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(59474)	0.134%	0.245%	0.130%	0.239%
X(5235)	0.222%	0.374%	0.214%	0.352%
X(49731)	0.228%	0.509%	0.211%	0.374%

For the location winner, the stored family-derivative RMS discrepancy is 0.0022842 and response-margin RMS discrepancy is 0.0044257. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `E2_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

5.10 E3: Three-dimensional plate charge centroid

Physical scenario and selector. Embed a thin triangular conducting plate in three-dimensional space and let a fixed total charge reach equilibrium.

$$\mu_* = \arg \min_{\mu} \iint |p - q|^{-1} d\mu(p) d\mu(q), \quad E3 = \int p d\mu_*$$

Derivation and algorithm. Use constant-charge triangular panels and exact self-panel potential integrals. Refine the panel subdivision and check charge positivity. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

Attractivity. Refined numerical evidence of nonattractivity: negative finite motions persist on positive-charge anisotropic meshes up to 3,180 panels. The finest halved-step ratio is -0.55181093. A continuum interval certificate remains open.

Distinguishing behavior. The thin longitudinal limit is a critical uniform-line asymptotic with horizontal mean zero; convergence can be logarithmically slow. Finite-panel densities near edges are illustrative, not pointwise certified singular coefficients.

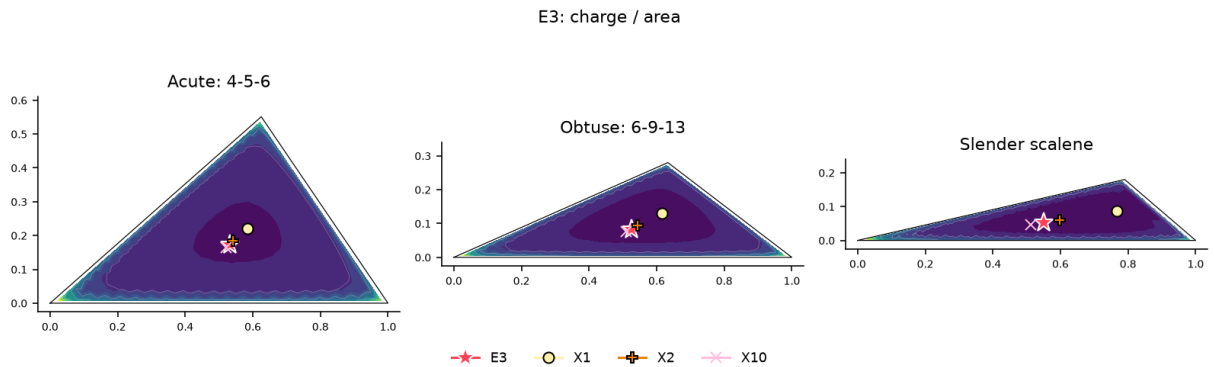
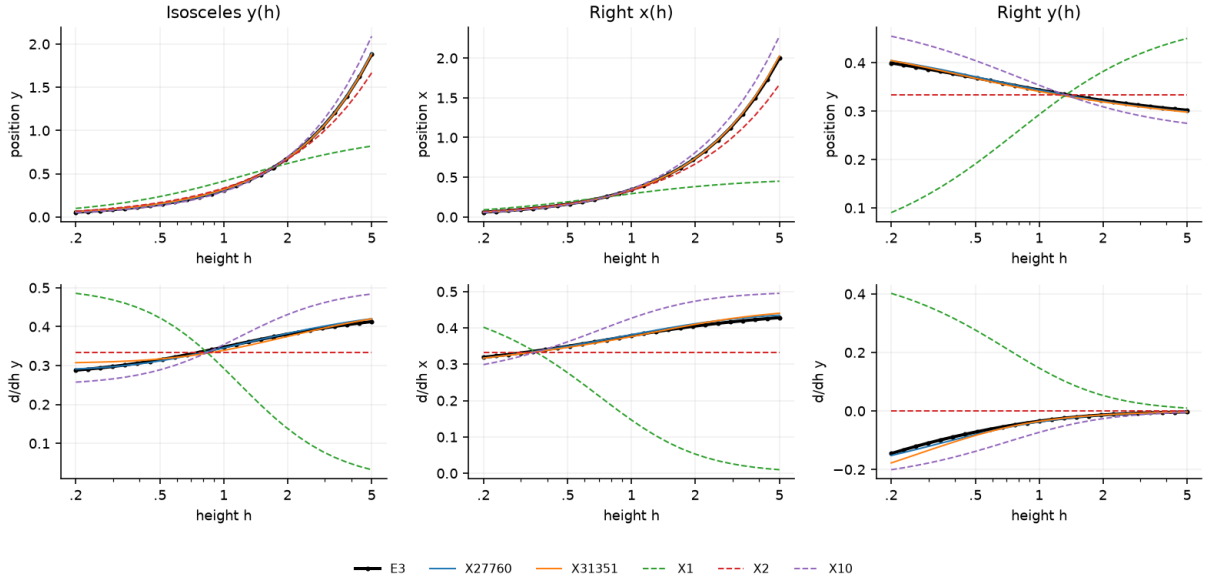


Figure 5.4: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

E3 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(27760)	0.268%	0.823%	0.436%	1.939%
X(31351)	0.380%	1.002%	0.404%	1.076%
X(46896)	0.521%	1.421%	0.792%	3.345%

For the location winner, the stored family-derivative RMS discrepancy is 0.0042283 and response-margin RMS discrepancy is 0.024585. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `E3_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

5.11 E4: Four-dimensional plate charge centroid

Physical scenario and selector. Take the charge equilibrium of a spatially uniform thickening, then send thickness/radius to zero. This limiting convention is part of the definition.

$$E4 = G = X(2)$$

Derivation and algorithm. Evaluate the exact area or perimeter first moment for the limiting center. The softened-kernel cutoff experiment is supplementary and is not a finite-radius physical conductor solve. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The limit lies strictly inside every nondegenerate triangle.

Attractivity. Strictly globally vertex-attractive through the exact uniform-thickness limiting identity with X2.

Distinguishing behavior. The zero-thickness set itself has zero Newtonian capacity in this ambient dimension. No finite-thickness error bound is asserted. The plotted density is the limiting uniform density.

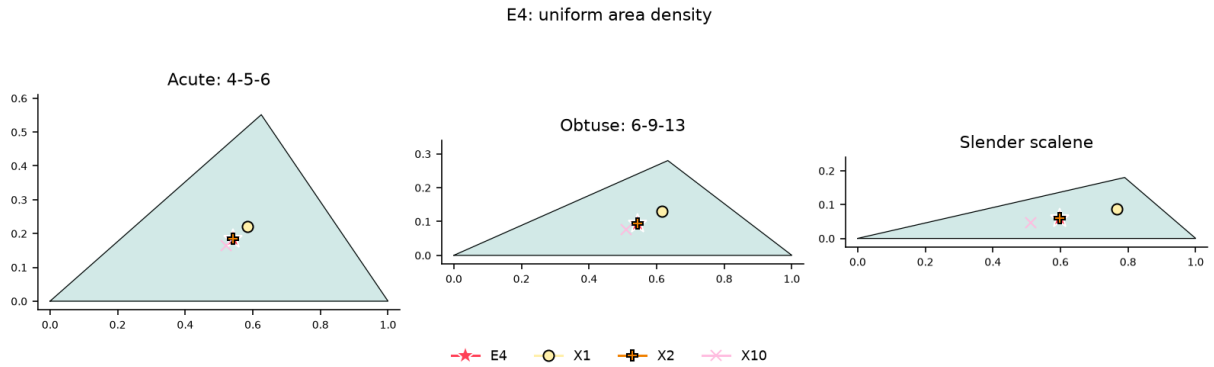
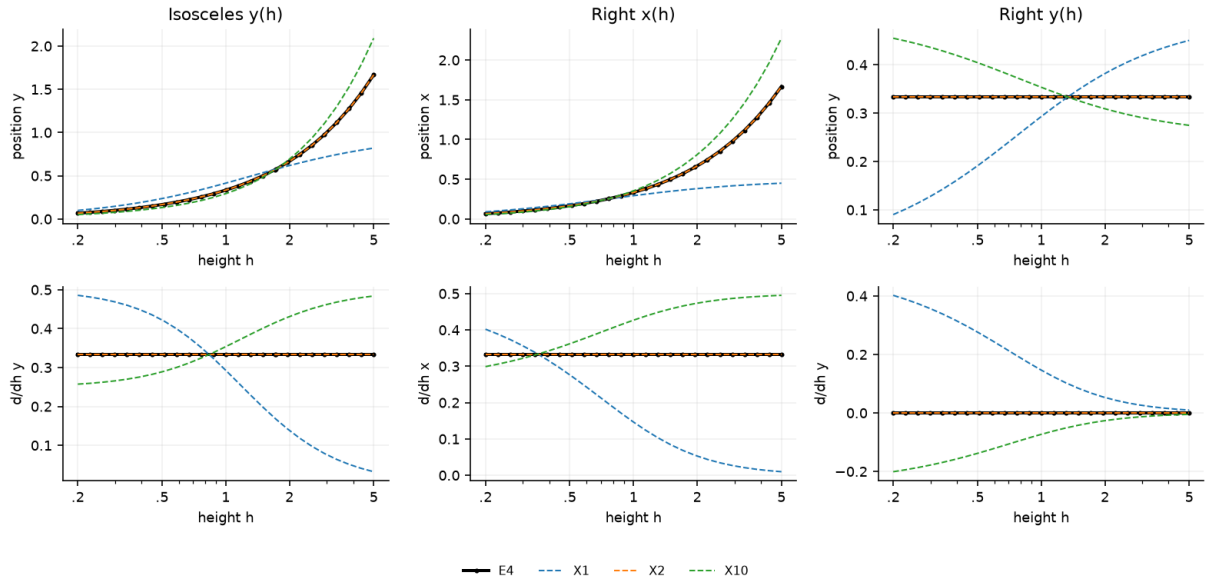


Figure 5.5: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

E4 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

This limiting center is exactly classical; no numerical closest-neighbor search is needed.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `E4_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

5.12 S3: 3-dimensional skeleton charge centroid

Physical scenario and selector. Take the charge equilibrium of a spatially uniform thickening, then send thickness/radius to zero. This limiting convention is part of the definition.

$$S3 = S = X(10)$$

Derivation and algorithm. Evaluate the exact area or perimeter first moment for the limiting center. The softened-kernel cutoff experiment is supplementary and is not a finite-radius physical conductor solve. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The limit lies strictly inside every nondegenerate triangle.

Attractivity. Strictly globally vertex-attractive through the exact uniform-thickness limiting identity with X10.

Distinguishing behavior. The zero-thickness set itself has zero Newtonian capacity in this ambient dimension. No finite-thickness error bound is asserted. The plotted density is the limiting uniform density.

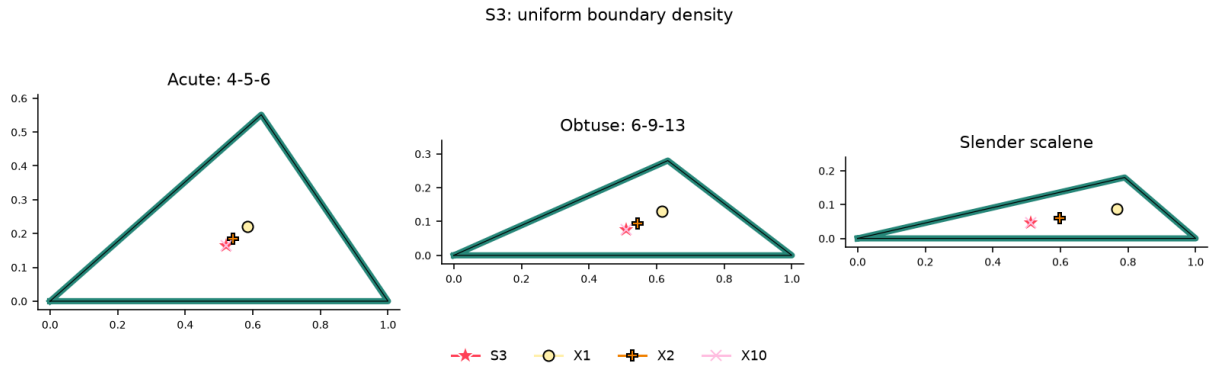
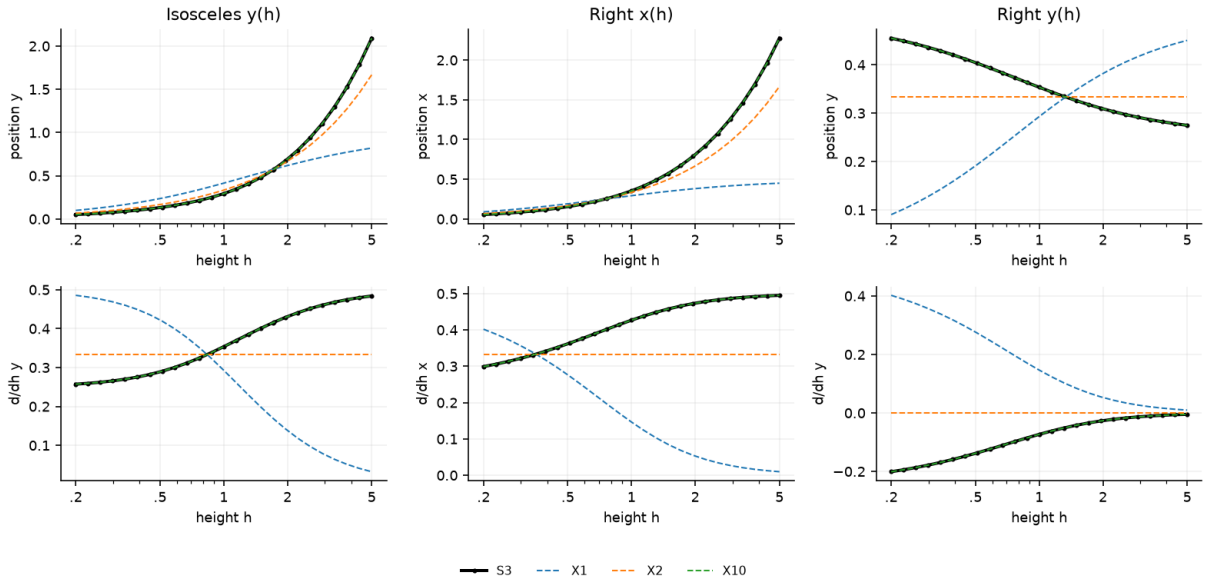


Figure 5.6: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures. A boundary density replaces an interior contour plot for this boundary-supported measure.

Fingerprints and comparison evidence

S3 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

This limiting center is exactly classical; no numerical closest-neighbor search is needed.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `S3_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

5.13 S4: 4-dimensional skeleton charge centroid

Physical scenario and selector. Take the charge equilibrium of a spatially uniform thickening, then send thickness/radius to zero. This limiting convention is part of the definition.

$$S4 = S = X(10)$$

Derivation and algorithm. Evaluate the exact area or perimeter first moment for the limiting center. The softened-kernel cutoff experiment is supplementary and is not a finite-radius physical conductor solve. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The limit lies strictly inside every nondegenerate triangle.

Attractivity. Strictly globally vertex-attractive through the exact uniform-thickness limiting identity with X10.

Distinguishing behavior. The zero-thickness set itself has zero Newtonian capacity in this ambient dimension. No finite-thickness error bound is asserted. The plotted density is the limiting uniform density.

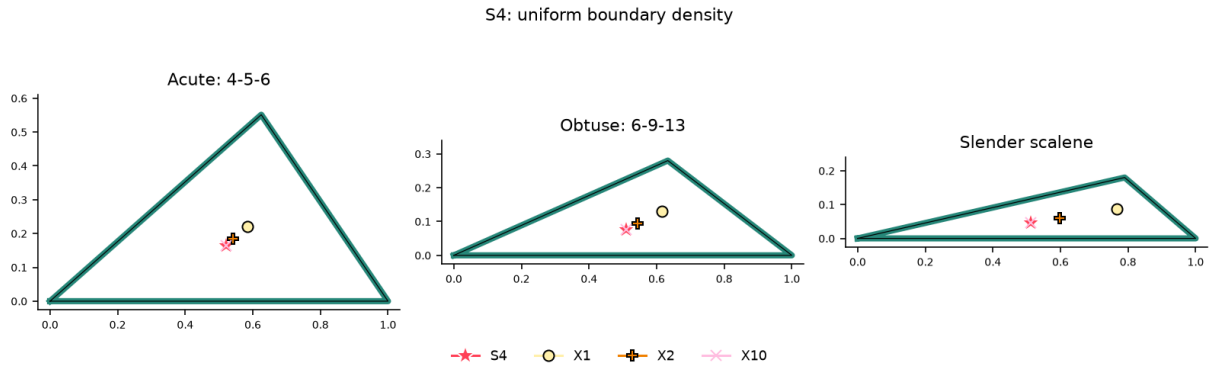
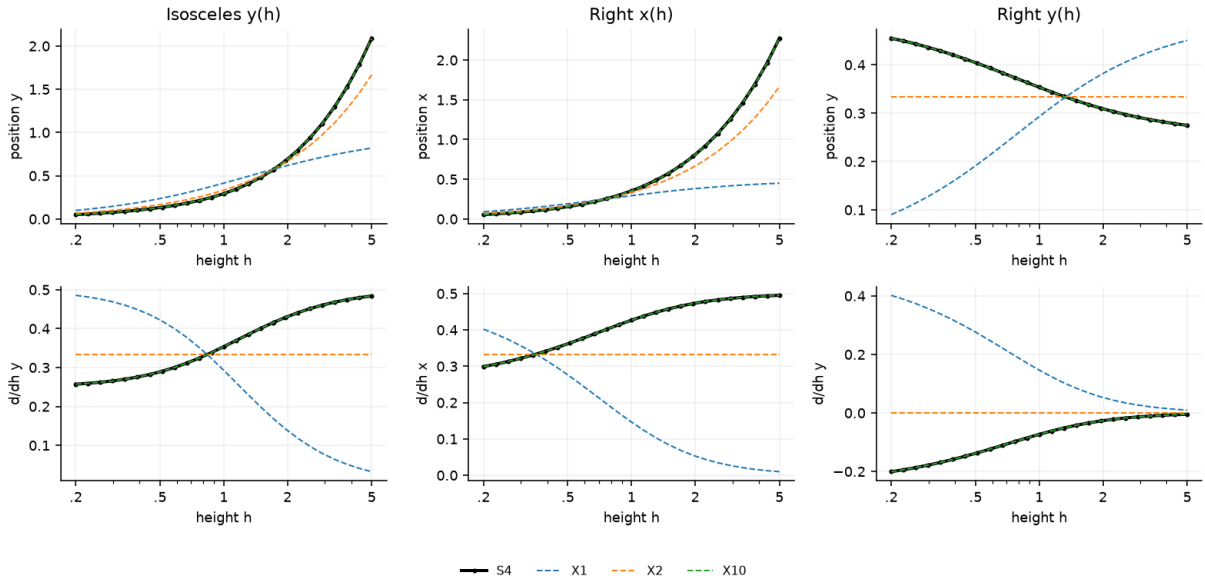


Figure 5.7: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures. A boundary density replaces an interior contour plot for this boundary-supported measure.

Fingerprints and comparison evidence

S4 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

This limiting center is exactly classical; no numerical closest-neighbor search is needed.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `S4_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

5.14 M1: Loop-field minimum

Physical scenario and selector. Pass a uniform steady current around the triangular wire and scan the normal magnetic field inside its plane.

$$M1 = \arg \min_{p \in T} \sum_e \frac{s_e / \sqrt{s_e^2 + d_e^2} + (l_e - s_e) / \sqrt{(l_e - s_e)^2 + d_e^2}}{d_e}$$

Derivation and algorithm. Minimize in barycentric coordinates, refine the stationary root, and use interval Krawczyk inclusion for the retained certificate. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Unique interior minimum; every side distance is at least $2/b(p_0)$ for any interior trial point p_0 , with b in the chapter normalization.

Attractivity. Not globally attractive: the earlier interval-certified negative finite vertex motion remains the rigorous witness.

Distinguishing behavior. At equilateral the exact response is $\text{diag}(4/9, 2/9)$. The center is implicitly algebraic, not proved transcendental. The plot uses log field to reveal contours despite boundary divergence.

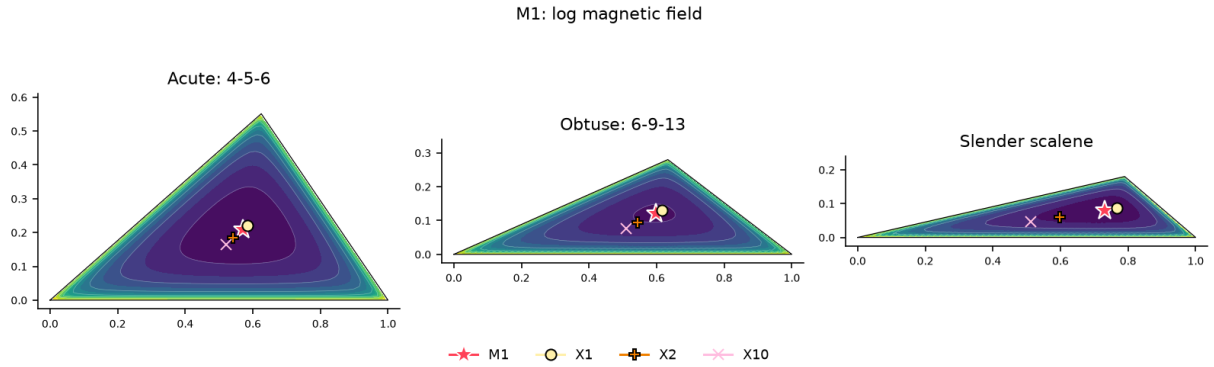
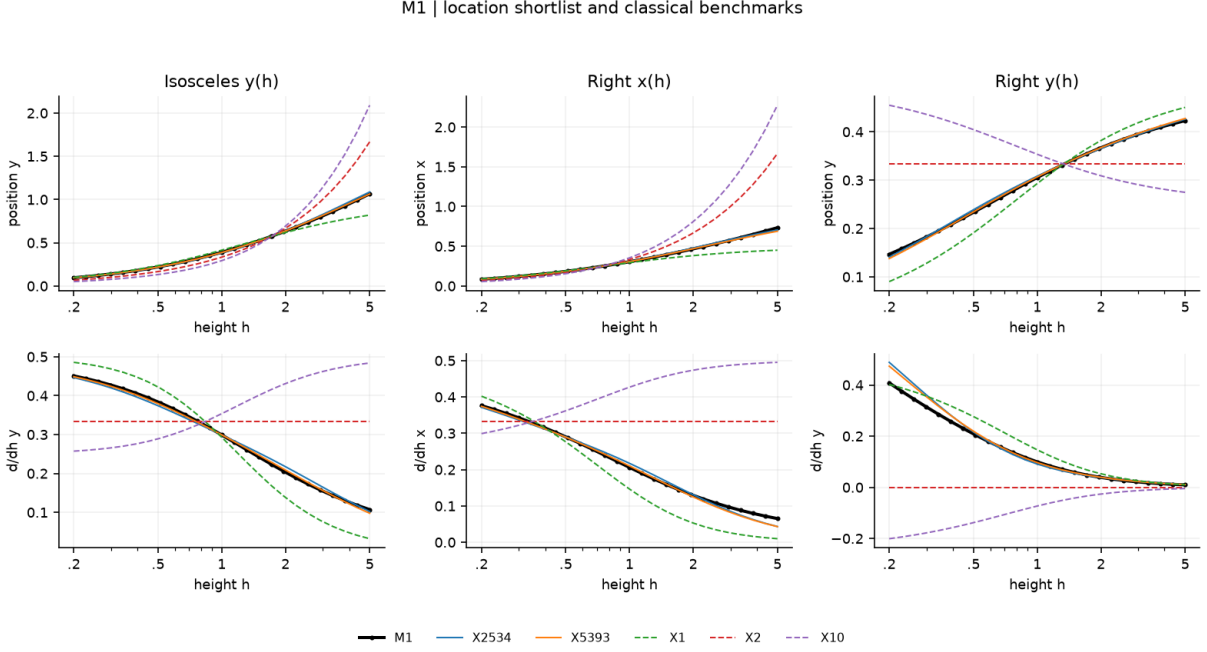


Figure 5.8: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(2534)	0.686%	1.839%	0.658%	1.441%
X(5393)	0.702%	1.864%	0.692%	1.579%
X(53841)	0.710%	1.832%	0.669%	1.390%

For the location winner, the stored family-derivative RMS discrepancy is 0.019438 and response-margin RMS discrepancy is 0.024371. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `M1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

Chapter 6

Fluids, membranes and temperature

6.1 F1: one Poisson equation, several laboratories

Consider steady unidirectional flow through a long straight duct with triangular cross-section. The velocity has only an axial component $u(x, y)$. In the low-Reynolds-number, fully developed regime, axial force balance reduces to

$$\mu\Delta u = dp/dz, \quad u|_{\partial T} = 0.$$

The no-slip wall supplies the Dirichlet condition. Dividing by the positive forcing scale gives $-\Delta u = 1$. The fastest-flow location is F1, the maximum of this torsion function.

The same equation follows for a uniformly tensioned membrane under a small constant transverse pressure: vertical force balance is $-T_0\Delta u = q$. A torsion stress function for a prismatic bar also satisfies a Poisson equation, although its maximum is not automatically the point of greatest shear stress; shear depends on its gradient. Finally, Dynkin's formula for Brownian motion with generator Δ gives $-\Delta\mathbb{E}_p\tau = 1$. F1 maximizes mean exit time. The common PDE explains these equivalences without identifying different physical observables by analogy alone.

Multiply by a zero-boundary test function v and integrate by parts:

$$\int_T \nabla u \cdot \nabla v = \int_T v.$$

The associated strictly convex energy is $\frac{1}{2} \int |\nabla u|^2 - \int u$. This gives the finite-element stiffness solve. The maximum principle yields positive interior values and zero boundary values. The known concavity structure of torsion on convex domains underlies its well-behaved interior maximum; finite discretization still needs a location search and refinement.

6.2 H: a fixed-time survival center

Start a homogeneous triangle at temperature one and hold every boundary point at zero. With diffusivity κ ,

$$\partial_t u = \kappa\Delta u, \quad u|_{\partial T} = 0, \quad u(p, 0) = 1.$$

Fix $\Delta = \kappa t/L^2$ and define $H(\Delta) = \arg \max u(p, t)$. The dimensionless choice is essential: holding physical time fixed while scaling the triangle would not give a scale-invariant center.

In eigenfunctions,

$$u(p, t) = \sum_j e^{-\Delta L^2 \lambda_j} \left(\int_T \psi_j \right) \psi_j(p).$$

These coefficients contain integrals of ψ_j , unlike the squared eigenfunctions in the thermal quantum density. For late times, factor out $e^{-\Delta L^2 \lambda_0}$ to avoid numerical underflow; that scalar does not move the maximum. Truncate the expansion, repeat with more modes, and separately refine the spatial mesh. Early-time fields can be extremely flat, so a small field error need not imply a small maximum-location error.

As $\Delta \downarrow 0$, the heat center approaches the incenter; as $\Delta \rightarrow \infty$, it approaches Q0m. At time zero the field is constant and there is no unique maximum. The identity

$$A^{-1}1 = \int_0^\infty e^{-sA}1 ds$$

relates torsion to the integral of heat fields. Taking a maximum does not commute with this integral. Thus F1 need not equal any specified H, and neither is Q0.

The sampled times are 0.01, 0.03, 0.1, 0.3, 1. A separate 6–9–13 heat path includes earlier times, but the 0.001 maximum was unresolved and is omitted from the converged path plot. It remains in the raw data with its error flag. For fixed positive Δ in a sufficiently thin triangle, the rapidly growing transverse eigenvalue gaps make ground-mode localization dominate; a simultaneous short-time/thin limit is a different regime.

6.3 Nonlinear tension: p-torsion maxima and means

Suppose the membrane's stored energy is proportional to the p th power of its slope. After choosing a displacement scale, the energy is

$$\mathcal{E}_p(u) = \frac{1}{p} \int_T |\nabla u|^p - \int_T u, \quad u|_{\partial T} = 0, \quad p > 1.$$

Its first variation yields $-\operatorname{div}(|\nabla u|^{p-2} \nabla u) = 1$. Strict convexity gives a unique field. Define $\operatorname{Mem}(p)$ by its maximum, and $\operatorname{MemMean}(p)$ by $\int pu(p)dp / \int u(p)dp$ (the integration variable in this expression is the spatial point). We use $p = 1.5, 3, 4$; at exponent 2 the mode is F1 and the mean is the zero-absorption density center.

The solver minimizes the discrete energy in a boundary-bubble basis. It uses a declared 10^{-18} addition to squared gradient norms to regularize differentiation at zero slope; quadrature and residual checks accompany it. Maxima for the lower exponents can be considerably more sensitive than means. Some reported maximum jets are unresolved and are not analytic constants.

In a narrow strip of height H , transverse integration of the solution scales as $H^{1+p/(p-1)}$. Integrating this power of the triangular width profile predicts the horizontal thin limit

$$\operatorname{MemMean}(p)_x \longrightarrow \frac{2p-1}{4p-3}x.$$

This is a leading strip asymptotic, not a certified remainder estimate. At large exponent the maximum approaches the distance-function maximum, X(1). All finite-exponent positive means are strictly interior, but none of those location statements proves vertex attractivity.

6.4 F1: Fluid torsion / linear membrane maximum

Physical scenario and selector. Measure maximum axial velocity in a triangular Poiseuille duct, maximum deflection of a uniformly tensioned loaded membrane, or maximal Brownian mean exit time.

$$-\Delta u = 1, \quad u|_{\partial T} = 0, \quad F1 = \arg \max_T u$$

Derivation and algorithm. Solve the quadratic FEM Poisson system with exact load integration and maximize the piecewise quadratic field. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive Dirichlet field has its maximum in the interior.

Attractivity. Negative numerical evidence, with unresolved refinement sensitivity. The inherited finite-element witness is negative at subdivisions 128, 192 and 256. Independent Ritz degrees 20 and 32 retain both negative steps, but degree 44 changes the half-step sign. The data do not supply a converged, validated continuum counterexample; global nonattractivity is not certified.

Distinguishing behavior. This is a maximum, whereas $\text{Abs}(0)$ is the mean of the same field. In thin triangles the maximum approaches the widest section. The heat integral identity relates fields but does not identify their maxima. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum. Also the shallow fixed-excess-area sheet minimum and the uniform-tension membrane maximum.

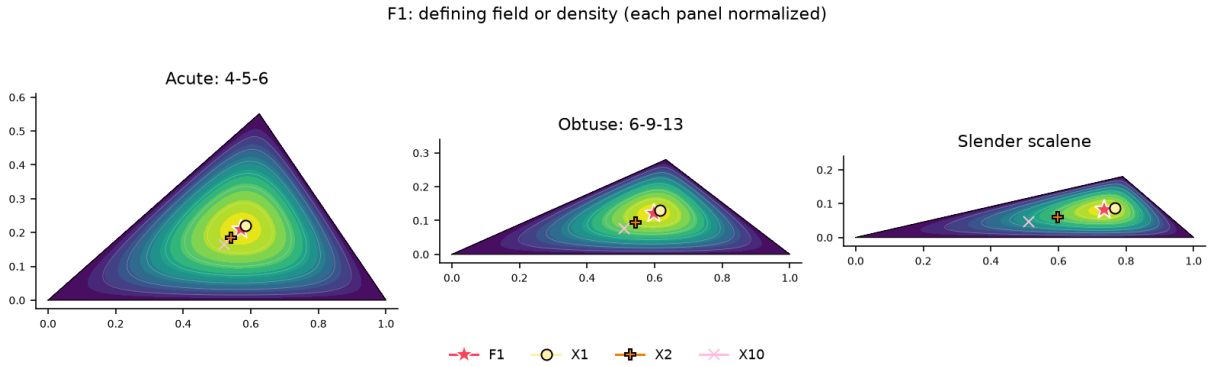
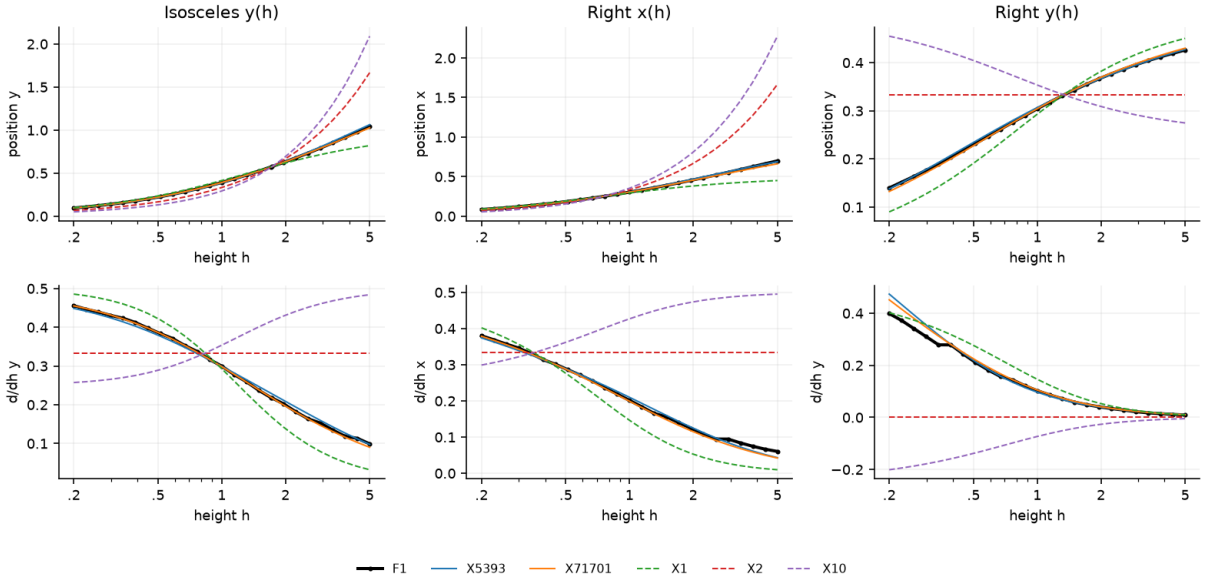


Figure 6.1: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

F1 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(5393)	0.422%	1.013%	0.405%	0.826%
X(71701)	0.507%	1.151%	0.514%	1.090%
X(2534)	0.546%	0.989%	0.514%	0.847%

For the location winner, the stored family-derivative RMS discrepancy is 0.017459 and response-margin RMS discrepancy is 0.049341. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `F1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.5 H0.01: Heat maximum at dimensionless time 0.01

Physical scenario and selector. Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.01) = \arg \max_T e^{-0.01L^2 A} 1, \quad A = -\Delta_D$$

Derivation and algorithm. Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive heat field is zero at the boundary at every positive time, so its maximum is interior.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

Distinguishing behavior. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.

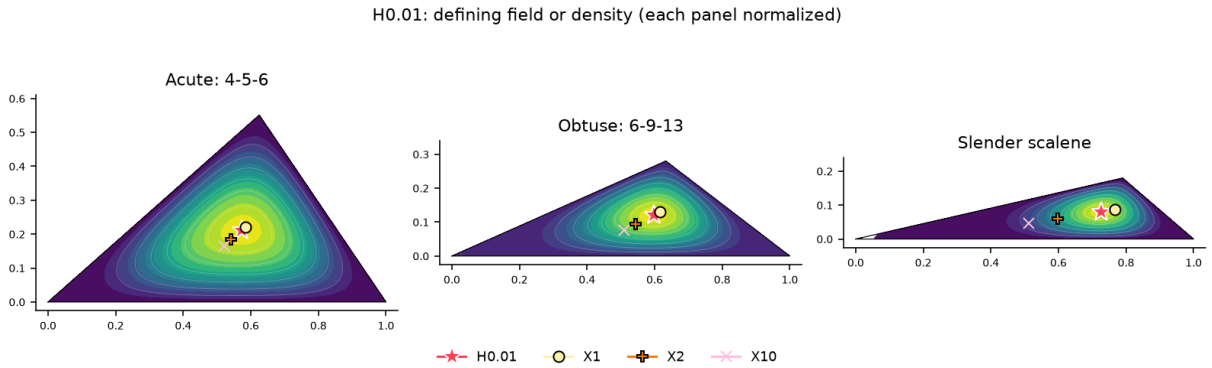
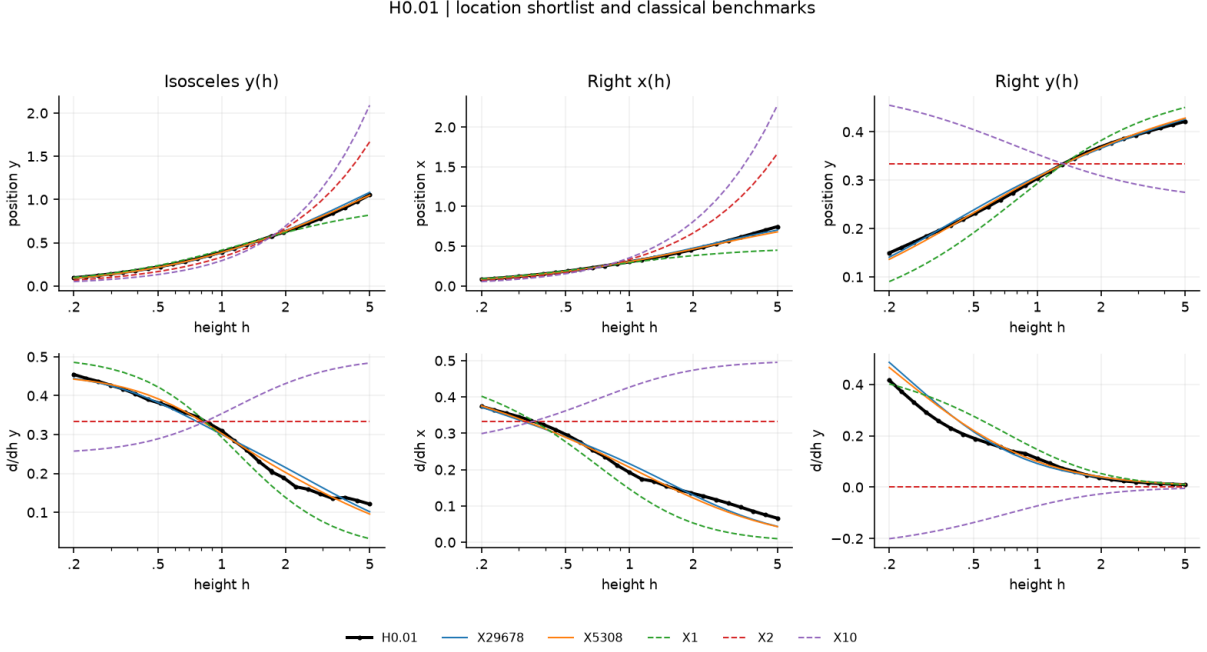


Figure 6.2: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(29678)	0.771%	1.997%	0.699%	1.367%
X(5308)	0.790%	2.438%	0.738%	1.592%
X(56032)	0.795%	2.682%	0.729%	1.547%

For the location winner, the stored family-derivative RMS discrepancy is 0.024125 and response-margin RMS discrepancy is 0.045068. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `H0.01_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.6 H0.03: Heat maximum at dimensionless time 0.03

Physical scenario and selector. Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.03) = \arg \max_T e^{-0.03L^2 A} 1, \quad A = -\Delta_D$$

Derivation and algorithm. Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive heat field is zero at the boundary at every positive time, so its maximum is interior.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

Distinguishing behavior. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.

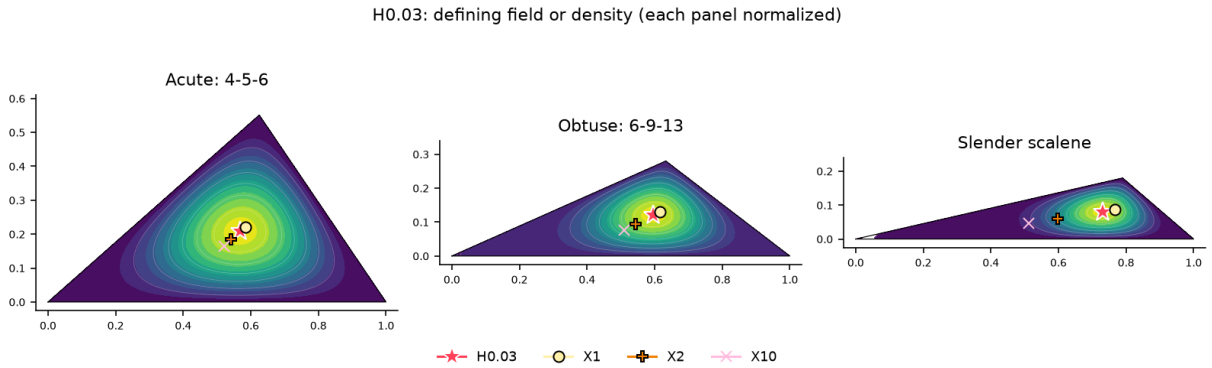
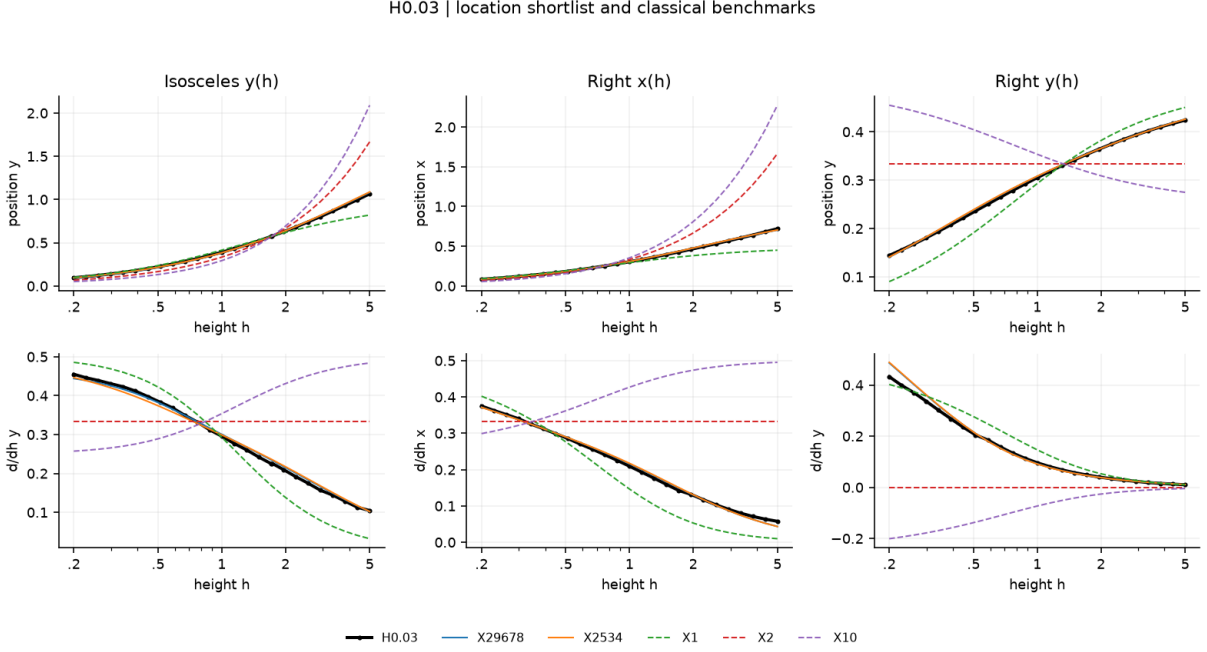


Figure 6.3: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(29678)	0.656%	1.997%	0.546%	1.274%
X(2534)	0.672%	1.706%	0.663%	1.498%
X(5393)	0.684%	1.730%	0.687%	1.518%

For the location winner, the stored family-derivative RMS discrepancy is 0.011922 and response-margin RMS discrepancy is 0.043986. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `H0.03_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.7 H0.1: Heat maximum at dimensionless time 0.1

Physical scenario and selector. Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.1) = \arg \max_T e^{-0.1L^2 A} 1, \quad A = -\Delta_D$$

Derivation and algorithm. Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive heat field is zero at the boundary at every positive time, so its maximum is interior.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

Distinguishing behavior. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.

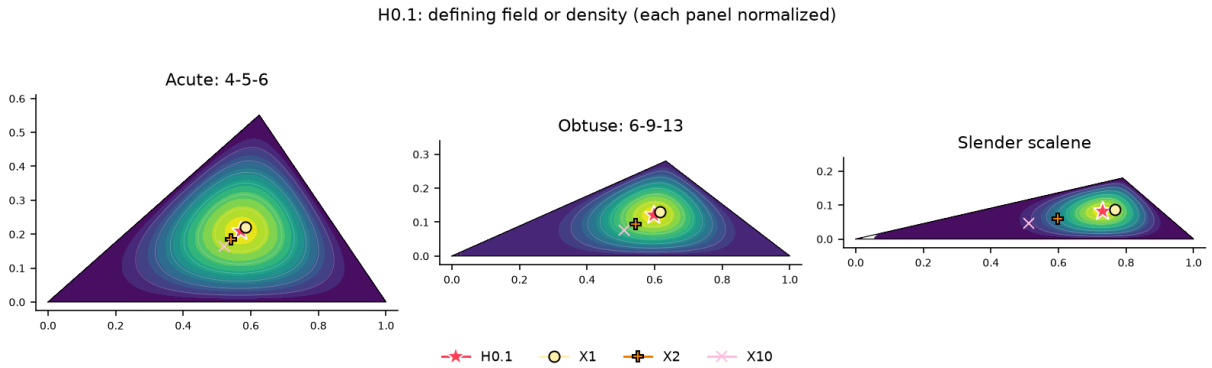
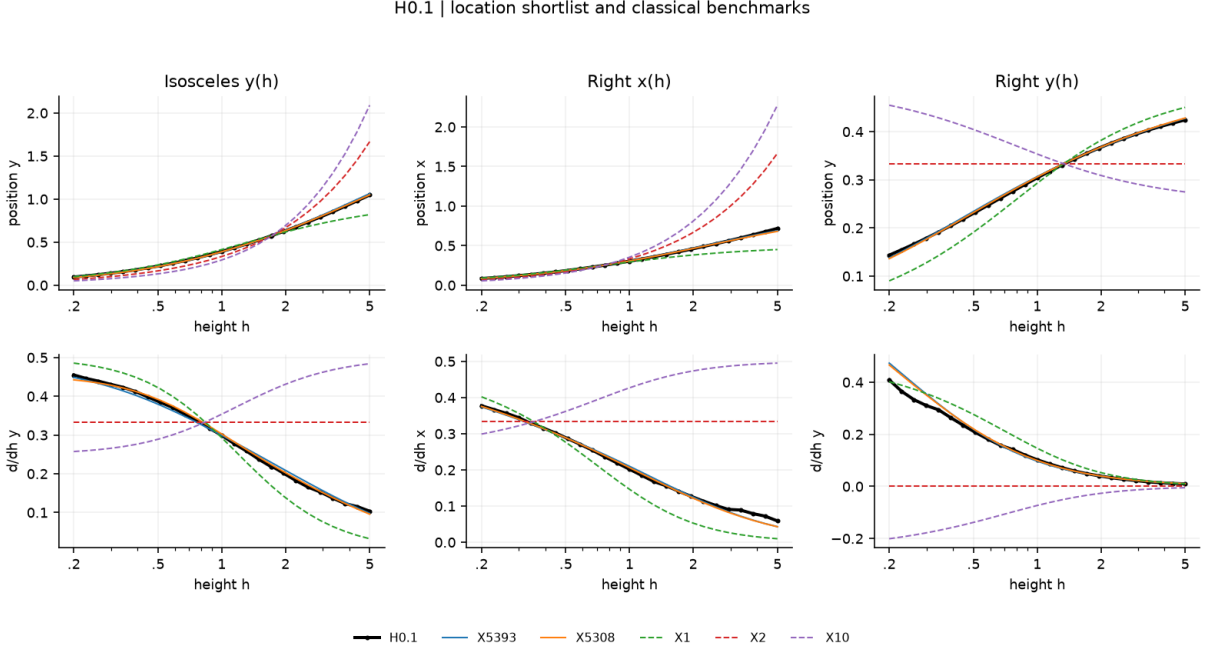


Figure 6.4: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(5393)	0.699%	1.730%	0.699%	1.518%
X(5308)	0.719%	2.438%	0.615%	1.375%
X(29678)	0.721%	1.997%	0.613%	1.274%

For the location winner, the stored family-derivative RMS discrepancy is 0.017266 and response-margin RMS discrepancy is 0.033028. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `H0.1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.8 H0.3: Heat maximum at dimensionless time 0.3

Physical scenario and selector. Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(0.3) = \arg \max_T e^{-0.3L^2 A} 1, \quad A = -\Delta_D$$

Derivation and algorithm. Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive heat field is zero at the boundary at every positive time, so its maximum is interior.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

Distinguishing behavior. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.

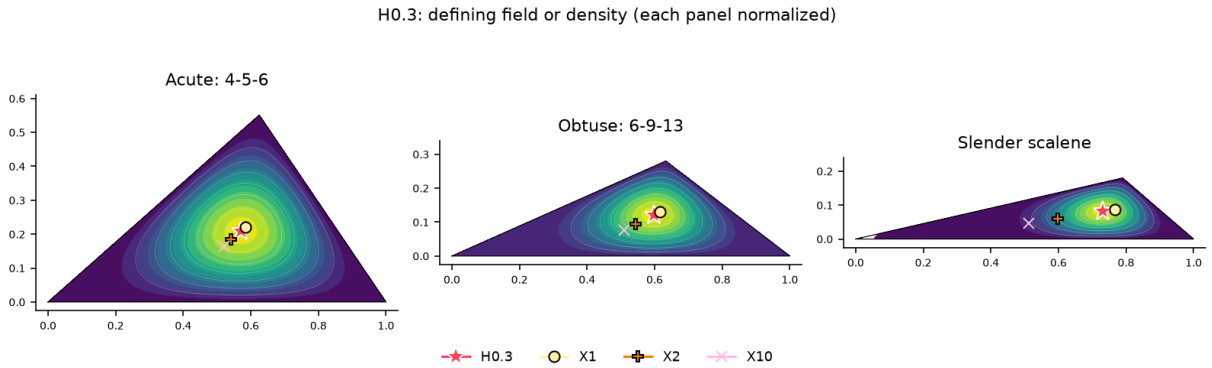
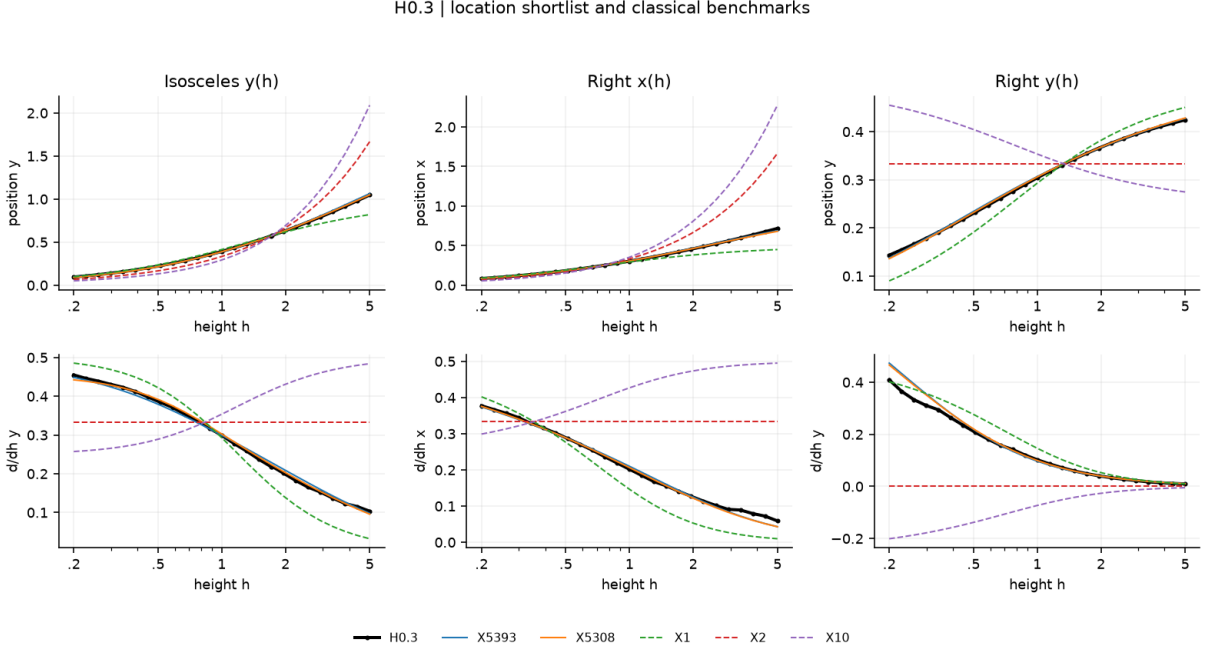


Figure 6.5: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(5393)	0.699%	1.730%	0.699%	1.518%
X(5308)	0.720%	2.438%	0.615%	1.375%
X(29678)	0.721%	1.997%	0.614%	1.274%

For the location winner, the stored family-derivative RMS discrepancy is 0.017284 and response-margin RMS discrepancy is 0.033033. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `H0.3_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.9 H1.0: Heat maximum at dimensionless time 1.0

Physical scenario and selector. Start with uniform temperature one, hold the boundary at zero, and locate the hottest remaining point at the specified dimensionless time.

$$H(1.0) = \arg \max_T e^{-1.0L^2A}1, \quad A = -\Delta_D$$

Derivation and algorithm. Use the Dirichlet spectral expansion with initial-data projection. Compare full and half spectra and refined meshes; factor out the ground exponential at late time. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive heat field is zero at the boundary at every positive time, so its maximum is interior.

Attractivity. The sampled heat-family finite witnesses have persistent negative responses on refined meshes; these are numerical, not interval-certified, counterexamples.

Distinguishing behavior. The family endpoints are $H(0^+) = X(1)$ and $H(\infty) = Q0m$. Keep L^2 shape dependent when differentiating. Late-time curve overlap with $Q0m$ is expected, not an identity at finite time.

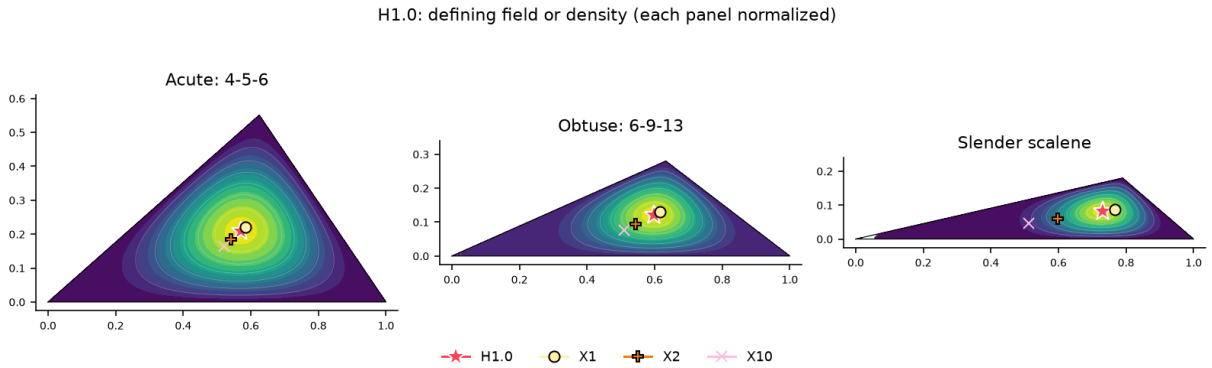
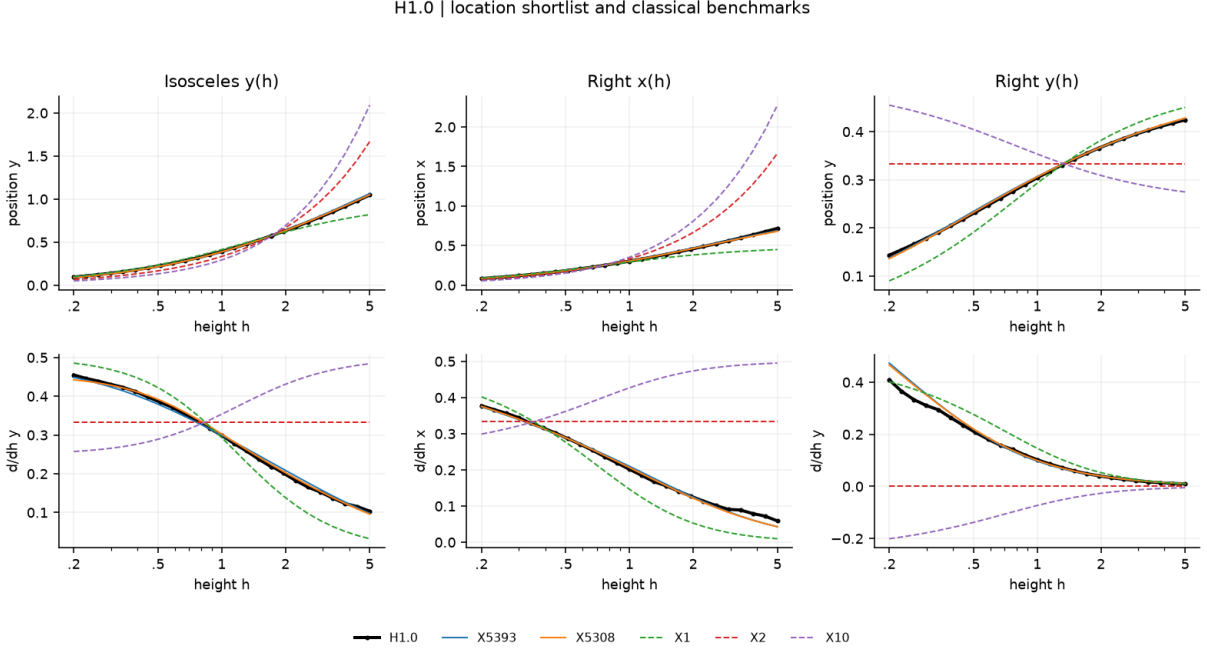


Figure 6.6: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(5393)	0.699%	1.730%	0.699%	1.518%
X(5308)	0.720%	2.438%	0.615%	1.375%
X(29678)	0.721%	1.997%	0.614%	1.274%

For the location winner, the stored family-derivative RMS discrepancy is 0.017284 and response-margin RMS discrepancy is 0.033033. These have their own normalizations and are not percentages of location error.

Resolution record. The joint electromagnetic/heat targets were met on 94/100 training and 37/40 holdout shapes. Exact limiting identities are separate from these numerical targets. Per-model mesh/panel changes are retained.

Reproducibility. The bundle directory `output/electro_heat/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `H1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.10 Mem1.5: Nonlinear membrane maximum, exponent 1.5

Physical scenario and selector. Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Locate the greatest transverse deflection.

$$-\operatorname{div}(|\nabla u|^{-0.5}\nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \arg \max u$$

Derivation and algorithm. Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Unique interior maximum by Borrelli, Mosconi and Squassina, arXiv:2201.12788, Theorem 1.1 (planar convex p-torsion, including triangles).

Attractivity. Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

Distinguishing behavior. The leading strip prediction for the mean is $C_x \rightarrow [(2p-1)/(4p-3)]x$, with exponent p fixed. No certified thin remainder is supplied.

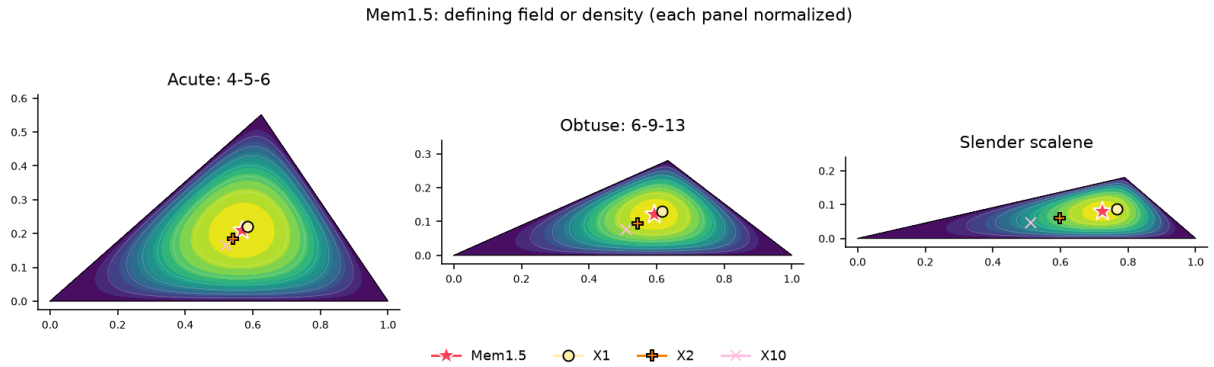
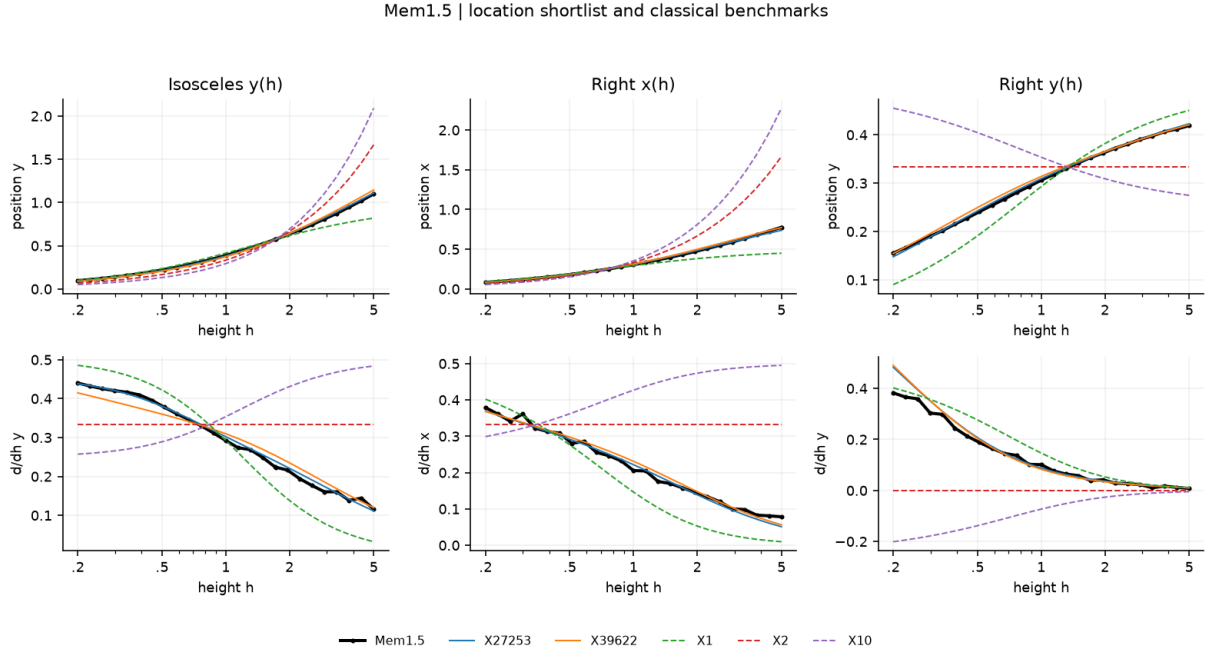


Figure 6.7: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(27253)	0.788%	2.545%	0.670%	1.582%
X(39622)	0.847%	2.277%	0.788%	1.611%
X(24936)	0.872%	2.649%	0.728%	1.891%

For the location winner, the stored family-derivative RMS discrepancy is 0.022755 and response-margin RMS discrepancy is 0.048015. These have their own normalizations and are not percentages of location error.

Resolution record. 89/100 training and 37/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 1.095% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Mem1.5_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.11 Mem3.0: Nonlinear membrane maximum, exponent 3.0

Physical scenario and selector. Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Locate the greatest transverse deflection.

$$-\operatorname{div}(|\nabla u|^1 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \arg \max u$$

Derivation and algorithm. Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Unique interior maximum by Borrelli, Mosconi and Squassina, arXiv:2201.12788, Theorem 1.1 (planar convex p-torsion, including triangles).

Attractivity. Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

Distinguishing behavior. The leading strip prediction for the mean is $C_x \rightarrow [(2p-1)/(4p-3)]x$, with exponent p fixed. No certified thin remainder is supplied.

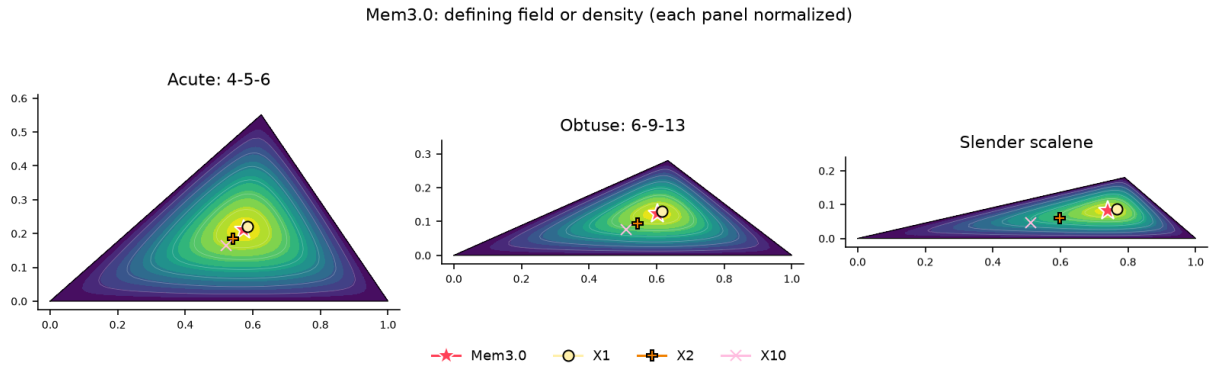
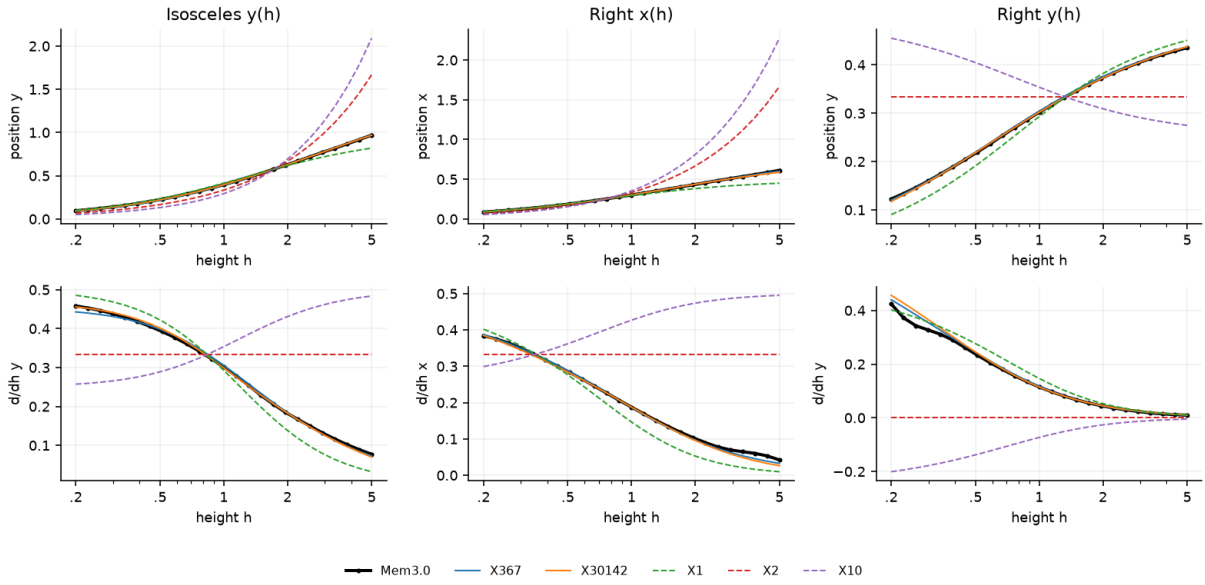


Figure 6.8: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Mem3.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(367)	0.333%	0.864%	1.007%	6.100%
X(30142)	0.401%	1.041%	1.050%	6.268%
X(29624)	0.406%	1.054%	1.064%	6.269%

For the location winner, the stored family-derivative RMS discrepancy is 0.011244 and response-margin RMS discrepancy is 0.051843. These have their own normalizations and are not percentages of location error.

Resolution record. 85/100 training and 32/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.552% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Mem3.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.12 Mem4.0: Nonlinear membrane maximum, exponent 4.0

Physical scenario and selector. Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Locate the greatest transverse deflection.

$$-\operatorname{div}(|\nabla u|^2 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \arg \max u$$

Derivation and algorithm. Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Unique interior maximum by Borrelli, Mosconi and Squassina, arXiv:2201.12788, Theorem 1.1 (planar convex p-torsion, including triangles).

Attractivity. Global vertex attractivity remains open. The retained maximum-response witnesses change sign or remain quadrature/degree sensitive. They are not accepted as continuum counterexamples.

Distinguishing behavior. The leading strip prediction for the mean is $C_x \rightarrow [(2p-1)/(4p-3)]x$, with exponent p fixed. No certified thin remainder is supplied.

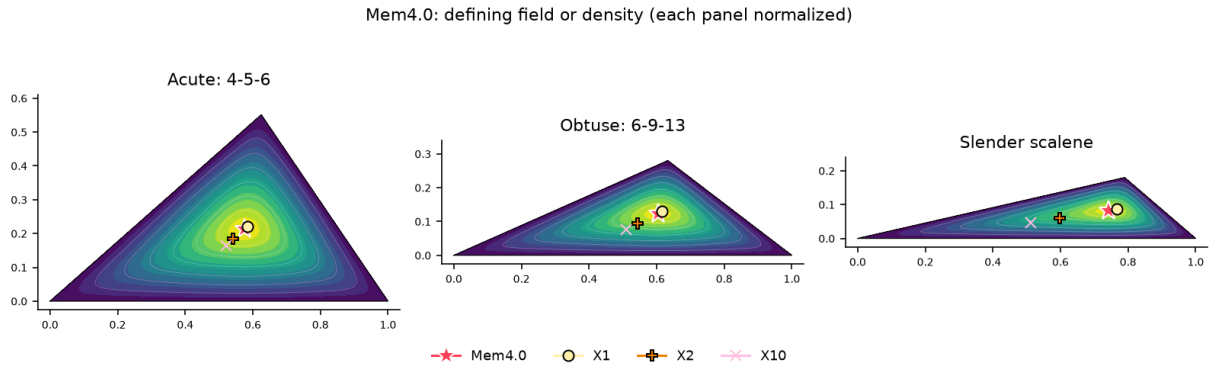
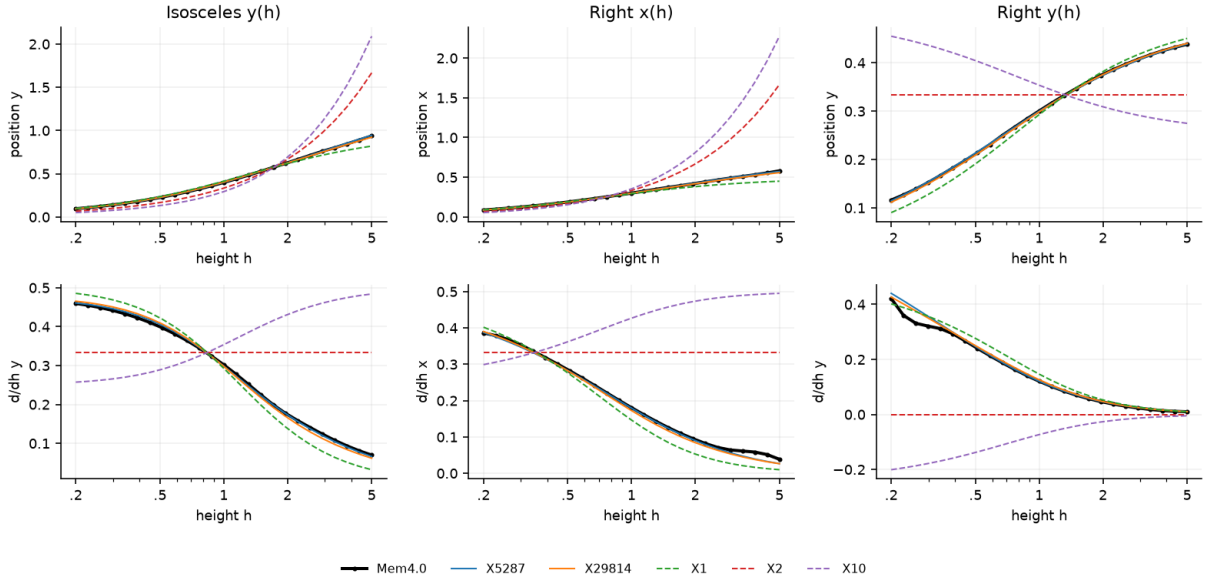


Figure 6.9: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Mem4.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(5287)	0.325%	0.966%	1.033%	6.322%
X(29814)	0.330%	0.990%	1.044%	6.325%
X(29624)	0.356%	0.960%	1.041%	6.322%

For the location winner, the stored family-derivative RMS discrepancy is 0.0135 and response-margin RMS discrepancy is 0.055773. These have their own normalizations and are not percentages of location error.

Resolution record. 76/100 training and 28/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.641% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Mem4.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.13 MemMean1.5: Nonlinear membrane density mean, exponent 1.5

Physical scenario and selector. Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Measure the deflection-volume centroid.

$$-\operatorname{div}(|\nabla u|^{-0.5}\nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \int pu / \int u$$

Derivation and algorithm. Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Positive density proves strict interiority of the mean.

Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Distinguishing behavior. The leading strip prediction for the mean is $C_x \rightarrow [(2p-1)/(4p-3)]x$, with exponent p fixed. No certified thin remainder is supplied.

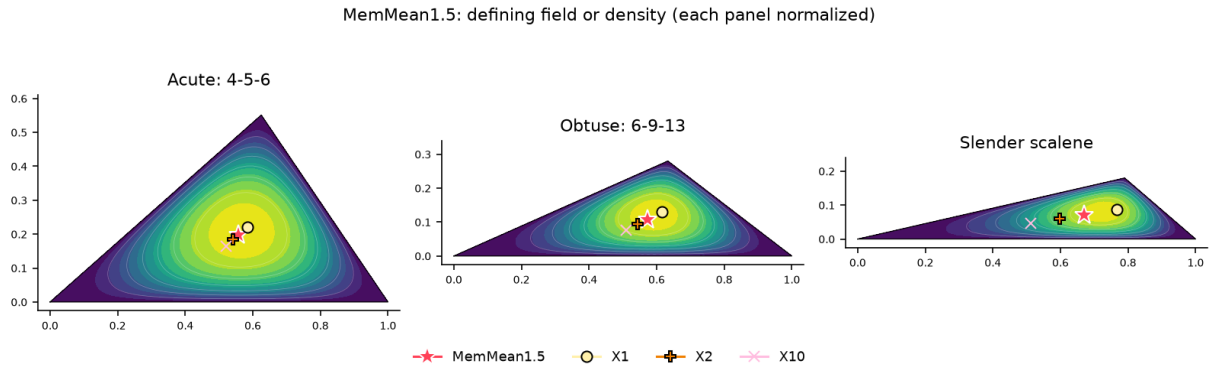
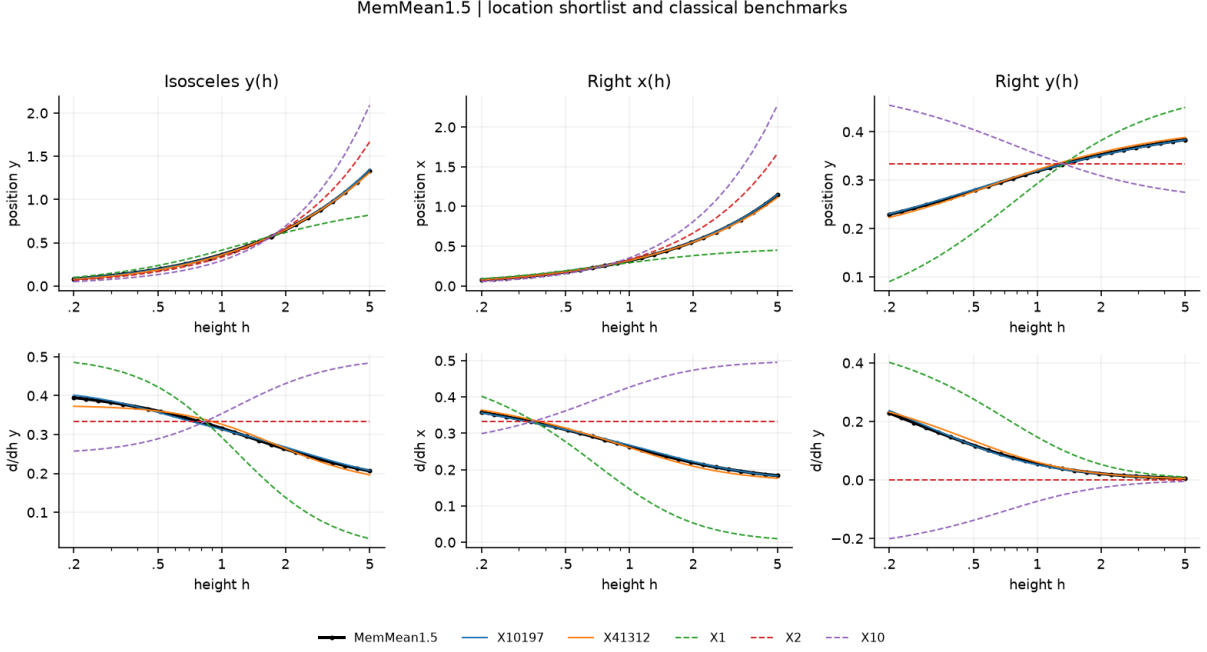


Figure 6.10: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(10197)	0.123%	0.233%	0.118%	0.231%
X(41312)	0.364%	0.661%	0.377%	0.605%
X(10180)	0.365%	0.723%	0.376%	0.706%

For the location winner, the stored family-derivative RMS discrepancy is 0.004039 and response-margin RMS discrepancy is 0.0075145. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 39/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.015% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `MemMean1.5_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.14 MemMean3.0: Nonlinear membrane density mean, exponent 3.0

Physical scenario and selector. Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Measure the deflection-volume centroid.

$$-\operatorname{div}(|\nabla u|^1 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \int pu / \int u$$

Derivation and algorithm. Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Positive density proves strict interiority of the mean.

Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Distinguishing behavior. The leading strip prediction for the mean is $C_x \rightarrow [(2p-1)/(4p-3)]x$, with exponent p fixed. No certified thin remainder is supplied.

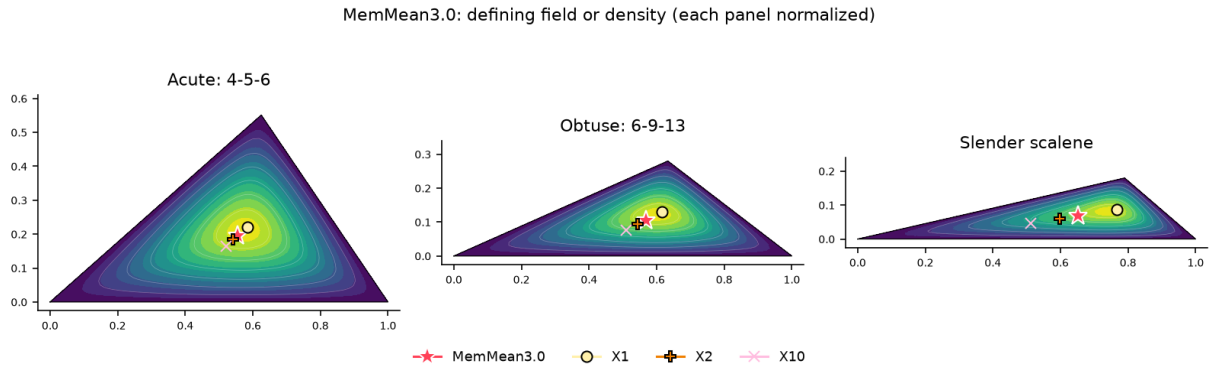
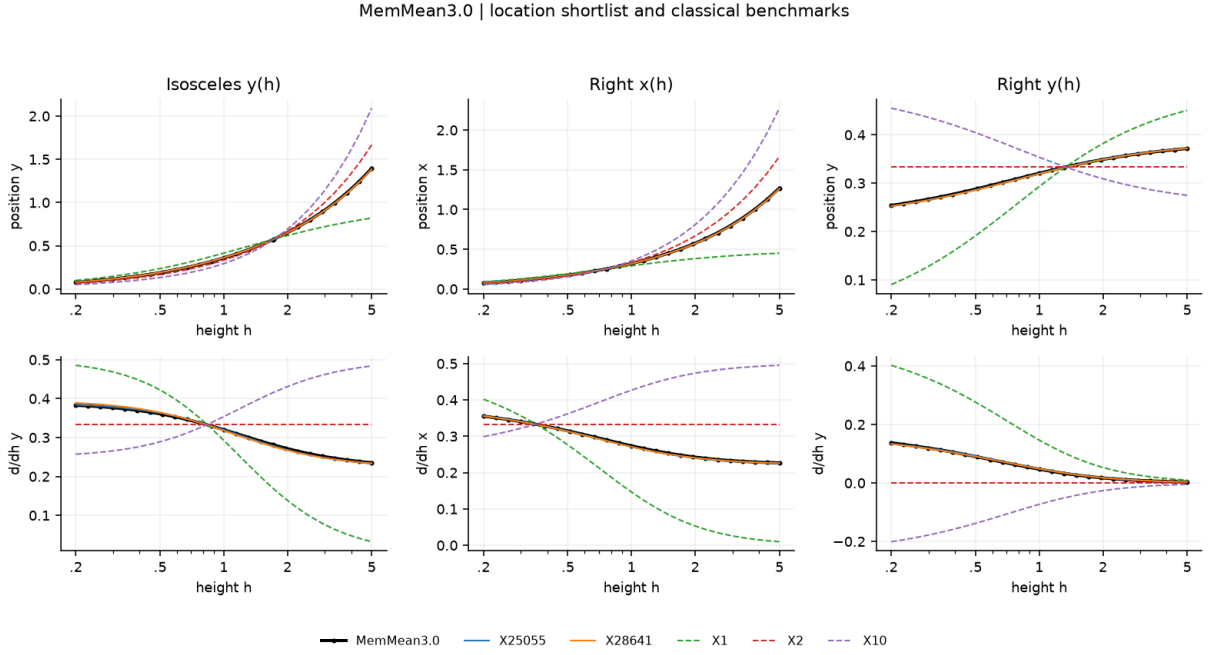


Figure 6.11: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(25055)	0.107%	0.191%	0.108%	0.186%
X(28641)	0.124%	0.187%	0.125%	0.179%
X(20153)	0.182%	0.349%	0.189%	0.495%

For the location winner, the stored family-derivative RMS discrepancy is 0.002297 and response-margin RMS discrepancy is 0.0062561. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 39/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.014% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `MemMean3.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

6.15 MemMean4.0: Nonlinear membrane density mean, exponent 4.0

Physical scenario and selector. Use an isotropic membrane energy proportional to the stated power of slope under uniform loading. Measure the deflection-volume centroid.

$$-\operatorname{div}(|\nabla u|^2 \nabla u) = 1, \quad u|_{\partial T} = 0, \quad C = \int pu / \int u$$

Derivation and algorithm. Minimize the convex discrete p-energy in a boundary-bubble basis, monitor original gradient residuals, and refine quadrature as well as degree. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Positive density proves strict interiority of the mean.

Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Distinguishing behavior. The leading strip prediction for the mean is $C_x \rightarrow [(2p-1)/(4p-3)]x$, with exponent p fixed. No certified thin remainder is supplied.

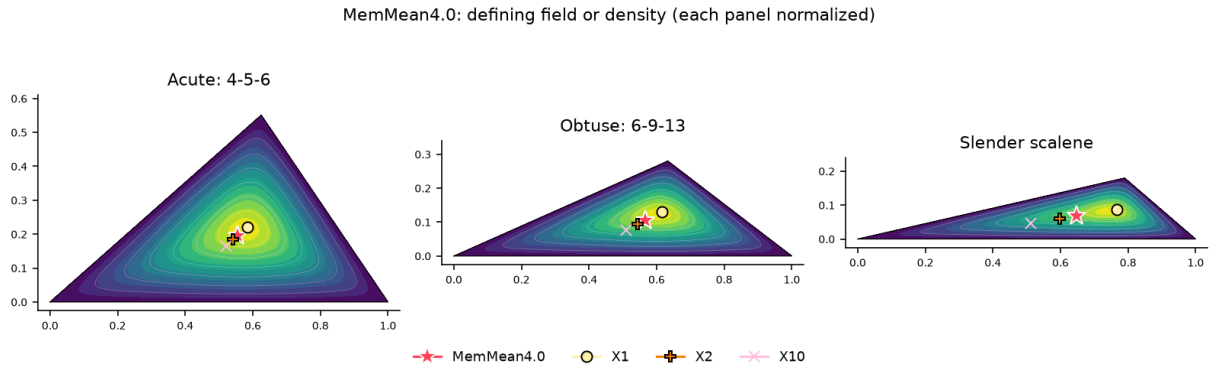
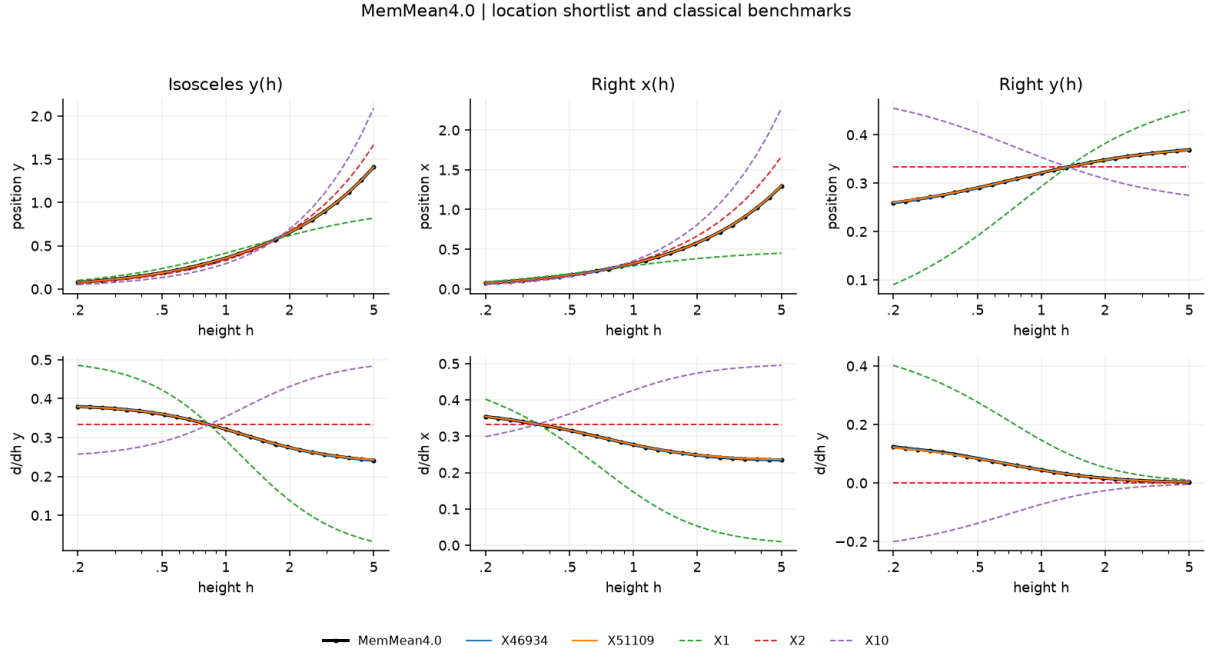


Figure 6.12: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(46934)	0.047%	0.079%	0.049%	0.080%
X(51109)	0.096%	0.203%	0.101%	0.197%
X(17397)	0.250%	0.392%	0.245%	0.383%

For the location winner, the stored family-derivative RMS discrepancy is 0.00093188 and response-margin RMS discrepancy is 0.0075691. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 39/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.020% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `MemMean4.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

Chapter 7

Elastic plates, buckling and fluctuations

7.1 The lowest point under gravity

A thin homogeneous plate resists bending, whereas a tension membrane resists slope. These yield fourth-order and second-order equations respectively. In linear Kirchhoff–Love plate theory, bending stiffness is $D_b = Eh^3/[12(1 - \nu^2)]$, with Young’s modulus E , thickness h and Poisson ratio ν . Uniform self-weight produces a transverse load $q = \rho hg$. For a clamped boundary,

$$D_b \Delta^2 w = q, \quad w = \partial_n w = 0 \text{ on } \partial T.$$

Take positive w downward. Multiplying by the constant D_b/q removes material parameters from the position of its maximum. The gravity plate center P_{grav} is therefore the maximizer of $\Delta^2 w = 1$ with clamped traces. This is the precise answer to “where is the lowest point?” in the homogeneous, small-deflection, clamped-plate model. A support that fixes height but allows rotation changes the boundary conditions and may change the answer.

The weak equation is $\int_T \Delta w \Delta v = \int_T v$ for $v \in H_0^2(T)$. The finite-dimensional basis uses the squared boundary bubble, ensuring both traces vanish. Uniformly loaded triangular clamped plates and boundary-function Ritz methods have a substantial earlier literature [12, 13]. A unique displacement field follows from coercivity. Uniqueness of the point of largest displacement is a separate question; the atlas reports the numerically selected global-mode candidate with this qualification.

7.2 What uniform heating can and cannot select

In a homogeneous flat plate with fully restrained in-plane boundary displacement and uniform temperature rise, the displacement $u = 0$ solves linear thermoelastic equilibrium. The thermal eigenstrain is uniform and creates uniform compression. In this model no special point has extremal density: the flat, uniform state has no spatially distinguished density maximum. After buckling, density per actual surface area depends on the in-plane displacement and surface Jacobian. That requires a nonlinear constitutive model and a branch choice.

Uniform isotropic compression N modifies the transverse equation to

$$D_b \Delta^2 w + N \Delta w = q.$$

The first critical value is the minimum Rayleigh quotient

$$\lambda_b = \min_{v \in H_0^2(T) \setminus \{0\}} \frac{\int (\Delta v)^2}{\int |\nabla v|^2}, \quad N_{\text{cr}} = D_b \lambda_b.$$

The associated mode satisfies $\Delta^2\psi = -\lambda_b\Delta\psi$. Bmode is the maximum of ψ^2 ; Bmean is its normalized first moment. Squaring removes the arbitrary sign of buckling up or down. At a multiple eigenvalue, a projector convention is needed for a canonical mean, just as in the quantum case. A numerical spectral gap is monitored, not assumed universally.

To compare shapes consistently before buckling, hold $\tau = N/N_{\text{cr}}$ fixed and solve

$$(\Delta^2 + \tau\lambda_b\Delta)w = 1, \quad 0 \leq \tau < 1.$$

Pheat(τ) is its downward maximum. We investigate $\tau = 0.5, 0.9$. The parameter changes with shape through λ_b ; derivatives include this dependence. It is not a fixed-temperature experiment. As $\tau \uparrow 1$, the first mode dominates if the uniform load couples to it. A post-buckled sheet without gravity can choose either sign and may have multiple stable states; it does not supply an unqualified lowest-point center.

7.3 Compliance and thermal variance

Apply a unit point force at p and measure deflection at that same point. If $G_b(x, p)$ is the clamped biharmonic Green function, the local compliance is $G_b(p, p)$. Its maximum defines Compliance. Point evaluation is continuous in the two-dimensional H_0^2 energy space, so this diagonal is finite; the analogous continuum membrane point variance has a different ultraviolet issue.

In a discretized elastic plate with stiffness matrix K , thermal equilibrium has covariance $k_B T K^{-1}$. If $f(p)$ is the vector of basis values at p , its displacement variance is

$$\text{Var } w(p) = k_B T f(p)^\top K^{-1} f(p).$$

Thus the largest-variance point coincides with the largest-compliance point [14]. The normalized first moment of this positive diagonal defines FluctMean. Changing temperature rescales the variance but not either selector. A plate sampled at many points with a scanning force probe is a direct demonstration. Smooth Ritz spaces converge comparatively slowly for a diagonal Green function, and FluctMean has the weakest resolution counts among the present plate means.

7.4 Energy density: an exact reduction

The bending-energy density, ignoring a common positive multiplier, is

$$e_\nu = (1 - \nu)|D^2w|^2 + \nu(\Delta w)^2 = (\Delta w)^2 - 2(1 - \nu) \det D^2w.$$

Define Energy as its first moment. The local contour depends on ν , but the center does not. For any affine function ℓ and clamped w , integration by parts gives

$$\int_T \ell \det D^2w = 0.$$

For instance, write $\det D^2w = \partial_x(w_x w_{yy}) - \partial_y(w_x w_{xy})$ and integrate; boundary terms vanish because the full gradient trace vanishes. The remaining first-derivative-weighted integrals vanish by another integration by parts. Thus the mass and first moments of e_ν equal those of $(\Delta w)^2$.

If $\Delta^2 w = q$ is constant, testing against ℓw gives

$$\int_T \ell (\Delta w)^2 = q \int_T \ell w.$$

The additional term $2 \int \Delta w \nabla \ell \cdot \nabla w$ vanishes by the clamped traces. Hence the energy centroid equals the deflection-volume mean and is independent of ν . Positivity of the energy supplies its

interior location certificate; this identity does not require assuming a positive biharmonic Green function. At finite Ritz order ℓw may leave the trial space, so equality of the two computed moments is a convergence check, not an automatic discrete identity.

In a local narrow clamped strip, w is proportional to $y^2(H - y)^2$. Its transverse integral scales as H^5 . This predicts $\text{Energy}_x \rightarrow 5x/7$ in the thin triangular limit, while the maximum approaches the widest section x . The two centers are therefore separated even at leading order in an asymptotic experiment.

7.5 Boundary conditions that remain separate constructions

The Navier plate has $w = \Delta w = 0$, so with $A = -\Delta_D$ its field is $A^{-2}1$. It can be computed by two Poisson solves. Positivity follows from two second-order maximum principles, but uniqueness of its maximum still needs justification. Vertex-supported plates require support size, rotation constraints and free-edge conditions; Poisson ratio can then affect shape. A pure tension membrane fixed only at isolated points in two dimensions is not the same coercive Dirichlet model. These conditional models are retained in the final inventory and must not be silently merged into Pgrav or F1.

7.6 Hanging-sheet restrictions and updated membrane evidence

The dedicated third chapter, *Membranes, Hanging Sheets, and Elastic Triangle Centers*, expands these models using the supplied hanging-chain/sheet variational manuscript and new 3–4–5 reference calculations. A sheet with only fixed boundary and total area has gravitational energy unbounded below: a fixed positive area can enter a cone whose radius tends to zero and depth tends to infinity. It defines no center.

Restricting the surface to three planar faces gives a different, exact experiment. For apex depth h and signed side distances $d_i(Q)$, lateral area is $\frac{1}{2} \sum a_i \sqrt{h^2 + d_i(Q)^2}$. The weighted mean of d_i is the inradius r . Jensen's inequality gives the unique minimum at $Q = X(1)$, with lateral area $s\sqrt{h^2 + r^2}$. A uniformly massive three-face sheet has mean depth $h/3$, so at fixed area its minimum gravitational energy selects the same incenter.

In the quadratic shallow-sheet model, fixed excess area means $\frac{1}{2} \int |\nabla w|^2 = E$. Cauchy–Schwarz against the torsion function gives $w = \sqrt{2E/\int u u}$. Its lowest point is F1. The tension-energy centroid equals the torsion-volume mean $\text{Abs}(0)$, by an affine first-moment identity. A uniformly shrinking membrane point-probe footprint instead selects the conformal-radius center E1 after removal of its common logarithmic divergence.

The maximum of the planar convex p -torsion problem is unique, including on triangles: Borrelli, Mosconi and Squassina, Theorem 1.1 and Remarks 1.2, 1.4, <https://arxiv.org/abs/2201.12788>. This settles the definition of $\text{Mem}(p)$ as a single point. It does not settle vertex attractivity or the numerical resolution of its shape derivatives. The dedicated chapter's new degree-20, 26 and 32 checks turn all five retained plate witnesses positive at both motion sizes. Those particular coarse negatives are excluded as counterexamples; global plate attractivity remains open. New reference values and all refinement inputs are in `output/membrane_plate_chapter/`.

7.7 Pgrav: Gravity-loaded clamped plate maximum

Physical scenario and selector. Uniform gravity or pressure deflects a homogeneous thin plate whose boundary height and slope are fixed.

$$\Delta^2 w = 1, \quad w = \partial_n w = 0, \quad C = \arg \max w$$

Derivation and algorithm. Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. An interior mode is numerically selected. Unique field or positive diagonal does not alone prove uniqueness of its maximum on every triangle.

Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Distinguishing behavior. This is the linear clamped model. Rotationally free boundaries, large deflections and nonuniform thickness change the problem.

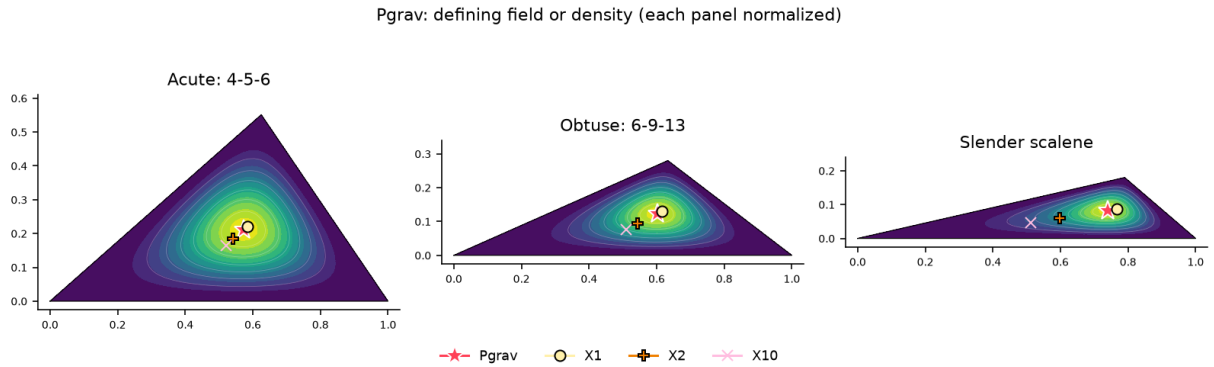
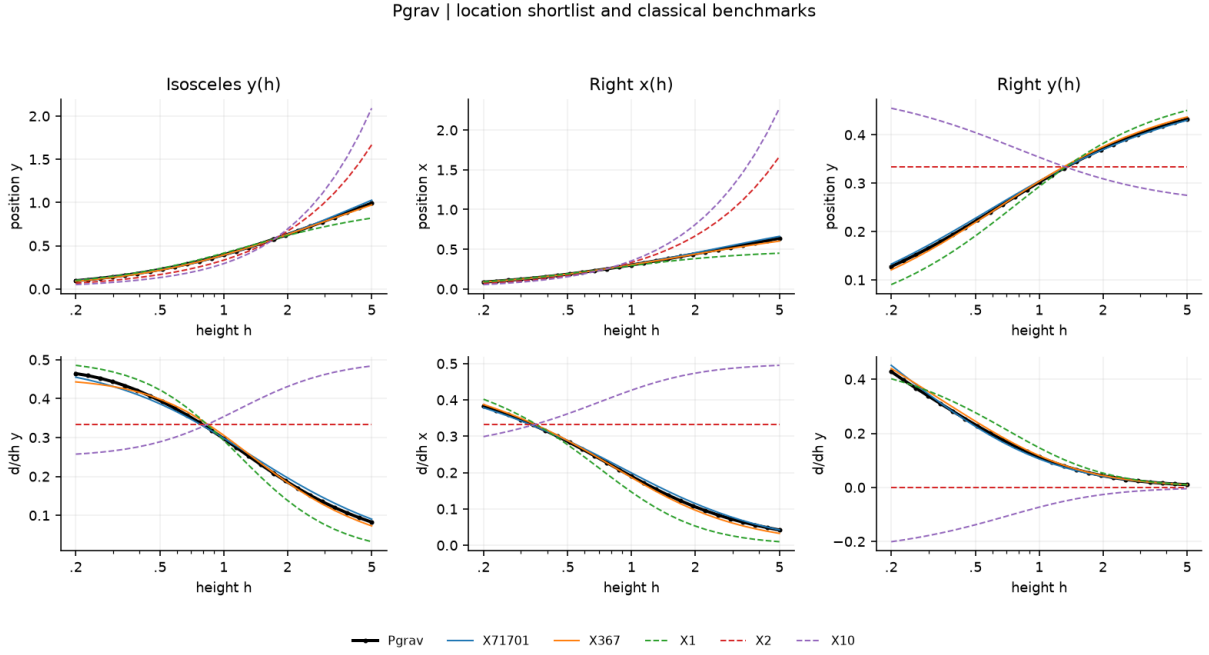


Figure 7.1: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(71701)	0.358%	0.657%	0.597%	2.963%
X(367)	0.434%	1.244%	0.652%	3.036%
X(53588)	0.446%	0.960%	0.692%	3.168%

For the location winner, the stored family-derivative RMS discrepancy is 0.0096384 and response-margin RMS discrepancy is 0.081832. These have their own normalizations and are not percentages of location error.

Resolution record. 92/100 training and 33/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.260% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Pgrav_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

7.8 Bmode: First buckling-mode maximum

Physical scenario and selector. Increase uniform compression to its first linear instability and locate the largest squared amplitude of the critical mode.

$$\Delta^2 \psi = -\lambda_b \Delta \psi, \quad C = \arg \max \psi^2$$

Derivation and algorithm. Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. An interior mode is numerically selected. Unique field or positive diagonal does not alone prove uniqueness of its maximum on every triangle.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Distinguishing behavior. At a repeated buckling eigenvalue the individual-mode construction needs a projector convention.

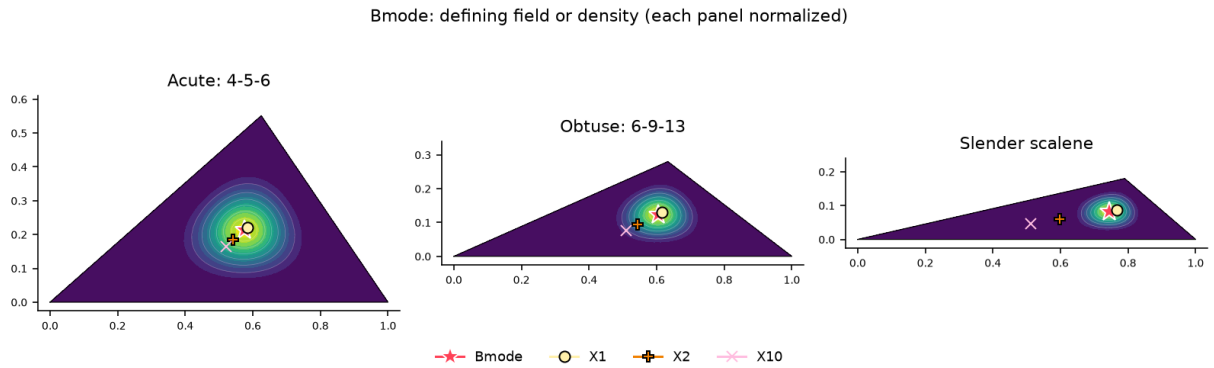
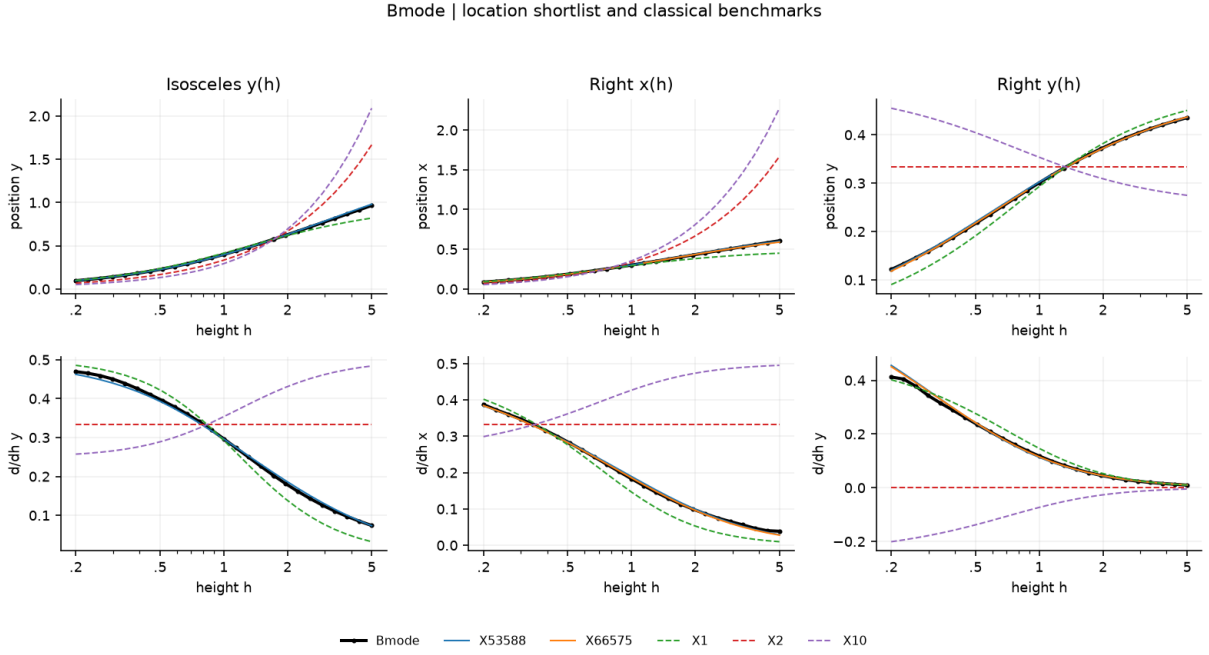


Figure 7.2: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(53588)	0.268%	0.642%	1.965%	9.702%
X(66575)	0.283%	0.661%	1.970%	9.703%
X(29624)	0.353%	1.049%	1.944%	9.383%

For the location winner, the stored family-derivative RMS discrepancy is 0.0097221 and response-margin RMS discrepancy is 0.072368. These have their own normalizations and are not percentages of location error.

Resolution record. 88/100 training and 34/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.303% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Bmode_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

7.9 Bmean: First buckling-mode expectation

Physical scenario and selector. Use the spatial first moment of the squared critical buckling mode, avoiding its arbitrary sign.

$$\Delta^2 \psi = -\lambda_b \Delta \psi, \quad C = \int p \psi^2 / \int \psi^2$$

Derivation and algorithm. Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Its positive defining density gives a strict-interiority certificate.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Distinguishing behavior. At a repeated buckling eigenvalue the individual-mode construction needs a projector convention.

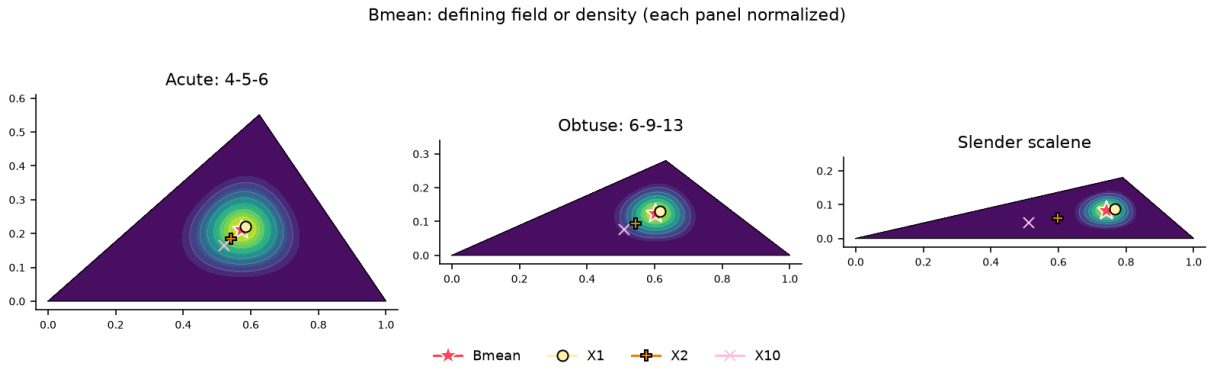
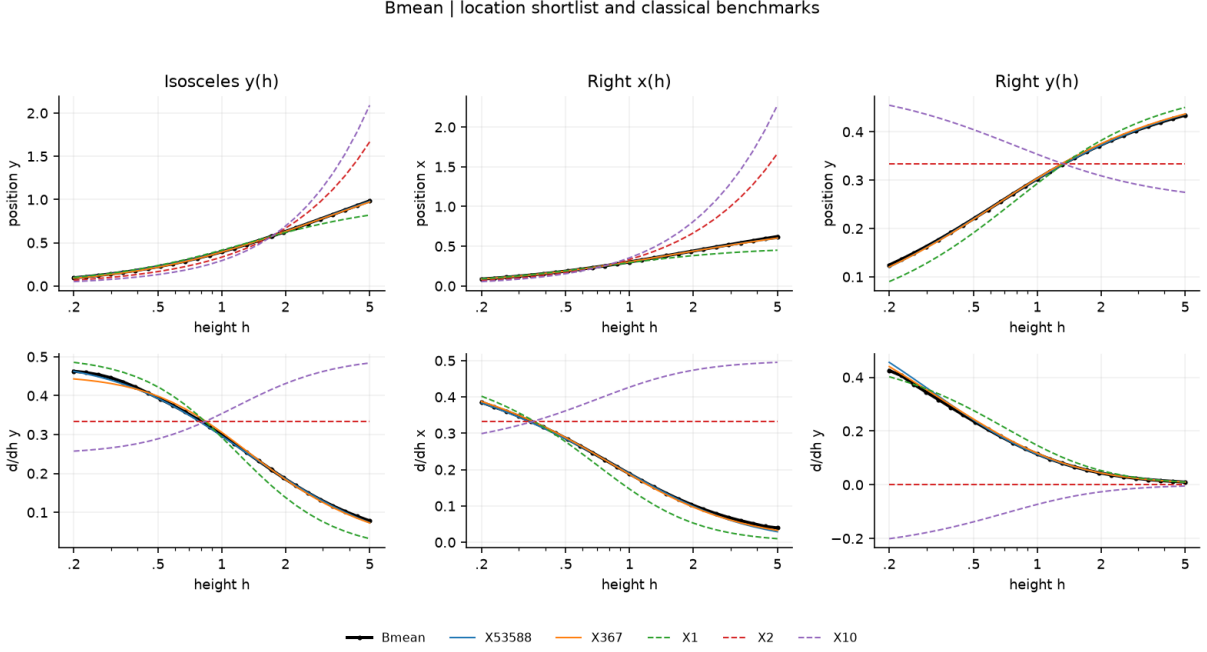


Figure 7.3: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(53588)	0.339%	0.832%	0.395%	1.303%
X(367)	0.366%	1.187%	0.379%	0.971%
X(66560)	0.395%	0.840%	0.446%	1.304%

For the location winner, the stored family-derivative RMS discrepancy is 0.0075984 and response-margin RMS discrepancy is 0.072244. These have their own normalizations and are not percentages of location error.

Resolution record. 89/100 training and 33/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.300% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Bmean_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

7.10 Compliance: Maximum point compliance / thermal variance

Physical scenario and selector. Scan a point load and measure displacement at the load point, or locate the greatest equilibrium thermal displacement variance.

$$C = \arg \max_p G_b(p, p)$$

Derivation and algorithm. Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. An interior mode is numerically selected. Unique field or positive diagonal does not alone prove uniqueness of its maximum on every triangle.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Distinguishing behavior. The inverse-stiffness diagonal converges more slowly than a uniformly forced smooth field; retain the per-center resolution counts below.

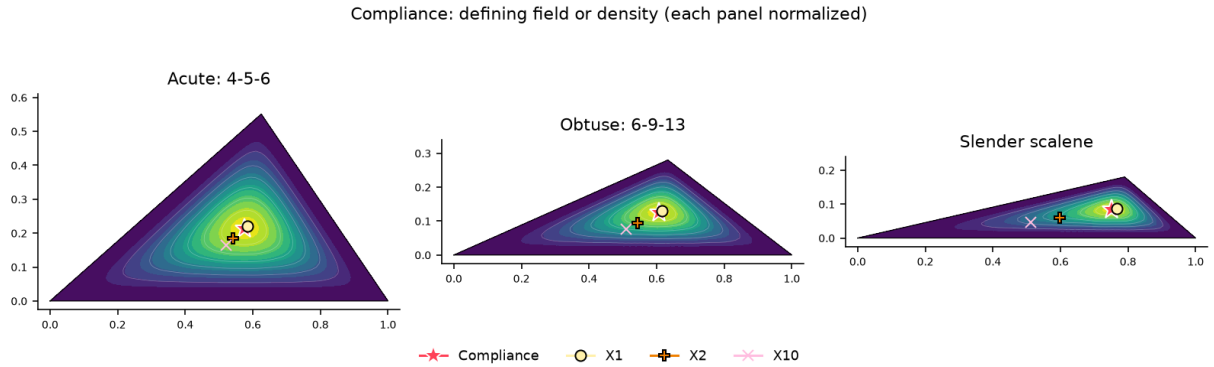
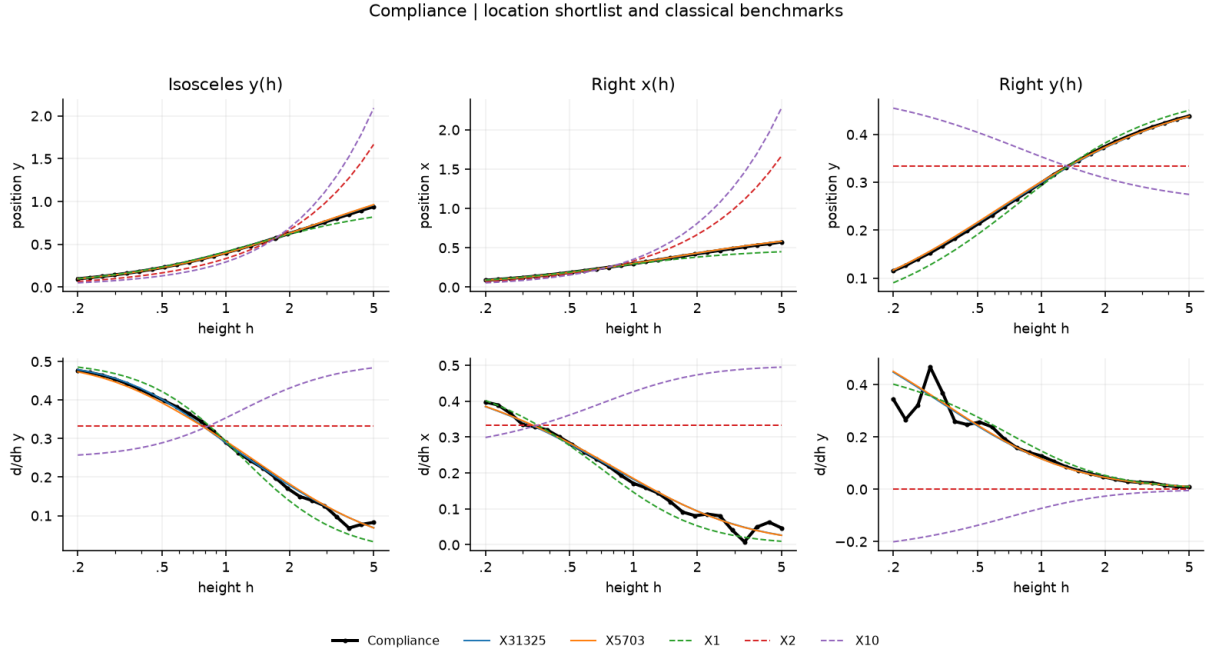


Figure 7.4: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(31325)	0.253%	0.606%	0.267%	0.848%
X(5703)	0.264%	0.590%	0.286%	0.848%
X(16015)	0.278%	0.441%	0.303%	0.801%

For the location winner, the stored family-derivative RMS discrepancy is 0.034909 and response-margin RMS discrepancy is 0.10768. These have their own normalizations and are not percentages of location error.

Resolution record. 80/100 training and 34/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.261% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Compliance_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

7.11 FluctMean: Thermal fluctuation-density centroid

Physical scenario and selector. Average position using the local plate displacement variance as a positive density.

$$C = \int_T p G_b(p, p) dp / \int_T G_b(p, p) dp$$

Derivation and algorithm. Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Its positive defining density gives a strict-interiority certificate.

Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Distinguishing behavior. The inverse-stiffness diagonal converges more slowly than a uniformly forced smooth field; retain the per-center resolution counts below.

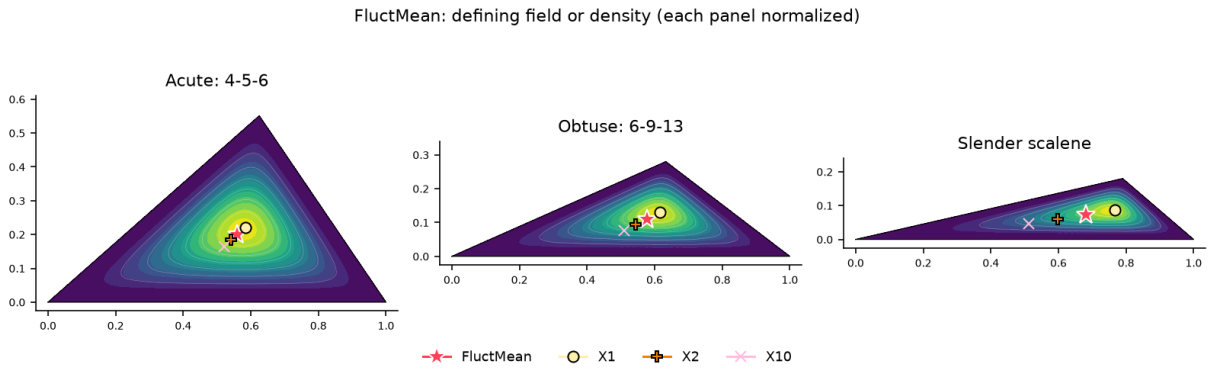
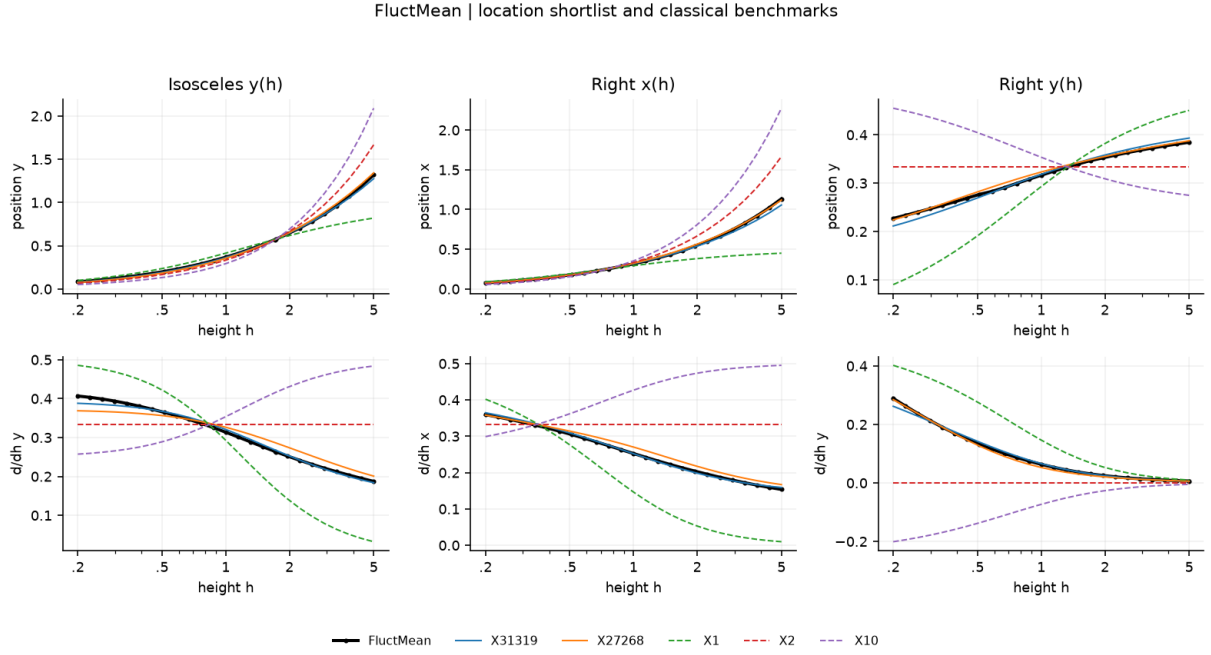


Figure 7.5: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(31319)	0.714%	1.573%	0.886%	2.110%
X(27268)	0.730%	1.932%	0.859%	2.235%
X(39738)	0.829%	1.830%	0.989%	2.070%

For the location winner, the stored family-derivative RMS discrepancy is 0.0077253 and response-margin RMS discrepancy is 0.03399. These have their own normalizations and are not percentages of location error.

Resolution record. 46/100 training and 18/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.538% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `FluctMean_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

7.12 Energy0.3: Bending-energy centroid

Physical scenario and selector. Use local bending energy under a uniform load as a spatial density; the plotted representative has Poisson ratio 0.3.

$$C = \int_T p e_\nu / \int_T e_\nu, \quad e_\nu = (1 - \nu)|D^2 w|^2 + \nu(\Delta w)^2$$

Derivation and algorithm. Use the clamped bubble-squared Bernstein Ritz basis. Assemble bending and gradient matrices; solve the load, generalized eigenvalue or inverse-stiffness problem and integrate or optimize the resulting field. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Its positive defining density gives a strict-interiority certificate.

Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Distinguishing behavior. The center is exactly independent of Poisson ratio and equals the deflection-volume first moment. The energy contour itself still depends on Poisson ratio.

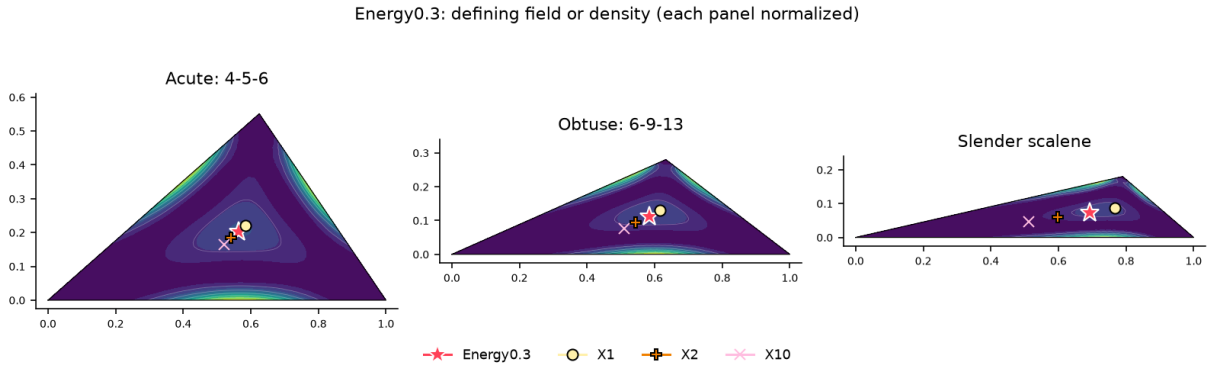
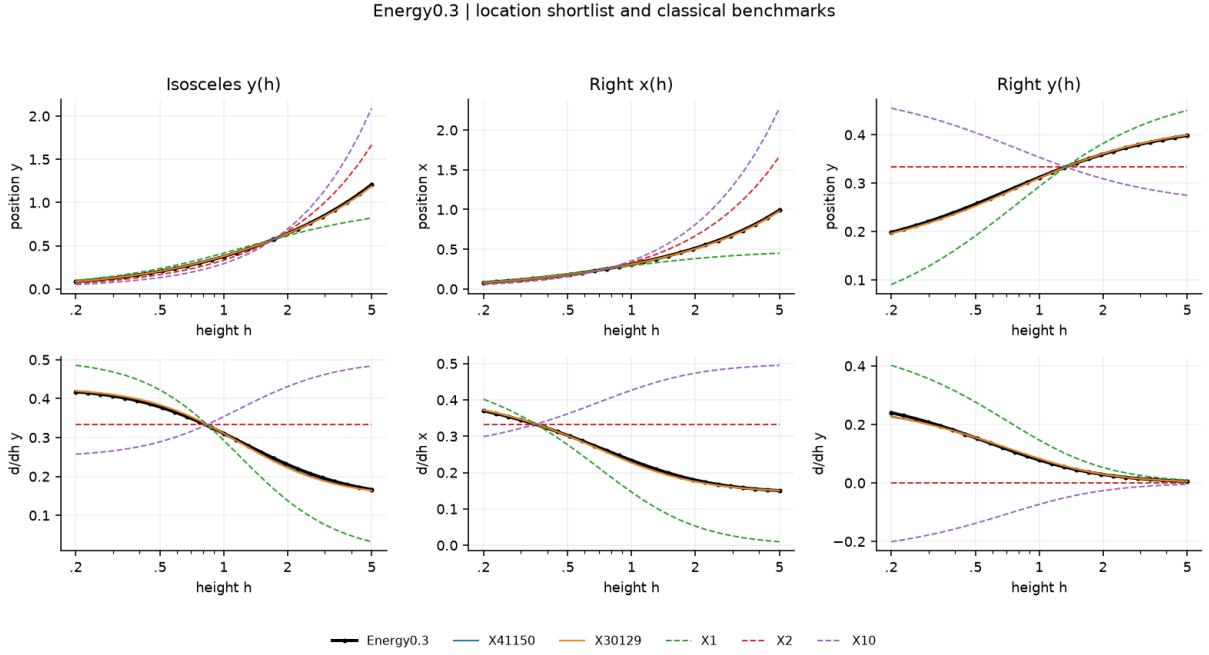


Figure 7.6: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(41150)	0.173%	0.329%	0.171%	0.306%
X(30129)	0.208%	0.397%	0.205%	0.369%
X(29823)	0.221%	0.412%	0.204%	0.364%

For the location winner, the stored family-derivative RMS discrepancy is 0.005177 and response-margin RMS discrepancy is 0.025689. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 38/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.022% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Energy0.3_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

7.13 Pheat0.5: Compressed, gravity-biased plate at fraction 0.5

Physical scenario and selector. Uniform heating produces compression; a weak downward uniform load selects a pre-buckling deflection. Fix the stated fraction of the shape-dependent critical compression.

$$(\Delta^2 + 0.5\lambda_b\Delta)w = 1, \quad w = \partial_n w = 0, \quad C = \arg \max w$$

Derivation and algorithm. Compute the first generalized buckling eigenvalue, solve the shifted bending system, and refine the maximum. Include the eigenvalue change in vertex derivatives. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The atlas selects the downward interior mode; global uniqueness of this selector is not established here.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Distinguishing behavior. The parameter is a fraction of critical compression, not fixed absolute temperature. Post-buckling branches and areal-density extrema are not solved by this linear equation.

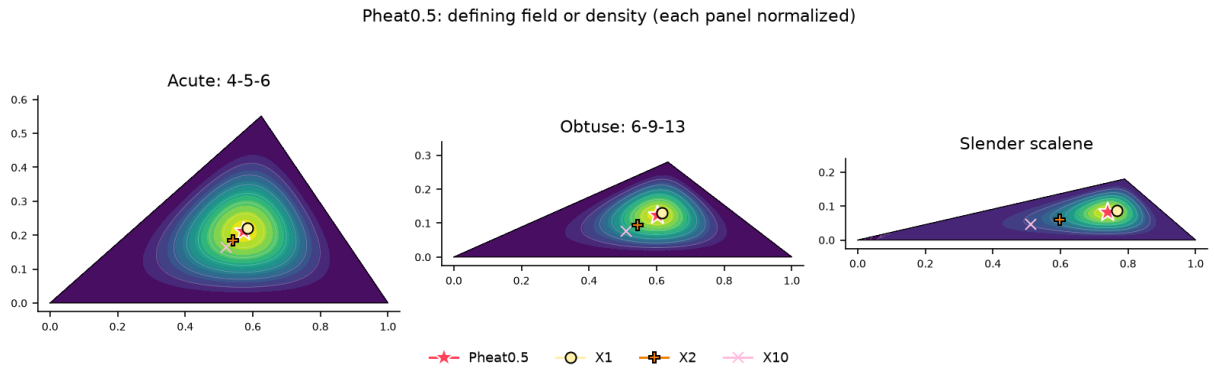
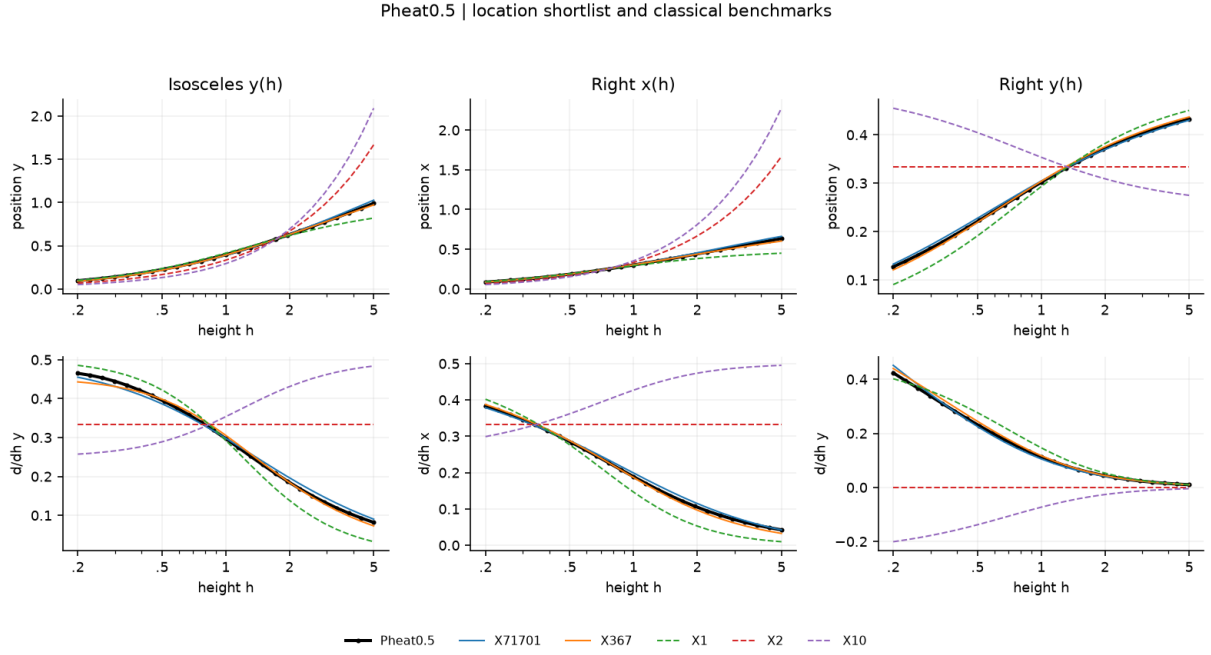


Figure 7.7: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(71701)	0.391%	0.719%	0.412%	1.037%
X(367)	0.416%	1.206%	0.431%	0.881%
X(53588)	0.418%	0.896%	0.465%	1.083%

For the location winner, the stored family-derivative RMS discrepancy is 0.010639 and response-margin RMS discrepancy is 0.080501. These have their own normalizations and are not percentages of location error.

Resolution record. 90/100 training and 33/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.301% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Pheat0.5_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

7.14 Pheat0.9: Compressed, gravity-biased plate at fraction 0.9

Physical scenario and selector. Uniform heating produces compression; a weak downward uniform load selects a pre-buckling deflection. Fix the stated fraction of the shape-dependent critical compression.

$$(\Delta^2 + 0.9\lambda_b\Delta)w = 1, \quad w = \partial_n w = 0, \quad C = \arg \max w$$

Derivation and algorithm. Compute the first generalized buckling eigenvalue, solve the shifted bending system, and refine the maximum. Include the eigenvalue change in vertex derivatives. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The atlas selects the downward interior mode; global uniqueness of this selector is not established here.

Attractivity. Global vertex attractivity remains open. The retained coarse negative witness is positive at all three new degrees 20, 26 and 32, with both the original and halved motion. That witness no longer supports nonattractivity; this does not prove global attractivity.

Distinguishing behavior. The parameter is a fraction of critical compression, not fixed absolute temperature. Post-buckling branches and areal-density extrema are not solved by this linear equation.

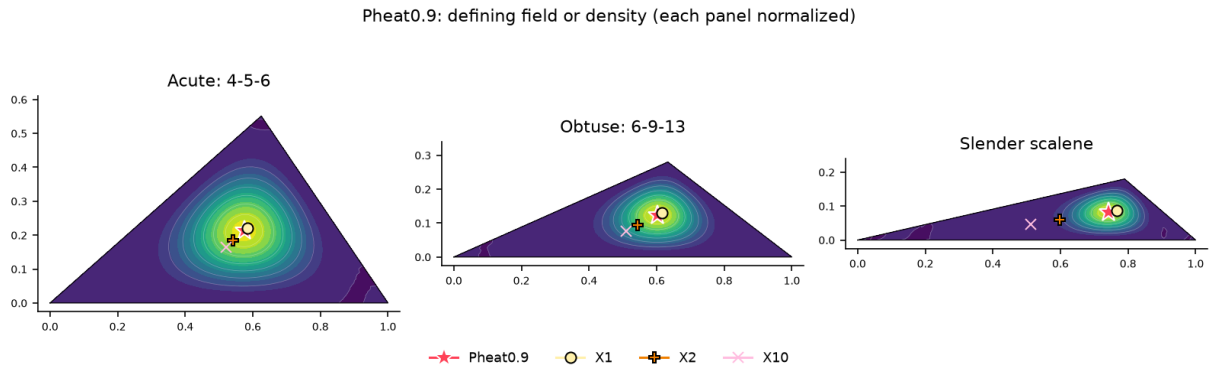
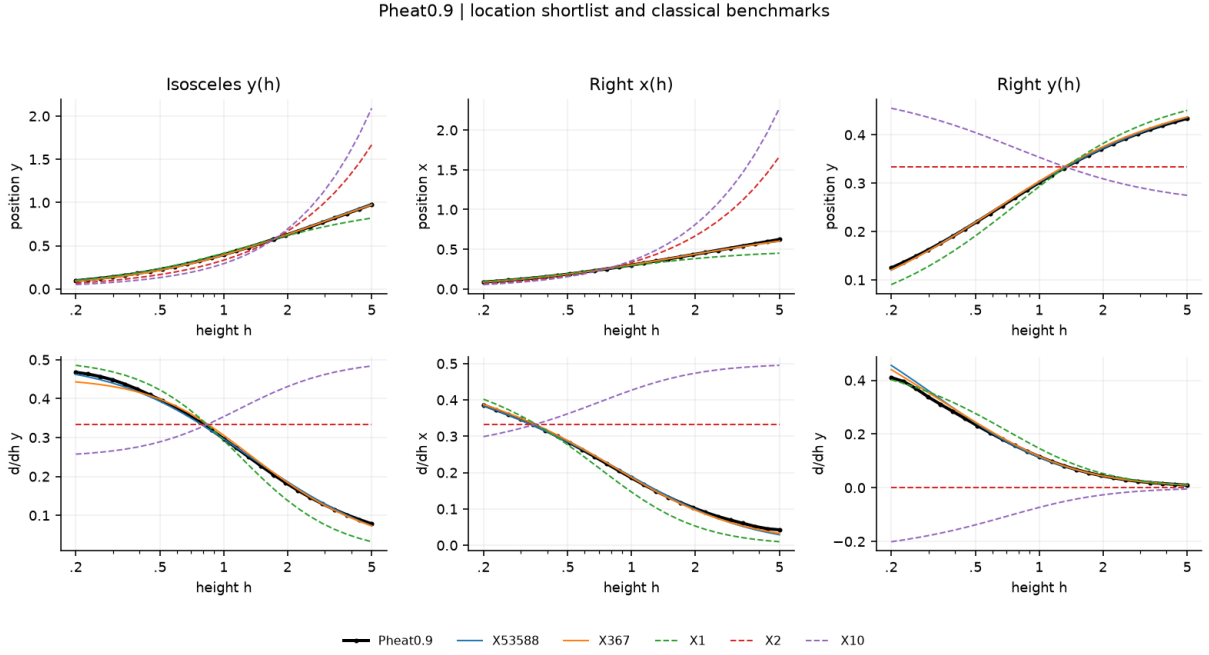


Figure 7.8: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(53588)	0.378%	0.855%	0.430%	1.266%
X(367)	0.389%	1.139%	0.399%	0.863%
X(30142)	0.440%	1.204%	0.455%	1.003%

For the location winner, the stored family-derivative RMS discrepancy is 0.010863 and response-margin RMS discrepancy is 0.067798. These have their own normalizations and are not percentages of location error.

Resolution record. 90/100 training and 33/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.322% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Pheat0.9_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

Chapter 8

Reaction, transport and population centers

8.1 Uniform production and bulk removal

Produce a substance at uniform rate in the triangle. Let it diffuse with coefficient D_c , disappear at rate kc , and be absorbed at the boundary. Conservation in a small region says that diffusive influx plus production balances removal. The steady equation is

$$-D_c \Delta c + kc = s, \quad c|_{\partial T} = 0.$$

After rescaling concentration and setting $\mu = kL^2/D_c$, the geometric equation becomes

$$-\Delta c + \mu c/L^2 = 1.$$

$\text{Abs}(\mu)$ is the normalized first moment of c . For $k > 0$ it is also the mean location of bulk disappearance events, since their density is kc and the constant cancels. At $k = 0$ there are no bulk events; the same formula remains the concentration or residence-density mean. It must not be described as a nonzero absorption-event distribution at that endpoint.

Testing against a zero-boundary function gives the matrix equation $(K + \mu M/L^2)c = f$. Positivity of the continuum solution follows from the maximum principle. For large μ , concentration is almost L^2/μ except in a boundary layer, so the normalized mean tends to G . At $\mu = 0$ the field is the torsion function; its mean is generally different from its maximum F1. The studied values are 0, 1, 10 and 100. In the thin limit at fixed μ , the transverse diffusion term dominates, predicting $\text{Abs}_x \rightarrow 3x/5$ from a strip integral proportional to H^3 .

8.2 A positive logistic equilibrium

Let a population diffuse and reproduce with local logistic law $rn(1 - n/K_0)$, while the boundary is lethal. With density scaled by carrying capacity and diffusion coefficient absorbed into r , stationarity is

$$-\Delta n = rn(1 - n), \quad n|_{\partial T} = 0.$$

Linearizing at extinction shows that growth competes with the first Dirichlet decay rate λ_0 . A positive branch exists when $r > \lambda_0$. We set $r = \kappa\lambda_0(T)$ with $\kappa > 1$ and define $\text{Bio}(\kappa)$ as the first moment of the positive equilibrium. This fixes the fraction above the persistence threshold while the shape changes. It is not a fixed dimensional growth rate.

Near threshold, the density is a small multiple of ψ_0 . Its mean approaches $\int p\psi_0 / \int \psi_0$, not Q0, which uses ψ_0^2 . At large κ the interior density approaches one and its mean approaches G . The

maximum principle bounds the positive branch by its carrying capacity. The continuum uniqueness and persistence theory is the appropriate foundation [15]; a solver returning the extinction solution does not define the requested center.

The numerical method minimizes the positive-branch energy, using a coercive $|n|^3$ extension for intermediate iterates. It then checks the residual of the original equation with n^2 , not just the optimizer's stopping criterion. Undershoots, residuals and mesh changes are retained. The values $\kappa = 1.1, 2, 10$ span near-threshold and more uniform regimes.

Bio(1.1) has a strong numerical attractivity counterexample independently reproduced with a quadratic finite-element solver. On the retained slender triangle, the normalized finite response is approximately $-0.10927, -0.10935, -0.10922$ at subdivisions 48, 80 and 128. This is much stronger evidence than one coarse Ritz sign, but it remains a numerical continuum assertion, not an interval certificate. Stability of the positive logistic equilibrium under population dynamics is compatible with this failure of vertex attractivity.

8.3 Abs0.0: Production–removal concentration mean, $\mu=0$

Physical scenario and selector. Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 0c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Derivation and algorithm. Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. Global vertex attractivity remains open. The retained coarse screen has no negative response for this entry, which is not a global positivity theorem.

Distinguishing behavior. At zero removal this is the torsion-density mean, not the torsion maximum; there are no bulk removal events at this endpoint.

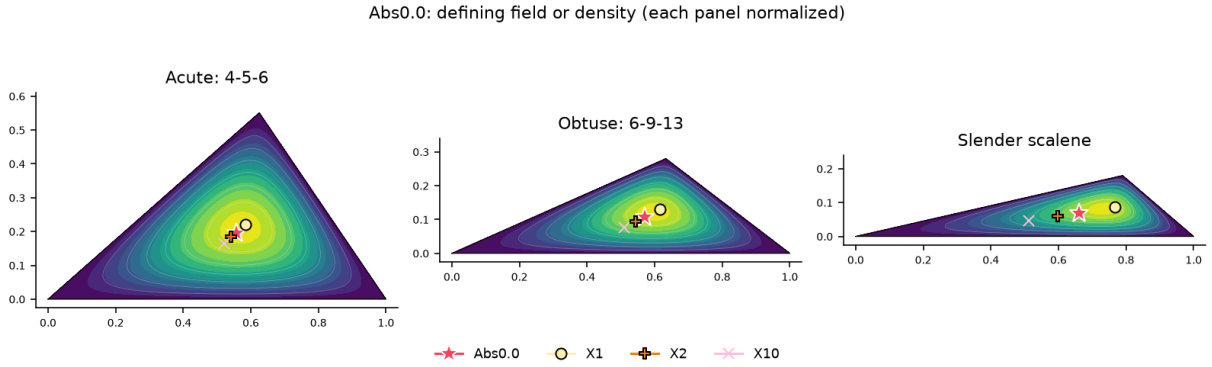
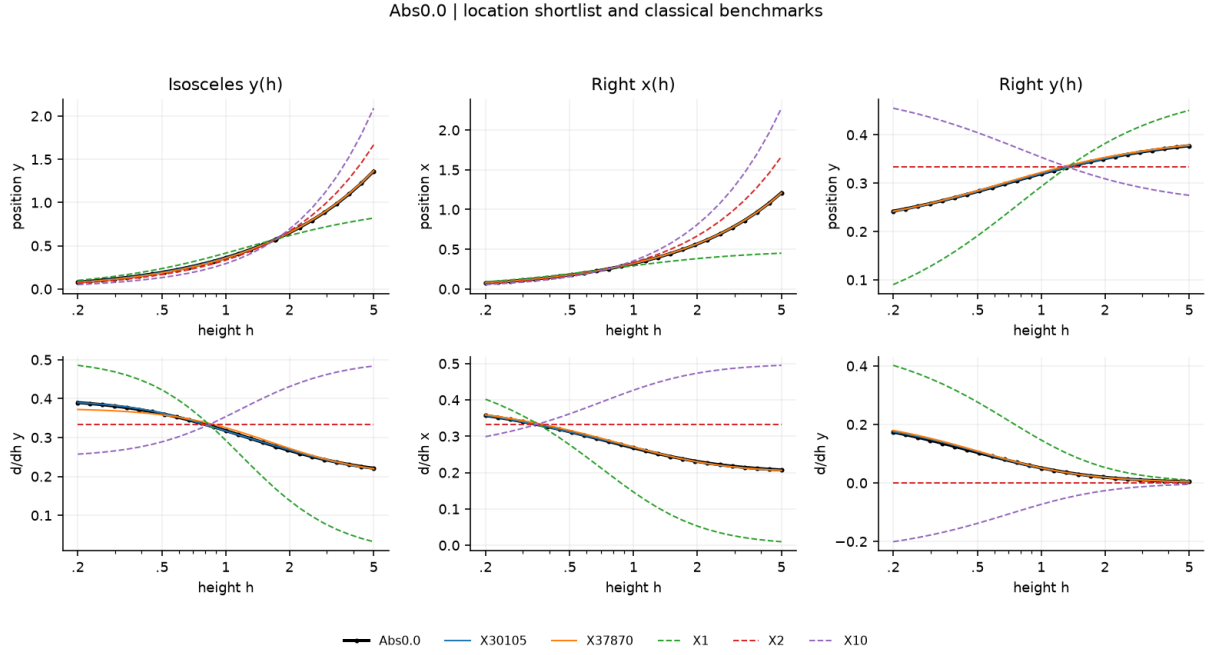


Figure 8.1: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(30105)	0.075%	0.270%	0.072%	0.210%
X(37870)	0.214%	0.370%	0.213%	0.370%
X(16831)	0.253%	0.608%	0.237%	0.554%

For the location winner, the stored family-derivative RMS discrepancy is 0.0015113 and response-margin RMS discrepancy is 0.0066162. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.008% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Abs0.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

8.4 Abs1.0: Production–removal concentration mean, $\mu=1$

Physical scenario and selector. Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 1c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Derivation and algorithm. Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. No negative response was found in the retained screen; global attractivity is open.

Distinguishing behavior. For positive removal the event-density centroid is the same. The large- μ limit is $X(2)$; fixed- μ thin strip asymptotics give horizontal limit $3x/5$.

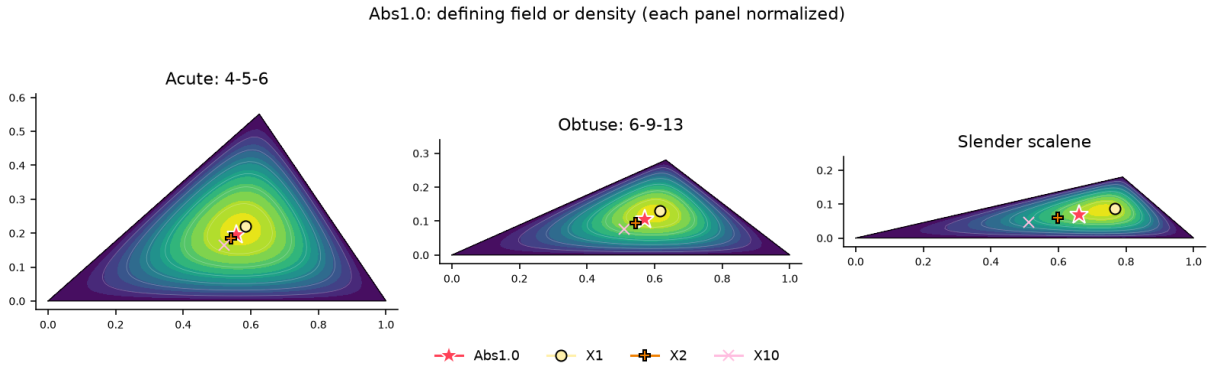
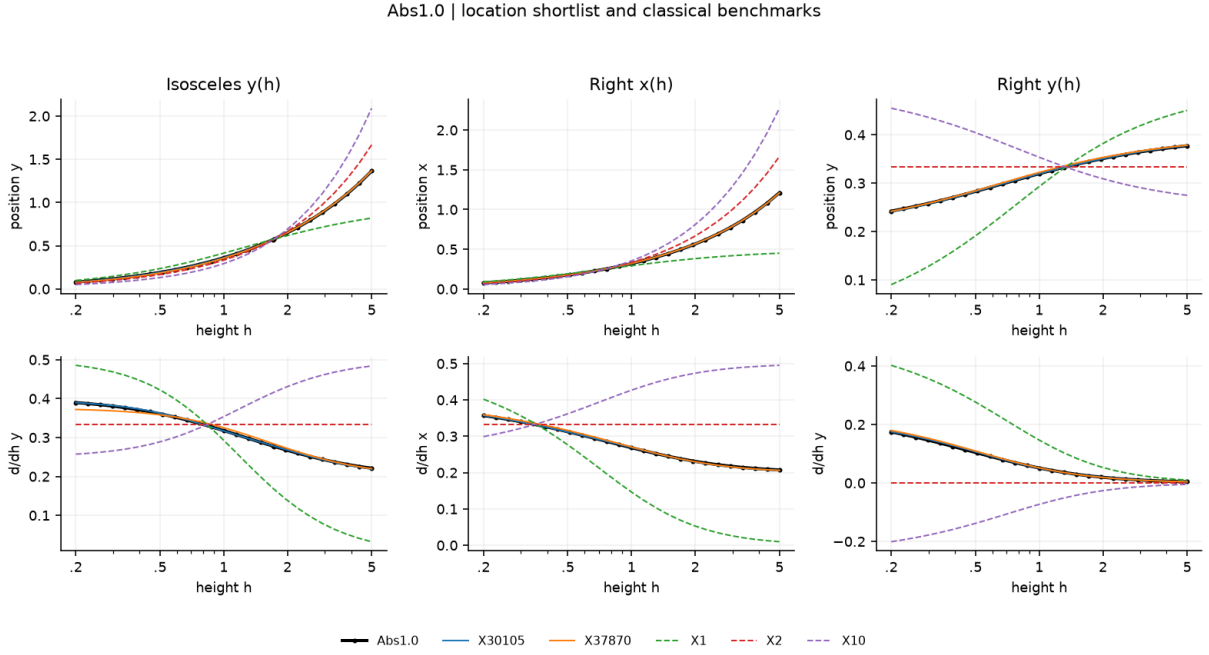


Figure 8.2: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(30105)	0.075%	0.269%	0.074%	0.209%
X(37870)	0.210%	0.367%	0.209%	0.367%
X(16831)	0.248%	0.608%	0.232%	0.552%

For the location winner, the stored family-derivative RMS discrepancy is 0.0014702 and response-margin RMS discrepancy is 0.0065978. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.008% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Abs1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

8.5 Abs10.0: Production–removal concentration mean, $\mu=10$

Physical scenario and selector. Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 10c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Derivation and algorithm. Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. No negative response was found in the retained screen; global attractivity is open.

Distinguishing behavior. For positive removal the event-density centroid is the same. The large- μ limit is $X(2)$; fixed- μ thin strip asymptotics give horizontal limit $3x/5$.

Abs10.0: defining field or density (each panel normalized)

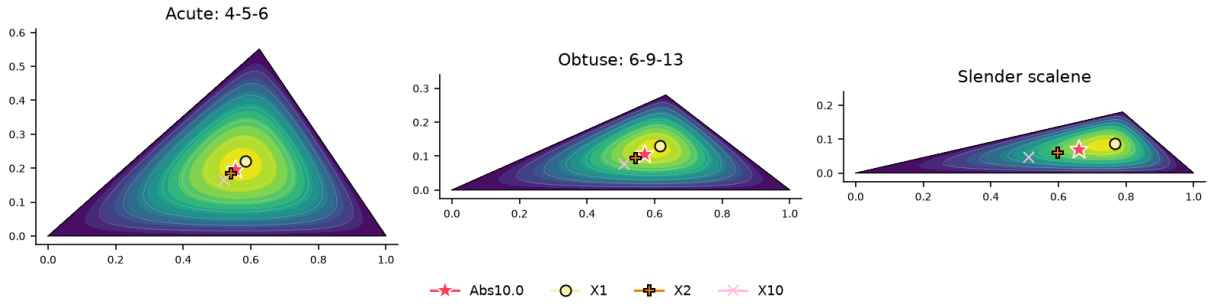
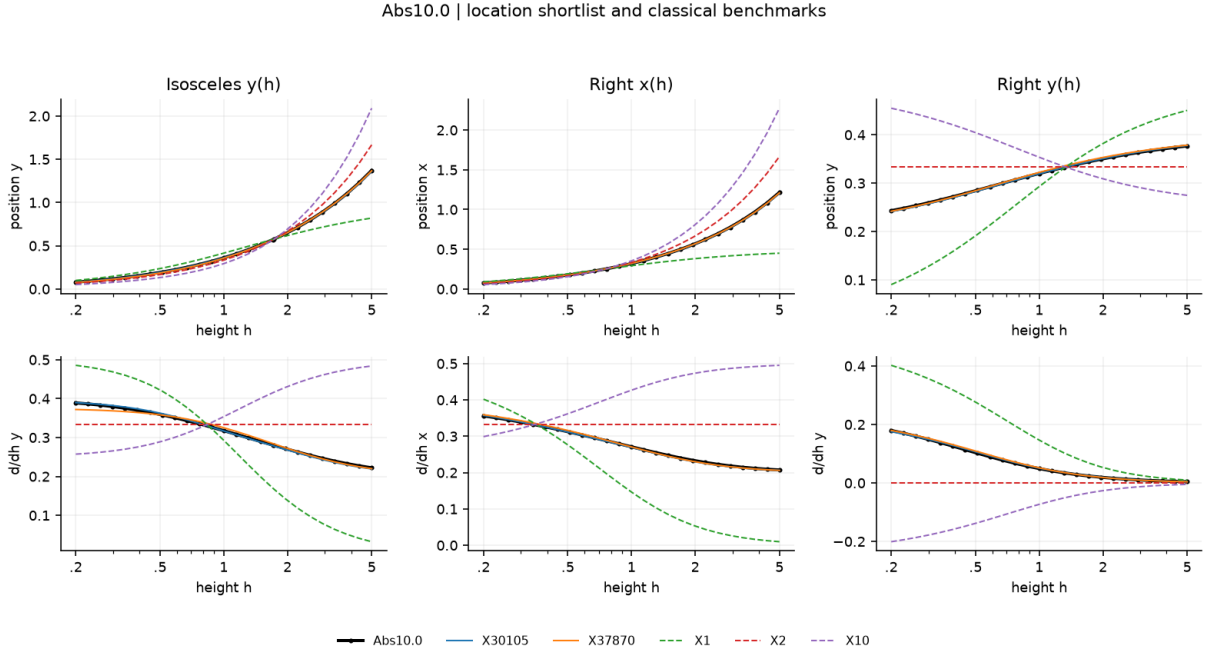


Figure 8.3: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

	ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(30105)		0.110%	0.268%	0.112%	0.196%
X(37870)		0.187%	0.342%	0.188%	0.343%
X(16831)		0.212%	0.606%	0.199%	0.535%

For the location winner, the stored family-derivative RMS discrepancy is 0.0025271 and response-margin RMS discrepancy is 0.0066771. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.008% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Abs10.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

8.6 Abs100.0: Production–removal concentration mean, $\mu=100$

Physical scenario and selector. Produce material uniformly, let it diffuse to an absorbing boundary and undergo uniform bulk removal. Take the concentration centroid.

$$-\Delta c + 100c/L^2 = 1, \quad c|_{\partial T} = 0, \quad C = \int pc / \int c$$

Derivation and algorithm. Solve the positive mass-shifted Dirichlet stiffness system, then integrate its first moment. Compare polynomial degree and original residual. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive continuum concentration gives strict interiority.

Attractivity. No negative response was found in the retained screen; global attractivity is open.

Distinguishing behavior. For positive removal the event-density centroid is the same. The large- μ limit is $X(2)$; fixed- μ thin strip asymptotics give horizontal limit $3x/5$.

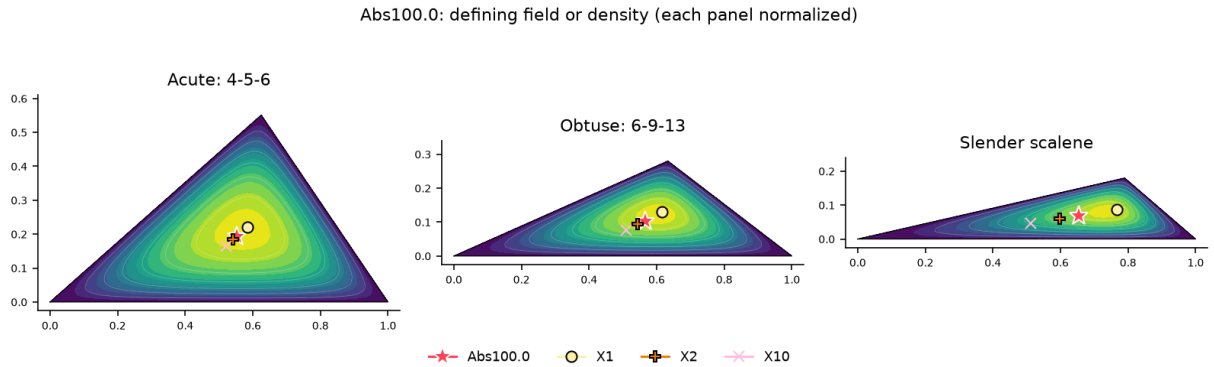
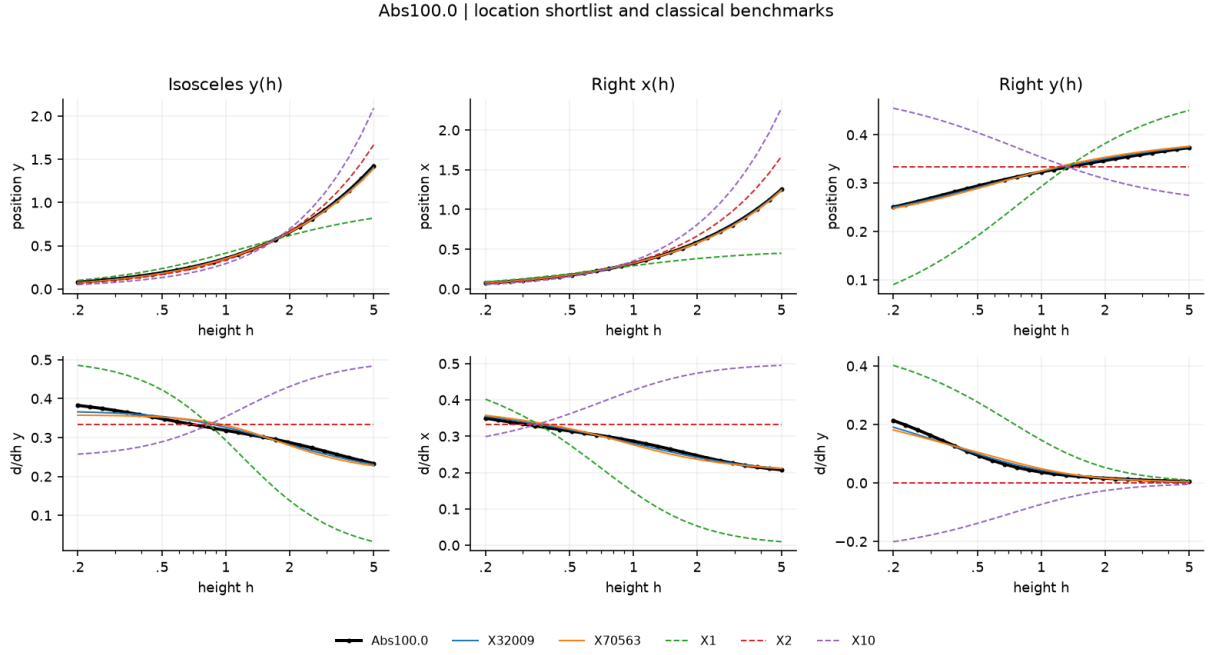


Figure 8.4: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(32009)	0.322%	0.889%	0.313%	0.756%
X(70563)	0.401%	0.787%	0.397%	0.670%
X(31336)	0.405%	1.124%	0.396%	1.031%

For the location winner, the stored family-derivative RMS discrepancy is 0.008608 and response-margin RMS discrepancy is 0.01452. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.008% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Abs100.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

8.7 Bio1.1: Logistic population mean, kappa=1.1

Physical scenario and selector. Let a population diffuse, grow logistically and die at the triangular boundary. Choose the positive persistent equilibrium, not extinction.

$$-\Delta n = 1.1\lambda_0 n(1 - n), \quad n|_{\partial T} = 0, \quad C = \int pn / \int n$$

Derivation and algorithm. Compute the first Dirichlet eigenvalue, minimize the positive-branch energy, then check the original logistic residual and integrate the density. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The unique positive continuum branch has a strictly interior mean.

Attractivity. A strong finite numerical counterexample is independently reproduced by quadratic FEM; no interval certificate is supplied.

Distinguishing behavior. Near threshold the mean tends to the first moment of ψ_0 , not ψ_0^2 . Large growth tends to $X(2)$. Fixed κ includes the shape dependence of λ_0 .

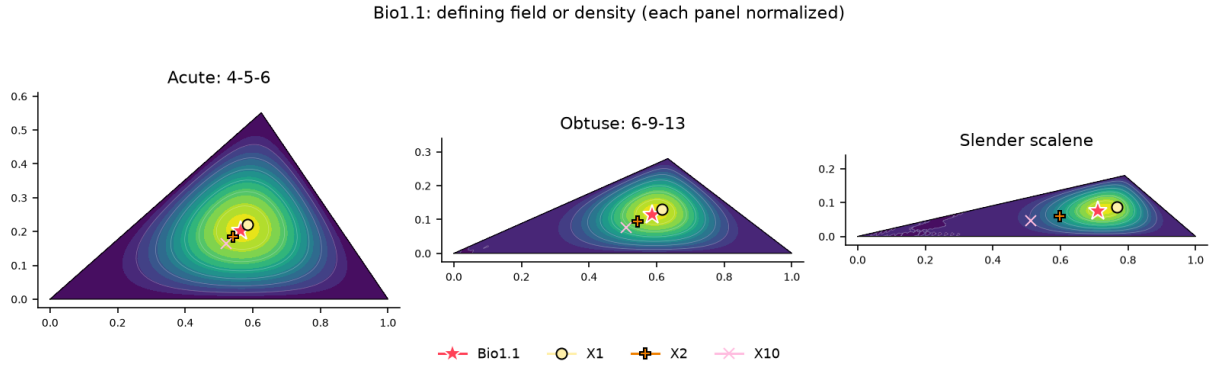
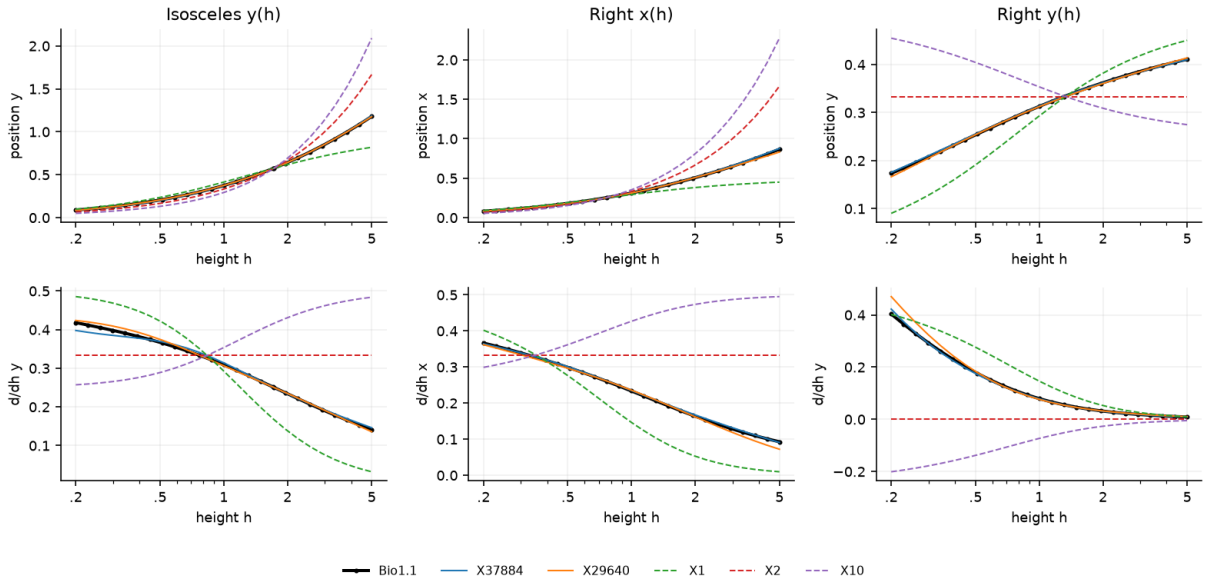


Figure 8.5: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Bio1.1 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(37884)	0.536%	1.534%	0.629%	2.110%
X(29640)	0.873%	2.945%	0.763%	2.395%
X(63628)	0.879%	2.473%	0.772%	2.390%

For the location winner, the stored family-derivative RMS discrepancy is 0.0067103 and response-margin RMS discrepancy is 0.056374. These have their own normalizations and are not percentages of location error.

Resolution record. 98/100 training and 39/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.143% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Bio1.1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

8.8 Bio2.0: Logistic population mean, kappa=2

Physical scenario and selector. Let a population diffuse, grow logistically and die at the triangular boundary. Choose the positive persistent equilibrium, not extinction.

$$-\Delta n = 2\lambda_0 n(1 - n), \quad n|_{\partial T} = 0, \quad C = \int pn / \int n$$

Derivation and algorithm. Compute the first Dirichlet eigenvalue, minimize the positive-branch energy, then check the original logistic residual and integrate the density. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The unique positive continuum branch has a strictly interior mean.

Attractivity. Global attractivity is unresolved; numerical screening is not a theorem.

Distinguishing behavior. Near threshold the mean tends to the first moment of ψ_0 , not ψ_0^2 . Large growth tends to $X(2)$. Fixed κ includes the shape dependence of λ_0 .

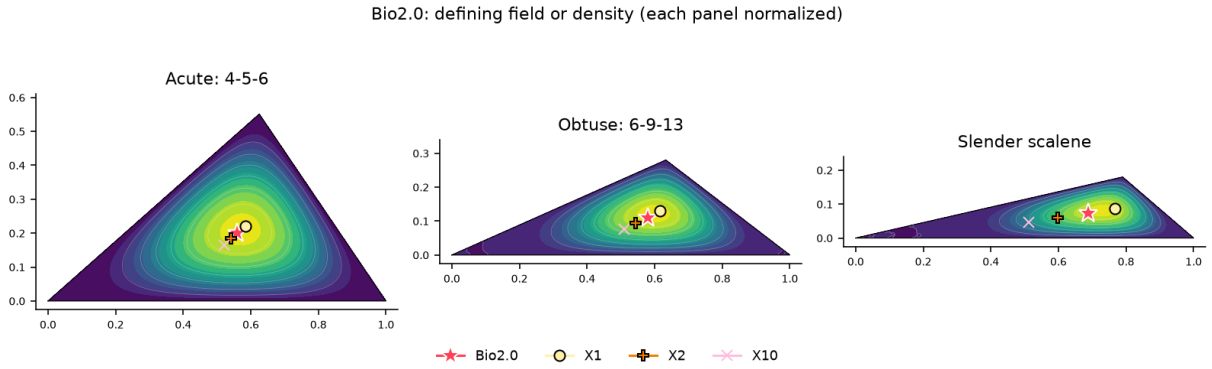
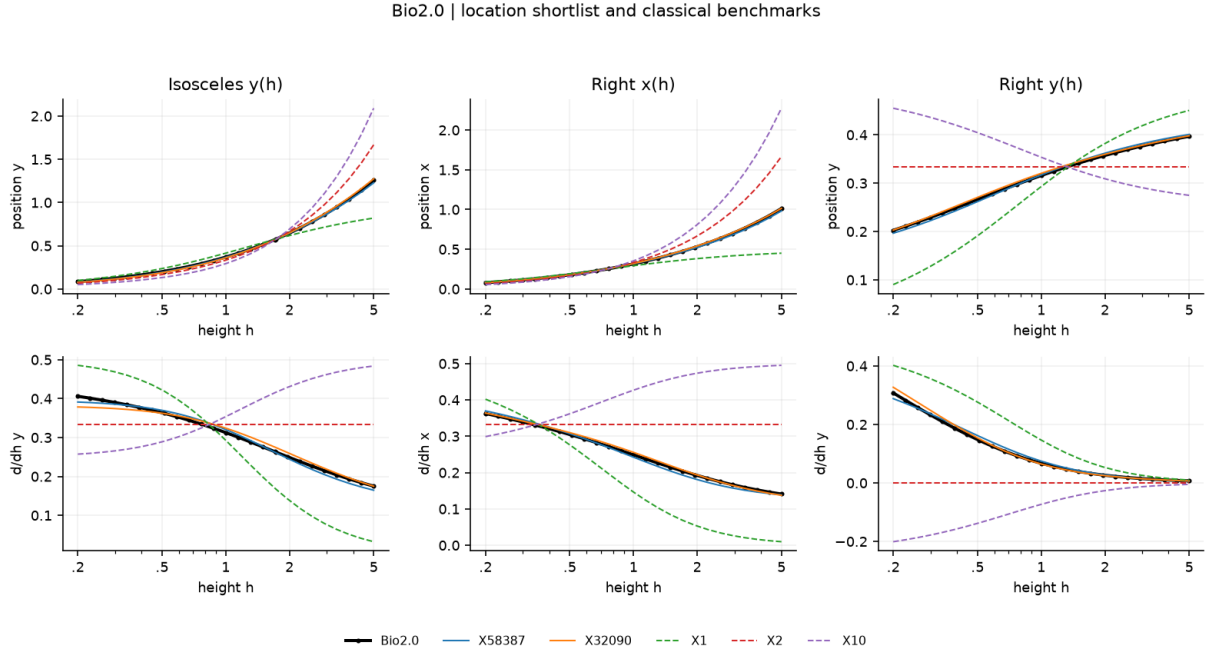


Figure 8.6: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(58387)	0.532%	1.610%	0.638%	1.926%
X(32090)	0.646%	2.056%	0.617%	1.481%
X(24857)	0.762%	1.543%	0.891%	2.599%

For the location winner, the stored family-derivative RMS discrepancy is 0.0095946 and response-margin RMS discrepancy is 0.024309. These have their own normalizations and are not percentages of location error.

Resolution record. 98/100 training and 37/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.070% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Bio2.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

8.9 Bio10.0: Logistic population mean, kappa=10

Physical scenario and selector. Let a population diffuse, grow logistically and die at the triangular boundary. Choose the positive persistent equilibrium, not extinction.

$$-\Delta n = 10\lambda_0 n(1 - n), \quad n|_{\partial T} = 0, \quad C = \int pn / \int n$$

Derivation and algorithm. Compute the first Dirichlet eigenvalue, minimize the positive-branch energy, then check the original logistic residual and integrate the density. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The unique positive continuum branch has a strictly interior mean.

Attractivity. Global attractivity is unresolved; numerical screening is not a theorem.

Distinguishing behavior. Near threshold the mean tends to the first moment of ψ_0 , not ψ_0^2 . Large growth tends to $X(2)$. Fixed κ includes the shape dependence of λ_0 .

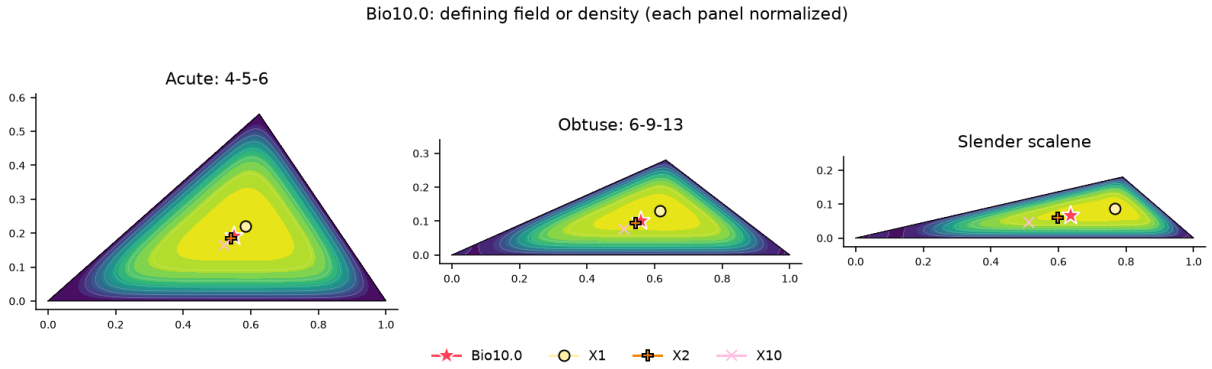
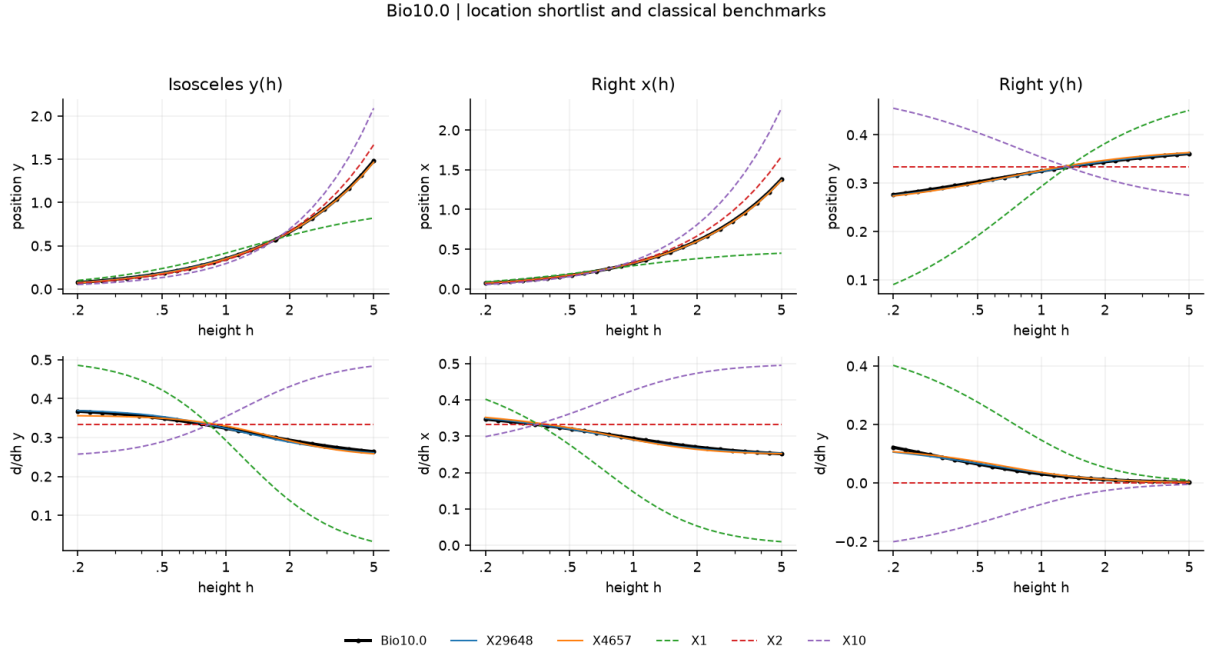


Figure 8.7: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(29648)	0.256%	0.647%	0.311%	1.005%
X(4657)	0.265%	0.549%	0.318%	0.981%
X(62648)	0.273%	0.762%	0.327%	1.029%

For the location winner, the stored family-derivative RMS discrepancy is 0.0048692 and response-margin RMS discrepancy is 0.007695. These have their own normalizations and are not percentages of location error.

Resolution record. 91/100 training and 36/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.055% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Bio10.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

Chapter 9

Diffraction, darkness and nonlinear imaging

9.1 Scaling before selecting a point

For a uniformly illuminated triangular aperture, scalar paraxial Fresnel propagation gives, up to a nonzero global factor,

$$A_F(p) = A_T^{-1} \int_T \exp\left(i\pi F|p - q|^2/L^2\right) dq, \quad F = L^2/(\lambda z).$$

The intensity is $|A_F|^2$. Fixing F gives a shape-defined experiment: changing wavelength requires moving the screen to $z = L^2/(\lambda F)$. This is wavelength independence under a dimensionless protocol, not achromatic behavior at a fixed physical screen. The scalar/paraxial approximation and uniform aperture phase are model assumptions [16].

9.2 Why an interior exact dark-point center is unavailable

Proposition 9.1. *For any nondegenerate triangle, any point p in its closed projection, and any finite $c > 0$, $\int_T e^{ic|q-p|^2} dq \neq 0$.*

Proof. By convexity, the intersection of each inward ray from p with the triangle is $0 \leq r \leq R(\theta)$. Polar integration gives

$$\int_T e^{ic|q-p|^2} dq = \frac{1}{2ic} \int_{\Theta} (e^{icR(\theta)^2} - 1) d\theta.$$

Its imaginary part is $\frac{1}{2c} \int_{\Theta} [1 - \cos(cR(\theta)^2)] d\theta$. This is nonnegative. On an open interval of rays meeting a straight side, $R(\theta)$ varies continuously and is not constant; it cannot take only the discrete radii for which $cR^2 \in 2\pi\mathbb{Z}$. The integral is therefore strictly positive. At a boundary point use the inward sector; the same argument applies. $\square$

This excludes an interior first-null center in this model at every finite Fresnel number, not merely at the numerically sampled values. Exterior zeros, phase-structured illumination, opaque-obstacle diffraction and vector/nonparaxial models are separate questions. A circular aperture has a constant radial distance at its center and does not satisfy the strictness step.

An intensity minimum on the closed triangle still exists, but it can occur at the boundary or in a tied set. Averaging discovered minima need not produce a dark point and need not find every minimizer. The data's DarkMin diagnostics are therefore not treated as canonical TC-valued objects.

9.3 Darkness-weighted centers

Normalize intensity by its mean over the triangle,

$$J_F(p) = \frac{|A_F(p)|^2}{A_T^{-1} \int_T |A_F|^2}, \quad C_{F,\beta} = \frac{\int_T p e^{-\beta J_F(p)} dp}{\int_T e^{-\beta J_F(p)} dp}.$$

The weight assigns more influence to darker locations without selecting an arbitrary member of a minimum set. It is strictly positive for finite β , so the center is unique as a moment, strictly interior and equivariant. The physical measurement can be implemented by mapping measured intensity through a calibrated exponential response and taking its spatial centroid. This is a declared response law; it is not an intrinsic zero of the optical field.

We study $F = 0.5, 1, 2, 4$ and $\beta = 1, 8$. Low β is a mild preference; high β emphasizes small dark regions and can strongly amplify shape response. As $\beta \rightarrow 0$, or as $F \rightarrow 0$ at fixed finite β , the center tends to G . Increasing β without bound may concentrate on boundary minima or multiple sites; continuity of the limiting selector is not presumed.

For a less single-frequency choice, average normalized intensity uniformly in $\log F$ over $[1/2, 2]$:

$$\bar{J}(p) = \frac{1}{2 \log 2} \int_{1/2}^2 J_F(p) \frac{dF}{F}, \quad C_{\text{persist},\beta} = \frac{\int_T p e^{-\beta \bar{J}(p)} dp}{\int_T e^{-\beta \bar{J}(p)} dp}.$$

This first averages intensity, then applies the darkness response. Averaging already-selected centers or averaging $e^{-\beta J_F}$ would define other families. The interval and response strength are part of the convention.

Smooth triangle quadrature evaluates the inner amplitude integral and then the moment integral. Phase oscillations require increasing quadrature with F ; a nearly zero intensity is not a certified zero of the complex amplitude. Equilateral derivatives can be computed by differentiating these finite-domain integrals after mapping to a fixed reference triangle. This avoids subtracting two independently located extrema.

The numerical equilateral apex Jacobians for $(F, \beta) = (1, 8)$ and $(2, 8)$ are respectively

$$\text{diag}(-0.22453347, 0.89120014), \quad \text{diag}(-1.57953177, 2.24619843).$$

Both have the required trace $2/3$ and a negative eigenvalue. Quadrature orders 20, 28, 36 and 44 give stable signs. The first loss of the horizontal sign occurs near $\beta = 6.13171$ at $F = 1$ and 2.32530 at $F = 2$. These are numerical sign/threshold results, not outward-rounded interval certificates. Stable finite witnesses also occur for strong $F = 4$ darkness and persistent darkness. A smooth strictly positive weight guarantees a good location, but not attractivity.

9.4 Nonlinear Gaussian fluorescence

Let the image irradiance be a Gaussian blur of the uniform aperture,

$$I_s(p) = \int_T \exp[-|p - q|^2 / (2s^2)] dq, \quad s = \sigma L.$$

Linear detection preserves G . Suppose instead that the emission or detector response is proportional to $I_s(p)^2$, and define its centroid over the full image plane. Expand the square as a double integral. Completing the square in the Gaussian shows that the product from source points q, r is centered at $(q + r)/2$, with weight $\exp[-|q - r|^2 / (4s^2)]$. Therefore

$$\text{Opt}(\sigma) = \frac{\iint_{T \times T} (q + r) e^{-|q - r|^2 / (4s^2)} dq dr / 2}{\iint_{T \times T} e^{-|q - r|^2 / (4s^2)} dq dr}.$$

Symmetry in q, r rewrites this as a moment of the positive aperture density $q \mapsto \int_T e^{-|q-r|^2/(4s^2)} dr$. That is the density shown in the atlas, not the full-plane fluorescence field itself. It gives an immediate strict-interiority certificate although the measured image extends outside the aperture. The studied widths are $\sigma = 0.1, 0.3, 1$. Both endpoint limits tend to G by localization and dominated convergence respectively; intermediate means can differ.

9.5 Exact one-dimensional optical thin limits

The longitudinal area density of the collapsing triangle is

$$h_x(t) = \begin{cases} (t+1)/(x+1), & -1 \leq t \leq x, \\ (1-t)/(1-x), & x \leq t \leq 1, \end{cases} \quad \int_{-1}^1 h_x(t) dt = 1.$$

Replace each normalized area integral by $\int h_x(t) dt$ and each squared separation by $(t-s)^2$; also $L^2 \rightarrow 2 + 2x^2/3$. For the rescaled vertical mean, use conditional height $h_x(t)/2$. Dominated convergence justifies these finite- F , finite- σ , finite- β limits. They provide cheap one-dimensional checks on the two-dimensional solvers, including a well-defined one-sided endpoint profile at $x = 1$.

9.6 Other illumination constructions already examined

The solid-angle field is $\Omega(p, h) = \int_T h(|q-p|^2 + h^2)^{-3/2} dq$. Its maximizer defines a height-parameter family. The exterior representation gives $\Omega(p, h) = 2\pi - hB(p) + O(h^3)$ on compact interior subsets. An exterior-half-plane bound excludes boundary escape of maximizing points for small h , so the small-height limit is M1. The large-height limit is G . It is not the incenter-to-centroid path sometimes guessed from clearance intuition.

A separate inverse-square illumination construction renormalizes the divergent integral $\int_T |q-p|^{-2} dq$ by subtracting its universal local logarithmic divergence. Polar integration leaves a multiple of $\int \log R_p(\theta) d\theta$. Its stationary point is the renormalized illuminating center, numerically investigated in the initial study. It is not the magnetic minimum. Its legacy ETC shortlist used a different pilot metric, which must not be merged with the later uniform-angle rankings.

Maximin illumination of the three vertices reduces, under its own optimized-height model, to the minimum-enclosing-circle center, with the acute/obtuse distinction described in Chapter 3. A total-internal-reflection fiber acceptance objective additionally depends on refractive-index ratio and whether only the first reflection or indefinite guidance is required. The original first-wall test does not certify later reflections. Finally, a Fraunhofer central bright direction is a direction of observation, not an intrinsic point in the aperture. These qualified constructions remain part of the inventory rather than being lost from the survey.

9.7 Dark0.5b1.0: Darkness mean, $F=0.5$, $\beta=1$

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_{0.5}(p)} dp}{\int_T e^{-1J_{0.5}(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

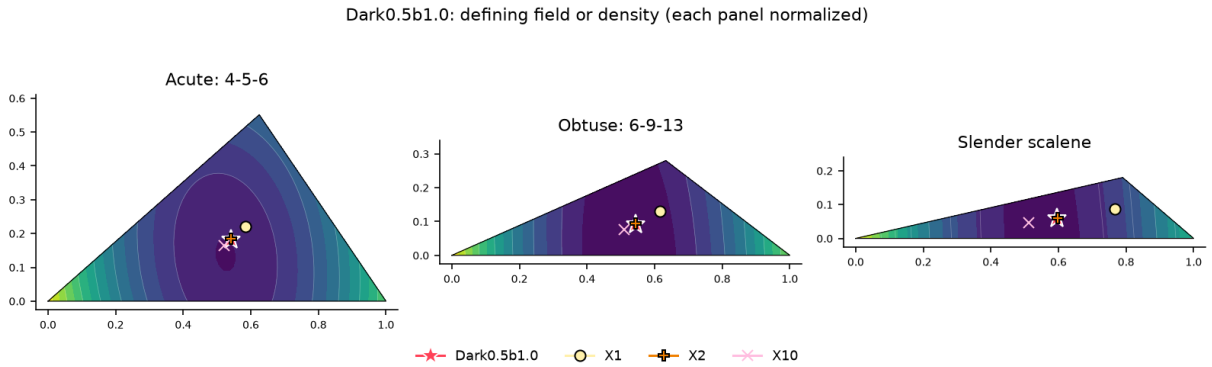
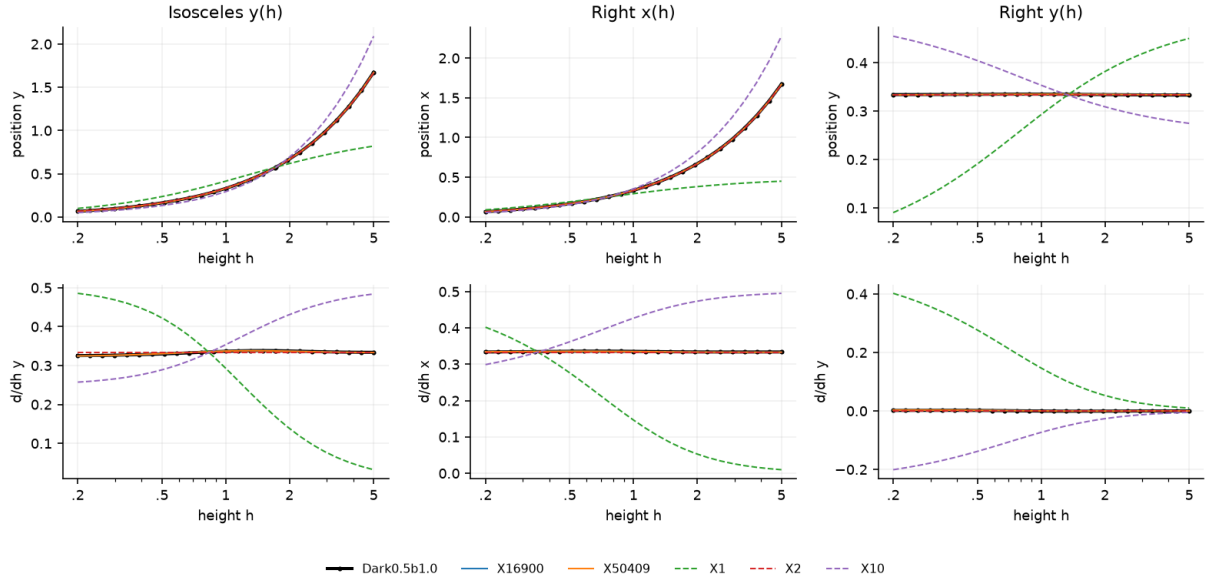


Figure 9.1: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Dark0.5b1.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(16900)	0.040%	0.066%	0.041%	0.067%
X(50409)	0.040%	0.066%	0.041%	0.067%
X(17514)	0.041%	0.067%	0.042%	0.067%

For the location winner, the stored family-derivative RMS discrepancy is 0.0007116 and response-margin RMS discrepancy is 0.00062516. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark0.5b1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.8 Dark0.5b8.0: Darkness mean, $F=0.5$, $\beta=8$

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_{0.5}(p)} dp}{\int_T e^{-8J_{0.5}(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

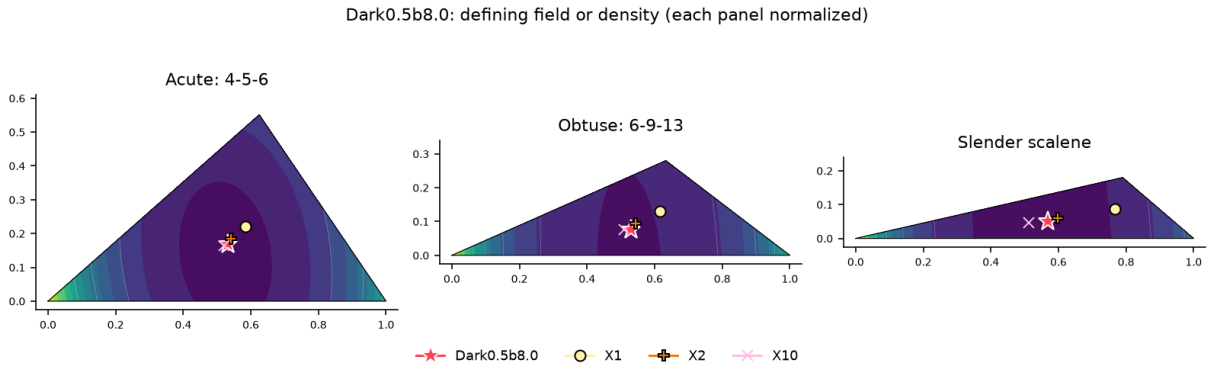
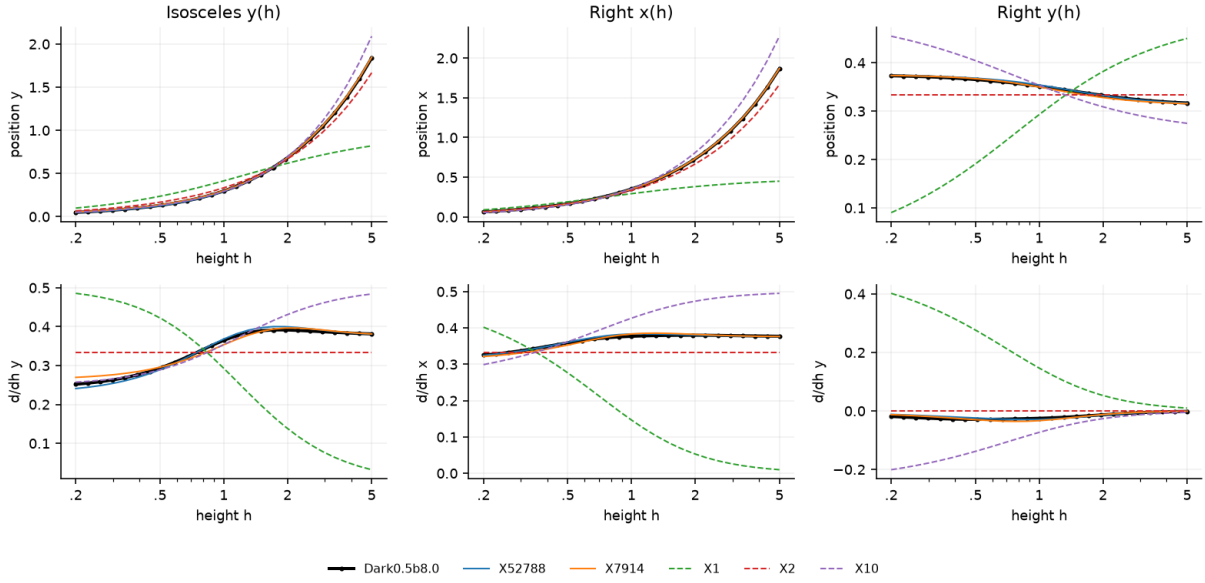


Figure 9.2: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Dark0.5b8.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(52788)	0.206%	0.297%	0.194%	0.287%
X(7914)	0.210%	0.360%	0.205%	0.360%
X(59214)	0.378%	0.673%	0.391%	0.674%

For the location winner, the stored family-derivative RMS discrepancy is 0.0058801 and response-margin RMS discrepancy is 0.0054872. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark0.5b8.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.9 Dark1.0b1.0: Darkness mean, $F=1$, $\beta=1$

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_1(p)} dp}{\int_T e^{-1J_1(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

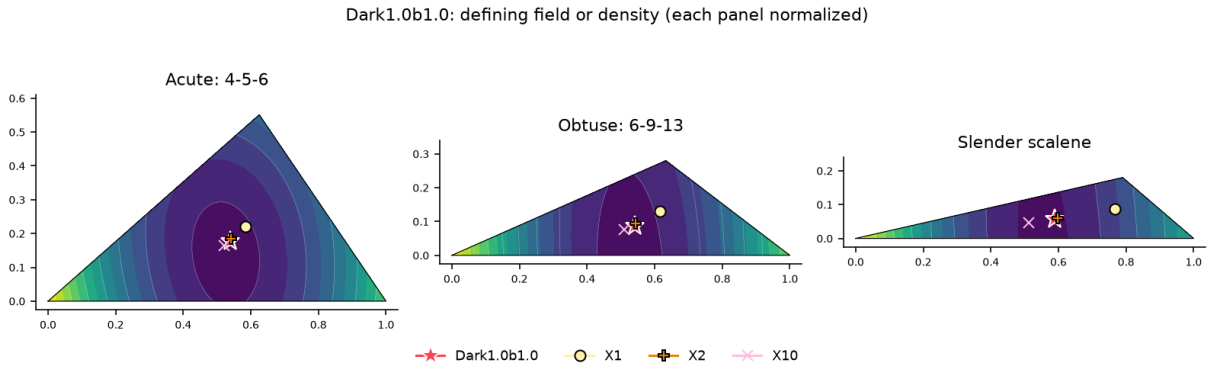
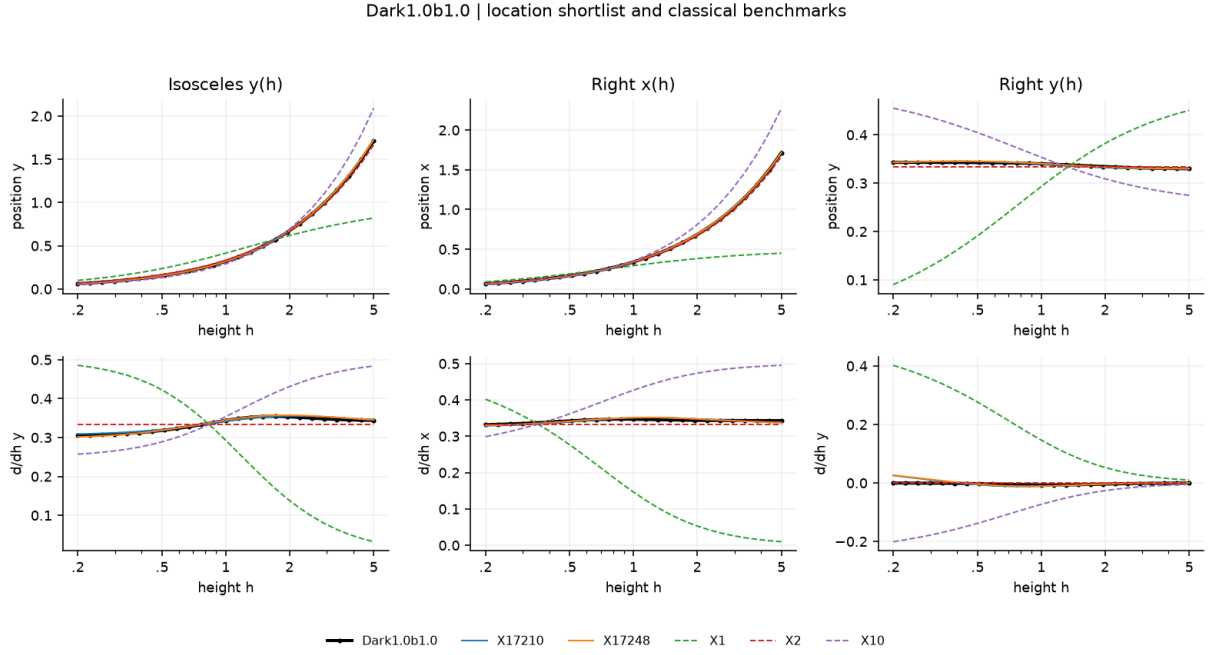


Figure 9.3: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(17210)	0.248%	0.724%	0.280%	0.888%
X(17248)	0.255%	0.724%	0.286%	0.888%
X(17306)	0.275%	0.779%	0.310%	0.906%

For the location winner, the stored family-derivative RMS discrepancy is 0.0067523 and response-margin RMS discrepancy is 0.0094687. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark1.0b1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.10 Dark1.0b8.0: Darkness mean, F=1, beta=8

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_1(p)} dp}{\int_T e^{-8J_1(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. Stable negative finite-response evidence; for F=1 and F=2 there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

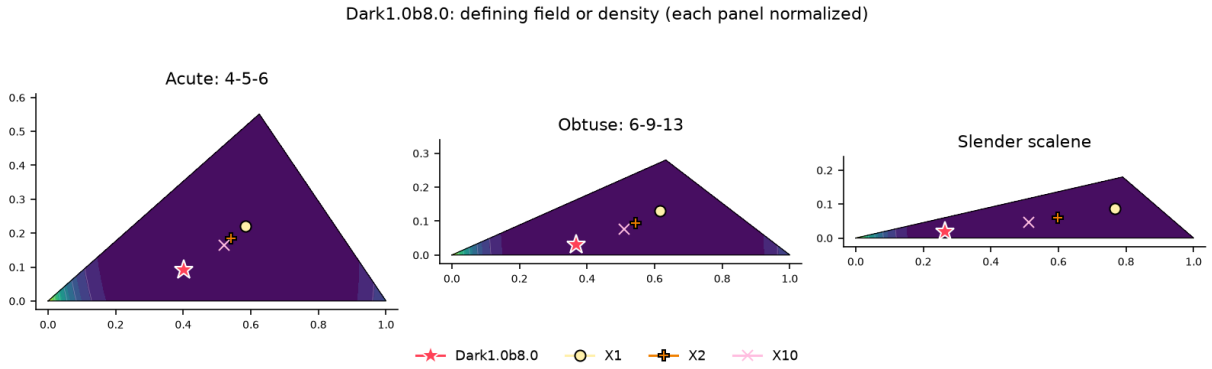
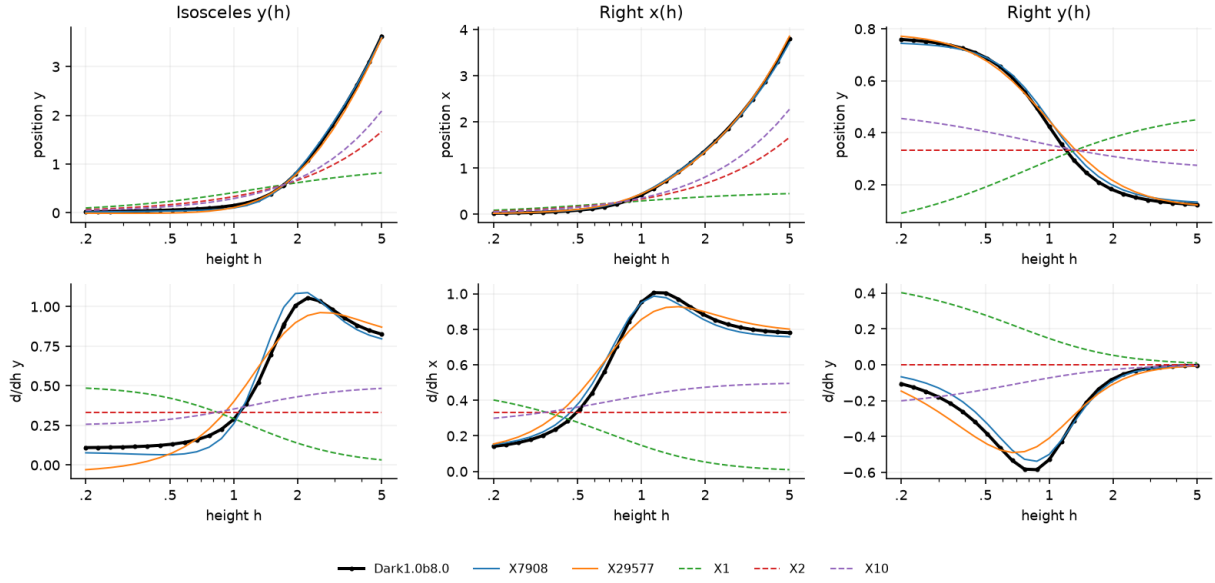


Figure 9.4: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Dark1.0b8.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(7908)	1.708%	2.602%	1.710%	2.537%
X(29577)	1.973%	3.292%	1.968%	3.166%
X(21712)	2.530%	5.018%	2.419%	5.040%

For the location winner, the stored family-derivative RMS discrepancy is 0.053359 and response-margin RMS discrepancy is 0.040955. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark1.0b8.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.11 Dark2.0b1.0: Darkness mean, $F=2$, $\beta=1$

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_2(p)} dp}{\int_T e^{-1J_2(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

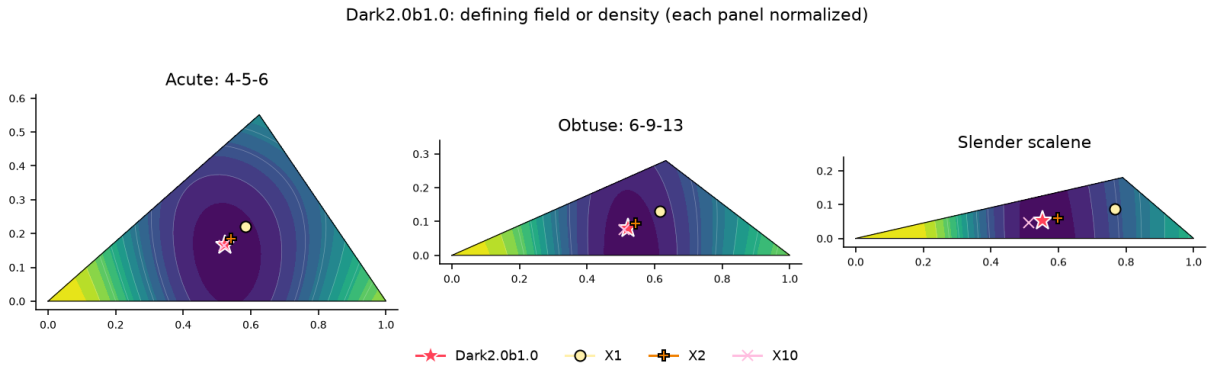
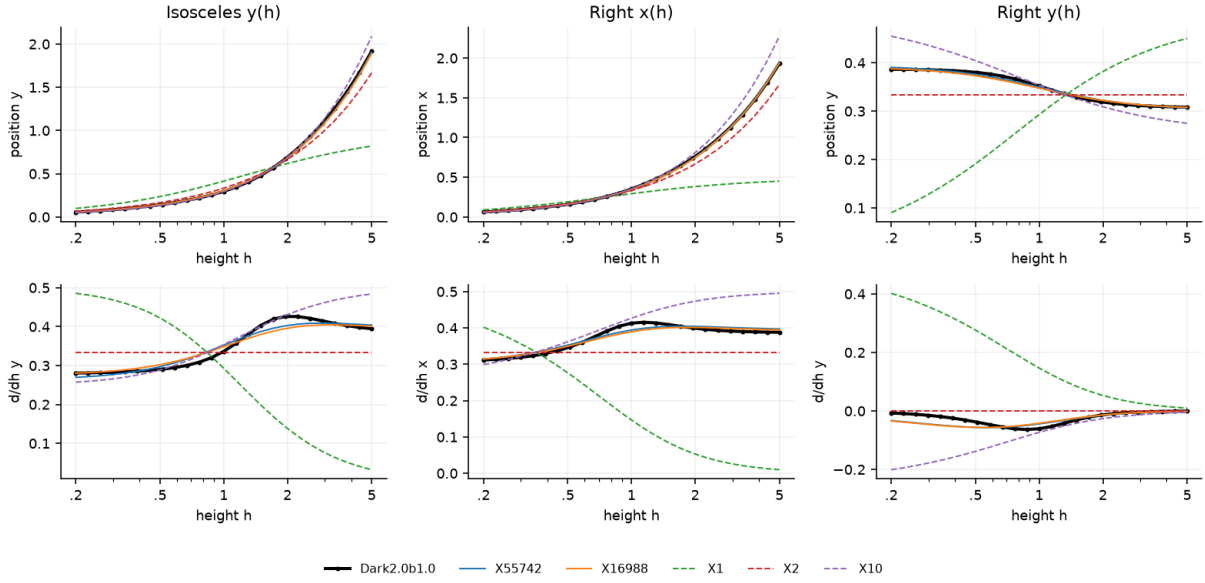


Figure 9.5: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Dark2.0b1.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(55742)	0.397%	0.667%	0.412%	0.676%
X(16988)	0.434%	0.872%	0.419%	0.855%
X(7944)	0.440%	0.673%	0.429%	0.668%

For the location winner, the stored family-derivative RMS discrepancy is 0.016252 and response-margin RMS discrepancy is 0.014069. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark2.0b1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.12 Dark2.0b8.0: Darkness mean, F=2, beta=8

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_2(p)} dp}{\int_T e^{-8J_2(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. Stable negative finite-response evidence; for F=1 and F=2 there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

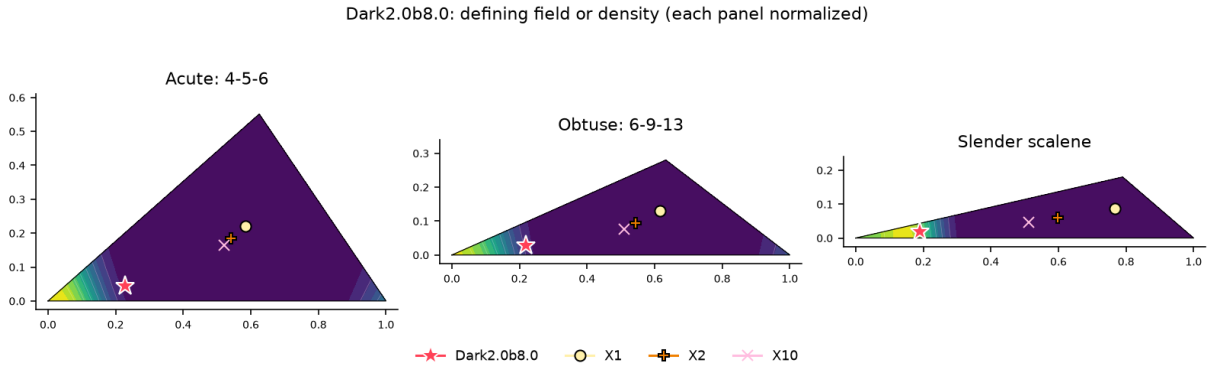
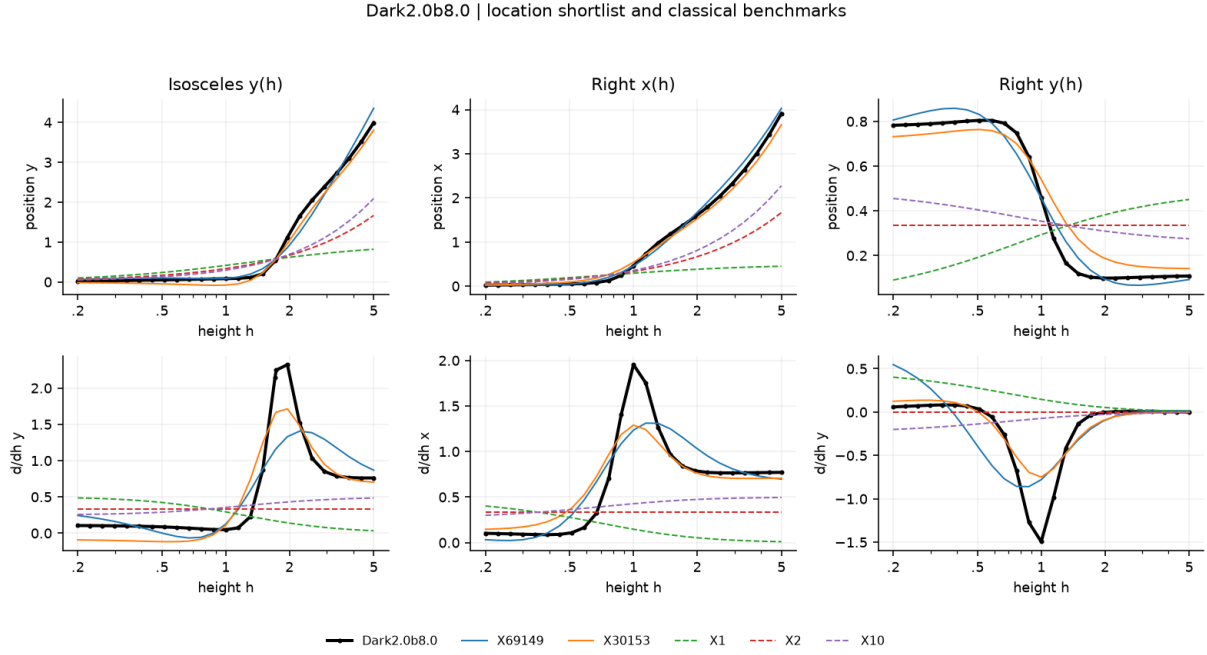


Figure 9.6: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(69149)	7.568%	16.819%	7.320%	14.708%
X(30153)	7.811%	12.205%	7.841%	12.126%
X(31000)	7.985%	18.297%	8.047%	16.942%

For the location winner, the stored family-derivative RMS discrepancy is 0.39207 and response-margin RMS discrepancy is 0.40571. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark2.0b8.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.13 Dark4.0b1.0: Darkness mean, $F=4$, $\beta=1$

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-1J_4(p)} dp}{\int_T e^{-1J_4(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. No negative response was found in the retained screen; a global proof is not supplied.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

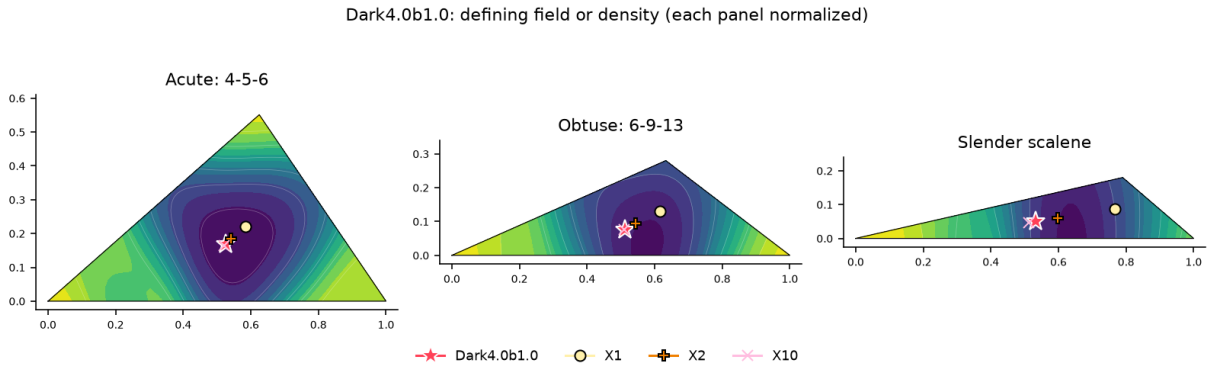
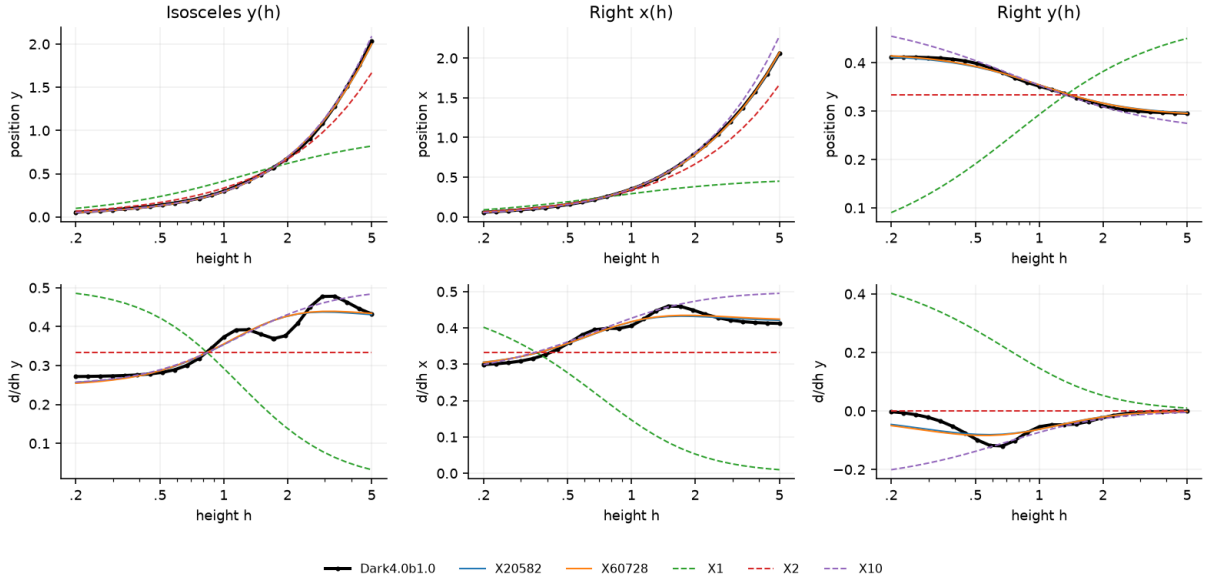


Figure 9.7: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: $X(1)$, $X(2)$, $X(10)$. Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Dark4.0b1.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(20582)	0.453%	0.862%	0.422%	0.756%
X(60728)	0.465%	0.871%	0.461%	0.776%
X(60643)	0.481%	0.920%	0.455%	0.928%

For the location winner, the stored family-derivative RMS discrepancy is 0.024647 and response-margin RMS discrepancy is 0.019894. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark4.0b1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.14 Dark4.0b8.0: Darkness mean, F=4, beta=8

Physical scenario and selector. Measure triangular-aperture Fresnel intensity at fixed Fresnel number, apply an exponential preference for darker positions, and take the weighted centroid over the aperture projection.

$$C = \frac{\int_T p e^{-8J_4(p)} dp}{\int_T e^{-8J_4(p)} dp}, \quad J_F = |A_F|^2 / \langle |A_F|^2 \rangle_T$$

Derivation and algorithm. Use nested triangle Gaussian quadrature for complex amplitudes and spatial moments, increasing order until both the center and its derivative stabilize. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. Strictly positive finite-beta weights prove a unique, strictly interior moment.

Attractivity. Stable negative finite-response evidence; for F=1 and F=2 there is also a negative equilateral derivative from differentiated quadrature. These signs are numerical, not interval certificates.

Distinguishing behavior. This is a darkness-weighted mean, not an exact dark point. Exact interior Fresnel nulls are excluded by Proposition 9.1. Wavelength independence requires adjusting screen distance to hold F fixed.

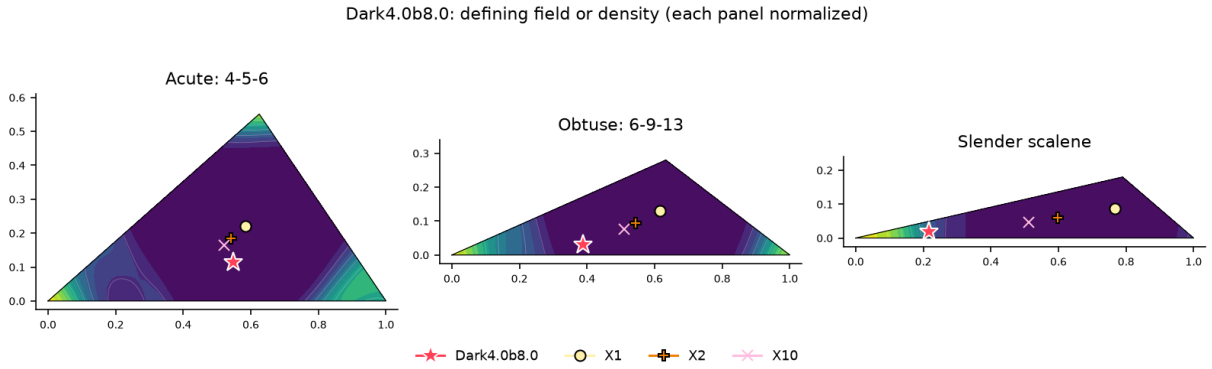
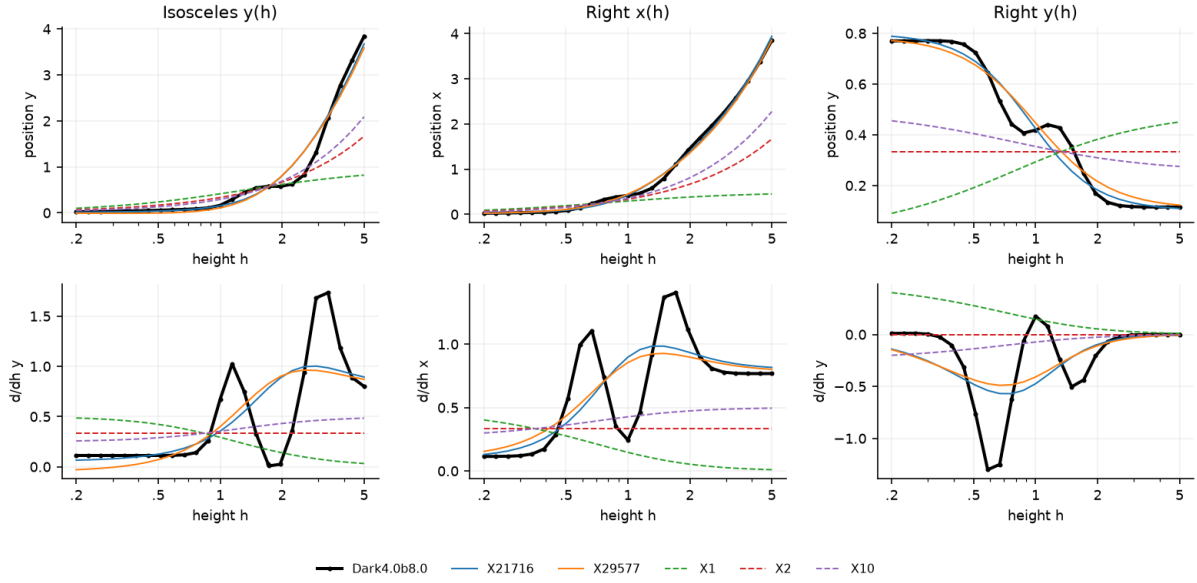


Figure 9.8: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Dark4.0b8.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

	ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(21716)		5.295%	18.091%	4.528%	13.025%
X(29577)		5.473%	18.143%	4.547%	12.730%
X(60637)		5.539%	12.718%	5.008%	10.551%

For the location winner, the stored family-derivative RMS discrepancy is 0.40521 and response-margin RMS discrepancy is 0.29202. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.006% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Dark4.0b8.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.15 Persist1.0: Log-F averaged darkness mean, beta=1

Physical scenario and selector. Average normalized intensity uniformly in log Fresnel number from one half to two, then apply the darkness response and take its mean.

$$\bar{J} = \frac{1}{2 \log 2} \int_{1/2}^2 J_F \frac{dF}{F}, \quad C = \int p e^{-1\bar{J}} / \int e^{-1\bar{J}}$$

Derivation and algorithm. Add Gauss quadrature in log F to the nested spatial integrals. Normalize intensity separately at every F before averaging. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive weight gives a unique strictly interior moment.

Attractivity. No violation in the retained screen; global attractivity remains open.

Distinguishing behavior. Averaging intensity first differs from averaging centers or exponential weights. Both the interval and response law are declared modeling choices.

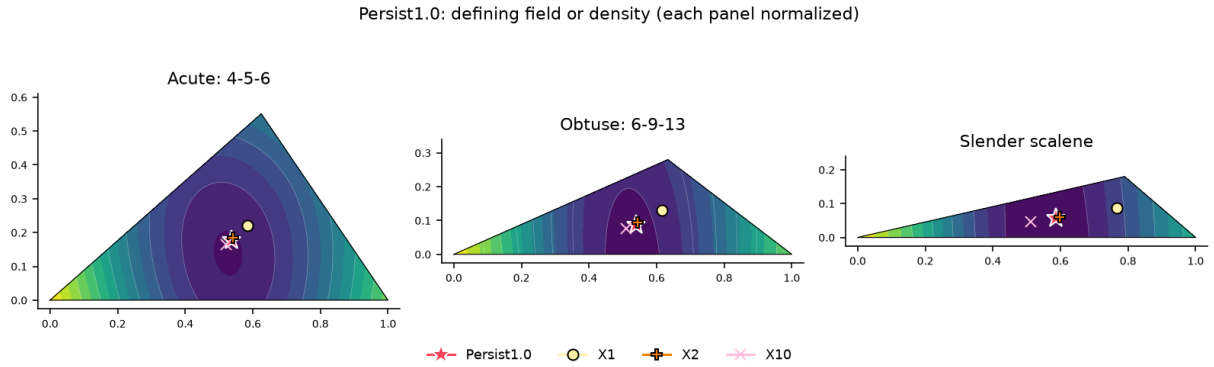
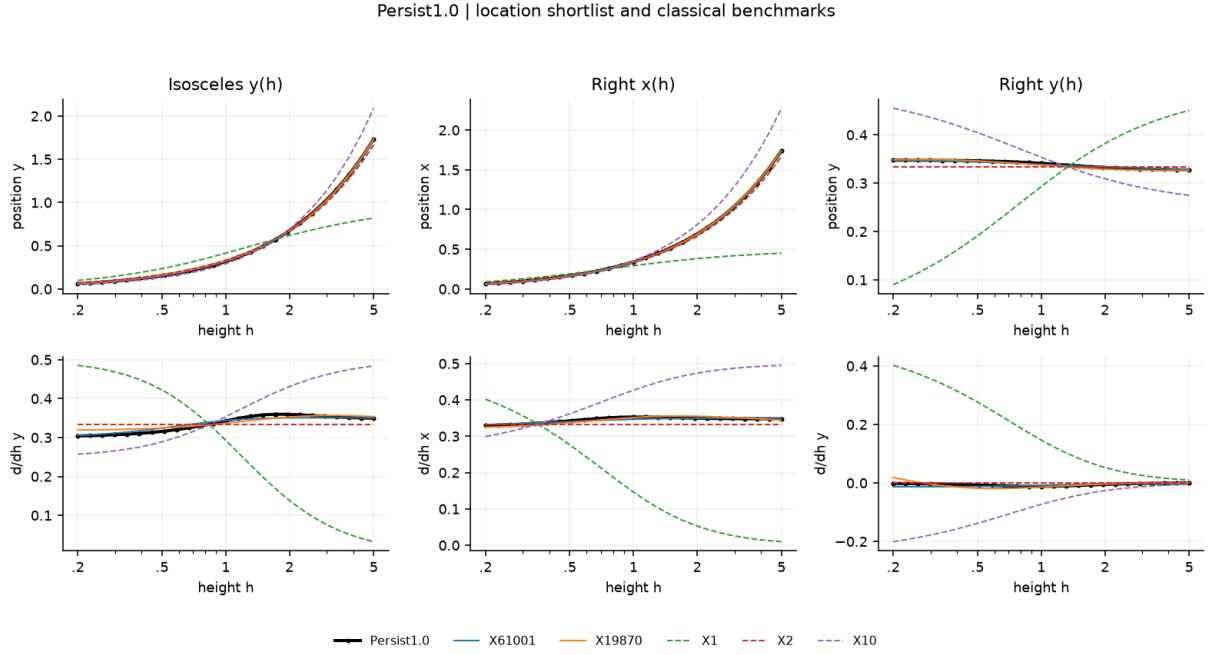


Figure 9.9: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(61001)	0.182%	0.271%	0.172%	0.272%
X(19870)	0.326%	0.931%	0.383%	1.263%
X(19872)	0.352%	0.564%	0.363%	0.564%

For the location winner, the stored family-derivative RMS discrepancy is 0.0062639 and response-margin RMS discrepancy is 0.0080263. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Persist1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.16 Persist8.0: Log-F averaged darkness mean, beta=8

Physical scenario and selector. Average normalized intensity uniformly in log Fresnel number from one half to two, then apply the darkness response and take its mean.

$$\bar{J} = \frac{1}{2 \log 2} \int_{1/2}^2 J_F \frac{dF}{F}, \quad C = \int p e^{-8\bar{J}} / \int e^{-8\bar{J}}$$

Derivation and algorithm. Add Gauss quadrature in log F to the nested spatial integrals. Normalize intensity separately at every F before averaging. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The positive weight gives a unique strictly interior moment.

Attractivity. Stable negative finite-displacement numerical witnesses; a validated continuum sign certificate remains open.

Distinguishing behavior. Averaging intensity first differs from averaging centers or exponential weights. Both the interval and response law are declared modeling choices.

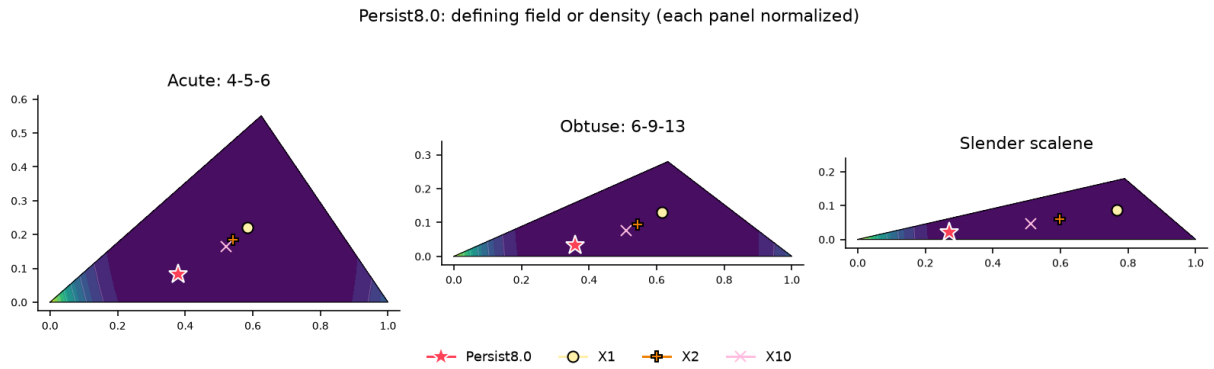
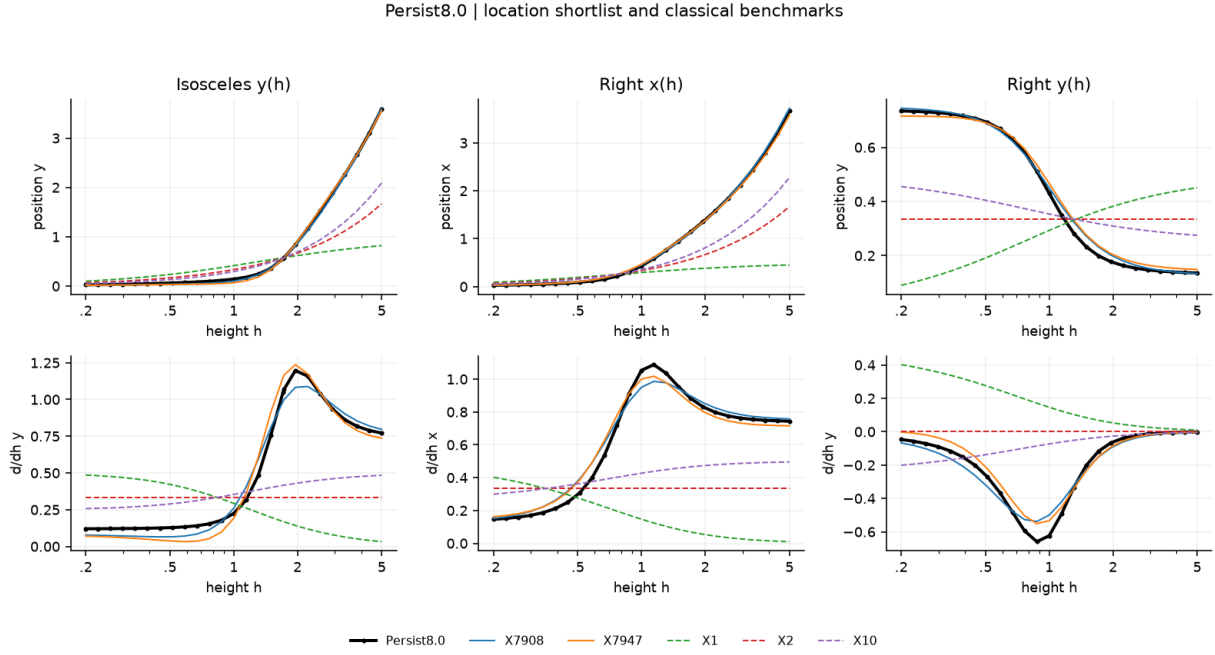


Figure 9.10: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(7908)	1.586%	2.602%	1.554%	2.526%
X(7947)	2.604%	4.436%	2.553%	3.506%
X(7895)	2.763%	4.746%	2.655%	4.550%

For the location winner, the stored family-derivative RMS discrepancy is 0.06426 and response-margin RMS discrepancy is 0.073805. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Persist8.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.17 Opt0.1: Nonlinear fluorescence mean, sigma=0.1

Physical scenario and selector. Blur a uniform luminous triangle by a Gaussian of width sigma times L and measure the full-plane centroid after quadratic intensity detection.

$$C = \frac{\iint (q + r) e^{-|q-r|^2/(4(0.1L)^2)} dq dr}{2 \iint e^{-|q-r|^2/(4(0.1L)^2)} dq dr}$$

Derivation and algorithm. Use the exact Gaussian pair reduction and smooth double triangle quadrature. This avoids truncating a full-plane image integral. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The pair formula is a positive mean of points inside the triangle and gives strict interiority.

Attractivity. No violation in the retained 100-shape screen; global attractivity is open.

Distinguishing behavior. Both very small and very large sigma tend to X(2). The contours show the equivalent positive aperture density, whose mean equals the full-plane quadratic fluorescence centroid.

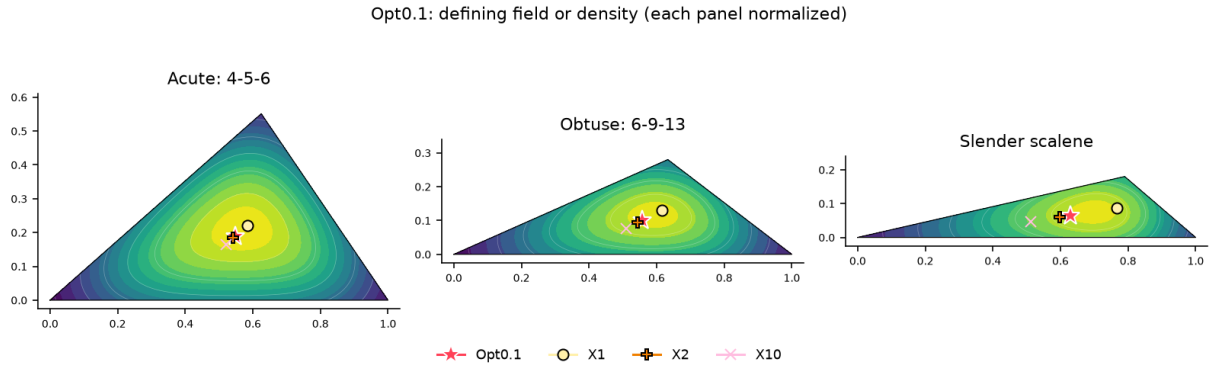
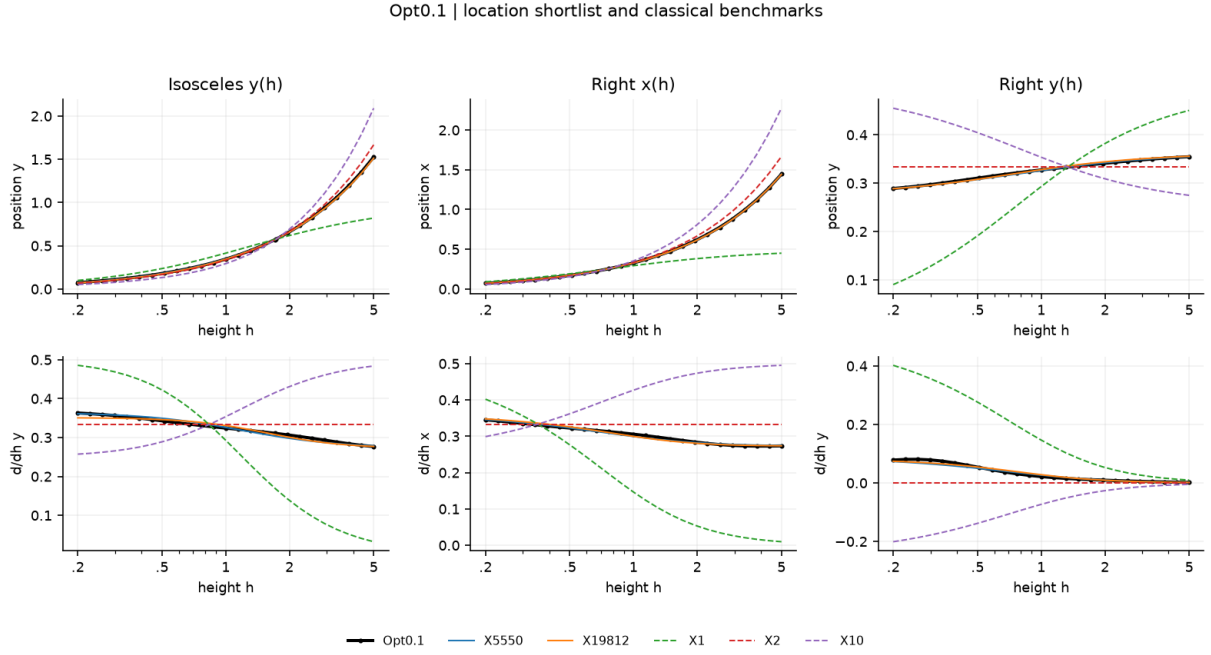


Figure 9.11: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(5550)	0.185%	0.400%	0.196%	0.586%
X(19812)	0.197%	0.426%	0.211%	0.594%
X(30598)	0.238%	0.451%	0.248%	0.594%

For the location winner, the stored family-derivative RMS discrepancy is 0.0060986 and response-margin RMS discrepancy is 0.011007. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Opt0.1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.18 Opt0.3: Nonlinear fluorescence mean, sigma=0.3

Physical scenario and selector. Blur a uniform luminous triangle by a Gaussian of width sigma times L and measure the full-plane centroid after quadratic intensity detection.

$$C = \frac{\iint (q + r) e^{-|q-r|^2/(4(0.3L)^2)} dq dr}{2 \iint e^{-|q-r|^2/(4(0.3L)^2)} dq dr}$$

Derivation and algorithm. Use the exact Gaussian pair reduction and smooth double triangle quadrature. This avoids truncating a full-plane image integral. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The pair formula is a positive mean of points inside the triangle and gives strict interiority.

Attractivity. No violation in the retained 100-shape screen; global attractivity is open.

Distinguishing behavior. Both very small and very large sigma tend to X(2). The contours show the equivalent positive aperture density, whose mean equals the full-plane quadratic fluorescence centroid.

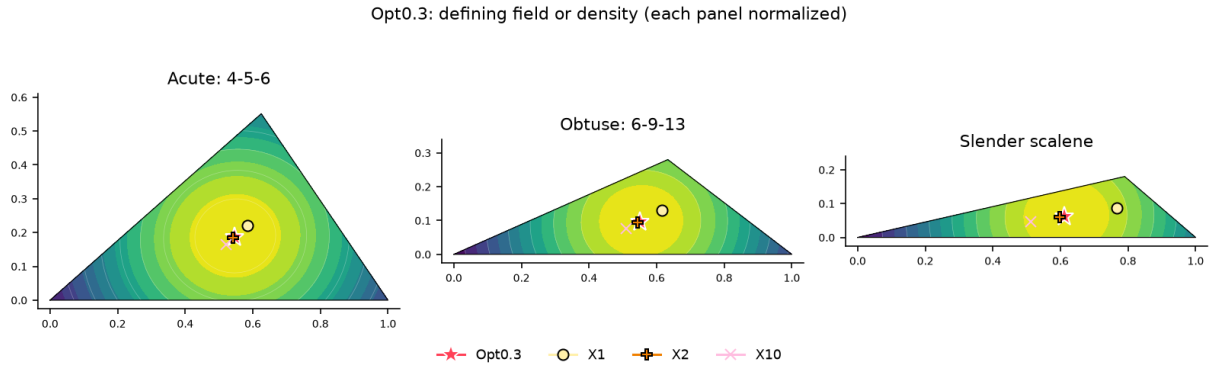
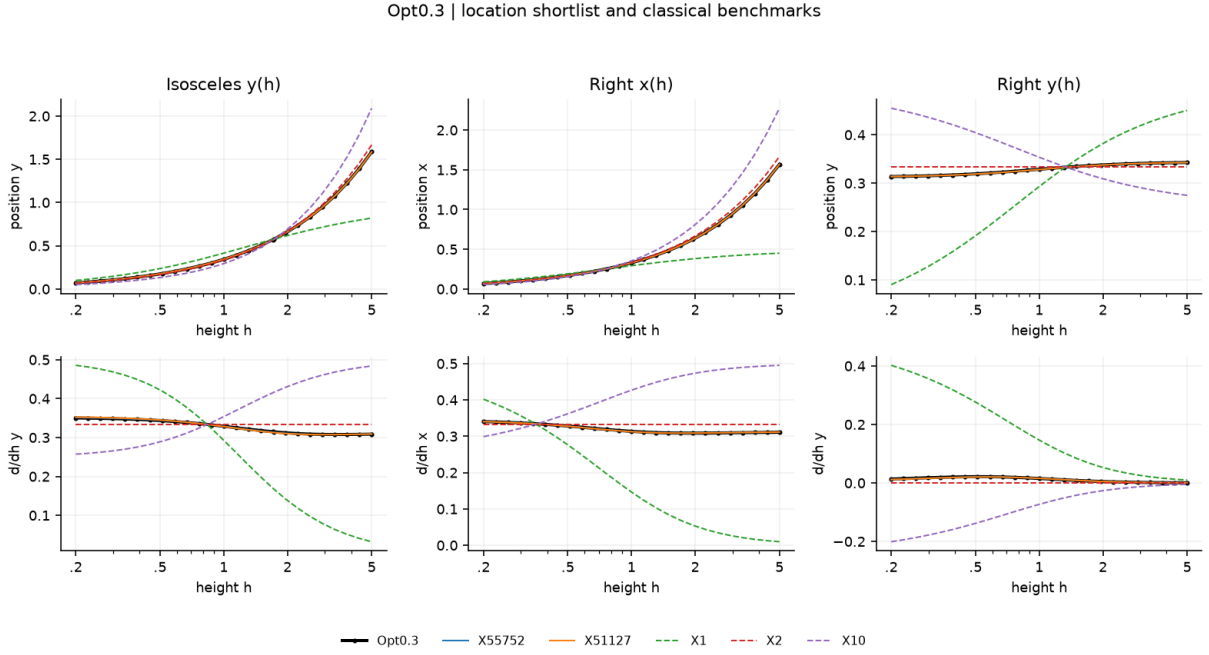


Figure 9.12: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(55752)	0.023%	0.034%	0.023%	0.033%
X(51127)	0.038%	0.064%	0.038%	0.064%
X(62940)	0.120%	0.204%	0.127%	0.205%

For the location winner, the stored family-derivative RMS discrepancy is 0.00082664 and response-margin RMS discrepancy is 0.00046494. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Opt0.3_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.19 Opt1.0: Nonlinear fluorescence mean, sigma=1

Physical scenario and selector. Blur a uniform luminous triangle by a Gaussian of width sigma times L and measure the full-plane centroid after quadratic intensity detection.

$$C = \frac{\iint (q+r) e^{-|q-r|^2/(4(1L)^2)} dqdr}{2 \iint e^{-|q-r|^2/(4(1L)^2)} dqdr}$$

Derivation and algorithm. Use the exact Gaussian pair reduction and smooth double triangle quadrature. This avoids truncating a full-plane image integral. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The pair formula is a positive mean of points inside the triangle and gives strict interiority.

Attractivity. No violation in the retained 100-shape screen; global attractivity is open.

Distinguishing behavior. Both very small and very large sigma tend to X(2). The contours show the equivalent positive aperture density, whose mean equals the full-plane quadratic fluorescence centroid.

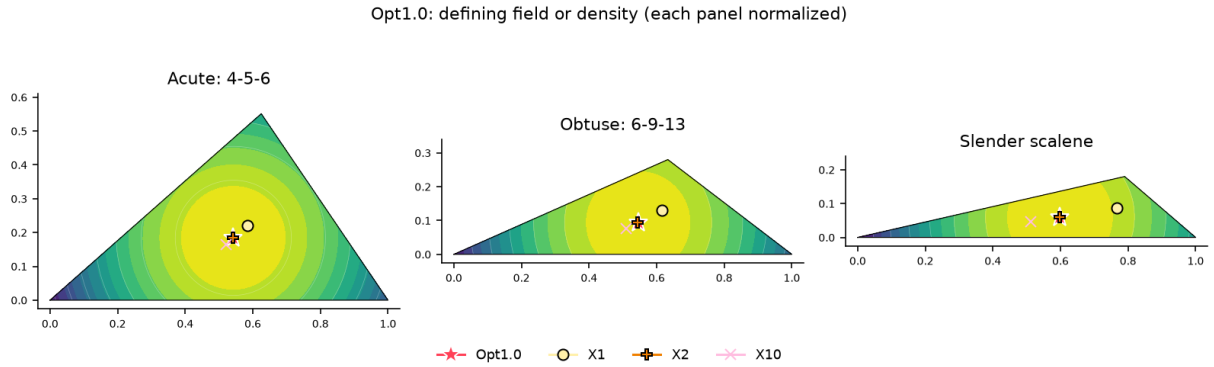
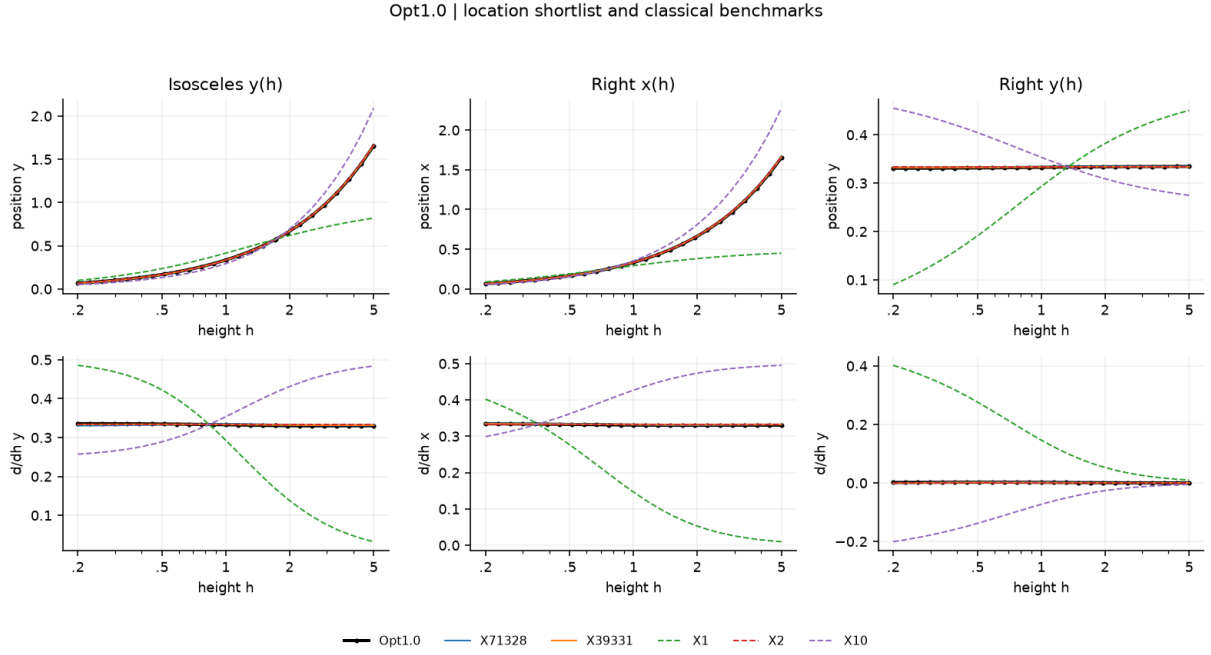


Figure 9.13: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(71328)	0.113%	0.223%	0.122%	0.253%
X(39331)	0.129%	0.242%	0.137%	0.265%
X(53004)	0.147%	0.264%	0.155%	0.267%

For the location winner, the stored family-derivative RMS discrepancy is 0.0022782 and response-margin RMS discrepancy is 0.0028032. These have their own normalizations and are not percentages of location error.

Resolution record. 100/100 training and 40/40 holdout triangles meet the retained successive-change target for this entry. The largest last training change is 0.000% of diameter. These are convergence diagnostics, not rigorous continuum error bounds.

Reproducibility. The bundle directory `output/mechanical_optical/` contains the 100-shape samples, held-out samples, thin and equilateral diagnostics, response tests, and top-ten ETC shortlist. The exact plotted numbers are also saved in `Opt1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.20 Illum: Renormalized inverse-square illumination center

Physical scenario and selector. Measure the renormalized inverse-square illumination received from a uniform triangular source, subtracting the universal local logarithmic divergence.

$$C = \arg \max_{p \in T} \int_{S^1} \log R_p(\theta) d\theta$$

Derivation and algorithm. Solve the equal edge-angle/area stationarity equations in positive barycentric coordinates. The illustrative contour uses angular ray quadrature, independently of the root equations. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The selected stationary root is checked to be interior. Root residuals alone do not certify a global uniqueness theorem.

Attractivity. Global attractivity remains unresolved. The new location scan is not a vertex-response certification.

Distinguishing behavior. This is distinct from solid-angle illumination and M1. The new common-angle location shortlist supersedes the earlier pilot ranking only for location, not for full fingerprint similarity.

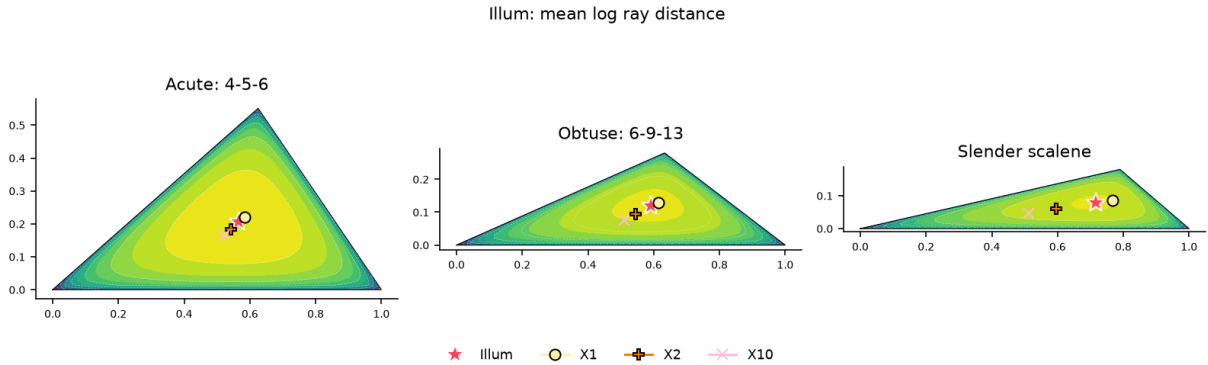
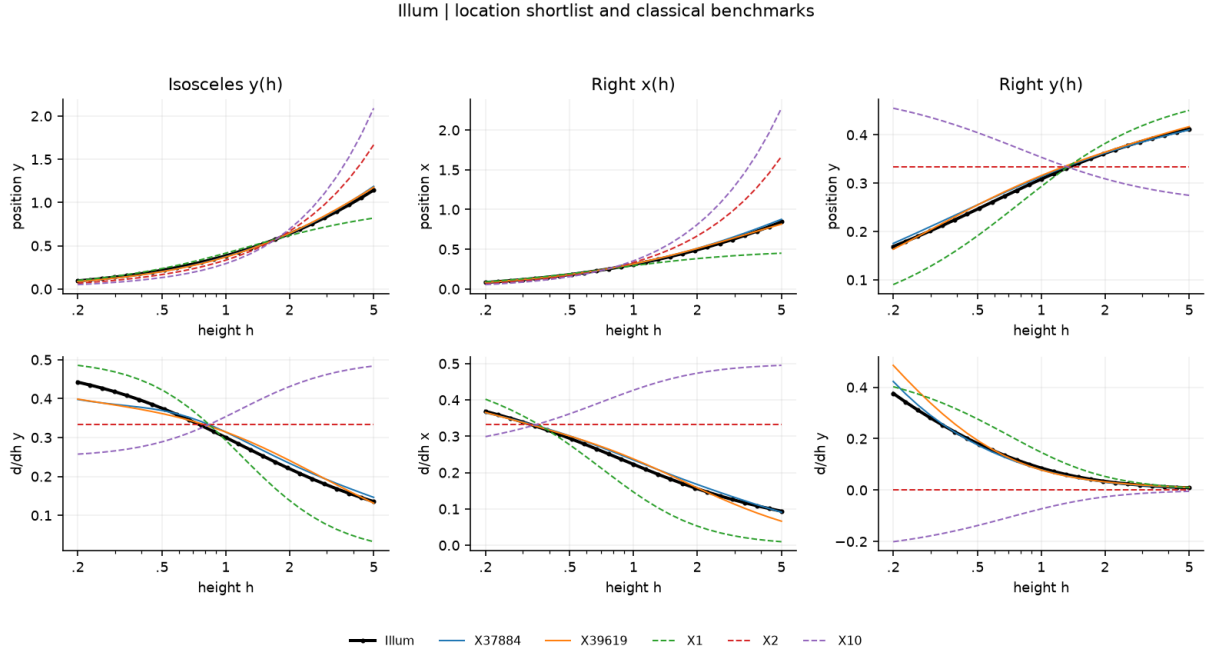


Figure 9.14: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(37884)	0.833%	1.753%	0.787%	2.118%
X(39619)	1.095%	3.069%	1.036%	2.663%
X(29622)	1.100%	4.177%	1.441%	5.577%

Resolution record. This supplementary floating-point root/optimizer study uses the same 100 training and 40 holdout shapes. It has no interval error bounds or full mesh-refinement campaign; close ETC ordering should be regarded as provisional.

Reproducibility. The bundle directory `output/illumination/` contains the 100-shape and held-out samples, family positions and derivatives, and top-ten ETC location shortlist. The exact plotted numbers are also saved in `Illum_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.21 Solid0.1: Solid-angle illumination maximum, height/L=0.1

Physical scenario and selector. Place an isotropic point source above the plane at a fixed dimensionless height and maximize the fraction of rays intercepted by the triangular receiver.

$$C = \arg \max_p \int_T \frac{0.1L}{(|q-p|^2 + (0.1L)^2)^{3/2}} dq$$

Derivation and algorithm. Evaluate the exact triangle solid-angle atan2 formula and optimize in barycentric coordinates. Hold height/L fixed when comparing different triangle shapes. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The numerical maximizer is interior. The projection onto a convex triangle cannot decrease this radial-kernel integral, so an outside global maximum is unnecessary; uniqueness still deserves separate analysis.

Attractivity. Global attractivity remains open here. A location comparison and family derivative curves are not a full response-sign test.

Distinguishing behavior. As height/L tends to zero the center approaches M1; as it tends to infinity it approaches X(2). These endpoint statements do not identify an arbitrary finite-height center with either endpoint.

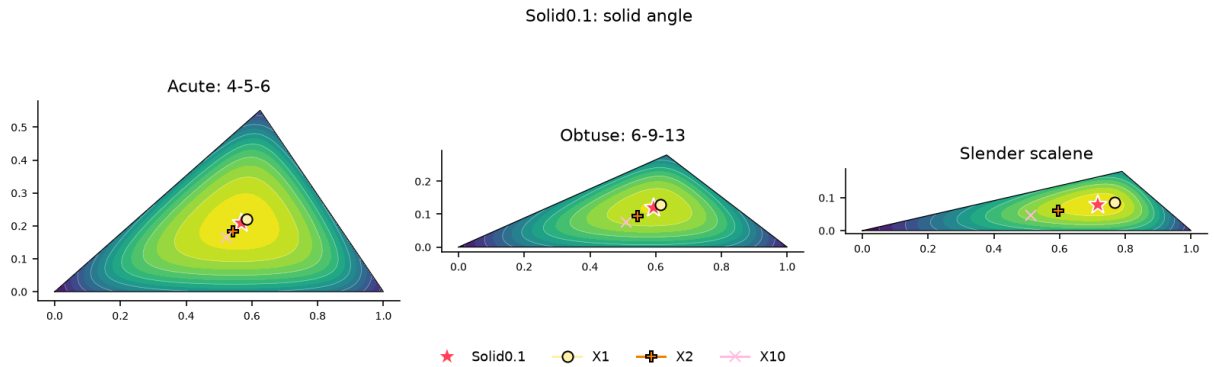
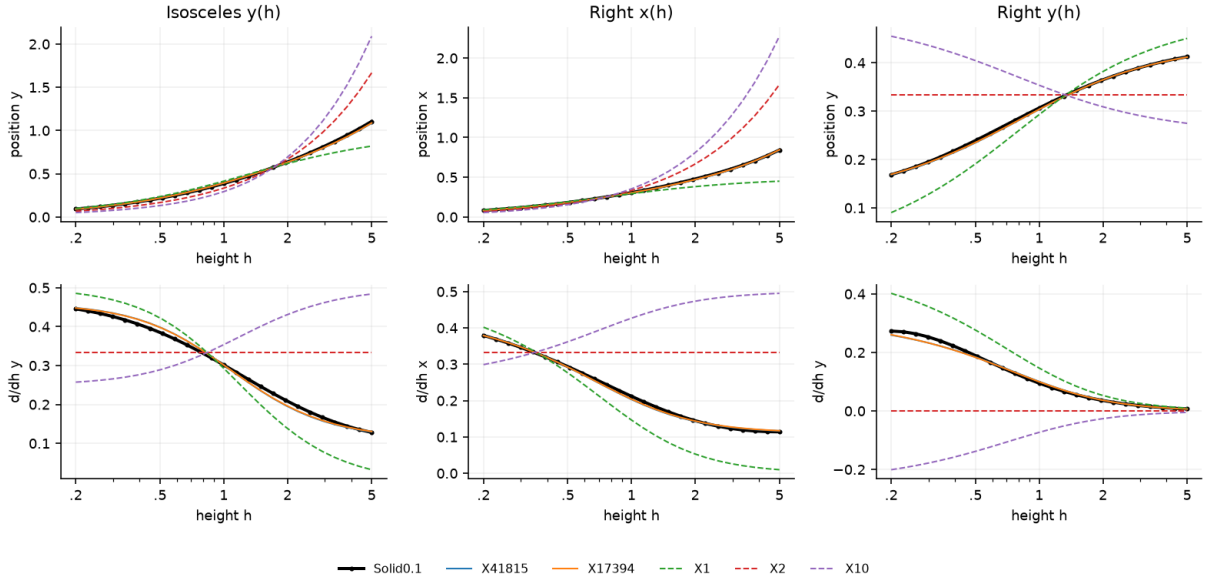


Figure 9.15: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Solid0.1 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(41815)	0.446%	1.283%	0.491%	1.573%
X(17394)	0.451%	1.334%	0.492%	1.567%
X(64667)	0.460%	1.439%	0.491%	1.572%

Resolution record. This supplementary floating-point root/optimizer study uses the same 100 training and 40 holdout shapes. It has no interval error bounds or full mesh-refinement campaign; close ETC ordering should be regarded as provisional.

Reproducibility. The bundle directory `output/illumination/` contains the 100-shape and held-out samples, family positions and derivatives, and top-ten ETC location shortlist. The exact plotted numbers are also saved in `Solid0.1_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.22 Solid1.0: Solid-angle illumination maximum, height/L=1

Physical scenario and selector. Place an isotropic point source above the plane at a fixed dimensionless height and maximize the fraction of rays intercepted by the triangular receiver.

$$C = \arg \max_p \int_T \frac{1L}{(|q-p|^2 + (1L)^2)^{3/2}} dq$$

Derivation and algorithm. Evaluate the exact triangle solid-angle atan2 formula and optimize in barycentric coordinates. Hold height/L fixed when comparing different triangle shapes. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The numerical maximizer is interior. The projection onto a convex triangle cannot decrease this radial-kernel integral, so an outside global maximum is unnecessary; uniqueness still deserves separate analysis.

Attractivity. Global attractivity remains open here. A location comparison and family derivative curves are not a full response-sign test.

Distinguishing behavior. As height/L tends to zero the center approaches M1; as it tends to infinity it approaches X(2). These endpoint statements do not identify an arbitrary finite-height center with either endpoint.

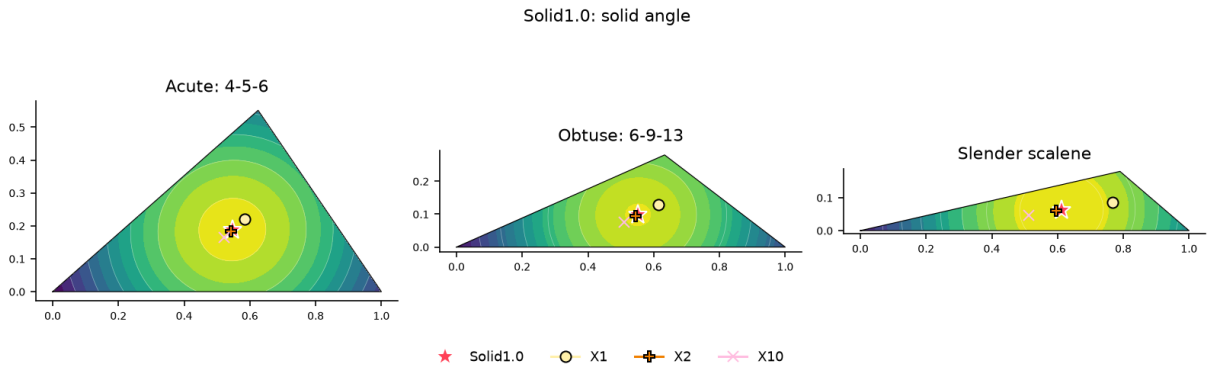
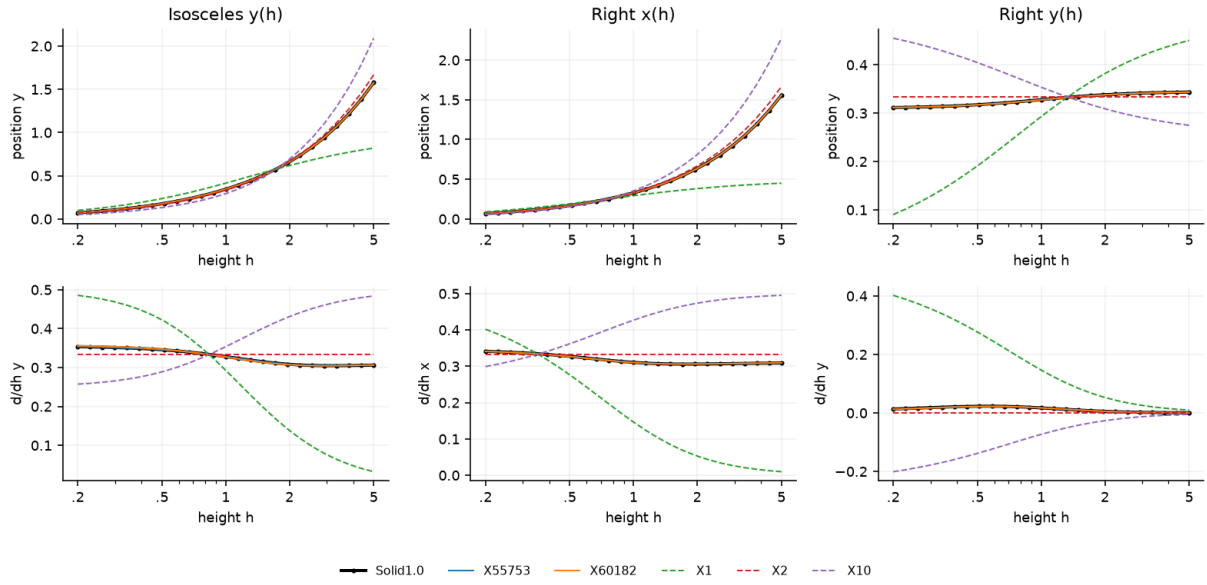


Figure 9.16: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence

Solid1.0 | location shortlist and classical benchmarks



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(55753)	0.011%	0.015%	0.011%	0.015%
X(60182)	0.034%	0.065%	0.036%	0.065%
X(51127)	0.147%	0.229%	0.153%	0.230%

Resolution record. This supplementary floating-point root/optimizer study uses the same 100 training and 40 holdout shapes. It has no interval error bounds or full mesh-refinement campaign; close ETC ordering should be regarded as provisional.

Reproducibility. The bundle directory `output/illumination/` contains the 100-shape and held-out samples, family positions and derivatives, and top-ten ETC location shortlist. The exact plotted numbers are also saved in `Solid1.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

9.23 Solid10.0: Solid-angle illumination maximum, height/L=10

Physical scenario and selector. Place an isotropic point source above the plane at a fixed dimensionless height and maximize the fraction of rays intercepted by the triangular receiver.

$$C = \arg \max_p \int_T \frac{10L}{(|q-p|^2 + (10L)^2)^{3/2}} dq$$

Derivation and algorithm. Evaluate the exact triangle solid-angle atan2 formula and optimize in barycentric coordinates. Hold height/L fixed when comparing different triangle shapes. The preceding family derivation explains how the physical balance or measurement leads to this equation and its boundary conditions.

Location and well-posedness. The numerical maximizer is interior. The projection onto a convex triangle cannot decrease this radial-kernel integral, so an outside global maximum is unnecessary; uniqueness still deserves separate analysis.

Attractivity. Global attractivity remains open here. A location comparison and family derivative curves are not a full response-sign test.

Distinguishing behavior. As height/L tends to zero the center approaches M1; as it tends to infinity it approaches X(2). These endpoint statements do not identify an arbitrary finite-height center with either endpoint.

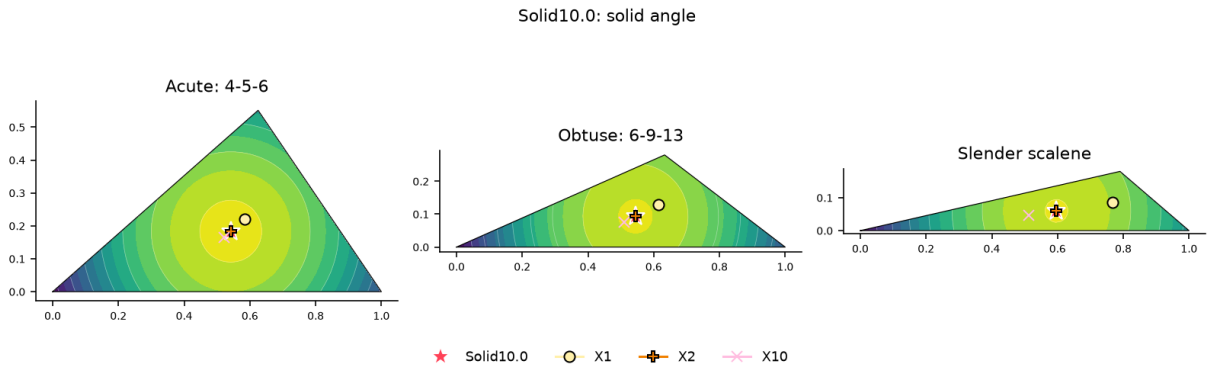
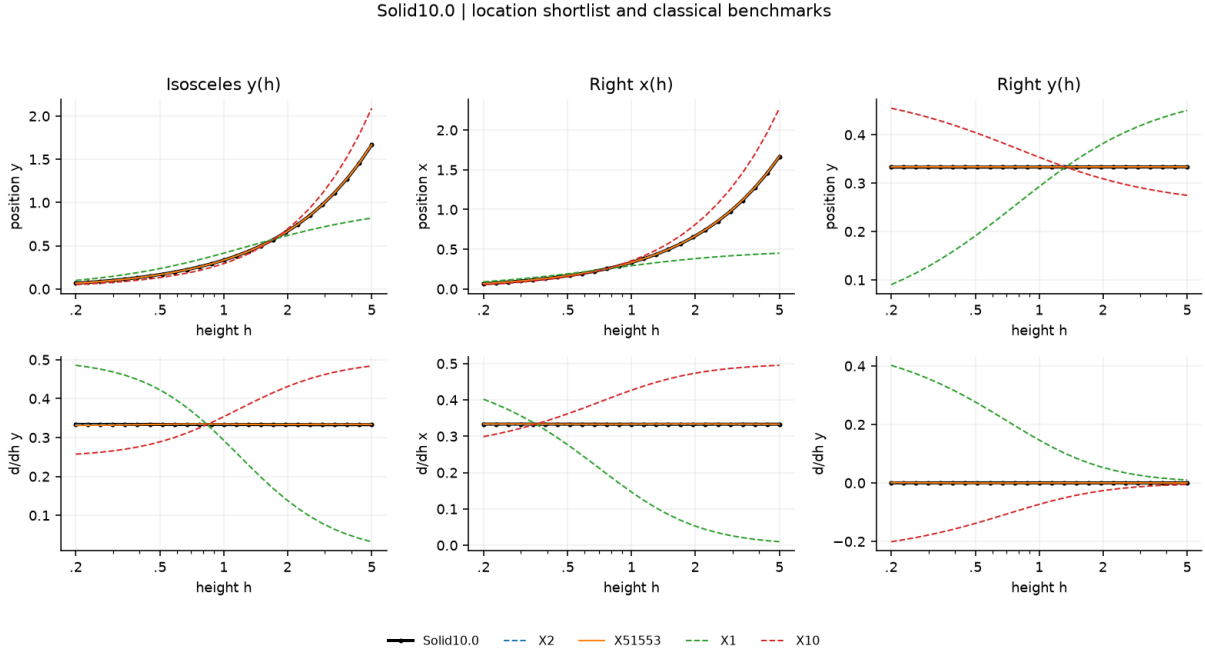


Figure 9.17: Field/density atlas in three scalene triangles. Red star: selected PITC; other markers: X(1), X(2), X(10). Each panel is separately normalized; these illustrative discretizations are not error enclosures.

Fingerprints and comparison evidence



Black curves show the investigated center. Colored curves show up to two location-shortlist ETC neighbors and the three classical benchmarks. The top row gives positions; the bottom row gives derivatives with respect to the family parameter h . Gaps denote undefined or excluded values. Crossing curves alone do not establish equality or attractivity.

ETC	Grid RMS	Grid max	Holdout RMS	Holdout max
X(2)	0.019%	0.028%	0.019%	0.028%
X(51553)	0.042%	0.061%	0.041%	0.058%
X(53004)	0.043%	0.089%	0.041%	0.085%

Resolution record. This supplementary floating-point root/optimizer study uses the same 100 training and 40 holdout shapes. It has no interval error bounds or full mesh-refinement campaign; close ETC ordering should be regarded as provisional.

Reproducibility. The bundle directory `output/illumination/` contains the 100-shape and held-out samples, family positions and derivatives, and top-ten ETC location shortlist. The exact plotted numbers are also saved in `Solid10.0_plotdata.json`. A close listed neighbor is an approximation on these samples, never an asserted ETC identity.

Chapter 10

Mass, charge and magnetic atlas extension

The September 30 audit reconciles the two dedicated chapters with this master catalogue. It adds eight entries, M2-M7 and L3-L4, and updates the inherited E1-E4, S3-S4 and M1 evidence. Read this chapter together with the original defining equations and the two expanded chapter PDFs.

The September 30 survey visits all 72,807 entries of the supplied ETC snapshot. The restricted evaluator retains 70,853 candidates finite on all 100 training shapes; the other 1,954 entries are excluded, not proved dissimilar. Each top-ten list is checked on 40 held-out shapes and compared separately by position fingerprints, derivative fingerprints and three-vertex response margins. Location RMS divided by triangle diameter is the ranking criterion. Thus this is a whole-snapshot computational survey, not an exhaustive symbolic classification or a proof of ETC nonmembership. E2 uses the independent exterior conformal evaluation. E3 retains the earlier resolved uniform-panel location grid; the new coarse strip-mesh grid is exploratory and excluded because it has poorer general-shape convergence. Its separate, finely resolved thin-triangle negative-motion experiment has positive charges at every level.

10.1 E1: Inner conformal / least-capacity center

Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.02086913).

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X53588	0.00253639	0.00246438	0.00530917
X66575	0.00293025	0.00297047	0.00670528
X66560	0.00403642	0.00399903	0.00582504
X29624	0.0042111	0.00389788	0.00830898
X8965	0.00435424	0.00433406	0.00686318

For X53588, position-fingerprint RMS / D is 0.00145761, derivative-fingerprint RMS is 0.00888637, and response-margin RMS difference is 0.0088342. The fingerprint comparison has 51 finite samples out of 51. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

10.2 E2: Outer conformal / planar charge centroid

Not globally attractive: interval-certified negative finite vertex motion (normalized dot product about -0.04665096), independently evaluated with the exterior Schwarz-Christoffel map.

Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X59474	0.00134526	0.00130542	0.0024001
X5235	0.00222067	0.00213914	0.00350119
X49731	0.00228472	0.0021109	0.00373791
X31494	0.00255483	0.00248814	0.00454243
X25960	0.00286186	0.00266665	0.00566459

For X59474, position-fingerprint RMS / D is 0.00152878, derivative-fingerprint RMS is 0.00348358, and response-margin RMS difference is 0.00444283. The fingerprint comparison has 51 finite samples out of 51. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

10.3 E3: Three-dimensional plate charge centroid

Refined numerical evidence of nonattractivity: negative finite motions persist on positive-charge anisotropic meshes up to 3,180 panels. The finest halved-step ratio is -0.55181093. A continuum interval certificate remains open.

Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X27760	0.00268211	0.00435905	0.0193914
X31351	0.00379642	0.0040384	0.0107557
X46896	0.00521082	0.0079198	0.0334464
X30915	0.00522478	0.00795365	0.0336552
X9780	0.00527946	0.00794309	0.0333402

For X27760, position-fingerprint RMS / D is 0.00237028, derivative-fingerprint RMS is 0.00583816, and response-margin RMS difference is 0.0245666. The fingerprint comparison has 51 finite samples out of 51. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

10.4 M1: Loop-field minimum

Not globally attractive: the earlier interval-certified negative finite vertex motion remains the rigorous witness.

Unique interior minimum; every side distance is at least $2/b(p_0)$ for any interior trial point p_0 , with b in the chapter normalization.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X2534	0.00686286	0.00657577	0.0144098
X5393	0.00702362	0.00691894	0.015792
X53841	0.00710349	0.00668813	0.0138951
X29678	0.00741535	0.00636905	0.014007
X39621	0.00766273	0.00714979	0.0120833

For X2534, position-fingerprint RMS / D is 0.00386736, derivative-fingerprint RMS is 0.0275604, and response-margin RMS difference is 0.0243488. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

10.5 M2: Vertex-current magnitude mean

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -1.61118595 and -1.61648215. No interval certificate is asserted.

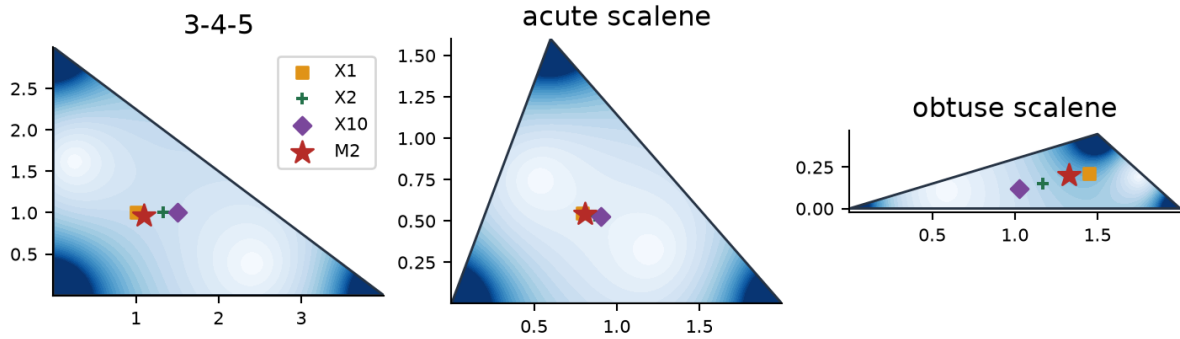
Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

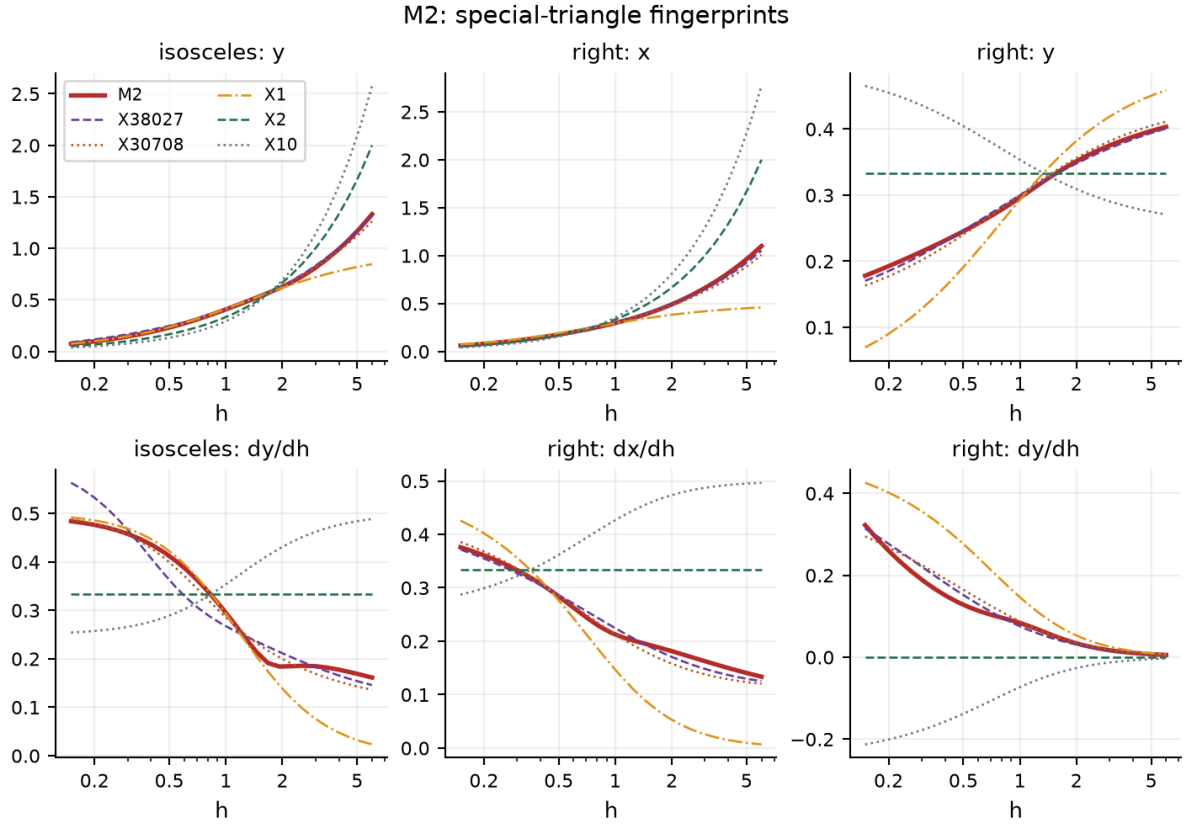
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X38027	0.00693143	0.0099179	0.0423536
X30708	0.0072812	0.0107841	0.0434288
X50293	0.00866959	0.0133356	0.055731
X29597	0.00909368	0.0121667	0.0434472
X3636	0.00939902	0.0138534	0.0557312

For X38027, position-fingerprint RMS / D is 0.00584481, derivative-fingerprint RMS is 0.0289519, and response-margin RMS difference is 0.0550474. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

Uniform axial vertex currents in an infinite triangular prism; take the interior field-observation mean. Weight is $|B|$ for M2/M4/M6 and $|B|$ squared for M5/M7. M3 instead requires equal circular core cutoffs and equals X360 exactly. Analytical segment/boundary kernels and corner-adapted quadrature; four-level finite-motion refinement.

M2 - Vertex-current magnitude mean





Field diagrams compare the reference 3-4-5 triangle and two further shapes with X1, X2 and X10. The fingerprint rows show positions and height derivatives with two ETC neighbors. The chapter retains the finite-height and equilateral comparisons for all seven magnetic selectors.

10.6 M3: Vertex-current energy limit

Strictly globally vertex-attractive: $M3 = X360$ exactly, with symmetric vertex-derivative eigenvalues (α plus or minus $\sin \alpha$)/ π .

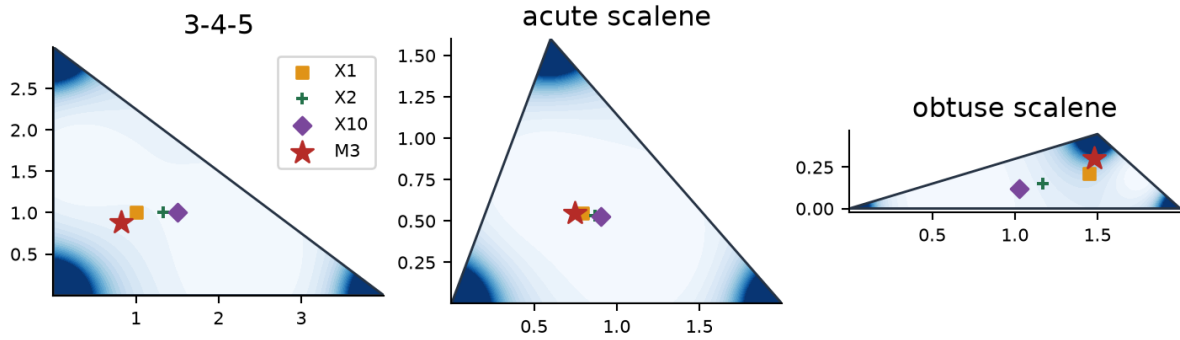
Strict interiority follows from the three positive angle barycentric weights.

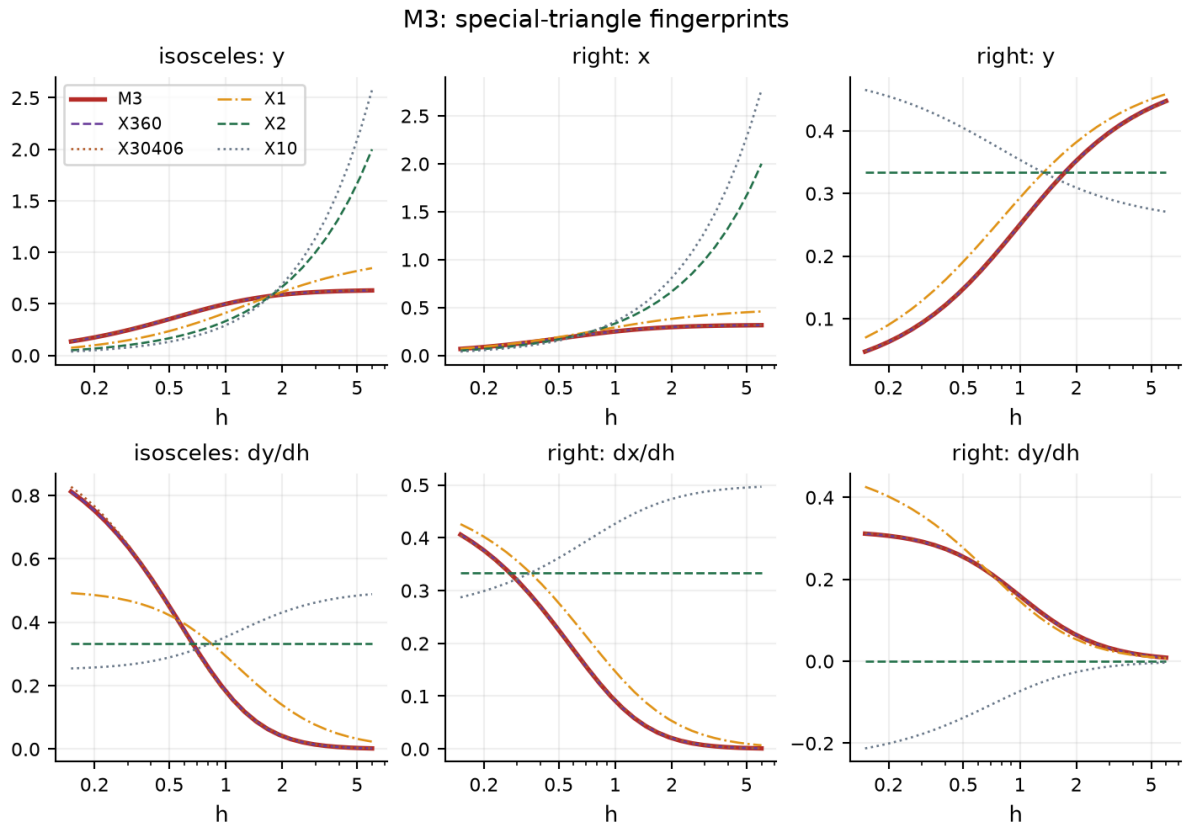
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X360	2.07319e-16	2.6653e-16	9.99219e-16
X30406	0.000861307	0.000797032	0.00182603
X53977	0.000918504	0.000924244	0.00139299
X30409	0.000955266	0.00094783	0.00162392
X8389	0.00114555	0.00114908	0.00172432

For X360, position-fingerprint RMS / D is 2.20496e-17, derivative-fingerprint RMS is 1.10773e-07, and response-margin RMS difference is 1.01909e-05. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

Uniform axial vertex currents in an infinite triangular prism; take the interior field-observation mean. Weight is $|B|$ for M2/M4/M6 and $|B|$ squared for M5/M7. M3 instead requires equal circular core cutoffs and equals X360 exactly. Analytical segment/boundary kernels and corner-adapted quadrature; four-level finite-motion refinement.

M3 - Vertex-current energy cutoff mean





Field diagrams compare the reference 3-4-5 triangle and two further shapes with X1, X2 and X10. The fingerprint rows show positions and height derivatives with two ETC neighbors. The chapter retains the finite-height and equilateral comparisons for all seven magnetic selectors.

10.7 M4: Surface-current magnitude mean

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.11778058 and -0.11804977. No interval certificate is asserted.

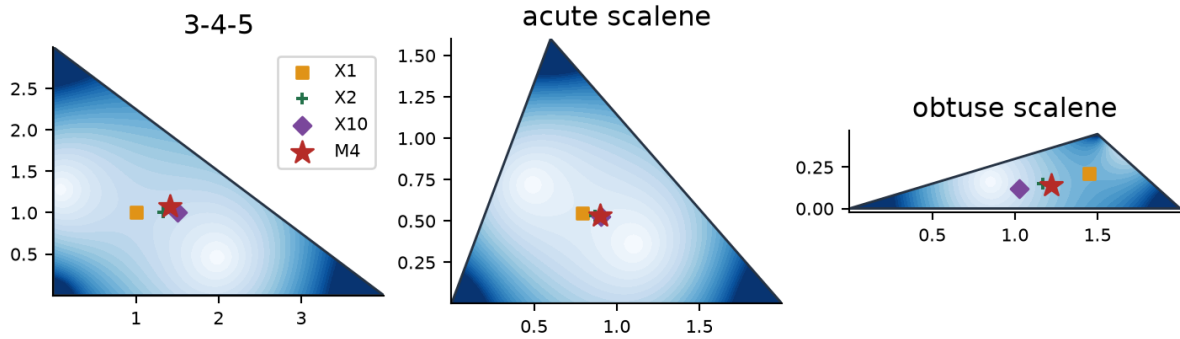
Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

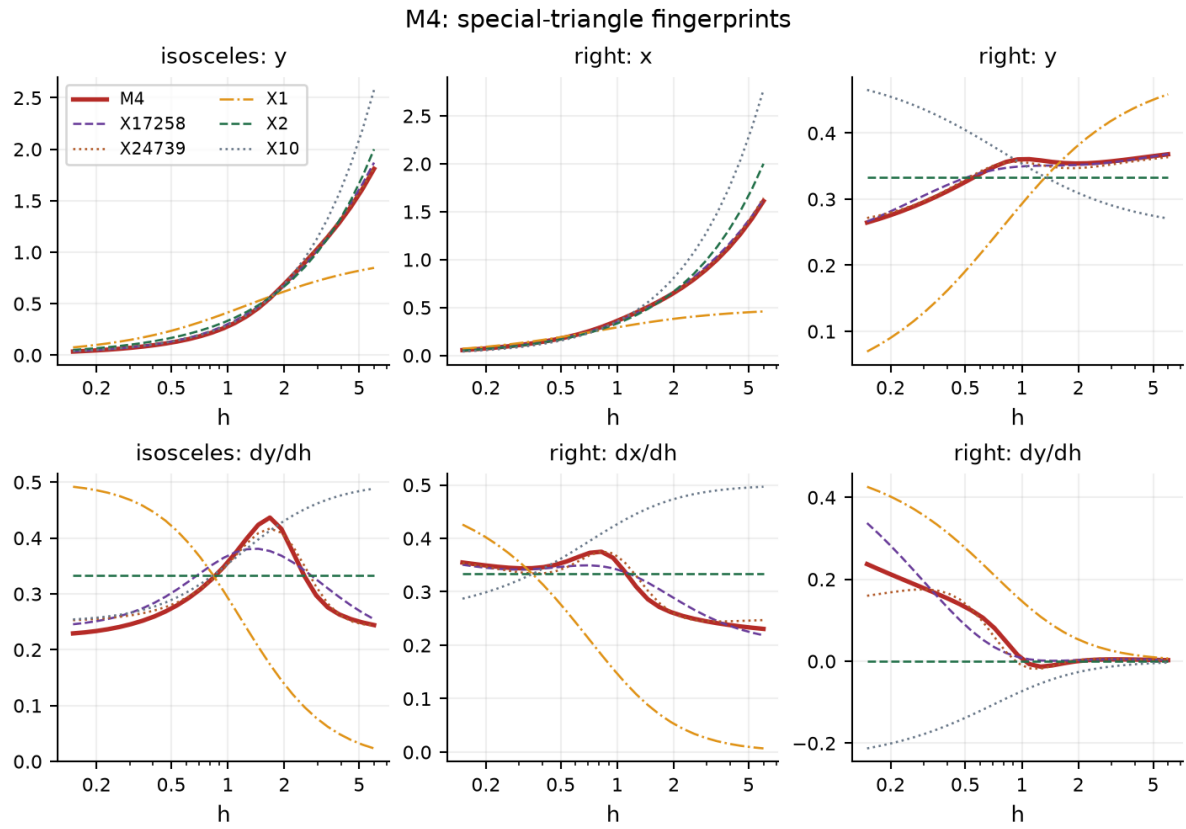
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X17258	0.00735564	0.00699618	0.0144227
X24739	0.00869396	0.0101256	0.0310545
X33947	0.00966709	0.0104356	0.0194045
X24658	0.0101628	0.0116685	0.0316963
X24657	0.0103155	0.0117293	0.0316964

For X17258, position-fingerprint RMS / D is 0.00687787, derivative-fingerprint RMS is 0.0344165, and response-margin RMS difference is 0.0308281. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

Uniform axial boundary currents in an infinite triangular prism; take the interior field-observation mean. Weight is $|B|$ for M2/M4/M6 and $|B|$ squared for M5/M7. M3 instead requires equal circular core cutoffs and equals X360 exactly. Analytical segment/boundary kernels and corner-adapted quadrature; four-level finite-motion refinement.

M4 - Surface-current magnitude mean





Field diagrams compare the reference 3-4-5 triangle and two further shapes with X1, X2 and X10. The fingerprint rows show positions and height derivatives with two ETC neighbors. The chapter retains the finite-height and equilateral comparisons for all seven magnetic selectors.

10.8 M5: Surface-current energy mean

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.72777353 and -0.72897714. No interval certificate is asserted.

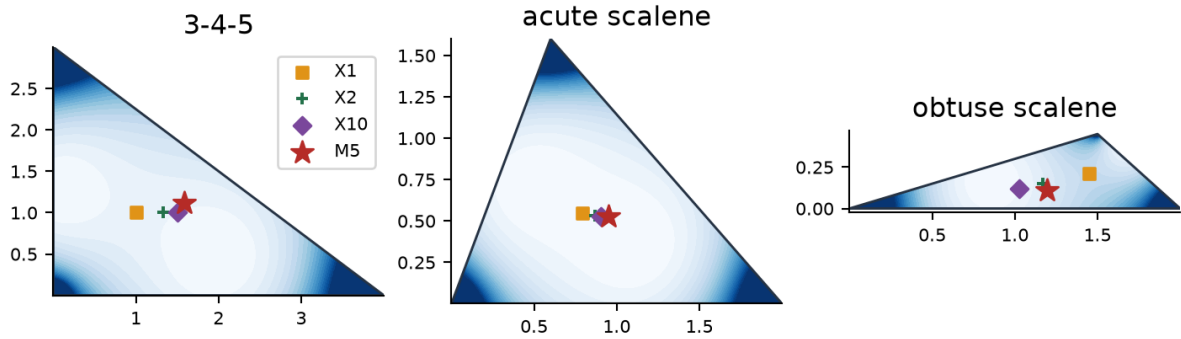
Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

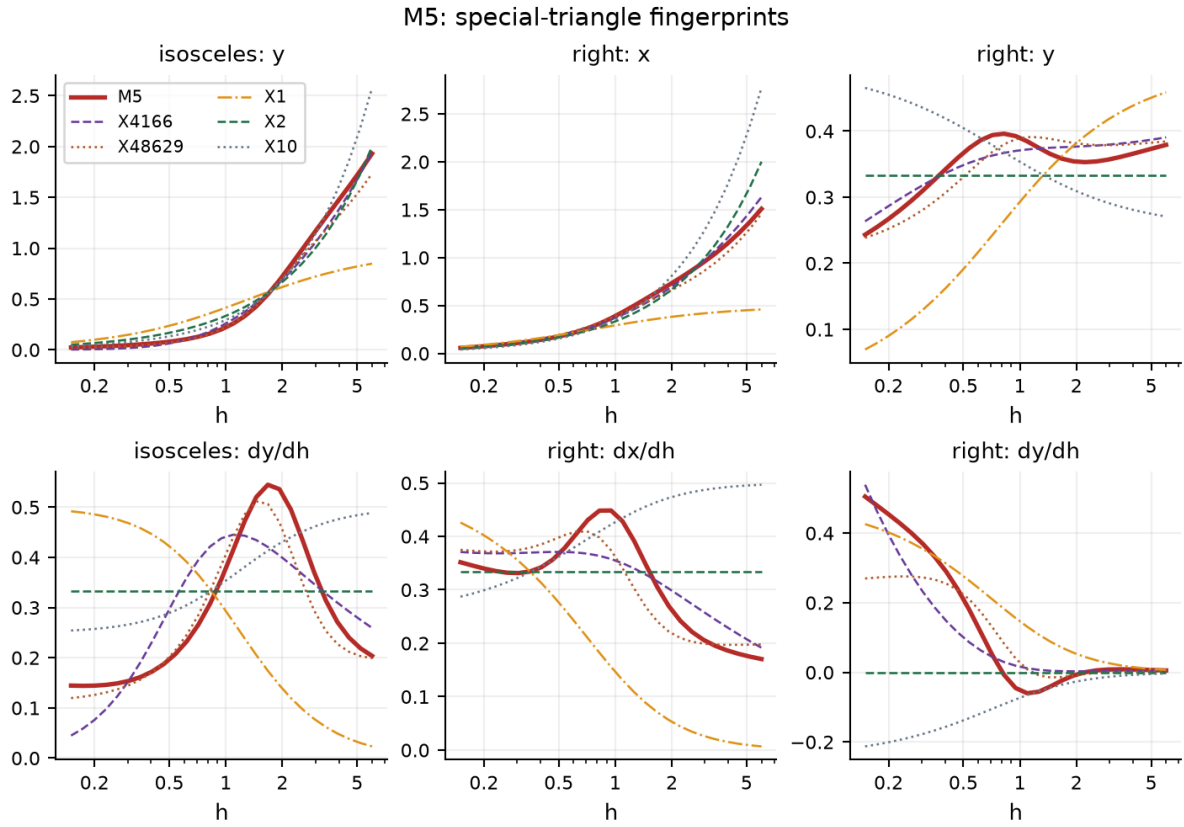
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X4166	0.0198646	0.0214742	0.056023
X48629	0.0217717	0.0228648	0.0538496
X33947	0.0237758	0.0251878	0.055156
X60544	0.0252782	0.026135	0.0560113
X70526	0.0255545	0.0312075	0.109718

For X4166, position-fingerprint RMS / D is 0.0179651, derivative-fingerprint RMS is 0.0813805, and response-margin RMS difference is 0.098876. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

Uniform axial boundary currents in an infinite triangular prism; take the interior field-observation mean. Weight is $|B|$ for M2/M4/M6 and $|B|$ squared for M5/M7. M3 instead requires equal circular core cutoffs and equals X360 exactly. Analytical segment/boundary kernels and corner-adapted quadrature; four-level finite-motion refinement.

M5 - Surface-current energy mean





Field diagrams compare the reference 3-4-5 triangle and two further shapes with X1, X2 and X10. The fingerprint rows show positions and height derivatives with two ETC neighbors. The chapter retains the finite-height and equilateral comparisons for all seven magnetic selectors.

10.9 M6: Volume-current magnitude mean

Global attractivity remains open. No negative response was detected in the original 100-shape screen, the 284-shape stress screen, or an additional 2,048-shape thin-triangle search; six low-margin cases were refined. These finite samples do not prove a global theorem.

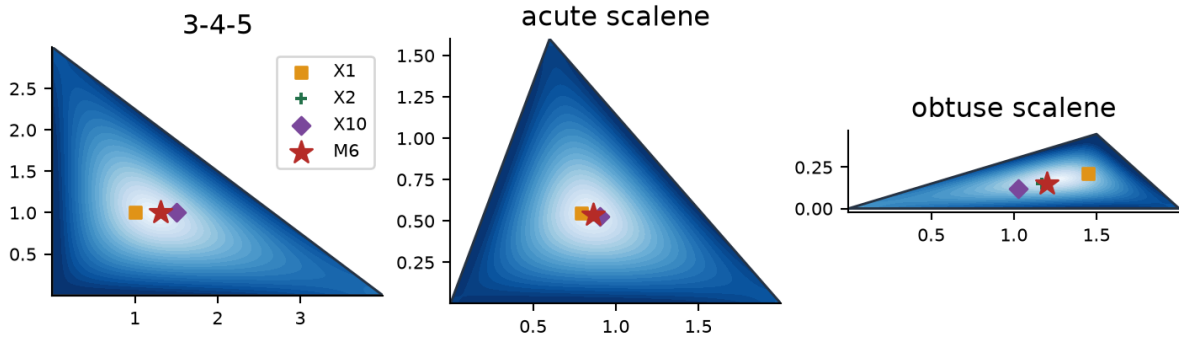
Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

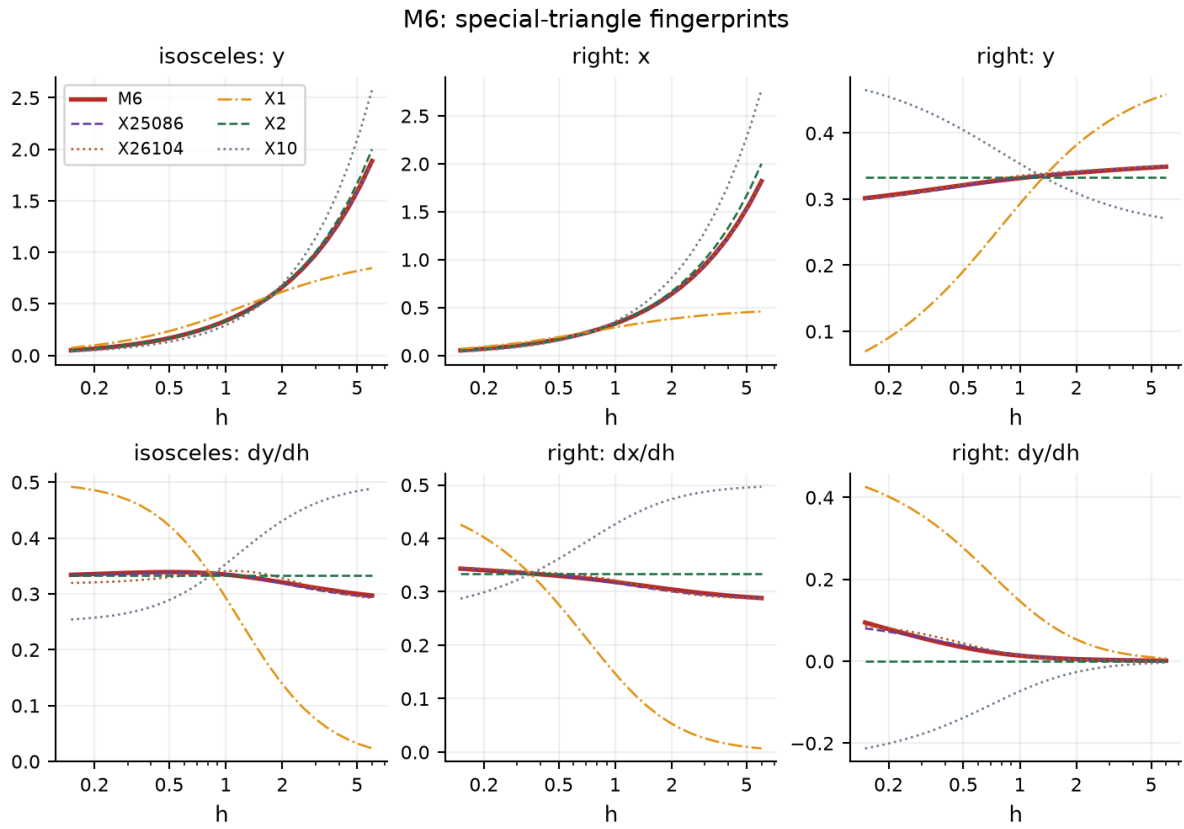
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X25086	0.00246473	0.00268194	0.00742478
X26104	0.00258123	0.00285751	0.00696481
X58859	0.00287241	0.0030829	0.00711097
X19935	0.00314996	0.00335865	0.00678586
X17325	0.00315741	0.00328295	0.00728227

For X25086, position-fingerprint RMS / D is 0.00125006, derivative-fingerprint RMS is 0.00350096, and response-margin RMS difference is 0.0122648. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

Uniform axial area currents in an infinite triangular prism; take the interior field-observation mean. Weight is $|B|$ for M2/M4/M6 and $|B|$ squared for M5/M7. M3 instead requires equal circular core cutoffs and equals X360 exactly. Analytical segment/boundary kernels and corner-adapted quadrature; four-level finite-motion refinement.

M6 - Volume-current magnitude mean





Field diagrams compare the reference 3-4-5 triangle and two further shapes with X1, X2 and X10. The fingerprint rows show positions and height derivatives with two ETC neighbors. The chapter retains the finite-height and equilateral comparisons for all seven magnetic selectors.

10.10 M7: Volume-current energy mean

Refined numerical evidence of nonattractivity: the order-256 finite-motion ratios are -0.10211640 and -0.10259846. No interval certificate is asserted.

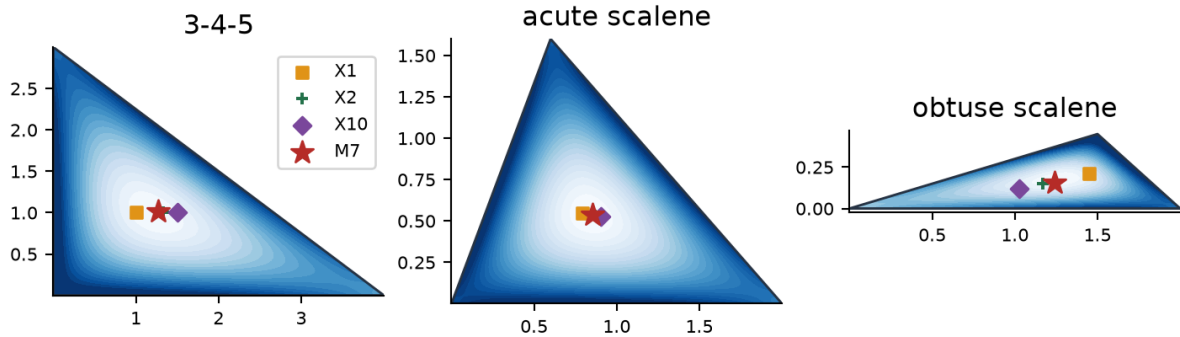
Strictly inside the triangle: a positive nondegenerate measure has positive barycentric first moments.

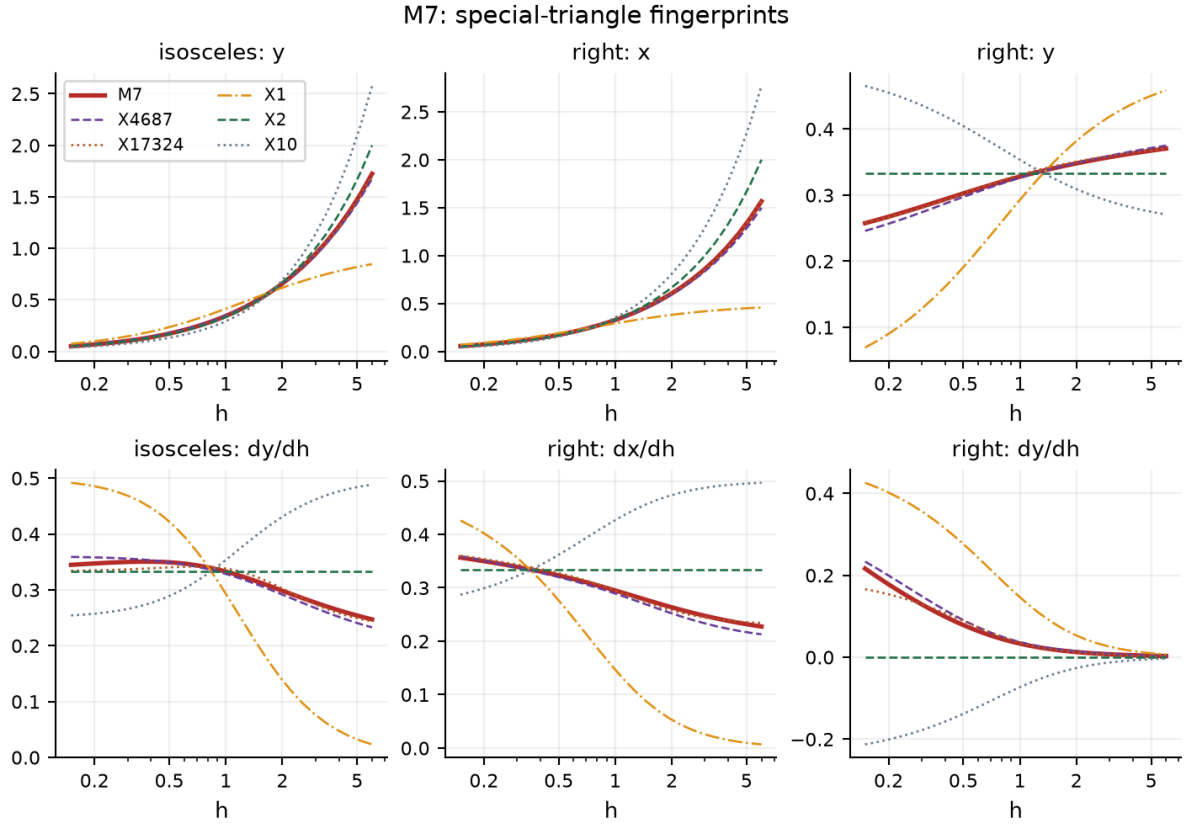
ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X4687	0.00574687	0.00577726	0.0120352
X17324	0.00614127	0.00799723	0.0286794
X53034	0.00621419	0.00725259	0.0205268
X25581	0.00640863	0.00663014	0.0136762
X40197	0.0065313	0.00793585	0.0253556

For X4687, position-fingerprint RMS / D is 0.00563704, derivative-fingerprint RMS is 0.0118469, and response-margin RMS difference is 0.0247311. The fingerprint comparison has 54 finite samples out of 54. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

Uniform axial area currents in an infinite triangular prism; take the interior field-observation mean. Weight is $|B|$ for M2/M4/M6 and $|B|$ squared for M5/M7. M3 instead requires equal circular core cutoffs and equals X360 exactly. Analytical segment/boundary kernels and corner-adapted quadrature; four-level finite-motion refinement.

M7 - Volume-current energy mean





Field diagrams compare the reference 3-4-5 triangle and two further shapes with X1, X2 and X10. The fingerprint rows show positions and height derivatives with two ETC neighbors. The chapter retains the finite-height and equilateral comparisons for all seven magnetic selectors.

10.11 L3: Three-dimensional prism Robin center

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01582971 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X66575	0.0021808	0.00208763	0.00460587
X53588	0.00262174	0.00243532	0.00413046
X16015	0.00376829	0.00389519	0.00819394
X29624	0.00380045	0.0032745	0.00812781
X59464	0.00384197	0.00388004	0.00733217

For X66575, position-fingerprint RMS / D is 0.00162041, derivative-fingerprint RMS is 0.0137814, and response-margin RMS difference is 0.00859114. The fingerprint comparison has 17 finite samples out of 34. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

10.12 L4: Four-dimensional prism Robin center

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01102191 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X66575	0.00233082	0.00212511	0.00388145
X16015	0.0031703	0.00319691	0.0060394
X59464	0.00333147	0.00331106	0.00587152
X5703	0.00342174	0.00336797	0.00740775
X53588	0.00344042	0.00322926	0.00627888

For X66575, position-fingerprint RMS / D is 0.00261203, derivative-fingerprint RMS is 0.014949, and response-margin RMS difference is 0.00828281. The fingerprint comparison has 17 finite samples out of 34. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

10.12.1 11.7 L3 and L4: complete finite-sample atlas extensions

For $n = 3$ or 4 , ground the boundary of the infinite prism T times $R^{(n-2)}$. Subtract the free-space singularity from its Dirichlet Green function at a point of the cross-section. Minimize the resulting Robin function, equivalently maximize harmonic radius. Translation invariance in the axial directions leaves a canonical point in T . L2 is E1. These selectors differ from the fixed-charge conductor centroids E2 and E3.

Fourier transformation in the axial coordinates reduces the problem to modified Helmholtz boundary problems in T . For each frequency k the planar kernel is proportional to $K_0(k \text{ times distance})$. Subtract its logarithmic self singularity analytically, solve the graded boundary system, and integrate in frequency with weight $k^{(n-3)}$. The new solver grades panels near both endpoints and source projections, scales frequency by the inradius, and optimizes inside the analytically certified region. Parameter m describes the grading density; merged panels are not simply m panels per side. Log-frequency quadrature covers -14 to $\log(30)$ in k times inradius. Cutoff and mesh errors remain numerical, not interval bounds.

The sample comprises 100 original shape points, 40 independent holdouts, 17 isosceles heights, 17 right-triangle heights, 25 thin configurations and two equilateral step sizes at two resolutions. All raw values, derivative steps and optimization diagnostics are retained.

10.12.2 L3: Three-dimensional prism Robin center

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01582971 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

The maximum training-grid change between grading parameters 10 and 18 was $0.000207269 D$; 75/100 changes were below $0.0001 D$. These changes are resolution diagnostics, not error bounds. At equilateral, the computed horizontal and vertical apex derivatives at grading 20 and step 0.005 are 0.4666520 and 0.2000183 . Symmetry forces zero mixed terms and trace $2/3$ in the smooth exact limit.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X66575	0.0021808	0.00208763	0.00460587
X53588	0.00262174	0.00243532	0.00413046
X16015	0.00376829	0.00389519	0.00819394
X29624	0.00380045	0.0032745	0.00812781
X59464	0.00384197	0.00388004	0.00733217

For X66575, position-fingerprint RMS / D is 0.00162041 , derivative-fingerprint RMS is 0.0137814 , and response-margin RMS difference is 0.00859114 . The fingerprint comparison has 17 finite samples out of 34. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

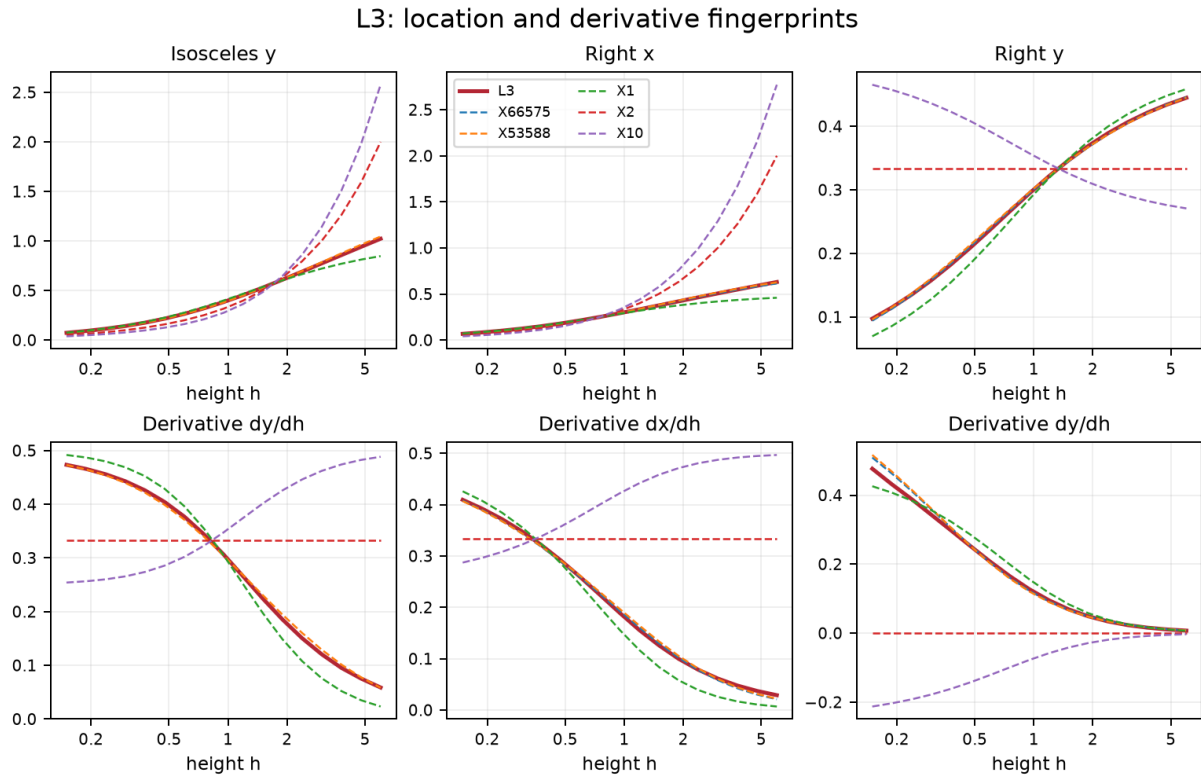


Figure. L3 position and derivative fingerprints, with its two leading ETC candidates and X1, X2, X10. The leading candidate X66575 is undefined in the supplied expression on the isosceles family, leaving only 17 of 34 fingerprint comparisons; its absent curve is intentional. No removable extension or identity is assumed.

10.12.3 L4: Four-dimensional prism Robin center

Refined numerical evidence of nonattractivity: the finite-motion ratio is -0.01102191 at grading parameter 48 and 128 frequency nodes. Four spatial/frequency refinements retain the negative sign; no interval certificate is asserted.

Every side distance is at least half the inradius: the center belongs to the triangle $I + (T-I)/2$.

The maximum training-grid change between grading parameters 10 and 18 was $0.000304191 D$; 84/100 changes were below $0.0001 D$. These changes are resolution diagnostics, not error bounds. At equilateral, the computed horizontal and vertical apex derivatives at grading 20 and step 0.005 are 0.4693248 and 0.1973406 . Symmetry forces zero mixed terms and trace $2/3$ in the smooth exact limit.

ETC candidate	Grid RMS / D	Holdout RMS / D	Holdout max / D
X66575	0.00233082	0.00212511	0.00388145
X16015	0.0031703	0.00319691	0.0060394
X59464	0.00333147	0.00331106	0.00587152
X5703	0.00342174	0.00336797	0.00740775
X53588	0.00344042	0.00322926	0.00627888

For X66575, position-fingerprint RMS / D is 0.00261203 , derivative-fingerprint RMS is 0.014949 , and response-margin RMS difference is 0.00828281 . The fingerprint comparison has 17 finite samples out of 34. These are separate diagnostics, not terms of an undisclosed weighted score. The response values are finite-difference screen estimates.

L4: location and derivative fingerprints

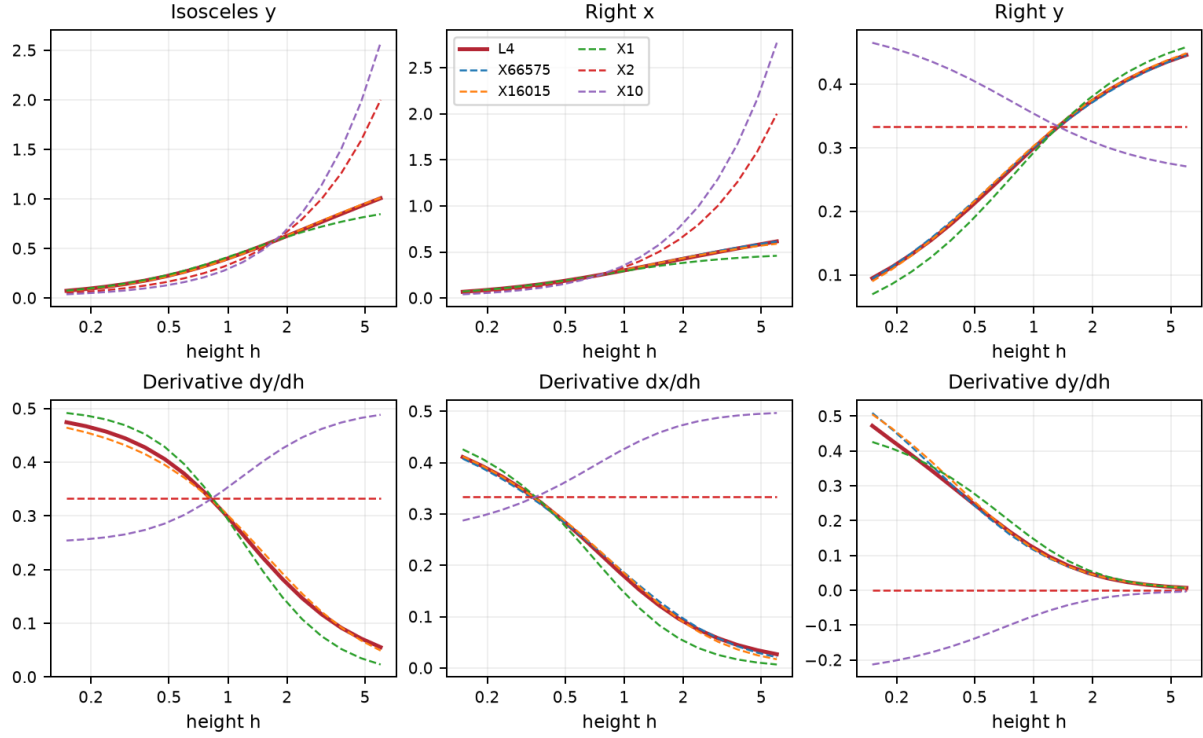


Figure. L4 position and derivative fingerprints, with its two leading ETC candidates and X1, X2, X10. The leading candidate X66575 is undefined in the supplied expression on the isosceles family, leaving only 17 of 34 fingerprint comparisons; its absent curve is intentional. No removable extension or identity is assumed.

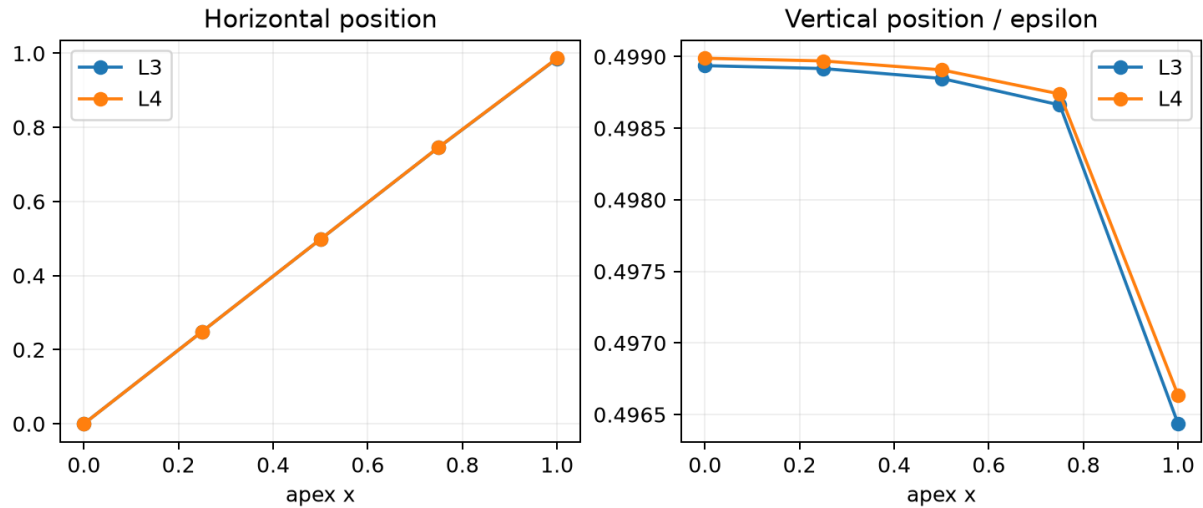


Figure. Finite-height collinear fingerprints at $\epsilon = 0.0125$ for L3 and L4, including the endpoint $x=1$. These are numerical thin configurations, not an asymptotic theorem.

The same refined witness is also negative for L6 (about -0.00254498). Its positive L10 value does not classify L10. No finite survey settles all dimensions n or every bounded-cavity thickness ϵ . Those parameter families retain separate open entries; they are not counted as fully surveyed centers.

10.12.4 Exact attractivity of X10 and its thin-conductor aliases

The uniform-wire centroid $W = X10$ is strictly vertex-attractive. This theorem was already in the supplied Attractive Centers IV, Section 4.1, and was missed in the earlier chapter. Here is the derivative calculation, so the conclusion does not depend on a positive numerical screen. Hold B and C fixed. Put $a = |B-C|$, $b = |C-A|$, $c = |A-B|$, $P = a+b+c$, $p = (B-A)/c$, $q = (C-A)/b$, and $u = p \cdot q$. Differentiate the length-weighted edge-midpoint mean. The symmetric part of its derivative with respect to A is

$$S_A(W) = \frac{b+c}{2P}I - \frac{bc}{2P^2}(pp^T + qq^T) + \frac{c(a+c) + b(a+b)}{4P^2}(pq^T + qp^T).$$

The directions $p+q$ and $p-q$ diagonalize this matrix. Their eigenvalues are

$$\mu_+ = \frac{2P(b+c) + (1+u)((b-c)^2 + a(b+c))}{4P^2}, \quad \mu_- = \frac{(b+c)(1+u)}{4P}.$$

Both are positive for a nondegenerate triangle. Integrating the positive quadratic form along any straight vertex displacement staying in the same nondegenerate half-plane gives a strictly positive finite-motion dot product. This proves the assertion globally on that domain. In particular the adopted uniform-thickness limits S3 and S4 inherit strict attractivity from X10. E4 inherits it from X2, whose derivative is $I/3$. A singular zero-thickness conductor with zero capacity is not itself being assigned an equilibrium distribution.

10.12.5 A location certificate for least-capacity centers

Let r be the triangle's inradius. Normalize the harmonic radius so a ball of radius r has harmonic radius r at its center. Domain monotonicity gives harmonic radius at least r at the incenter: the ball of radius r is contained in the infinite prism. On the other hand, the prism is contained in each side half-space. Reflection in the plane boundary gives harmonic radius exactly twice the distance d to that plane. A point maximizing harmonic radius must therefore satisfy

$$r \leq r_\Omega(L_n) \leq 2d_i(L_n), \quad d_i(L_n) \geq r/2.$$

The identical argument with conformal radius proves the planar E1 assertion. Thus all these centers lie in the closed inner parallel triangle obtained by shifting every side inward by $r/2$: the homothetic copy $I + (T-I)/2$. The proof applies to every infinite prism dimension, independently of a numerical mesh. For the bounded product cavity T times a ball of radius η , replace r on the lower side by $\min(r, \eta D)$; the infinite-prism bound cannot be imported unchanged into a thin bounded cavity.

10.12.6 What the two new interval certificates prove

For E1, the incomplete-beta conformal map is evaluated by a hypergeometric series with outward-rounded complex intervals. In the retained disk $|z| < 1$, successive coefficient ratios are below one. The remaining tail after 201 terms is bounded by the last term times $|z|/(1-|z|)$. A 65-digit calculation encloses the finite-motion ratio in $[-0.020869126, -0.020869125]$.

For E2, the charge centroid equals the constant Laurent coefficient of the exterior map. An independent three-prevertex Schwarz-Christoffel construction, interval Taylor integration and explicit endpoint bounds enclose its finite-motion ratio in $[-0.046650959, -0.046650958]$. Both intervals lie strictly below zero. Decimal inputs, before/after coordinate enclosures, tail bounds and executable verifiers are supplied. These certificates disprove global vertex attractivity; they

do not assert that every triangle or every direction has negative response. The full exterior-map derivation is in the audit companion's certificate proof note.

10.13 Scope and the next physical discipline

M6 remains mathematically open. E3, M2, M4, M5, M7, L3 and L4 still need interval validation of their numerical negative witnesses. The continuous bounded-cavity thickness family and all higher prism dimensions cannot be classified by the L3/L4 samples. The E3 coarse strip-grid trial and unresolved uniform-mesh stress witnesses are retained but excluded as evidence. The separate refined anisotropic E3 witness has positive panel charges at every level.

The user selected membranes and solid elastic laminae as the dedicated third chapter, now supplied with this update. Heat conduction, diffusion and Brownian motion remains a proposal for a later chapter. It joins heat maxima $H(\Delta)$, heat-content centroids, the Poisson/torsion selector F1 and mean exit times, before linking the diagonal heat kernel to Robin centers and the late-time heat field to the quantum ground state. A mean weighted by ground-state amplitude differs from the quantum probability mean weighted by its square.

The update directory contains every finite-motion input, certificate enclosure, refinement record, top-ten shortlist and the independent verification script. The certificate proof note explains the exterior Schwarz-Christoffel construction and its rigorous quadrature tails.

Chapter 11

Cross-family conclusions and research directions

11.1 An identity map of the catalogue

The same point can have several physical realizations, and the same field can yield several points. The following relations are exact within their stated conventions:

Construction	Identification
Uniform area moment	X(2)
Uniform perimeter moment	X(10)
S2, logarithmic boundary charge	E2
E4, uniform-thickness limit	X(2)
S3 and S4, uniform-radius limits	X(10)
Linear membrane / Poiseuille mode	F1
Mem(2)	F1
MemMean(2)	Abs(0)
Heat, positive-time limit at zero	X(1)
Heat, late-time limit	Q0m
Thermal quantum mean, low temperature	Q0
Thermal quantum mean, high temperature	X(2)
Plate variance maximum	Plate compliance maximum
Uniform-load bending-energy mean	Deflection-volume mean
Solid-angle maximum, small height	M1
Solid-angle maximum, large height	X(2)

Limits are not equalities at arbitrary finite parameter values. E4/S3/S4 also require the specific thickness convention; a nonuniform cross-section may bias the leading projected density. Closely spaced curves for F1, M1 and Q0m do not merge these different definitions.

11.2 Thin fingerprints across disciplines

The horizontal limits below refer to $(-1, 0)$, $(1, 0)$, (x, ϵ) with fixed $0 \leq x < 1$. They give compact diagnostic distinctions.

Center/family	Horizontal limit	Evidence type
X(2), E4	$x/3$	Exact geometry/declared conductor limit
X(10), S3, S4	0	Exact geometry/declared conductor limit
E0, E1	x	Conformal expansion
E2	0	Segment logarithmic equilibrium
E3	0	Critical thin-plate asymptotics
M1, F1	x	Localization/narrow-domain analysis
Fixed low quantum states	x	Transverse-energy localization
Fixed positive-time H	x	Ground-mode thin regime
Energy	$5x/7$	Leading strip prediction
Abs, fixed finite removal	$3x/5$	Leading strip prediction
MemMean(p)	$(2p-1)x/(4p-3)$	Leading strip prediction
Finite optical parameters	1D weighted integrals	Dominated-convergence reduction

Rescaled vertical coordinates further separate centers with the same horizontal limit. E0 has ratio $2/3$, E1 has $1/2$, X(2) has $1/3$, and X(10) tends to $1/4$. The endpoint $x = 1$, eigenvalue transitions and simultaneous parameter limits require their own analysis. Raw thin data and family-specific plots remain in the companion studies.

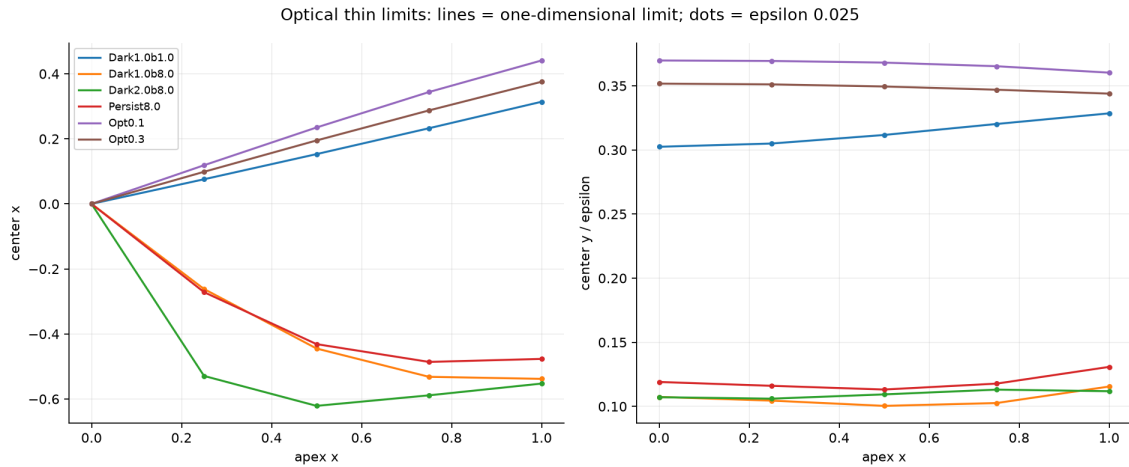


Figure 11.1: Optical thin limits: one-dimensional limiting integrals compared with finite-thickness numerical triangles. This is a cross-check of two different numerical representations, not an interval error certificate.

11.3 Attractivity: what can currently be said

Exact classical identities give strict attractivity for X(2), E4, X(10), S3, S4 and M3=X(360). M1, E1 and E2 have interval-certified negative finite witnesses. E3, M2, M4, M5, M7, L3 and L4 have refined numerical negative evidence, as detailed in the September 30 extension. The quantum expectations, F1, sampled heat maxima, near-threshold biomass and strongly weighted darkness have numerical negative evidence of differing reliability. Q0m has a weak, mesh-sensitive candidate witness. The new mechanics chapter also records a half-step sign change at high Ritz degree for the F1 witness, so its continuum attractivity conclusion remains unresolved. No negative response in a finite screen is not a positive theorem for M6, E0, concentration means, fluorescence means or weak darkness.

The most important unresolved tasks are not more decimal digits at one ordinary triangle. They are validated shape derivatives or finite-motion enclosures, coverage of thin and nearly symmetric regimes, and proofs of uniqueness for selectors defined by a maximum. The strong

optical equilateral signs are attractive targets for interval quadrature because their margins are large. The magnetic proof demonstrates the desired standard: a unique exact root is enclosed, then an interval calculation certifies a strictly negative finite dot product.

11.4 Location beyond mere interiority

The E0 barycentric hexagon and the quantum and magnetic inner parallel triangles are explicit, nontrivial regions established by their defining models. The project's standard attractive-center chest is a different kind of result: it applies only after the required global attractivity and extension hypotheses have been established [3]. It cannot be imposed on every stable physical equilibrium. Testing membership numerically is informative, but failing to find an exterior sample does not certify membership throughout shape space.

For an implicit maximum, a practical certificate strategy is to enclose the field gradient and Hessian on boxes, discard boxes whose gradient cannot vanish, prove negative/positive definite curvature near the retained extremum, and compare boundary or other candidate values. For a positive moment, interval quadrature plus a rigorously bounded PDE/density approximation is required. An interval integration of an uncertified approximate field alone would certify the wrong problem.

11.5 Accuracy of symbolic surrogates

Compact shape-equivariant surrogates are useful for exploration and interactive demonstrations. Some held-out RMS errors, as percentages of diameter, are about 0.0503% for Q0m, 0.0607% for Q0, 0.0282% for Energy, 0.026% for Abs(0), 0.0134% for MemMean(3), and 0.0034% for Opt(1). These summarize sampled positions, with the numerical targets' own error still present. They are not uniform approximation theorems.

A single small ansatz performs poorly for several excited states and for highly concentrated darkness: the $F = 2, \beta = 8$ optical fit has a held-out error of roughly 5.26%. Spectral branches, local dark structures and near-degenerate shapes are mechanisms, not random nuisances that can always be removed by a higher polynomial degree. A useful next step is region-aware fitting constrained by exact endpoint behavior, followed by independent validation. Differentiating a good position fit does not produce a certified attractivity model.

11.6 Constructions retained with qualifications

S2. The skeleton in two-dimensional logarithmic electrostatics has exactly the E2 equilibrium measure. Its field/density atlas, fingerprints, nearest ETC results and attractivity status are those in Section 5.9; it is not a second distinct numerical center.

Quantum projector means. Full repeated-eigenspace densities are canonical; individual basis states are not. A dedicated projector-family ETC/response study remains open.

Thermal quantum expectation. A well-defined heat-kernel-diagonal mean interpolating between Q0 and X(2). It is described mathematically in Chapter 4; no dedicated 100-shape scan is claimed.

Navier plate. The maximum of $A^{-2}1$ is a distinct supported-plate selector. A dedicated refined atlas and mode-uniqueness proof remain open.

Vertex-supported plates. Support-pad size, free-edge conditions and rotational constraints must be fixed. A family of such experiments is possible, but a single shape-only center is not determined by “supported at the vertices.”

Uniformly heated density extrema. Uniform restrained linear thermoelastic compression has no distinguished density point. A nonlinear buckling/density model requires a constitutive and branch specification.

Stress and first failure. Curvature, principal stress, strain energy and flaw-weighted failure probability are different fields. Their maxima may lie on the boundary or be nonunique. The uniform-load energy *mean* is already a well-defined entry; it does not stand for all stress extrema.

Current-filament collapse. Attraction without a regularizing constitutive law does not define a stable, unique, shape-only endpoint. Finite conductivity, pressure or self-inductance constraints would need specification.

Principal axes and Steiner foci. Covariance eigenlines are line/tensor observables through the area or perimeter centroid, not additional single points. An unordered pair of foci requires a convention before it can be treated as a point selector.

Optical nulls, TIR acceptance and central bright directions. The nonexistence, branch and extra-parameter issues are recorded in Chapter 9. They are part of the systematic classification rather than missing numerical values to be filled with arbitrary choices.

The inventory deliberately separates “well defined but not yet fully scanned” from “not defined without further conventions” and from “proved impossible under the stated assumptions.” This preserves the earlier proposals without overstating the amount of completed numerical investigation.

11.7 Demonstrations that communicate the distinctions

1. Cut the same scalene shape as a filled plate and a wire frame, then balance both. Overlay $X(2)$ and $X(10)$. A third junction with side-normal springs demonstrates $X(6)$.
2. Show heat, torsion and ground-state density in the same triangle. Let the heat slider move H , and display separately Q_{0m} and Q_0 so that mode and expectation cannot be confused.
3. Animate Brownian paths with exit-side colors and a survival-time histogram. E_0 equalizes colors, F_1 maximizes mean lifetime, and H selects the greatest survival probability at the chosen time.
4. Scan an in-plane triangular current loop and an elevated illumination source. Plot the magnetic minimum and the solid-angle maximum as source height decreases; the two approach one another.
5. Drag a vertex and draw both its displacement and the center displacement. Display their signed dot product and uncertainty. Replay the interval-certified M_1 witness as a concrete distinction between physical stability and vertex attractivity.
6. Compare linear and quadratic detection of the same blurred triangular image. Linear centroids remain at G ; nonlinear fluorescence shifts and returns toward G at both width endpoints.

7. Replay billiard particles and their time averages separately, with SD, RMS and tail quantiles. Include the periodic exception so a slowly converging cloud is not presented as a universal theorem about every trajectory.

Chapter 12

The reproducibility companion

The archive retains the project layout: `research/` contains algorithms, `output/` contains data and study reports, and `output/article/` contains editable LaTeX, catalogue metadata, plot data and figure sources. The final PDF is also included. A top-level README gives short commands for checking the bundle, rerunning the magnetic certificate, regenerating the new atlas, and rebuilding the manuscript. No proprietary Wolfram kernel is required for the supplied importer.

12.1 What is and is not a certificate

Item	Meaning
Magnetic interval JSON and verifier	Enclosed roots, strict Krawczyk inclusion, and negative finite-motion dot product. Requires the accompanying analytic uniqueness argument.
Positive-moment proposition	Analytic strict-interiority statement under positivity/support hypotheses.
E0, quantum and magnetic bounds	Analytic region certificates with numerical checks separated.
Mesh, degree and quadrature changes	Diagnostics of convergence; not rigorous continuum enclosures.
Checksums and data validation	Integrity and structural reproducibility; not mathematical proof of a PDE approximation.
ETC shortlist and symbolic fit errors	Finite-sample comparisons; neither identities nor uniform bounds.

12.2 Reproduction levels

The first level checks file hashes, parses datasets and verifies the catalogue/figure count. The second runs small deterministic scientific checks and the existing interval certificate. The third regenerates the figure atlas from archived refined family data and freshly computed illustrative fields. The fourth reruns full 100-shape studies, refinements and all-entry ETC scans. This final level is substantially more expensive and requires the externally supplied ETC snapshot. Each level has a different purpose; a fast smoke test is not advertised as a complete rerun of every study.

The database path is configurable through `PITC_ETC_DB`. Its original filename and SHA-256 provenance are recorded in `database_import.json`. The bundle does not redistribute the 72,807-entry database or third-party PDFs. The compiled expression cache is also excluded. Published

plots and ranking identifiers can be inspected without it; recomputing the ETC curves or scan requires the local snapshot.

12.3 Current coverage and limits

The quantum joint targets were met on 99/100 grid and 39/40 validation shapes. The electromagnetic/heat joint targets were met on 94/100 and 37/40. For all thirty-four latest variants simultaneously, the counts are 43/100 and 18/40; this is driven by difficult plate diagonals and some maxima, not uniform failure of every model. The individual result sheets give the more informative per-center counts. Unresolved samples remain in the rankings and their flags remain available, so small differences between close-ranked entries should not be overinterpreted.

The contour atlas uses three diameter-normalized scalene triangles: sides 4–5–6, sides 6–9–13, and vertices $(0, 0)$, $(1, 0)$, $(0.79, 0.18)$. Ritz degree is 10; FEM subdivision is 35 with 72 spectral modes; optical quadrature order is 30; the E3 illustration has 484 panels. These figures explain what is being measured. They do not replace the higher-resolution study data or certify uniqueness of a plotted mode. Conformal contours cover a sampled upper-half-plane patch; white regions inside the outline mean unplotted parameter coverage, not vanishing fields.

12.4 Attribution and publication status

The manuscript was prepared from the user’s supplied research draft, the local project calculations and the references listed below. Derivations supplied here are not claims of priority without a separate literature search. The bibliographic sources establish background models and selected theorems; the numerical tables are this project’s archived experiments. Author names, affiliations, acknowledgements and submission metadata remain to be assigned by the researcher. This document has not been submitted or published automatically.

Appendix A

Numerical equilateral jets and surrogate coefficients

The coefficients below use $C_x = \alpha x + \beta xy + O(\rho^3)$ and $C_y = 1/\sqrt{3} + \gamma y + \delta x^2 + \eta y^2 + O(\rho^3)$. The form and trace relation $\alpha + \gamma = 2/3$ follow analytically from symmetry. Except for the exact classical and M1 linear values discussed in the text, these numbers are fitted numerical diagnostics, not certified constants or remainder bounds. Five decimal places identify the stored fit, not its accuracy. Mem(1.5) and Mem(3) maximum jets are omitted because of unresolved quadrature sensitivity. Individual degenerate quantum-state jets are not defined.

Entry	α	γ	β	δ	η
E0	0.54200	0.12467	-0.22990	-0.06576	-0.05471
E1	0.46398	0.20269	-0.17012	-0.02808	-0.04735
E2	0.24698	0.41969	0.09546	0.02705	0.02280
E3	0.29239	0.37428	0.05015	0.01038	0.01326
M1	0.44445	0.22222	-0.14255	-0.02495	-0.03920
F1	0.45062	0.21605	-0.17407	-0.01453	-0.05318
H0.01	0.46327	0.20340	-0.22142	-0.00182	-0.07319
H0.03	0.44326	0.22341	-0.15189	-0.01925	-0.04421
H0.1	0.45034	0.21633	-0.16361	-0.01953	-0.04803
H0.3	0.45043	0.21624	-0.16398	-0.01942	-0.04819
H1.0	0.45043	0.21624	-0.16398	-0.01942	-0.04819
E4	0.33333	0.33333	-0.00000	-0.00000	-0.00000
S3	0.25000	0.41667	0.10825	0.01804	0.03007
S4	0.25000	0.41667	0.10825	0.01804	0.03007
Pgrav	0.45941	0.20725	-0.16238	-0.02754	-0.04470
Bmode	0.46538	0.20130	-0.17151	-0.02883	-0.04757
Bmean	0.45864	0.20803	-0.16529	-0.02587	-0.04647
Compliance	0.46557	0.20109	-0.17568	-0.02716	-0.04871
FluctMean	0.40209	0.26458	-0.08976	-0.01466	-0.02503
Energy0.3	0.42065	0.24601	-0.11457	-0.01834	-0.03208

Entry	α	γ	β	δ	η
Pheat0.5	0.46034	0.20633	-0.16378	-0.02815	-0.04513
Pheat0.9	0.46343	0.20324	-0.16835	-0.02843	-0.04655
Abs0.0	0.38998	0.27669	-0.07437	-0.01187	-0.02083
Abs1.0	0.38967	0.27700	-0.07395	-0.01181	-0.02071
Abs10.0	0.38709	0.27958	-0.07046	-0.01132	-0.01971
Abs100.0	0.37223	0.29444	-0.05071	-0.00833	-0.01413
Bio1.1	0.41653	0.25014	-0.11252	-0.01579	-0.03223
Bio2.0	0.40450	0.26217	-0.09551	-0.01387	-0.02721
Bio10.0	0.36724	0.29943	-0.04448	-0.00712	-0.01245
Mem4.0	0.47007	0.19659	-0.18533	-0.02561	-0.05288
MemMean1.5	0.39306	0.27361	-0.07895	-0.01380	-0.02377
MemMean3.0	0.38612	0.28054	-0.06891	-0.01246	-0.02040
MemMean4.0	0.38371	0.28296	-0.06564	-0.01080	-0.01828
Dark0.5b1.0	0.32857	0.33810	0.00063	0.00381	-0.00106
Dark0.5b8.0	0.27520	0.39146	0.03653	0.03208	0.00149
Dark1.0b1.0	0.31227	0.35439	0.00974	0.01337	-0.00121
Dark1.0b8.0	-0.22452	0.89117	1.00671	-0.02027	0.34236
Dark2.0b1.0	0.24694	0.41973	0.09630	0.02667	0.02321
Dark2.0b8.0	-1.57876	2.24539	5.15947	-0.92467	2.02932
Dark4.0b1.0	0.29717	0.36950	0.00779	0.02744	-0.00656

Entry	α	γ	β	δ	η
Dark4.0b8.0	0.65757	0.00910	-0.85796	0.14826	-0.33561
Persist1.0	0.30681	0.35985	0.01749	0.01422	0.00109
Persist8.0	-0.40405	1.07071	1.42017	-0.07157	0.49734
Opt0.1	0.35618	0.31049	-0.02955	-0.00501	-0.00818
Opt0.3	0.35215	0.31452	-0.02319	-0.00470	-0.00616
Opt1.0	0.33598	0.33068	-0.00309	-0.00075	-0.00078

A.1 Six-coefficient positive-weight approximations

These are the softmax coefficients for mechanics, reaction and optics. Their order is $(A_0, B_0, A_1, B_1, A_2, B_2)$. The baseline and power convention are specified by `mech_opt_fit.features` and in the machine-readable model record. They are empirical shape functions, not exact formulas for the physical centers. The quantum and electromagnetic ansatz coefficients are supplied separately because their parameterizations differ; substituting one family's coefficients into another family's formula is invalid.

Entry	A_0	B_0	A_1	B_1	A_2	B_2
Pgrav	-20.7639	29.9845	43.5307	-67.0012	-26.1315	41.0649
Bmode	-17.4369	24.6801	36.9702	-55.7894	-22.4230	34.5802
Bmean	-19.6833	28.1229	43.6556	-66.7308	-27.1299	42.3265
Compliance	-8.9265	12.1704	14.9248	-22.3952	-7.8070	12.1894
FluctMean	44.2519	-71.4584	-146.6847	236.6317	114.5669	-184.1910
Energy0.3	-0.5962	0.8777	0.9597	-1.9429	-0.5109	1.0816
Pheat0.5	-20.3793	29.3572	42.7270	-65.6243	-25.6691	40.2735
Pheat0.9	-21.5475	31.1866	46.8884	-71.8555	-28.6802	44.7920
Abs0.0	-0.8789	1.3210	1.3581	-2.7878	-0.6399	1.4502
Abs1.0	-0.8749	1.3131	1.3336	-2.7584	-0.6304	1.4370
Abs10.0	-0.8392	1.2420	1.1094	-2.4904	-0.5316	1.3047
Abs100.0	-0.3315	0.2216	-1.7093	1.3824	1.4437	-1.4498
Bio1.1	10.6867	-12.6265	-23.4953	34.4550	18.0753	-27.6389
Bio2.0	5.7083	-5.9105	-9.0472	13.1182	6.2123	-9.6675
Bio10.0	1.1427	-0.4624	-0.9461	0.9229	0.7360	-0.9485
Mem1.5	-63.9734	102.4547	147.5985	-245.0950	-90.5389	151.9068
Mem3.0	-25.0341	36.2942	59.7590	-91.0377	-38.4366	59.4048
Mem4.0	-23.6463	33.8529	59.0255	-88.4783	-38.6672	58.8259
MemMean1.5	-2.3791	3.0286	3.4708	-5.9961	-1.6242	3.1545
MemMean3.0	-0.3223	0.5026	0.6391	-1.2211	-0.3722	0.7324

Entry	A_0	B_0	A_1	B_1	A_2	B_2
MemMean4.0	-0.2128	0.3264	0.5596	-0.9654	-0.4092	0.7118
Dark0.5b1.0	-0.0116	0.0066	0.8466	-1.6348	-0.1870	0.5353
Dark0.5b8.0	0.4408	-1.6408	5.3961	-10.5301	-1.4944	4.1156
Dark1.0b1.0	0.2308	-0.6381	1.7346	-3.3595	-0.1348	0.7012
Dark1.0b8.0	19.4523	-46.2043	-55.0574	110.2549	26.9971	-66.0474
Dark2.0b1.0	2.6869	-6.2527	-5.6649	11.4367	3.3367	-7.9624
Dark2.0b8.0	43.7451	-94.2830	-175.8297	347.4186	139.0010	-300.6291
Dark4.0b1.0	2.4976	-6.3593	0.4813	-0.0934	-6.5345	10.0431
Dark4.0b8.0	-8.0677	7.2632	142.5459	-277.2225	-155.8255	302.4536
Persist1.0	0.5000	-1.2507	0.7357	-1.3910	0.2853	-0.3383
Persist8.0	26.0232	-58.4595	-81.8704	163.6807	50.2483	-114.7867
Opt0.1	-0.9595	2.9035	4.2009	-7.8413	-2.7541	4.8560
Opt0.3	-0.5642	1.5032	1.3477	-2.6145	-0.4587	1.1459
Opt1.0	-0.0723	0.1917	0.1365	-0.2674	-0.0502	0.1274

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